A Song of Space and Time

A Song of Space and Time

A novel

GARY LECKEY

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Dedicated to Tina. If you don't quit, you can't lose. Love conquers all.

[Epigraph: Gary to supply, with its source and any permission needed]

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HOW TO READ THIS BOOK

This is a novel built on evidence. After the prologue, every chapter is one story told in three layers, and the layers always come in the same order.

Belfast, by day

A researcher in Belfast sets the evidence for one place and one time side by side. He reads each source, keeps what he can defend, and settles on a single year, the *found year*. The found year is the book's own device: the year at which the evidence he trusts overlaps best. It is never offered as a proven date, and he writes his strongest doubt beneath it. He works with a method he calls the Core, which lays rhythms over one another to see whether patterns appear. The Core finds patterns. It does not prove anything, and the book never asks it to.

That night

He dreams his way into a library where a great book called the Ledger keeps itself. It opens at his found year, the page turns, and the dream is a scene in that place and time. The people in the dream, and what they say, are invented. The setting, the objects and the customs around them follow the record as closely as it allows.

The master page

When the dream is over the Ledger writes up what it showed, in two hands. *In ink* goes what the record supports. *In pencil* goes what is tradition, guesswork or his own reading, including the found year itself.

The same distinction runs through the notes at the back of the book. For every chapter they say which statements fall into which of four kinds:

Attested
: a source, a find or a measurement says so, and the notes give it.

Tradition
: later writers or long-held belief say so. It may be true, but nobody can check it against the evidence.

The author's reading
: an inference the author has drawn from the evidence and offers for the reader to weigh. These are marked as readings, never as findings.

Invented for the story
: characters, speeches, small events and anything else made up to carry the story. The notes say where.

The found years, with the doubt that goes with each, are gathered in a table after the notes. Where the record is thin, the book and the notes say so.

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PROLOGUE

The Granite Sky

Belfast, and the Antrim coast one summer in the late 1990s: the found year, about sixty million years ago



Illustration 1. *A boy with the stone at the waterline on the Antrim coast.*

Belfast, by day

"RIGHT, HOW OLD ARE YE, then?" Gary said to the stone.

The stone said nothing, which was what it had said for years. It sat on the kitchen windowsill in Belfast between a jar of pens and a plant that would not die. It sat easily in a closed fist, flattened a little, and grey. He had picked it off a beach up the Antrim coast one summer at the tail end of the 1990s, aged six or so. It was the first thing he had ever kept.

He had left school in Belfast with a handful of exams and gone straight out to work. He washed pots in a restaurant kitchen, filed for a firm of solicitors in London, counted stock under strip lights and minded other people's paperwork on both sides of the border. In the

end he stood under cranes at Kilroot in a hard hat, talking loads down by hand. In the evenings he taught himself to make a computer do what he asked it. And in the hours after that he read through the old stories and kept finding the same journey in them. It went out to the edge of the known world and came back with something small and plain. He wanted to know when: not roughly, but when.

So he asked the machines first, because they were quick.

He typed in a request for everything on how old the ground was, papers and not websites.

"The Earth formed about 4.54 billion years ago," one of his AI models answered. It added that the oldest known mineral grains were zircon crystals from the Jack Hills in Western Australia, about 4.4 billion years old, and that some researchers argue they record early liquid water.

"Argue, or know?" Gary said.

He went and read the paper himself, and the authors were straighter than the summary: they thought so. They said plainly they were not sure. He wrote *not sure* in the margin and liked them for it, then read on through the rest of the old things on his list. The Acasta Gneiss in the far north of Canada, about four billion, among the oldest whole rock on the surface of the world. The stromatolites still building their slow reefs at Shark Bay, whose ancestors had made the air fit to breathe. He checked every line against the paper it came from, because a machine that has read everything will tell you a thing with great confidence that nobody ever wrote.

"None of that's you, mind, that's only the shelf you're sitting on," he told the stone.

He held the stone up to the laptop's wee camera, turning it in the light off the screen like a jeweller with a ring. He knew rightly it was daft, even while he was doing it.

"Grey, fine-grained beach stones on the Antrim coast are most often basalt," the machine said all the same. "The Antrim lavas are roughly sixty million years old."

"Most often is grand. Now show me the lavas," Gary said.

Dark grey, nearly black when wet and pale as a winter sky when it dried: that was how the coast's basalt behaved, black lava laid down over white chalk. The Geological Survey and a 2010 study of the lavas gave him numbers he could hold. The Lower Basalt, 62.6 million years ago, give or take three hundred thousand; the Tardree rhyolite, 61.3; the Upper Basalt, 59.6. The whole pile laid down inside five or six million years while the Atlantic was opening.

He laid the dates over one another like tracing paper held up to the kitchen light, the lavas on the bottom and the papers on top. Where the lines crossed there was one answer, a few million years wide and honest about it. *About sixty million years ago*, he wrote in the notebook. Underneath, because he meant to do it this way every time from now on, he wrote the same words again with a question mark hooked on the end.

"Unless you're not from here at all, and never were," he said.

That was the doubt, and it was bigger than the number: nobody had ever tested the stone. The ice that once covered the north of Ireland had carried stones over the water from Scotland and dropped them on every beach. Some of those erratics were hundreds of millions of years older than any lava in Antrim. The only way to be sure was to cut it. He was not going to cut it.

He set the stone back on the sill and the page in the drawer. There was no good way to tell anybody on a building site that you'd been up till two arguing with a rock, so he told nobody.

That night

He went to bed with the number in the drawer. Sleep took him not into the dark but into a library he had never been in and knew at once: shelves climbing high out of sight, the air salt and dusty, and away in the stacks somewhere the Ledger, a great book that kept itself. A page lifted and turned over with no hand on it, and at the top

was the same figure he had left in the kitchen, with its question mark. Then the page was a beach, and a boy standing where the stones began, and nobody saw him but his father.

• • •

There were ten thousand stones on the beach, and Gary wanted one.

He was six or so that summer, in the late 1990s, a wee lad from a working-class street in Belfast. He had never in his life met so many of anything that was not people. The stones lay in a long grey bank between the grass and the water, round ones and flat ones and ones shaped like a loaf. Low down they were wet where the tide had been at them, and higher up the sun had left them dry and pale. When the sea came in it pulled at them, and when it went out they rattled after it, a sound like a crowd at a match breathing out all at once. He stood right at the brink of it in his good shoes, which his mother had warned him to keep dry. He could feel the cold coming up through the soles.

"Pick one, so," his father said behind him. "Go on. They're free."

When Gary turned round, his father was standing on the shingle with his feet apart and his hands in his jacket pockets, squinting at the water like a man trying to work out what it was for. They had come out of Belfast that morning by the lough shore. The big chimney of the power station at Kilroot stuck up out of the fields like a pencil stood on its end. After it came the coast road, bend after bend above the sea with the windows down and the radio on, and Gary's face squashed against the glass at every one of them.

"Is it still there?"

"The sea? It's still there," his father had said. "It was there before Belfast and it'll be there after it, so you can stop checking."

He had not stopped checking the whole way up. At every gap in the hedges he was up on his knees on the back seat for the grey line of it, and every time there it was, flat under the sky, right where he had left it.

His mother had her shoes off the minute the car stopped and was away down to the water on her own. Now she was walking along the edge of it slowly with her coat open and her shoes hanging off two fin-

gers, stopping now and again to look out at nothing in particular. Further out the sea had gone the colour of a wet slate roof. Beyond it lay a long low shape of land, blue with the distance.

"That's Scotland," his father said.

"It is not."

"It is so. You could near shout over to it."

Gary did not believe him, because Scotland was on the map at school, a different colour. It did not seem right that you could just see it, sitting there across the water like a neighbour's house across the street.

"Half the stones on this beach came from over there," his father said. "The ice brought them, years and years ago. Before there was anybody."

"Which half?"

"Now you're asking."

"How do I know which one?" he said.

"Which what?"

"Which stone's mine. There's that many."

His father took the question seriously, and got down on his hunkers beside him with his forearms across his knees. He looked along the beach for a good while, which was one of the things Gary liked about him, though he couldn't have told you so then. "You'll know," he said in the end. "It'll be the one that doesn't want to be left."

Gary went down the bank with the stones sliding and clacking under his trainers, his arms stuck out like a tightrope walker's. Near the bottom it flattened, the water running in and out between the pebbles in thin hissing sheets, and the stones down there were darker and smoother and shone as if somebody had polished them. He crouched. A white one with a hole through it went back down, and so did a red one like a slice of ham. Then a wave came further than the rest and went right over his shoes, and the cold of it knocked the breath out of him, and still he didn't move. There in front of him, as the water sucked back through the gravel, was a stone that was neither white nor red nor anything at all.

It was grey, hen's-egg size and a bit flattened, so that it sat in the hollow of his palm as if the hollow had been made for it. There was nothing on it at all, no stripes, no spots, no holes. Wet, it was near enough black. He watched it dry in his hand and the dark went out of it from the edges in, patchy, like a pavement after a shower, till all that was left was the plain soft grey of a winter sky. It was cold, colder than the sea, and then, slowly, it wasn't.

He closed his fingers round it.

• • •

And the page went back.

He did not move, and neither did the beach. But the years ran out from under it like the tide going out, fast, and took things with them: the houses first, then the road, then the chimney at Kilroot, then the fields. He was still standing there on the stones with the grey one shut tight in his fist, and round him, a piece at a time, the coast was coming undone.

First there was ice, grey-white and higher than any building in Belfast. It came down out of the north grinding over the land like a scourer over a burnt pot, carrying stones from the black cliffs and the white ones and the hills of Scotland across the water and rolling them against each other till they broke. It took thousands of years to draw back again. It went unevenly, a bit here and a bit there, and dropped what it had carried in heaps along the shore for the sea to sort out after.

Before the ice there was fire, back when the ground split along the coast and the ocean was opening. Lava came welling up through the cracks and ran out over the land in sheets, red going to black, layer on layer; where it cooled slowly it cracked into columns like the mud at the bottom of a dried-up puddle. Under it lay chalk as white as a new sheet. It was made of the shells of creatures too small to see, drifting down through a warm sea in their billions, and knots of hard black flint grew in it. The boy watched the black come down over the white.

He knew those cliffs; he had passed them that morning on the coast road, black on top and white underneath, and his stone had been in the black.

And under the chalk, further down than he could see, older and older: a warm bay where low grey humps of living slime stood in the shallows, building themselves out of sand and light, a fingernail at a time. A banded grey rock on the bank of a far northern river. A red ridge where grains smaller than salt had held on through everything. They did not look like much, and that was the first thing the page showed him: the oldest things never do.

Then the tide came back in, and the years with it, and the sea went to work. Every tide picked the stones up and dropped them and rolled them, and every winter storm threw them up the beach and dragged them down again. The corners came off and the edges came off, until what was left was smooth. Nobody polished it; everything did. And when people came, a moment ago in the long count, they walked along shores like this one and bent and picked up stones like this one. Some chipped the flints into tools, and some stood the big stones up on end, and some, nobody knows how many, just liked the feel of one in their hand and kept it.

• • •

"Well?" his father said, and it was only the beach again.

Gary came back up the bank with his shoes squelching and his hand held out flat. His father bent down over the stone and studied it for ages, in case there was something in it a quicker man would miss.

"That's a good one," he said. "That's a grand one, son."

"It's only grey. It's not even a proper colour."

"Aye. Well." His father straightened up and put his hands back in his pockets and stared out at the water. "Grey's the colour of most things that last."

Gary could think of nothing to say to that. So he said nothing, which his father seemed to take for the right answer. They stood together up on the crest of the bank, the two of them, with the sea wind flattening the grass at their backs. After a while his mother came

back along the shore towards them with her shoes in her hand. She stopped below them where the stones ran out and gazed up at the pair of them. Gary held out the stone for her to see.

She came up the stones with one hand holding her coat shut against the wind, took it off him, and rolled it over in her fingers, rubbing her thumb across it to see would the colour come back. It had dried pale, plain as the step at the back door.

"Where did you get that, love?" she said.

"Down there by the water. It was by itself."

"It was not by itself. There's ten thousand of them."

"It was by itself," Gary said.

His mother looked at him and laughed, not a big laugh, only a breath through her nose with the wind taking most of it. His father heard it anyway, and turned round on the stones and smiled like somebody had told him good news.

"And what's it for?" she said.

Gary looked from the stone in her hand along the bank to all the others, thousands on thousands of them lying wet in the wash and rattling under every wave that came in. He thought about it properly before he answered.

"It's not for anything," he said. "It's just for keeping."

She gave it back to him and closed his fingers round it with her own. Her hand was cold from the water. "Then you keep it," she said.

They ate their sandwiches in the car with the doors open, and drank tea out of a flask that tasted of the flask. The light went down over the sea and turned it silver and then pink. On the way home Gary lay along the back seat with his head on his mother's coat. He could hear his father humming under his breath, an old tune with no words, and the tyres on the road, and further off the sea. He had the stone in his fist, warm now from his hand. It would stay warm as long as he held it. He held it all the way down the long dark road into Belfast, until the street lights came on round him and he was asleep.

The master page

The page did not close at once. He was in the library once more, the book spread out and the Ledger near at hand, like a foreman who stands with you while you sign for a load. It had columns, as his own notebooks did: what was known, what was only told, what was only hoped. It wrote in two hands. What the evidence carried went down in ink. What only he believed went down in pencil, and it stayed in pencil until something better came.

In ink it wrote the lavas, 62.6 to 59.6 million years, with the papers beside them, and the zircons with their *not sure* left in. In pencil it wrote the stone. *Antrim basalt, about sixty million years: by colour and by feel, never tested; possibly an erratic from over the water.*

The beach's part was the page. Before he could lay one place against another he had to learn to read a thing as a page with a date on it, and to write the date with its question mark, and to say out loud what would make it wrong. Every other page in the book would stand on that.

Some day he would take the patterns to the Core, the framework he had taught himself to build: many rhythms laid over one another, something steadying them, and the past coming back after a delay to shape what comes next. It could not have told him how old a stone was. The Core does not get a vote.

And his mother's question was still open on the page. *What's it for?* At six he had answered it without looking up: it's just for keeping. Nearly thirty years on he had nothing better to say, only more of the same: a drawer of printouts, a notebook gone soft at the corners, and the habit of putting the known in one column and the hoped in another. You keep the thing that doesn't want to be left. The sky was the same: nobody needed it written down, and people wrote it down anyway. You keep it honestly, and when the test goes against your best guess you cross it out in the same plain hand you wrote it in.

Then the Ledger turned to a page further on and wrote the next line in ink before he could ask. Not twenty miles west along that same coast, on a bluff above the River Bann, people with trowels in the

1970s had found the rings of round huts at Mount Sandel, with hazelnut shells in the hearths and the small bones of salmon. Under it, in pencil, a question he had never thought to ask: how did a family with no calendar know when the salmon were coming?

He woke with the light coming grey over the roofs towards the lough, and on the windowsill the stone was pale and dry again, its colour gone back to nothing much.

PART I

Origin

The Standing Stones

CHAPTER 1

The Salmon Camp

The River Bann, at Mount Sandel: the found year, 7700 BCE



Illustration 2. *Ness and Orrin on the bank above the weir, with the huts on the bluff.*

Belfast, by day

GARY STARTED WITH MOUNT SANDEL and asked the machine for a date.

"Mount Sandel, on the River Bann near Coleraine, is usually dated to about 7000 BCE, making it one of the earliest known settlements in Ireland," the machine said.

"Usually, by who?" Gary said.

That was the figure on his own map, the one from the books he had grown up with. He made the machine give him the papers and read them himself, because his AI models had twice handed him a confident line that nobody had ever written. In the 1970s, on a field marked down for houses, Peter Woodman's team had found the rings of round huts on the bluff above the river: post-holes leaning inwards. So the

posts had been bent together at the top, huts about six metres across with a hearth scooped into the middle, rebuilt more than once on nearly the same ground. Then in 2009 Bayliss and Woodman had taken the radiocarbon dates again and modelled them together. He checked every one of their dates by hand against the lab numbers.

"Seven thousand's wrong. Hut-building between 7790 and 7635 cal BC, probably inside one or two generations," he told the machine.

The machine said he was right and thanked him for the correction.

"Don't thank me. Fix your books."

The map had been out by seven hundred years. He had believed the map, so he wrote that down too.

The hearths were the best of it, since the soil on the bluff is acid and whatever did not go through a fire is gone. In the hearths and pits were burnt hazelnut shells, crab-apple pips, the seeds of water lilies and the bones of wild pig. There were small burnt fish bones as well: salmon and sea bass, eel and flounder. Woodman had read them as a calendar, salmon and bass in summer, eels going down in autumn, nuts in autumn, and argued somebody was on the bluff for much of the year. Others asked whether it was a place people came back to, again and again.

He built it up the way you'd build a load on a pallet, heavy stuff at the bottom. The huts gave him a window, and the food gave him a year inside the year. The river was the clock either way.

"No calendar and no tide tables, so how did a family on that bluff know the salmon were coming?" he said to the room.

The machine offered him a paragraph on Mesolithic seasonality, and he closed it. He fished when he could, and he knew from that, not from any paper, that a river keeps its own time. Salmon come up with the water rising after rain. The run has signs a man learns by watching, and they change from river to river. That was a fisherman's sense, not evidence, and he marked it as his own: it would go in the dream, not in the notes.

The flint was evidence of a harder kind. The tiny blades at Mount Sandel, for arrows and fish spears, were mostly the flint that weathers out of the white chalk under the black basalt along the north coast. That chalk was a day's walk from the bluff, and he had stood under those cliffs as a wee boy without knowing what was in them.

He wrote *7700 BCE* and, under it, the same again with a question mark hooked on the end, because a number without one looked like it was showing off.

Under it, in plain words, he wrote down the doubt as well. The window was a hundred and fifty years wide, made by a statistical model from one site, and 7700 was the middle of the window, not a find. Nobody could say which spring a girl first learned the salmon. The people were thinner still, for no bones of the Mount Sandel families survive. A handful of Irish hunter-gatherers from other places, read for their DNA in 2020, looked like an island people a long time apart. There were perhaps a few thousand on the whole island, finding husbands and wives a long way off. How the first of them came over the water, nobody knows.

He wrote the year in the notebook with the window beside it, all hundred and fifty years of it, and went and made beans on toast.

That night

The library smelled of river water. The Ledger was nearer the door than it had been, as though it had come to meet him. The page it held open had the number from the notebook at the top, *7700*, with its question mark in pencil. The page turned, and it was spring on a bluff above brown water. There was hazel in catkin, smoke going up through a turf roof, and a girl going down a steep path behind an old man. He stood on the bank as a boy, unseen.

• • •

It was the first spring of the new hut. The hazel was in catkin and the river ran brown and loud with the end of the snow from the hills, and Ness's grandfather said it was time she learned the salmon.

She was fourteen winters, or near it, born in the moon when the eels go down to the sea. Her mother had been dead since the middle of the winter. Ness still said things like *her mother had always said*, out loud, to anyone who would listen. Then she would stop, and feel the hole open in front of her like a step in the dark that was not there.

The new hut stood on the bluff above the river, where you could see a long way both ways, down towards the sea and up towards the lough. The old one stood a little along the bluff, on nearly the same ground. Its posts had gone soft, the way posts do in a lifetime. The men had cut new ones in the autumn, young straight trunks as thick as Ness's arm. They set them in a ring that took her eleven of her own strides to cross. They bent them in at the top and lashed them, and laid hides and turf and reed over them until it was as snug as the inside of a fist. In the middle, in a hollow scooped out of the floor, was the hearth. It had not been let go out since the day they moved in. A new house, her grandfather said, has to learn to be warm, the same as a child has to learn to walk.

They had moved in a little before the dark of the year. Her mother had swept the floor with a bundle of birch twigs and laid the first fire herself, kneeling, with her hair tied back. Then the cough that had been in her chest since the autumn got worse instead of better. She lay by that same fire through the short days with her head on Ness's knee, and one morning before it was light she stopped. Ness had been the one holding her, and she had only known that the breathing she was listening to had become a different kind of quiet.

They carried her down to the place on the bank where the family's dead were put. They laid her with her stone axe and a necklace of shells and a handful of hazelnuts in a little bark box, because she had always been the one who found the best trees. Her grandfather did not weep; he stood a good while looking at the river. At last, without looking at Ness, he put out his hand and closed it round hers. They stood like that until the light went.

That had been in the winter, and now it was spring. He said it was time she learned the fish.

• • •

Orrin was the oldest person Ness had ever known. His hair had gone the colour of ash right through. His hands had knots in them like an old hazel. He went down the steep path to the river sideways. He had a stick of blackthorn, black and shining from his grip. But he could still see a fish under brown water better than any man of the family. He could still sit half a day at the weir without moving.

"The young men think fishing is pulling things out of water," he said, as they went down. "It isn't. A child can pull a fish out of a trap. Fishing is knowing when. If you know when, the fish come to you. If you don't, you can stand in the river till your legs go blue and go home with nothing but a cold."

"Then how do you know when?" Ness said.

"Ah." He stopped on the path to get his breath, and looked at her the way you look at somebody who has picked the right stone out of a heap. "That's the whole of it. That's what you're coming down here to learn, and it'll take you longer than it took me, because you ask more questions."

The weir was a fence of woven hazel that ran out from the bank at a slant. There the river narrowed over a bed of stones, and the water piled up and went white. At the end of the fence, where the water ran fastest, there was a gap. In the gap was a long basket of willow with its mouth upstream, so that whatever swam into it could not easily find its way out. There were smaller traps along the bank for the eels, and on a flat stone near the water the place where the fish were gutted.

Orrin lowered himself onto a mossy rock above the weir, very slowly, with a long breath out at the end of it. Ness sat on the grass beside him. Below them the river went by, brown and fast and full of the hills.

"Tell me what you see," her grandfather said. Ness saw brown water and white water, and the weir with the river leaning on it. "It's high," she said. "Higher than last moon. The snow's going, up in the hills."

"Good." The snow water was cold, he said, and thick with earth, and the salmon did not like it. They waited down in the mouth, where the salt water and the sweet water mixed. They hung there with their noses up the river and thought about it. He pointed downstream with his stick, towards the mud banks and the sea beyond them. "You'll get the odd one now. Big ones, the brave ones, or the foolish ones. But that's not the run. The run is later."

He told her about the signs. When the blackthorn was white on the far bank, the first big salmon came up, but few. When the swallows came back and began to skim the water in the evenings, more. Then the river went down after the snow and ran clear. When a good long soaking rain came in the hills after that, and the river rose again, quick, not cold this time, you watched. They liked to come on the rising water, he said, the way a man likes to walk with the wind at his back. When the elder was in flower the river would be full of them, silver and blue, coming up through the white water so thick you could hear them. The smoke of the fish fires would hang over the bluff for days.

Ness asked what came after.

After, he said, they went up to the lough and further, and rested, and went thin and red. In the autumn they lay in the gravel of the little rivers and made their young, and most of them died there. "And on the dark autumn nights the eels go down. That's your moon. Your mother's moon."

"Why do they come back?" she said, after a while. "The salmon. If they're going to die."

"Because this is where they were born," her grandfather said. "They go away to the sea when they're as long as your finger, and nobody knows where they go. Nobody. And then one year something in them turns, and they come back. Not to any river. To this one." He

tapped his chest with two knotted fingers. “My grandmother used to say every fish that comes up the river is carrying the way home in its chest.”

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When the sun was high they hauled the willow basket up onto the stones, and there was nothing in it but a stick and a drowned leaf. “There,” he said. “Now you know something. Most of fishing is learning that today’s not the day.”

Coming back along the bank, he stopped at a little spit of shingle where the flood had washed out a scatter of stones. He bent with a grunt and picked up a dark grey knob of flint, white-skinned on the outside like a bird’s egg rolled in chalk. “That came from the shore,” he said. “Down where the white cliffs are, past the mouth. Your mother had a good eye for flint. She’d walk the shore after a storm and come back with her bag full.” He gave it to her. “You keep that. You’ll learn to work it next winter, by the fire, and make the wee blades for the arrows and the fish spears.”

Ness put it inside her tunic against her skin. It warmed there, slowly, and by the time they reached the foot of the bluff path she had stopped noticing it was there.

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That night they roasted the last of the winter’s hazelnuts in the embers. Her aunt Teva made a broth with the dried meat of a pig the men had killed in the autumn. There were the dark shrivelled crab apples from the store, sour enough to make your eyes water. Somebody had hung a string of water-lily seeds from the lough to dry above the hearth. They turned slowly in the warm air.

“The run’ll be early this year,” said one of the men, licking his fingers. “The water’s warm already.”

“It’ll be when it’s ready,” Orrin said.

“You said that last year.”

“And I wasn’t wrong.”

“You were late.”

"The fish were late," Orrin said. "I was there on time." And Teva laughed so hard she had to put the broth down.

When the men had gone to their sleeping places, Ness spoke into the dark. "Grandfather."

"Mm."

"Were you here when the river was frozen?"

The fire ticked, and she heard him shift on his bed of hides. "Somebody's been telling you stories."

"Mother. She said you remembered the cold years."

"I remember some cold years, when I was smaller than you," he said at last. "You could walk out on the lough. But your mother might have meant the older thing."

Ness waited.

"My grandmother's stories. About when the ice walked."

Ness lifted her head. "The ice walked?"

That was what she said, he told her. A long, long time ago, before anyone, the whole land was under the ice, higher than the hills. It moved slowly, the way a man walks when he is carrying something heavy, and it pushed the stones before it. Then the sun grew strong and the ice went back, a little every year. The trees came, and the birds, and the deer and the pigs, and after a long time the people came. He paused. "Over the water, she said. In boats. Out of the sunrise. Following the birds, maybe, or the fish."

"Is it true?"

"I wouldn't know. I wasn't there." She heard the smile in his voice. "But look at the stones on the shore. Every one of them rounded off, every one of them come from somewhere else. My grandmother said the ice brought them and the sea polished them. I've never seen a better reason for so many stones."

Ness lay back down and put her hand on the flint inside her tunic. It was warm now, warm as she was.

"Mother would have known," she said.

"Your mother," her grandfather said gently, "would have said, *go to sleep, the both of you,* and she'd have been right."

• • •

She went down to the river every morning after that. She put her hand in the water and said whether it was colder or warmer, and looked at the height of it against the stake he had told her to watch. She looked at the blackthorn on the far bank, just beginning to show white at the tips. When his knees were bad she went down by herself and came back and told him everything. He listened with his eyes closed and asked her questions she could not always answer.

On the ninth morning, or the tenth, there was a fish in the trap.

It was the biggest living thing she had ever held in her arms that was not a child. It was a salmon as long as her leg and as heavy as a seal pup, clean bright silver along its sides, with a blue back and a scatter of small black spots like seeds. When she and her cousin Bel hauled the basket up onto the stones it flung itself about with a force that knocked her backwards. Bel ran for the men. Ness knelt on the wet stones with her arms round the basket and looked at its eye, which was round and black and full of the river. She thought: *carrying the way home in its chest.*

The men killed it with a club and carried it up the bluff. Her grandfather came out of the hut last, slowly, on his stick, and looked at it for ages and then at her.

"The first one," he said. "One of the brave ones. Or the foolish ones." He put his knotted hand on top of her head, just for a moment. "Your mother caught the first one, the year she was your age."

Ness had not known that. She had to go and stand behind the hut for a while, by herself, with her face turned to the wall.

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The mist came on a still grey morning in the second half of that moon, when the blackthorn was fully out and the first swallows had come back to the river.

Ness and her grandfather were at the weir. He was telling her about the stake line below the white water, where the men would stand with their spears when the run came. You had to lead the fish when you threw, he said, because the water bent the light and the fish was never quite where it looked to be.

"It's like most things," he was saying. "What you see isn't quite where it is. You have to aim for where it's going to be. Your mother was good at that. Better than me. She'd stand there, still as a heron, and you'd think she wasn't watching at all, and then—"

He stopped.

Ness looked up at him, but he was not looking at her. He was looking downriver, towards the mouth. His face had gone quite still, the way a deer's face goes still when it lifts its head from the grass.

She followed his eyes.

The mist was coming up the river the way the tide comes, low and steady, filling the valley from the sea end first. The mud banks at the mouth went soft and then white and then were not there at all. This was sea fog, grey-white and thick, and it moved along the water at a walking pace. It brought the smell of the sea with it, salt and weed and something cold and clean underneath. It swallowed the far bank and the alders at the bend. The sound of the white water seemed to go quieter in front of it, the way voices go quiet when someone comes into a hut.

Her grandfather did not speak or move. He sat with his stick across his knees and his head a little on one side, and he was listening.

She listened too, and she heard the river, and the mist, or thought she did: a kind of soft hush that was really only the river sounding different. She heard a gull, somewhere out in the white, crying once and then not again. And she heard nothing else at all.

The mist went over the weir and over them, cold on her face and wet in her hair. The whole world shrank to a ring of grey: the rock, the grass, the white water just below, and nothing beyond it. Her heart was thumping. She took hold of a fold of her grandfather's cloak, the way she had done when she was small.

He did not look down at her. After a while, a long while, his hand came off the stick and found hers and held it, the way it had on the riverbank in the winter. His grip was warm and dry and very strong.

Then, as quietly as it had come, the mist began to thin, pale and then paler, until she could see the far bank like a shape through a skin, and then the sun, low and white as a shell. The white water was loud again. Somewhere up on the bluff a child was shouting, and a crow answered.

Her grandfather let out a long breath through his nose and let go of her hand. He went on in exactly the voice he had been using before. "—and then she'd throw, and she'd have it. Every time. So you aim for where it's going to be."

Ness stared at him.

"Grandfather," she said.

"Mm."

"What was that?"

"Sea fog. It comes up the river in the spring, when the water's cold and the air's warm."

"No," she said. "I mean—what were you listening to?"

He looked at her then, with eyes the pale grey of river water, very clear. For a moment she thought he was going to tell her. Then something in his face closed over, gently. It was the way you would draw a hide across a doorway on a cold night, not to keep anyone out but to keep the warm in.

"My grandmother," he said slowly, "used to say that the mist has people in it. That when it comes up from the sea, it comes from somewhere a long way off, further than the boats go, and sometimes it carries a voice. Not often. Once or twice in a life, maybe. Most people never hear anything at all." He picked up his stick and turned it in his hands. "She said if you ever hear your name in it, you're to stand still and listen, and not be afraid. That's all she said."

"Did you hear your name?"

Her grandfather got up off his rock, very slowly, one hand on the stick and the other on her shoulder. He looked down the river, where the last of the fog was lifting off the water in rags.

"I heard the river," he said. "Same as you. Come on, now. Your aunt will have the fire up, and I'm old, and my feet are wet."

He started up the path, sideways, one foot and then the other. Ness stood a moment longer with her hand inside her tunic, round the flint, which was warm as her own skin. She looked down the river towards the place where the mist had come from, and listened as hard as she had ever listened to anything.

She heard the white water at the weir, and a swallow, and then another. She heard, far off down the valley, the slow sea.

Then she turned and went up the path after him and caught him up. She put her arm under his to help him with the steep part, and he let her.

The master page

The page stayed open after the mist had gone. The Ledger sat at his elbow and wrote, and he watched its two hands.

In ink: the huts on the bluff, 7790 to 7635 cal BC; the hazelnuts and water-lily seeds and crab apples in the hearths; the burnt bones of salmon and eel. In ink, too, the line back to the beach where the book began, because the flint in the girl's tunic came out of the same white chalk under the same black rock as the shore where he had picked up his stone. In pencil: the year, *7700*, the middle of a window. In pencil: blackthorn, swallows and elder as the signs of the run, which was a fisherman's guess and his own. And forward, on a page not yet reached, a plain in the south of Iberia where people with sheep would cut notches in a stick: the Ledger wrote that in pencil and drew no line between them. Nothing joins a salmon camp on the Bann to a stone on the Alentejo except that both were counting.

What the Bann added was the season. Before anyone set a stone on a sunrise, people knew the year by what the river did, and a calendar was a list of signs held in an old man's head and handed to a girl. Everything later in the book would be built on top of that.

When the time came he would take the season to the Core, the thing he had taught himself to build, with its three working parts: cycles stacked on cycles, a hand that steadies them, and an echo that arrives late. It could not tell him what a grandfather heard in a fog, and he would not ask it to.

The girl had asked it on the path: *then how do you know when?* The answer was the list in the old man's head, blackthorn and swallows and the river rising warm after rain, and it was handed on the only way it could be, by going down to the weir every morning and being wrong. Cold feet on the stones, the old man shaking his head, the walk back up with nothing in the basket. Then one morning the river is different and you're there for it. Ness had learned that you track it so you know when the salmon are coming and the family eats, and she had learned the other half too: when you hear something in the fog that you cannot prove, you stand still and listen, and you say no more than you heard.

On the next page the Ledger had left a gap the shape of a girl in a lambing pen. He put his hand flat on it, half expecting it to be warm, and got paper, and the smell of sheep.

CHAPTER 2

The Stones at the Edge of the Plain

The Alentejo, near Évora: the found year, 4800 BCE



Illustration 3. *Telo on the long rope as the great stone is raised.*

Belfast, by day

THE PHOTOGRAPH ABOVE THE DESK showed a hooked line pecked into grey granite, like a shepherd's crook. Gary had been looking at it for weeks.

He typed *Almendres, west of Évora*, and asked it when. The machine said the Almendres Cromlech dated to around 6000 to 5500 BCE, which made it one of the oldest megalithic monuments in Europe, older than Stonehenge by two thousand years.

"Says who? Give me the paper that says it, not a page that says a paper says it," he told the screen.

It gave him a confident page with no source at the bottom. He closed it and went to the papers himself, and read them through, and checked every date by hand. About ninety-five granite stones in rings and ellipses on a slope among cork oaks, opening towards the east, some of them carrying crooks and circles and little cup-marks. Mário Varela Gomes, who dug there, set out its phases. First, small rings of stones, perhaps three, from the end of the sixth millennium BC, when the first farming people were settling the plain. Then, in the fifth millennium, the great double ellipse of bigger stones built on to them, then use and reuse into the early third.

"And the radiocarbon?" he asked.

"Radiocarbon dates place the—"

"Show me one. One date, from one lab, on one bit of charcoal from under those stones."

It could not, because there were none; the archaeoastronomers who later measured the region's enclosures said so in plain print. There are no direct radiocarbon dates for these stones. The chronology comes from what was found in the ground and from comparing the shapes.

So he took it a sheet at a time, the way the solicitors had taught him to build a bundle, oldest at the back. Sheet one: the small old rings, five thousand at the very earliest. Sheet two: the great ring added in the fifth. Sheet three was the scene he already had, and the scene had made its own claim without asking him. There was a little ring already standing that nobody remembered building, and a man who wanted a great one beside it. If the old ring was beyond memory, it was generations old, and the great ring was only beginning. That put the morning in the early fifth millennium, not the sixth.

He turned to the stones themselves, the local granite of the plain, the grey boulders that still lie among the cork oaks like sleeping cattle. The biggest at Almendres stand taller than a man. Nobody needed to bring them from far, but they needed hands, and rope, and a reason, and a few days when the shepherds were not up on the summer grass. A scene could carry that and a paper could not, and he marked it as his own.

4800 BCE, he wrote, and pencilled the question mark in after it. His map had been seven hundred years early, and he wrote that down too.

Then the line: Pimenta and Tirapicos, measuring twelve of the Alentejo enclosures, found they face east, into the arc where the sun and moon rise, more often than chance would give. The best fit in their model was a full moon around the autumn harvest, and the sample was too small to choose for certain. Almendres itself is usually read as facing the equinox sunrise.

"Almendres faces the equinox sunrise," the machine said.

"Whose equinox, theirs or ours?"

The machine said it meant the astronomical equinox, when the sun rises due east and the day and the night are near enough equal.

"They hadn't got one," he said. "Equinox is a modern word. They'd a sun coming up, and a hill, and a lot of patience."

He wrote it down so he would not forget it. A man on that hill who counted the days from one standstill of the sun to the next and halved them would land on a middle day a day or two from the astronomical equinox, not exactly on it. Nobody on that hill needed it exact; they needed it the same every year.

And then the people: in 2019 a study of ancient Iberian genomes found that the first farmers carried ancestry that had come, over many generations, from Anatolia, and that local hunter-gatherer ancestry came into their families as well. Down the Tagus at Muge, the hunters had heaped their shells into middens and buried their dead in them. A grandmother's joke about shell-people and boat-people would not be evidence. It would be a fair way to say what the evidence said.

The doubt he wrote twice: no radiocarbon, so the year leaned on finds and shapes and one excavator's phasing; a new date could move it centuries either way. And an alignment is an orientation read by a person, which is easy to make say what you want.

The machine offered that the same crook turned up in Brittany, carved on standing stones around the Gulf of Morbihan. This, it said, suggested a maritime route between the Alentejo and Brittany.

"Source. A paper, mind, with a name on it."

It gave him a travel blog.

"Catch yourself on," he said, and wrote the crook in the margin in pencil, and left it there. A shared sign is a pattern, and a pattern is a question. He did not mention the year to anyone that day. He looked at the crook above the desk instead.

That night

The library was warm that night, and smelled of dry grass and sheep. The Ledger had a page open on a lectern near the window. At the top, pencilled, stood *4800*, its hook of doubt beside it. A thin line ran across the page from one edge to the other, like a horizon. When he looked down again the lectern was a hurdle, and the page was a lambing pen under oak trees with a ring of thorn round it and a girl asleep against an old ewe. He had a boy's knees, and they were cold, and nobody looked his way.

• • •

The lambs came early that year, before the rains had properly finished, and Telo slept with them.

She did not mind, for the pen was a ring of thorn and stacked stone at the bottom of the slope, out of the wind. On cold nights she lay in the straw with her back against the warm flank of the old ewe her father called Grandmother, and the lambs fell asleep in a heap against her belly. Her father had gone up to the high ground with the rest of the flock and her two older brothers, to find the first grass. At the gate of the pen he had put his hand on her head. "You're the one I trust with the little ones," he had said. She was twelve. She had waited all her life for somebody to say a thing like that. Now she found she was frightened of it, and lay awake listening for foxes.

The long house stood a little way up from the pen, wattle and daub under a roof of river reeds. By day the slope was full of noise: children, the dogs, the women grinding barley on flat stones by the door. At night there was only the pen and the sheep breathing and, up on the hill, a small fire that burned long after every other fire was out.

That was her uncle's fire. Brannok sat up there most nights now, and most mornings before it was light, on a big flat boulder at the top of the slope with a sheepskin round his shoulders, and watched the east.

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The country was all low hills and long slopes, rolling away under holm oak and cork oak. The trees were dark and round-headed and spaced out across the grass as if somebody had planted them, which nobody had. Everywhere among the trees there were boulders of grey granite lying in the grass like sleeping cattle, some as small as a sheep and some as big as a house. Telo's people said that if you dug down far enough anywhere you would come to the grey stone, which was the bone of the world.

Telo's grandmother said their people had not always been there, even if the boulders had. Long ago, in her grandmother's time or before it, the people had come along the coast in boats. They brought sheep and goats tied in the bottoms and seed corn in skin bags, and pots with patterns pressed into the clay with the edge of a shell. They had found other people already there, who lived on shellfish and deer and heaped the shells of their eating into hills along the rivers. Some of those had gone away, and some had stayed and married in. "So there's a bit of the shell-people in all of us, and a bit of the boat-people, and a lot of the sheep," her grandmother said.

That was what Brannok was doing too, Telo thought, up on his boulder: throwing something over his shoulder, one day at a time, and waiting to see what it would make.

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He was her mother's brother, and he had a bad leg from a fall among the rocks, set crooked. He no longer went up to the high grass. Some of the men called him the Watcher, and not kindly. The old people did not laugh; they brought him a share of the meat, and asked him in low voices when to sow and when to burn the old grass off the slopes.

"What does he do up there all night?" Ilun said once, at the grinding stones.

"Counts," Arro said. "He's soft in the head."

"He counts so you'll have grass to put your sheep on, and you'll mind your mouth about your elders," Telo's grandmother said, without looking up from the barley.

"But when he's dead, who'll count?" Telo said.

Her grandmother stopped grinding. "Now that you'd have to ask him," she said.

He had a stick, a long straight rod of olive wood, smooth and dark, covered from end to end with little cuts made with the edge of a flint, close together, in rows. It was a tally, and Telo had asked him once what the cuts were.

"Days," he said. "One cut for every morning I've sat up there and watched it come."

"All of them?"

"All the ones I've watched, since you were no bigger than that lamb."

She had looked at the stick, which was nearly as long as she was tall and cut on all four sides, and could not think how many days that was.

In the winter just past he had taken her up the hill for the first time. They had sat together on the boulder under his sheepskin, which smelled of sheep and smoke and him, until the sky over the far hills began to go grey.

"Now," he said. "Watch where it comes." The grey went to yellow and then to a thin bright red along the edge of the land. The hills went black against it, and then there was a spark on the line of them, so bright she had to look away, and the sun was up.

"Where did it come?" Brannok said. "Be exact. What's it next to?"

She squinted. Far off along the hills the ridge dipped, and a single tall tree stood on the skyline. The sun had come up a little to the right of it. She said so.

"Good." He took a flint from the bag at his belt and made a small cut at the end of the last row. "Tomorrow it'll come up in the same place, and the day after. It stops twice in the year, at the cold end and the hot end. For a few days it rises in the same place every morning, as if it's thinking. And then it turns round and starts back the other way."

"Back where? It's just the sun. It comes up where it comes up."

"Along the hills. Towards the left, a little every day, past the tree, past the dip, past the big rise like a sleeping dog, until the long days come, and then it stops again, far over there." He pointed a long way along the ridge. "And thinks again. And turns. And comes back."

Telo had seen the sun come up every day of her life and never once noticed that it came up in a different place. She felt dizzy, as though the boulder had shifted.

"Does everybody know that?"

"Everybody's seen it," her uncle said. "Not everybody's looked."

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All that winter she went up with him when she could. He showed her the stakes. They were straight hazel rods driven in at intervals of a few paces in a line running down the slope towards the east, each with a cut a hand's width from the top. If you sat in the smoothed hollow on the boulder that he said his own backside had worn there, and looked along the tops of the stakes, they lined up into one and pointed at the far horizon like a finger.

"That's where it'll come, on the middle day," Brannok said.

"There's no middle to the days. They just keep coming."

He showed her the stick, and a cut deeper than the others, with a little cross beside it, a long way back along one side. "That's the day it stopped at the hot end, last summer." He ran his thumb along the cuts, round the end and up the next side, to another deep cut with a cross. "And that's the day it stopped at the cold end, this winter. I've counted between them, these six years, and it comes out the same every time, or as near as makes no matter. Halve that count and you

come to a day in the middle, when it'll rise halfway between the two places where it stops, and the day and the night will be near enough as long as each other."

"How do you know?"

"Because I've watched it," he said. "Six times. Three on the way up and three on the way back. And then I want to set a stone there. On that line. So that when I'm gone, and you're gone, anybody can sit where I'm sitting and wait for the morning when the sun comes up over the head of the stone, and know it's the middle of the year, and the time to take the flocks up. They won't need a stick. They won't need me."

Telo looked at him, and in the first grey light his face was lined and tired. His crooked leg was stuck out stiff in front of him on the rock.

"I'd still want you," she said.

He had no answer to that. After a moment he put his arm round her inside the sheepskin and pulled her in against him for a breath or two before he let her go.

• • •

He had not been the only one thinking about stones.

Up the slope, on a gentle rise below the hilltop where the trees stood back and the ground fell away towards the east, there were stones standing already. They made a little ring, a dozen or so, none higher than a man, grey and lichened and leaning. The old people went there at certain times of the year to do certain things Telo was not old enough to be told about. Nobody remembered who had put them up: some said the first boat-people, and some said the shell-people, and some said they had grown there.

What Brannok wanted was bigger: a great ring below the old one, on the slope facing the sunrise, open on its east side, with its first stone set on his line. "A stone that size will take every pair of hands on the hill," Arro's father said at the fire, "and the hands have better things to do."

"Like what?" said Telo's grandmother. "Lying in the sun?"

The old people had listened, and in the end they said yes, after the lambing, before the men took the flocks up. The men grumbled and said yes too.

He chose the stone himself, a long grey boulder lying among the oaks, a man's height and half as much again. "Why that one?" Telo said. "There's bigger."

"It's sound. It's got no cracks in it. And it's got a good flat foot that'll sit down in the ground and not rock." He slapped the flank of it the way her father slapped a ram. "And it's taller one way than the other. When it stands, it'll stand like a person."

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They moved the stone in five days, and it was the hardest thing Telo had ever seen people do.

First the men dug round it with antler picks and levered it up with long oak poles, a little at a time, until they could get a sledge under it. The sledge was a frame of oak trunks lashed with rope and rawhide. Then they laid a track of logs along the slope and pulled, six ropes twisted from bark and the long fibres of the plants the women grew by the river. Brannok walked in front calling the pull. "*And*," and they leaned, and the stone did not move. "*And*," and with a groan from the wood the sledge slid forward a hand's breadth. The boys ran to carry the logs round to the front. A hand's breadth. And again.

Telo was not allowed on the ropes, so she carried water. On the fourth afternoon the sledge came out from among the oaks onto the open slope. She stopped on the path and looked at the great stone and the people bent and straining in the low sun, and her uncle limping ahead with his arm raised. She thought: I will remember this.

The hole was ready, on his line, a sloping trench with a deep pit at the end. Brannok had spent half a morning on his knees by it with his olive rod and three stakes. He sighted back up towards his boulder and moved the stakes a finger's width this way and that.

On the fifth morning, they ran the stone down the trench with its foot over the pit and its head on the slope above, and stopped. Everybody went quiet, because this was the part where people got killed.

• • •

Brannok came to find her at the spring, where she had gone for water. "Leave that," he said. "Come with me." He took her up past the stone to the top of the slope. There one long rope lay out on the grass by itself, running up from the head of the stone to a stout oak stake, once round it and then loose.

"This is the long rope," he said. She did not know what it was for, and shook her head.

"When the stone starts to come up, the men down there will pull too hard. Everybody does. And a stone that size, once it gets going, wants to go right over on its face, and break, or break somebody." He put the end of the rope into her hands. "You let it out round the stake as it comes up, a bit at a time, and you watch the stone, not me. And when it's upright, you hold. Not one finger further. You hold till I tell you."

Telo looked at the rope, and at her cousins Arro and Ilun at the bottom of the slope, who were sixteen and fourteen and much stronger than her, looking up at her with their mouths open.

"Why me?" Telo said.

"Because anybody can pull," her uncle said. "Any fool with shoulders can pull. I need someone who knows when to hold." He put his hand on her head, the way her father had done at the gate of the pen. "I've watched you with the lambs. You don't panic. You wait. You'll do."

• • •

Below her, the men lifted the head of the stone with long oak levers and jammed timbers under each gain. The foot slid down into the pit, and they built a crib of timbers under it until it lay at a slant like a man getting out of bed. Then Brannok shouted "*Ropes*," and the haulers took the strain.

Telo wrapped the long rope once more round the stake and leaned back on it. She could feel the stone at the other end, the way you feel a fish on a line, a great slow weight, alive, listening.

"*And*," her uncle shouted.

The haulers pulled, and the rope went tight and began to slide round the stake, rough and hot through her palms. She let it slide, slowly, and the stone came up, and up. The whole hillside seemed to lean with the men. She felt the moment when the weight changed and the stone stopped being pulled up and began to want to fall forward, and she held.

The rope bit into her hands as she dug her heels in and leaned back with all her weight. The stake creaked, and down the slope two of the haulers went over in a heap. The stone stood there trembling, one finger past upright, and then, because she held, it came back. It settled. It stood.

"*Hold*," Brannok said. He had stopped shouting, and was standing at the foot of it with both hands flat on the grey side, the way her father steadied a ewe at lambing. "*Hold*, Telo. Good girl. Hold."

She held. She held while the men ran in with the packing stones and rammed them down round the foot with the ends of poles, thump and thump, and earth on top and more stones. She held while her arms shook and her hands went from burning to numb. When Arro came up and offered to take the rope she shook her head without looking at him. She held until her uncle had walked all round the stone three times and leaned his whole weight against it, and it did not move.

Then he looked up the hill at her and nodded, and she let go.

She sat down very suddenly on the grass. Her hands lay open in her lap, red and raw, and she could not close them. Down the slope somebody had started a song. Brannok came up to her, swinging his bad leg, and lowered himself down beside her and took her hands and turned them over. Then he took a little pot of sheep fat from his bag and rubbed it into her palms with his thumbs, the way her mother did with the lambs when their feet cracked.

Before they filled the last of the hole he sent her down to drop something in it. “Something of yours, so the stone knows who held it,” he said. She picked up a small plain grey pebble from the edge of the spring and dropped it in among the packing, and it was gone. That night, by the fire, with the meat going round, Brannok sat down beside her. “Tomorrow’s the middle day. By the stick. If I’ve got it wrong I’ll have to dig it up and drag it two paces to the right, and nobody will ever help me.”

“You’ve not got it wrong,” Telo said.

“Then sleep,” he said. “I’ll wake you.”

• • •

He woke her in the black dark before the birds, in the straw of the lamb pen where she had gone back to sleep because she had promised her father. The old ewe lifted her head and put it down again. It was cold and the stars were very bright. They went up the slope without a light, slowly, because of his leg and her hands, and she held onto his belt.

The stone stood on the slope below the old ring like a person waiting for somebody. They went past it and up to the boulder. Brannok pulled her in beside him, inside the sheepskin, and they waited.

She heard the first bird, and then all of them. The sky went grey over the hills, and then yellow, and then that thin bright red along the edge. Below them the stone came slowly out of the dark until she could see the lichen on it and the head of it standing up against the sky. And she saw then what he had done.

From where they sat, the stone stood in front of the gap between the dog-backed rise and the one beyond it. Its head was framed there like something set in a doorway, and the brightest part of the sky was just behind it.

He had not set it to face the sunrise on any day. He had set it to face the one sunrise in the middle, between the two stoppings, from this one seat on this one boulder, where someone would sit, long after him, and wait.

“Uncle,” Telo said.

"Hush."

The gold came up behind the stone and filled the gap. For a moment the stone was a black shape with light round its edges like a hand held up before a fire. Then the sun broke out over the top of its head, so bright that Telo covered her eyes. When she took her hands away it was sitting on the head of the stone like a bird on a post, and Brannok was crying.

She had never seen a man cry, except her grandfather at her grandmother's burying. He made no noise, but sat with the tears running into his beard and his hands on his olive rod, looking at the stone.

"There," he said. "There. It's right."

Telo leaned against him, and they watched the sun come clear of the stone and the hills. The shadow of the stone came slowly up the slope towards them, long and thin, until it pointed straight back at the place where they sat.

"It'll do that every year," Brannok said. "On this one morning, and the one in the autumn when the sun comes back the other way. After me. After you." He made a cut on the olive rod at the very end of the last row, deeper than the others. Beside it, very carefully, he made a small mark she had never seen him make, a hooked line like a shepherd's crook.

"What's that one, with the little crook beside it?" she said.

"That's you," said her uncle. "That's for the one who held the rope."

• • •

Over the years that followed they raised more stones beside that first one, a few each spring, in a great ring below the old one. Telo held the long rope for most of them, until her hands were hard. Brannok pecked a crook into the side of one of the stones with a hammerstone, a patient job that took him most of a summer, and a ring beside it like the sun.

He died when she was a grown woman with three children of her own. It was in the autumn, on the day after the second middle day of that year, as if he had waited to see it. On the next middle day in

spring, before it was light, Telo went up the hill alone with the olive rod and sat in the hollow he had worn. She watched the sun come up out of the gap and sit on the head of the stone she had held. Then she made a cut at the end of the last row, and went down to take the flocks up to the summer grass.

She had not once in her life been more than two days' walk from that slope. She knew only that the stone stood, and that it was right, and that she had held it.

The master page

The Ledger left the line on the page after the girl had gone down the hill, and he sat with it a while before it wrote anything.

In ink: the stones on the slope west of Évora, the small old rings and the great ellipse built beside them; the crooks and circles pecked into the granite; the enclosures of the Alentejo facing east into the rising arc. In ink, the first farmers and the shell-eaters becoming one people over many generations. In pencil: the year, *4800*, with no radiocarbon under it. In pencil: the equinox, because a day found by halving a count of notches lands a day or two from the true one, and because the best model in the measurements leaned towards a harvest moon as much as the sun. Backwards, the Ledger drew a line in ink to the river, because the girl on the bluff and the girl in the pen had both been taught the year by an old person who watched. Forwards, across the water to the north, it wrote the crook on the Breton stones in pencil and joined nothing, the way you'd leave a gap in a wall you weren't sure would take the weight.

The Alentejo's part was the line. On the Bann the year was a list of signs. Here it was fixed to the ground: a seat, a row of stakes, a stone on the horizon, so that anyone could sit where the watcher sat and know the day without him. A tally on a stick could be lost down a rabbit hole. Nobody was losing a boulder.

In time he would take the line to the Core, the framework he had built for setting one rhythm on top of another: the sun's year, the steadying hand of a count checked over and over, and a memory that keeps its own appointments, every spring and every autumn. It could

show him where one rhythm sat on another. It could not tell him what Brannok said to a girl on a boulder. That belonged to her, and to the scene.

The girl had asked the hard part of it at the grinding stones: *when he's dead, who'll count?* The stone was the answer, and the stone was still there, lying among the cork oaks for tourists to lean on. You track it so you know when to take the flocks up and the grass will be there. And you check it six years running before you trust it, which is the bit nobody puts on the postcards. Anybody can pull on a rope. Somebody has to know when to hold.

Up the coast to the north, on a page he had not reached, the same hooked sign stood on a heath above the sea, and two young people were arguing in the wind.

The Ledger inked the last word on the Alentejo page, *hold*, and laid its pen across the line like a stake driven in to mark it.

CHAPTER 3

Eleven Hundred Stones

Carnac and the Gulf of Morbihan, Brittany: the found year, 4500 BCE



Illustration 4. *The Ménec rows at dawn.*

Belfast, by day

GARY ASKED HOW MANY STONES there were at Carnac, and the machine said the alignments held more than three thousand standing stones.

"And at Ménec?"

"About eleven hundred, in eleven rows."

"About. Good man, that's the first honest word you've said all week," he said.

The Ménec rows run across the heath in Brittany from a ring of stones at the western end, where the tallest stand taller than a man. They run eastward for about a kilometre, the stones dwindling as they go. Kermario and Kerlescan carry the lines on further east. Nobody

agrees on the count to the last stone. Gary liked that, because a place where nobody can count the stones is a place that has kept something back.

His map said *c. 4500 BCE*, and for once he expected it to be wrong, because standing stones have nothing in them to count the carbon of. The machines brought him every figure on the internet, from four thousand to six thousand five hundred. He threw most of it out once he had read where it came from. Then he read the papers himself, line by line, and checked their numbers by hand against the tables.

Three sheets went down on the table, one over the next, like dockets on the spike in the first kitchen he ever worked. The first was the great mound of Saint-Michel above Carnac, where the important dead were laid with axes of Alpine jade and beads of green stone. Its first phase was modelled to 4782 to 4594 cal BC by Bettina Schulz Paulsson in 2019. The second was a dig published in 2025 at Le Plasker, two kilometres west of the Ménec rows, where a line of standing stones could at last be dated. It gave between 4600 and 4300 cal BC, with a tomb of about 4700 built over an older hunters' hut. The third was the green stone itself: variscite, and its chemistry traced much of it to the mines at Pico Centeno near Encinasola in the far south-west of Spain, worked from about 5200 BC. Somebody had carried that stone up the Atlantic coast.

Where the three overlapped, the mound standing, the rows going up, the green stone arriving from the south, was the middle of the fifth millennium. He wrote *4500 BCE*, and because the rows themselves had never been dated, he said the question mark out loud before he put it in.

"Where does the sun come up at midsummer at Carnac and at Évora, in 4500 BCE, with the tilt of the Earth as it was then?" he asked.

The machine gave him two numbers, and he did not trust them. So he worked it through with a pencil and a calculator, and got about thirty-one degrees north of east at Évora and about thirty-seven at

Carnac, nine degrees of latitude further north. The winter sunrise swung out the same amount the other way. The middle of the year sat due east in both places, near enough.

"So the further north you sail, the wider the sun walks, and the middle stays put. Dead on," he said. Anyone who sailed that coast and watched every morning would see it. That was not a pattern he had imposed. It was geometry. He put it in ink and let the dream have it.

The same 2019 paper made a bigger claim, from more than two thousand dates: that the great stone tombs of Europe began here, in north-west France, and spread by sea. Others think they began in several places at once. He wrote both down and, for once, took neither side.

The doubt went in plain: Ménec itself is not directly dated. Le Plasker is a neighbour, and neighbours can be centuries apart. The green beads came from the south-west of Spain, not from the Alentejo, so the woman in the dream comes from the right direction and not from any proven place. And the song that names a girl from another page is his, not the past's.

The boats were harder, because logboats of that age and older have been found in France, hollowed trunks at Noyen-sur-Seine and Bercy. But nothing that survived could have carried a crew up the open Atlantic. "Could a hide boat have done it?" he asked.

The machine said no hide boat of that age had been found.

"Aye, I know. Hide rots, and that's the trouble with the best evidence," he said.

A boat of hide on a light frame could have done it, and nobody has found one that old. He gave the sister a logboat and the stranger a hide boat, and wrote beside it that the first was evidence and the second was need.

Then the water: the Gulf of Morbihan was partly dry land when the first stones went up; the twin rings on the islet of Er Lannic now stand half drowned. That was in the papers and in every grandmother's story along that coast. It was the one thing in the chapter he did not have to put a question mark beside.

He shut the laptop and went out for a walk, and did not say anything about any of it.

That night

That night the library had a wind in it, and sand on the floor between the stacks. The Ledger had moved to the far end, by a tall window over grey water. The page it held had *4500* at the top in pencil. Wind got under it and lifted it, and when it came down it was a heath above the sea, long rows of standing stones walking east, and two young people arguing at the top of their voices. He was at the end of the rows with a boy's height and a boy's cold ears, and the wind paid him as much heed as they did.

• • •

Ael and Enora had been born on the same night, in the same storm, a quarter of a moon too early. Their mother liked to say that they had come out arguing.

"Ael first," she would say. "Head first, the way you're meant to, quiet as you like, looking round him as if he was measuring the place. And then her, a long while after, the wrong way round, kicking and roaring fit to bring the roof in. I said to the women, that one's going to sea."

She had been right, for at nineteen Enora was the best builder of boats on that whole coast, from the river at Étel to the mouth of the great inland sea. She knew it, which did not make her easier to live with. Ael was the one who raised the stones. He had worked on the rows since he was twelve, carrying, then digging, then setting. Now that the old setter's hands were too bad to hold a lever, it was Ael the people looked at when they wanted to know where a stone should go. He was slow and careful and did not like to be hurried. She was quick and sure and did not like to wait. They had shared a boat until they were twelve, and a quarrel, more or less, ever since.

That spring the quarrel was about the rows.

"Who are the rows for, then?" their mother asked, the first night it boiled over at the fire. "The sun, or the boats?"

Neither of them answered her, because each of them thought the answer was so plain that only a fool would ask.

• • •

The rows began in the west, at the ring.

The ring was old: an oval of stones set shoulder to shoulder, enclosing a space of short turf where the people met at the turning points of the year. From its eastern side, running away across the heath towards the sunrise, went the rows. There were eleven of them, side by side, more or less, like the ridges a giant might scratch with a digging stick. By the ring the stones were taller than a man, some of them twice as tall, grey granite blotched with lichen. As the rows went east the stones grew smaller, until at the far end, where the work had got to, they were no higher than a child. From the ring you saw them dwindle away like people walking off down a road.

And at the head of it all, a little to one side, there was the Old One: a single slender stone, more than twice a man's height, that had been standing there before anybody thought of the rows. Nobody knew who had raised it, and the children said it walked down to the sea on midsummer night to drink.

There were more than a thousand stones in the rows by then, a thousand and some, though no two people who counted them ever came out at the same number. Ael said there were eleven hundred, near enough. Enora said there were as many as there were fools to haul them. They were both proud of them, and neither of them would have said so to the other if you had paid them in jade.

The trouble was the next row.

• • •

"It goes this way," Ael said.

They were standing at the eastern end of the work, in the wind. The heath ran away brown and purple on either side, all gorse and heather and turf cropped by the cattle. In front of them, where the next stones would go, was a line of stakes that Ael had driven in at dawn.

"Along the line of the sun," he said. "Like the old setter showed me. You stand in the ring on the morning of the longest day, and you watch where it comes up, and the rows are meant to follow it. They always did. They're meant to point at it."

"They don't, though," said Enora.

"They do, near enough."

"They don't. They bend. Look at them." She swung her arm back along the lines. "They follow the rise. Whoever laid them out had the sense to follow the ground and not go up hill and down dale after something in the sky."

"That's because they were laid out badly. That's what I'm trying to put right."

"That's because they were laid out *well*." She took him by the arm and turned him to face the south, where the sea lay flat and grey and shining beyond the dunes. "When you're out there in a boat, coming in along this coast after three days from the river mouths, with the light going, what do you see?"

He did not answer.

"You see the stones," Enora said. "The Old One first, on the skyline like a mast, and then the ring, and then the rows, running along the land, along the shore, pointing you home. I've come in on a grey evening with the fog coming down and seen them, and known where I was, and wept like a baby, because I knew I wasn't going to drown. That's what they're for. Not just for the sun. For the people on the water. Put your next row running off inland after your midsummer sunrise and you'll break the line."

"Nobody coming in from the sea is going to see one row."

"I will."

"You're one boat."

"I'm one boat now," Enora said.

They stood glaring at each other with the stakes between them until the haulers along the rows began to laugh. Enora shouted at them so fiercely that they went away.

"The old setter said sun," Ael said, when they had gone.

"The old setter's hands are gone, and his eyes are going, and he's never been out past the point in a boat in his life."

"He's seen more sunrises than you've had hot meals."

"So has the Old One," said Enora, "and you don't ask it where the stones go."

• • •

That night their mother sent them both out of the house, because she could not hear herself think. They went down to the shore and sat on the upturned hull of Enora's newest boat. It was a log of oak as long as four men, hollowed out with fire and adze over the whole of the winter. It still smelled of smoke and new wood from the burning.

They did not speak for a while. The tide was out, and away to the east, beyond the headland, lay the opening into the great inland sea, which their grandmother said had been half dry land when she was a girl. There was a ring of stones over there on a low hill near the water, and every year the sea came a little closer to it. The sea reached everything in the end, she said. So you might as well build where you could see it coming. Out past the headland a bank of fog sat on the water, and neither of them looked at it for long.

"I'm not wrong," Ael said at last.

"Neither am I."

"We can't both be right."

Enora was quiet, and then she laughed, short, through her nose, the way their father had. "Mother says that's the trouble with the pair of us. We've never once thought we could."

• • •

The boat came on the third morning, out of the south.

Enora saw it first, because Enora was always looking at the sea. She straightened up from the hull she was scraping with the adze in her hand, and then she began to run.

By the time Ael got down from the rows half the village was on the beach. The boat was coming in through the shallows, eight paddlers bending and lifting together. It was not like any boat Ael had seen. It was long and low and narrow, made of hides sewn together and stretched over a frame and greased black. It sat on the water as light as a gull. In the bow a woman was standing, with one hand on the high curved stem, watching them come.

The paddlers ran it in and hauled it up the sand, their faces burned dark and their hands cracked, and one of them sat down straight away with his head between his knees. The woman walked up the beach towards the people, and stopped, and lifted both her hands to show that they were empty.

She was older than their mother, perhaps, with grey running through her black hair. Round her neck she wore a string of beads of a deep clear green, like the sea in a cove when the sun is on it. They were so green that the children pushed forward to look, and their mothers pulled them back.

She said something that nobody understood. Then she said it again, slower, in another kind of speech, a coast speech, the kind the traders used along the shore, and this time some of the older people nodded.

One of the old men said she was asking whose the stones were, the ones on the hill. She had seen them from the sea, two days off, and had come to see them.

• • •

Her name was Sela, and she had come from a country of hills and oak trees and dry summers, in the south beyond the great bay. It was more than two moons of paddling up the coast, putting ashore each night where she could and asking at every landfall who lived further north. The paddlers were her sister's sons and their friends. The green beads came from her own hills, where people dug the green

stone out of the ground and polished it and traded it up and down the coast. Some of it, she found, came this far: she had seen the same green round the neck of a woman on the beach.

All of this came out slowly, in the coast speech and in gestures and drawings in the sand. And on the second morning, when she had slept and eaten, she asked to be taken up to the stones.

Ael took her, and Enora came too, though neither of them had asked the other.

• • •

She walked along the rows very slowly, touching the stones, looking back at the ring and up at the Old One. At one of the big stones near the ring she stopped and stayed. She put her hand flat on a place where somebody, long ago, had pecked a little carving into the granite: a hooked line, like the crook a herder carries.

She said something, very softly, in her own speech.

"What is it?" Enora said.

Sela traced the hook with one finger, then drew it in the air, and pointed south, and said a word, and pointed to herself.

"She says they have that at home," Ael said slowly. "On the stones at home."

Then she did a thing neither of them expected. She sang, low, with her hand on the stone, a short song that went round and round with a beat in it like paddling. They could not understand the words except for one, at the end of every round, which sounded like *Telo*.

When she had finished she turned to them and began, with a great deal of effort, to explain.

• • •

They went down to the beach for the rest of it, because the sand was smooth there and you could draw with a stick and rub it out with your foot. Some of the old people sat on the dunes to listen. The old setter came, leaning on a boy, and lowered himself onto a log.

In her country, Sela said, there were stones too, not rows but rings. There was a great ring on a slope, among the oaks, facing the morning, and smaller ones, and single stones on the hills. All of them were so old that the people who had raised them were only names in the songs. And the song she had sung was one of those. It was about the first stone of the great ring. It told of a man with a crooked leg who had counted the days, and a girl who had held the rope.

"Counted what days?" Ael said.

She showed them. She drew a line in the sand for the horizon, and the sun coming up at the cold end of the year, far to the right, and at the hot end, far to the left. She drew a notched stick and made the motions of cutting a notch each day and counting, out loud, on and on. Then she drew a line down the middle between the two ends of the sun's walk, and put a stone on it.

"The middle day," Ael said. He was leaning forward, with his elbows on his knees. "When the day and the night are the same."

The old setter made a noise on his log. Ael looked round, and the old man was nodding, slowly.

"That's ours," Ael said. "That's what we do. We stand in the ring and watch. The longest day and the shortest, and the middle days. That's the count. That's how we know when to—" He stopped. "How has she got it? How has she got our count?"

"Maybe it's not ours," said Enora. "Maybe it's hers."

Sela was watching them, and she had understood, or understood enough. She shook her head. Then she smiled and touched her chest and pointed at Ael's, and drew a line in the sand between the two of them, a long line, wavering, the way a shore wanders.

"Neither," said the old setter suddenly, in his cracked voice. They all turned to look at him. "Both. She means it came along the shore. Like her. Like the green stones." He coughed, and spat, and wiped his mouth. "My grandfather used to say the count came from the south. I never believed him. I thought it was one of his stories."

• • •

Then Ael, who had been staring at Sela's drawing with his lips moving, said, "But it's wrong."

They all looked at him.

"Her drawing. It's wrong. Look." He knelt in the sand and took the stick from Sela's hand, gently. "Her sun stops here at the hot end, and here at the cold. But ours goes further. Further round to the left in the summer, further round to the right in the winter. I've watched it every year since I was twelve. Her two ends are too close together."

He drew his own sun's walk outside hers, so that her little walk sat inside his like a small bowl inside a large one, and sat back on his heels, frowning.

Sela began to laugh. She laughed until she had to sit down. Then she took the stick back and pointed at the middle line, the line of the middle day, which ran straight up the drawing between both walks and was the same line for both.

Then she pointed north, out past the islands to the open water, and drew another walk, larger still. An old woman on the dunes who had traded down the coast in her youth put her words into theirs.

The further you went towards the cold, Sela said, the wider the sun walked. She had seen it herself, every morning of the voyage. But the middle day was the middle day everywhere, and the middle line never moved.

Ael sat very still.

"Then the old setter's midsummer line," he said slowly. "The one he taught me. The one I've been trying to make the rows follow."

"Where did he get it?" Enora said.

They both turned to look at the old man on his log. He was quiet so long that Ael began to fidget. At last he lifted his shoulders and let them fall.

"From my teacher," he said. "Who had it from his. Who had it, I always heard, from the south." He laughed, a dry wheezing laugh with no breath behind it. "So I've been teaching you a southern sun, boy, all these years. And you've been trying to lay it down on a northern hill."

• • •

It was Enora who saw it first, the answer. She would say so for the rest of her life. Ael would never once deny it, which their mother said was the most surprising thing about the whole business.

She drew the ring at the head of the rows, and the rows running off from it along the shore, bending with the land. Then, standing in the drawn ring, she drew a line out to Sela's middle-day sunrise. She drew another out to Ael's own midsummer sunrise, the northern one he had watched with his own eyes, and looked up at them.

"The rows don't have to point at it," she said. "The ring does. You stand in the ring to count. You always did. The rows are for walking, and for the people on the water, and for the dead, and for whatever else they're for. Let them follow the land. Let them be a road. And set the count in the ring, where you stand. Set it right, with your own sun, the one you've watched. Not the old man's. Not hers. Yours."

Nobody said anything.

"And if you want the rows to lean a bit towards the summer as they go," said Enora, "the way they already do, then let them lean. It's a long way to the end. Nobody's going to fall over."

Ael studied the drawing. He looked at Sela with the green beads at her throat, and at the old setter with his eyes closed, and then at his sister.

"That's what you said in the first place," he said.

"It is."

"You didn't say the bit about the ring."

"I didn't know the bit about the ring," said Enora. "She did." And she held out the stick to Sela, who took it, and smiled, and drew a small hooked line in the corner of the sand, a herder's crook, and put the stick down.

• • •

They laid the next row the following spring, along the rise of the land, bending a little, leaning a little towards the summer. The stones grew smaller as they went, and Ael set them. Enora watched from the sea and told him when a stone looked wrong from the water, and sometimes he moved it.

The old setter's name was Konan, though the young ones had mostly forgotten that. In the ring he had a new stone raised that summer on the line of the northern midsummer sunrise as Ael had watched it. He was carried up to sit with his blind eyes turned to the east on the longest morning, and asked Ael to tell him, out loud, exactly where the sun came up. "Good," he said. "That's yours, now." He died before the winter, in his sleep, with his family round him. They laid him under one of the great mounds on the high ground where the important dead were put, with a green stone axe that had come from very far to the east, beyond any coast, and a few of Sela's beads.

Sela stayed the summer and taught Enora how to sew hides over a frame with sinew so that the seams did not leak. She showed her how a hide boat could ride over a wave that would fill a logboat of oak. Enora taught her how to burn out a log. They came up to the fire at night hoarse from arguing about hulls. When the middle day of autumn was past and the weather began to turn, Sela's paddlers dragged their boat down the sand. She gave away the last of her green beads, one to each of the twins, and one to their mother.

On the beach, before she got into the boat, she sang the song again. Then she said something slowly in the speech of the village, which came out strange and careful. The old trading woman said it was a thing they said in her country when someone went away. "She's asking what she's to carry back."

The twins looked at each other.

In the end it was Ael who answered. He went up the dunes to where the haulers left their tools. He came back with a small round hammerstone, grey and plain, worn smooth by a lifetime of pecking at granite, and closed Sela's fingers round it.

"Tell them," he said, "the sun walks wider here. Tell them the middle line is the same. Tell them we're still counting."

Sela nodded, and got into the bow, and the paddlers pushed off through the shallows. The boat went out through the low surf and turned south along the coast, light as a gull, and was gone behind the headland.

Enora stood at the edge of the water until long after it was out of sight, and Ael came and stood beside her.

"She'll not come back," Enora said.

"No."

"Somebody will, though," Enora said. "Somebody always does. That's what the rows are for." In the wet sand at her feet, where the next wave would take it, she drew a little hooked line like a herder's crook, and a long wavering shoreline beside it, running away to the north. Then she turned and looked to where, on the skyline above the dunes, the Old One stood against the evening. The long grey rows ran east along the land, getting smaller as they went.

"Come on," she said. "I've a hull to finish. And you've stones to count."

"Eleven hundred," said Ael.

"As many as there are fools to haul them," said Enora, and took his arm, and they went up the beach together, arguing, into the dusk.

The master page

The Ledger kept the page open while the tide came in over the drawing in the sand, and then it wrote.

In ink: the rows at Ménec, eleven of them, and the neighbouring line at Le Plasker set up between 4600 and 4300 cal BC; the mound of Saint-Michel with its Alpine jade and its green variscite from the mines of south-west Spain; the sun's walk wider at Carnac than on the

Alentejo, which is only geometry; the Gulf of Morbihan drowning its own stone rings. In pencil: the year, *4500*, because Ménec itself has never been dated. In pencil, and smaller: Telo's name in a stranger's song. Nothing in the ground links a girl on a slope near Évora to a ring at Carnac, except the crook pecked into stones in both places and the green stone that came up the coast. The Ledger drew that line faint, and did not ink it. Forward, it wrote the next page in pencil too: passage tombs built on the coasts of Ireland, which some think came the same way, by sea.

Carnac gave it the sea-road. The season had been kept on a river and the line had been fixed to a hill. Here, for the first time on the shelf, the knowledge moved: carried in boats along a coast, with the green stone and the axes and the shape of a tomb, and checked against a new sky wherever it landed. A count that travels has to be reset at every landfall, the way you'd put your watch right coming off the boat at Cairnryan.

Some day he would take the sea-road to the Core, his framework of overlapping cycles, a steadying hand, and the past that circles back. It might show him where the same pattern came back at different places along a coast. It could not tell him who sang a song on a beach. Nobody could, now.

The mother had put it at the fire: *who are the rows for, the sun or the boats?* Both, it turned out, once the count was set in the ring and the rows were let follow the land. The boy had wanted it for the boats, so you can come home in a fog and know where you are. Gary liked the middle line best, the one that came out the same for both of them while only the ends moved. And Enora had the last word, as she usually did: set the count with your own sun, the one you watched, not the old man's and not the stranger's, and whatever travellers bring you, try it on your own hill first.

Further along the shelf the Ledger had marked a ridge above a bend of a river in Ireland, and a slot over a door where the winter sun would be let in.

By then the tide had reached the drawing in the sand, the rows and the ring and Enora's crooked arrow, and it took them all back without a sound.

CHAPTER 4

Winter Light at the River Boyne

Brú na Bóinne, the bend of the Boyne: the found year, 3200 BCE



Illustration 5. *Siadal in the chamber as the winter light lies along the floor.*

Belfast, by day

GARY ASKED HOW LONG THE light stayed in the chamber at Newgrange. The machine said that on the mornings around the winter solstice, sunlight entered through the roof-box and lit the chamber floor for about seventeen minutes. He asked who had measured it, and the machine named a tourism website, so he sighed and went to O'Kelly.

Newgrange sits on a ridge above a bend of the Boyne, in the bend the old texts call Brú na Bóinne. Knowth is a short walk to the west and Dowth to the east. The mound is about eighty-five metres across, ringed by ninety-seven kerbstones. A passage about nineteen metres long runs into it from the south-east, climbing as it goes, to a cross-shaped chamber under a corbelled roof. Over the door is the slot, the

roof-box that Professor M. J. O'Kelly of University College Cork cleared in 1963. On the morning of 21 December 1967 he went in alone and watched the light come up the passage. Gary read O'Kelly's account himself, twice, before he let any machine summarise it.

Then the dates: his AI models gave him a dozen numbers, from 3000 to 3500, and half of them came from other summaries. He went past them to the excavation report and checked the lab numbers by hand.

He asked where the dates came from, to see what it would say, and it said organic material within the tomb.

"From the putty, so write that down," he said.

When the builders laid the roof slabs of the passage they packed the joints with burnt soil and sea sand to keep the rain out. They pecked grooves along the top of the slabs to carry water off. O'Kelly sent two samples of that caulking to the laboratory at Groningen. GrN-5462-C came back at 4425 radiocarbon years, plus or minus 45; GrN-5463 at 4415, plus or minus 40. Calibrated, both sit around 3200 BC, with a spread of a couple of centuries either side.

He started with the roof, because that was where the dates were. The caulking went in when the roof went on, and the roof-box is part of that roof. He had talked loads down by hand at Kilroot. You did not set a lid on until what was under it was right. And the light only works if the roof-box was planned with the passage, because the floor rises. Without a slot above the door, the low midwinter sun would hit the floor halfway along and never reach the chamber. So the date of the putty was, as near as he could get, the date of the first winter the light could have gone in.

3200 BCE, he wrote. He made the question mark small, smaller than usual, because the putty was good evidence and he wanted it to know that.

The sea and the mountains were in it too. The white quartz at the entrance came from the Wicklow Mountains, far to the south. The grey rounded granite cobbles came from the shores across Dundalk Bay to the north. Most of the big passage slabs were greywacke from the coast near Clogherhead, probably brought round by sea and up

the river. How the quartz was laid is still argued over. O'Kelly rebuilt it as a sheer white wall. Others think it lay as a spread before the door, and that the wall is a 1970s idea built in concrete. Gary put the white stones heaped about the doorway in the dream, which both sides could live with, and wrote the argument in his notes.

Then the genome: in 2020 a team led by Lara Cassidy read the DNA of a man whose bones lay in the right-hand recess, the most richly carved. His parents had been first-degree relatives, brother and sister or parent and child. The authors read that, with his place in the tomb and his distant kin at Carrowmore and Carrowkeel, as the sign of an elite that kept its line inside the family. Gary wrote the finding in ink and the reading in pencil. A family's choices five thousand years ago were not his to judge.

The doubts went last: the calibrated range is wide, and the caulking dates the roof, not the people. And the light no longer does what it did. The Earth's tilt has changed a little since, and the sunrise with it.

"So did the beam go further in back then, when it was new?" he asked.

"Most calculations suggest it reached further into the chamber."

He wrote *most* in too, because it was the honest word.

He looked up how you would see it now: on an ordinary day the guides take visitors into the chamber, switch off the lights and let an electric lamp by the door play at being the sun. Only a few people, drawn by lottery, can be inside on the real mornings.

"What are the odds of getting in?" he asked.

"Tens of thousands apply each year, and a few dozen names are drawn."

"Aye, well, I'll stick to the lamp," he said, and put the kettle on, still thinking of the real sun creeping up that passage in the dark.

That night

The library was dark that night, and very cold. The Ledger stood at the end of a long narrow aisle between the stacks. The page it held had *3200* pencilled at the top, with its small hook. A thin bright line

ran across it from the door end. He walked towards it and the aisle got lower, and somewhere along it the stacks turned to slabs and the floor to a roof under a low sky, with a young woman lying flat on it, feeling the joints with her fingers. By then he was crouched by the kerb of the cairn at a boy's size, and no one turned.

• • •

The first winter Siadal worked on the great mound, she carried earth. The second winter she carried sand. The third winter she learned to cut the grooves. The fourth winter, which was this one, she was the one the others came to when the roof leaked.

It did not leak now, and that was the point. She lay on her back on top of the passage, in the half-dark, with her cheek almost against one of the great roof slabs. Her fingers worked along the joint where it met the next, feeling for any gap. There was none. Every crack was packed tight with the putty she had spent three winters learning to make. It was burnt earth from the hearths ground fine and mixed with sand from the seashore, worked with water until it was stiff as dough, then pressed in with the thumbs and smoothed. Along the top of each slab ran a shallow groove she had pecked herself with a quartz point, no deeper than a fingernail. Whatever rain got down through the cairn would run along it and off the side, instead of dripping onto whoever lay below.

"Is it dry?" somebody called up from the end of the passage.

"It's dry," Siadal called back.

"Are you sure?"

She sat up, carefully, because there was not much room between the roof slabs and the cairn stones heaped above. She looked down the slot towards the light and the young man's face peering up at her, and smiled.

"It'll be dry," she said, "when your grandchildren's grandchildren are dust. Go and tell her that."

• • •

People had been building on the ridge above the bend of the river, where her mound stood, for as long as anyone remembered. The river came down out of the west, wide and brown and full of salmon in their season. It swung round in a great loop under the ridge, so that the land inside it was nearly an island, low and rich and green, and the cattle grew fat.

There were mounds all along the ridge now, old ones grown over with grass and new ones raw with clay. A short walk to the west another great mound was going up, nearly as big as this one, and to the east a third. The people of the river said that the mounds were the houses of the dead, and that was true. But they were also the places where the year was kept.

This was the biggest of them all, and her mother had worked on it, and her mother's father. It was a hill made by hands, river stones and field stones with layers of turf and clay between, as high as seven tall men standing on each other's shoulders. Round the foot of it ran a ring of huge kerbstones like the rim of a basket, some carved with spirals and lozenges and wavering lines like water. When the low winter sun came across them the carvings seemed to move.

And through the middle of it went the passage, running from the doorway on the south-east back and up into the dark heart of the hill.

• • •

Siadal had been inside it many times. The passage was longer than any house, twenty paces and more from the door to the chamber. It was walled with great upright slabs set edge to edge and roofed with more laid across, so narrow that two people could not pass. And it was not level, which was the thing everybody noticed the first time: the floor rose as it went. You went in at the door bent double and walked

up into the hill, turning sideways in places where the slabs leaned. Then, quite suddenly, the walls opened and the roof went up into the dark. You were standing in the chamber.

"It'll never reach the back," one of the haulers had said, the first winter, sitting on the kerb with his bread. "The floor climbs. The sun comes in low. Any child can see it'll hit the floor halfway and stop."

"Then any child can go and tell Liath she's wrong," Siadal had said. "Go on. I'll hold your bread for you."

He had not gone; nobody ever went.

Inside, the mound was shaped like a person lying with arms outspread. The passage was the body and the chamber the head, with a little room opening off it on each side and a third straight ahead. In the little rooms lay great shallow basins of stone. The roof was built of slabs laid one over another, each reaching a little further in, round and round and up, until a single great stone closed the top. The walls were covered with carvings, and on a stone in the far room three spirals wound together into one.

Siadal had sealed every joint in the passage roof herself, from the doorway to the chamber. On nights when she could not sleep she went over it in her mind, joint by joint, the way other people counted cattle.

She could not sleep often this winter, for she was carrying her first child, due at the turn of the spring. She lay at night with her hand on her belly and felt it turn like a fish, and counted the joints.

• • •

Cuan was away at the shore.

He was a quarryman, and a good one. Most of the great slabs in the passage had come from the coast to the north, a day's walk away. There the grey rock lay in the cliffs in flat beds like a pile of hides, and broke away under the sea's battering in long straight pieces. When he had found one, his crew worked it free with wedges and levers, dragged it down to the water at low tide and lashed it under a raft of

logs. When the tide floated the raft they paddled it down the coast and up the river on the flood, slowly, over days, to the landing place under the ridge.

He had brought other stones too: rounded cobbles of white quartz that glittered in the sun, from a shore far to the south below the mountains. There were grey speckled ones too, round as eggs, from the beaches under the mountains to the north across the bay. They had come in boats, many boatloads over many years, traded for cattle and axes and good flint. Now they lay heaped about the doorway, white and grey together, so that on a bright day the whole face of the hill seemed to shine.

He had gone back to the coast in the autumn for the last one. It was the stone for the top of the little window above the door, and it had to be long enough to span the gap and flat on its underside. He had searched for it for two moons. It was in place now, over the roof-box, with its face carved in a row of crosses like stitching. He had gone back to the shore to help his brother bring in a stone for one of the other mounds. He would not be home until after the turning of the year.

"I'm sorry," he had said, at the door of their house, with his pack on his back. "I'd stay if I could."

"I know."

"You'll go in? On the morning?"

"If she asks me."

"She'll ask you." He had put his big cracked hands on either side of her face and looked at her. "Tell me about it after. Every bit. You know I'll not believe it till you tell me."

• • •

The old woman who would do the asking was called Liath. Her hair had long since gone past grey to white. She was the one who had judged the angle.

Nobody knew how old she was, and her eyes were milky now, though she could still see light and shapes. She sat most days wrapped in skins on a flat stone at the top of the bank near the entrance, facing south-east.

But for more years than anyone could count, Liath had walked up onto the ridge every midwinter morning in the dark. She sat on this same stone and watched the sun come up.

"Any fool knows where the sun comes up," she used to say. "Over there, over the low hills across the river, where at midwinter it rests a few days before it turns back. That's not the question. The question is the exact place, and the exact height, and how it looks from deep inside a hill that isn't built yet."

So she had driven stakes into the ground, long ago, when the mound was only a ring of kerbstones and a pile of earth. They ran in a line from where the chamber would be towards the place where the sun stopped, and she moved them a finger's width at a time, winter after winter. When the passage slabs went up, she had stood at the end of them with a cord stretched from the chamber floor out through the mouth of the passage. She had the men raise and lower it until she said stop. And she had found what nobody had thought of. Because the floor of the passage climbed into the hill, the low midwinter sun coming in at the door would strike the floor halfway along, and never reach the chamber at all.

It would have to come in higher, over the door, through a slot in the roof the width of a hand. Then it could run straight down the passage at the level of the chamber floor. "Like water," she had said, so the story went, "finding its level."

That had been four winters ago, when Siadal had been a girl of fourteen with a basket of earth on her back. She had stopped to listen. She had seen the old woman stand at the end of the passage with the cord in her hand and her milky eyes turned towards the low sun, and say, very quietly, "There. That's where it'll come."

• • •

On the evening before the shortest day, Liath sent for her.

Siadal went up through the frost in the last light. The workers had gone down to the houses by the river. The great heap of stones stood against the darkening sky with the doorway black in its white face. Above the doorway, under the carved lintel Cuan had brought from the shore, the roof-box, a dark slot like a half-closed eye.

Liath was on her stone, wrapped in her skins, holding her hands out to a small fire a boy had lit for her.

"Sit," she said.

Siadal sat on the frozen grass beside her, awkwardly, with the child heavy in her.

"You sealed the roof," said Liath.

"Yes."

"Is it dry?"

"It's dry."

Liath nodded. "Good. If the roof's not dry the dead don't rest." She was quiet for a while. "They've put him in, the high one. Did you know?"

Siadal knew, because the bones of the high man had been carried into the chamber at the autumn feast. They were laid in the right-hand room, whose stones were carved more richly than any other. His family were the greatest on the river, and had kin in the other great mounds right across the country, to the far coasts. People said other things about that family too, in low voices. They said that they married close, closer than other people would, to keep it all among themselves.

"Tomorrow," said Liath, "the sun will come in. If I've judged it right. And somebody has to be inside to see it."

"Not the high family?"

"They'll be outside, in their best cloaks, and watch it go in at the window, and say it's going in to him. And it is. But somebody has to be inside, to see where it goes. To see if it reaches the back."

"Why me?"

Liath turned her milky eyes towards her.

"Because you sealed the roof," she said. "Because you know every stone of that passage by touch, in the dark. And because," and she put her dry, light hand on the swell of Siadal's belly, "you're carrying somebody who'll still be here when all of us are under the grass, and somebody ought to tell them."

Siadal looked at her.

"Won't you go in yourself?" she said. "You've waited for it longer than anyone."

"Look at me," she said. "I can't walk to the river. Do you think I could crawl up that passage? No. I'll sit here, on my stone. That's my place. I'll see it from here. You go in."

• • •

The stars were still out over the river, hard and bright.

They went up in the dark, Siadal and two old men of the high family who kept the door, with a small lamp of fat in a stone dish. They took the blocking stone from the door, and Siadal went in alone.

She went up the slope into the hill on her hands and knees, with the lamp in one hand. She knew the walls the way she knew her own house. There was the slab that leaned in at the top, and the narrow place where you turned your shoulders. Overhead was the joint she had sealed on the first day of the third winter, when her fingers were so cold she could not feel them. She put up her hand and touched it as she passed. Dry.

At the end she came up into the chamber and stood, slowly, one hand on the wall, and lifted the lamp. The stone under her hand was grey and cold, and dry, the way she had made it.

The roof went up into the dark, and the little rooms opened round her with their basins dim in the shadows. In the right-hand room, under the richest of the carving, there was a heap of bones, with beads and a stone ball and a pin of antler laid among them. She did not go near them. She put her free hand on her belly and said under her breath the words her mother had taught her to say to the dead, so they would know she meant no harm.

Then she blew the lamp out.

• • •

The dark was complete. In a house at night there was always something, the red of the fire, a line of grey round the door, but here there was nothing. She held her hand in front of her face and could not see it. The cold came up through her feet. The hill stood over her, all the weight of stone and turf and clay that her mother had carried, held up by the roof that she herself had sealed. She heard her own breath, too loud, and the child turning over inside her, and nothing else.

She sat down on the floor of the chamber with her back against the stone of the left-hand room, facing the passage, where she knew it was though she could not see it. She waited.

She waited and waited, thinking about Cuan on the cold shore, and about the child, and about the old woman outside in the frost. She got cold, and began to wonder whether the sun had come up already and missed the window by the width of a finger.

And then, far off down the passage, there was a grey.

It was so faint that at first she thought she had imagined it. Then it was a line, thin as a thread, gold, lying along the floor of the passage far down by the door.

She stopped breathing.

The line moved, coming up the passage towards her along the floor, slowly, the way a tide comes in over flat sand. As it came it grew brighter and wider, from a thread to a cord to a band the width of her hand, gold and then white-gold, as straight as a stretched rope. The walls came out of the dark on either side of it, glittering faintly. It came past the leaning slab and the narrow place and into the chamber. She drew her feet in as it passed, the way you would from water, and it did not stop. It went on across the chamber floor, straight to the back, and up onto the stone at the back of the far room, and lay there glowing.

And the whole chamber lit up.

Not from the beam, but from what it struck: the light bounced off the floor and the back stone and filled the chamber with a dim gold glow. She could see everything. The roof went up in its rings of stone, and the spirals and lozenges stood out sharp, like fresh cuts. She saw the basins, and the bones of the high man with the light falling softly on them. She sat with her hands pressed to her mouth and the tears running over them, and watched the band of light lie on the back stone. Then, very slowly, it began to narrow.

And it was then, sitting in the gold light at the heart of the hill, that she understood.

She thought of the old woman four winters ago with the cord in her hand, saying *That's where it'll come*. She thought of the stakes, moved a finger's width at a time for longer than Siadal had been alive. And she knew, the way you know a thing all at once, that the old woman had never expected to be here for this. She had judged the angle of the light for a morning she thought she would never live to see, and had done it anyway, and done it right. Then she had walked back to her stone and sat down and waited, not for herself but for whoever would be sitting here, in the dark, when it came.

The band of light narrowed to a thread and drew back down the passage the way it had come, and was gone. The dark came back, but it was not the same dark. Siadal sat in it a while with her hand on her belly, where the child had gone quite still, as though it had been listening too.

Then she got up and found the wall with her hand. She went back down the passage towards the door, on her hands and knees, touching every joint in the roof as she went.

• • •

The light outside was so bright after the dark that she came out into the open blinking. There was frost on everything, and the low sun stood just above the hills across the river, red-gold, exactly where it should be, with the valley full of mist below it. The high family were

all there in their best cloaks, talking at once and some of them weeping. Somebody took her arm and asked her something. She did not answer, because she was looking for the old woman.

Liath was on her stone at the top of the bank, wrapped in her skins, her face turned to the sun. For one terrible moment Siadal thought she had died there in the night, waiting. Then the old woman turned her head.

Siadal went up the bank as fast as she could with the child in her, slipping on the frost, and knelt beside the stone and took her hands. They were not as cold as she had feared.

"Did it come?" said Liath.

"It came."

"All the way?"

"All the way to the back," said Siadal. "It lay on the back stone like a hand, and the whole place lit up, all the carving, and it stayed as long as it takes to milk a cow, and then it went back out the way it came." She found she was crying again. "You were right, to a finger."

Something in the old creased face loosened and smoothed, the way the ground softens after a frost.

"I saw it go in," Liath said. "From here. A little gold line. I can still see that much. I thought I'd be dead long before, and it'd be some girl not born yet sitting in there. I'd made my peace with that. And then it was you. Somebody I could ask."

"You could have gone in," Siadal said. "We'd have carried you."

"No," said Liath. "It wasn't for me. I only had to know it would come. You had to see it." She laid her hand, very lightly, on Siadal's belly. "Tell this one, when it's old enough. Tell it you sat in the dark in the heart of the hill on the shortest day, and the sun came in and found you. Tell it it'll come again next year, whether there's anybody there to see it or not. That's enough."

"I'll tell it," said Siadal.

"Good," said the old woman, and turned her face back to the sun, and closed her eyes, as the mist lifted off the river and the frost began to melt on the white stones round the doorway. The great hill stood over them both, dry to the last joint, holding the year.

• • •

Liath lived until the spring, long enough to hear that Siadal's child had been born on the morning of the middle day. It was a girl, small and loud and whole. The old woman had the baby put in her arms, and said that it had a good grip and would make a quarryman. She died a few days after, in her sleep. On the next midwinter morning, and every one after that for as long as Siadal lived, somebody sat on the old woman's stone in the dark before dawn and watched the sun go in at the window.

Siadal told the child, as she had promised. The child told her own children, mouth to ear, the way such things go on. She knew only that the sun had come in and found her, and that the roof was dry, and that an old woman who could not see had been right to the width of a finger.

The master page

The Ledger let the page lie open a long time, the way the light had lain on the back stone, before it wrote.

In ink: the caulking dates, 4425 and 4415 radiocarbon years, about 3200 BC; the roof-box, cleared in 1963; O'Kelly alone in the chamber on a December morning in 1967, watching the thread of gold come up the passage towards him, as she had waited, more than five thousand years before. In ink, the quartz from the south and the cobbles from the north and the slabs brought by sea. In ink, the man in the right-hand recess, and what his genome said about his parents. In pencil: what it meant, an elite keeping its line in the family; that belongs to the authors, and they said so carefully. In pencil: how far the beam reached in her day, because the tilt has changed and the calculations differ. Back along the shelf, the Ledger drew a faint line to Carnac, in

pencil: the passage tomb is an Atlantic shape, and some think the idea came by sea. Forward, it wrote in ink the next page's fact, a tomb in Orkney whose passage takes the midwinter sun at its setting, and in pencil the thought that the two were kin.

The Boyne's part was the aperture. The season had been read off a river, the line fixed to a hill, the sea-road carried along a coast. Here the sky was let into a room, through a slot the width of a hand, on one morning of the year, so that the dark itself told the date. And the page added a second thing, gently: a family that kept the year also kept its line, and the two were bound together in the same stone.

In time he would take the aperture to the Core, the framework he had built for stacking cycles, steadying them, and letting the past arrive late; and a beam that comes back to the same stone every midwinter is that idea made of light. But it is a way of looking. It proves nothing, and he had learned to say so before anybody asked.

The hauler on the kerb had asked the only question that mattered, the way a man asks it with bread in his hand: would it ever reach the back? It reached the back. She had done it because an old woman asked her to, and because the child would need to be told. Somebody had judged the angle for a morning they would never see, and judged it right, and then gone home and kept the roof dry for whoever would be sitting in the dark after them. Gary had worked beside men like that. You never saw their names on anything.

The next page was already lifting in the draught: islands at the top of the map, and a page where the dead and the living kept the same kind of house, and somebody was counting the dark days to midwinter.

Long after, a man in Belfast would sit up late with his ledger open beside him and the old woman on her stone in his head. He woke at the table in the small hours with the lamp on, and the kitchen was cold, and for a moment he did not know where the door was.

CHAPTER 5

The Orkney Winter

Skara Brae, and the lochs of Stenness, Orkney: the found year, 2900 BCE



Illustration 6. *Gwen in the passage with the knotted cord, as the midwinter sunset light runs down it.*

Belfast, by day

THE FIRST THING GARY WROTE at the top of a clean page was always a date. The second was the same date with a question mark. He had learned that on night stock-takes years back, walking the aisles with a handset under the strip lights. You count what is on the shelf, not what the sheet says should be there.

Today the top of the page was empty, and down the margin he wrote five words: *who, what, where, when, how*. The sixth he left off. The stone on the windowsill kept its counsel too.

“Every published date for the Stenness sites: Skara Brae, Barnhouse, the Stones of Stenness, Maeshowe. One table. Flag where they disagree,” he told the machines.

They were quick, and wrong often enough that he checked every line against its paper.

The first question was *when*, and Skara Brae on the Bay of Skaill answered first: stone houses with stone beds and stone dressers, lived in on and off from before 3300 BCE to after 2500. Barnhouse, on the shore of the Loch of Stenness: founded about 3100, and by the 2016 models its last great hall abandoned in the earlier twenty-ninth century. The Stones of Stenness beside it, raised between 3000 and 2900, in use while the village was.

He asked about Maeshowe. The machine said it was built around 2800 BCE, and when he asked who said so, it named Historic Environment Scotland.

He looked it up: Historic Environment Scotland *tentatively* suggested about 2700. A 2017 review called the chronology uncertain: the radiocarbon came from peat in the ditch. The dates were in conflict with each other and with the layers they came from.

"Tentatively. You left out the best word in the sentence," he said.

He stood the three dates side by side, like three men on a lift who all have to say *ready* before it goes. If the tomb was begun while the village still had smoke in its roofs and the ring stood with its hearth, there was one narrow place where all three overlapped: the last generation of Barnhouse. *2900 BCE* went at the top, and straight under it, in smaller writing, the case against: if the people who put Maeshowe at 2700 are right, the village is cold when the tomb begins, and his guess is two hundred years early. If he was wrong, he wanted it written down where he could find it again.

For *who*, in 2022 geneticists read twenty-five people buried in north-west Orkney in the Bronze and Iron Ages. Most carried new ancestry from the steppe beyond the Black Sea, as across Britain. But on the father's line most of the men still carried the old Orkney lineage of the tomb-builders for at least a thousand years. The mothers' lines, though, were mostly new.

"So the sons kept the farm and the girls came over on the boat," he said.

"That is a simplification of the authors' interpretation."

"Aye. That's what it's for."

For *what*, *where* and *how*, grooved pottery appears first at Barnhouse and spreads across Britain and Ireland. Maeshowe follows the cross-shaped plan of the Boyne tombs, with the same care for the midwinter sun. And the Orkney vole, by its DNA, came from the coast of what is now Belgium about five thousand years ago.

"How does a vole cross the North Sea?"

"Most likely as a stowaway in fodder or bedding."

"In the hay, then. Wee stowaway," he said.

The sea had been crossed often enough that the mice came too.

Then came *19 years, Orkney?*, underlined twice in his old notes. In 1973 Alexander Thom and his son called the Ring of Brodgar a lunar observatory, following the moon's rising point along the horizon over about 18.6 years. Clive Ruggles and others found that the precision those claims needed did not stand up. And nineteen was another number altogether, the cycle after which the moon's phases fall back on the same days of the sun's year. He wrote *18.6 is not 19*, and under it, *don't marry them just because they're neighbours*.

The last entry came from the other end of the world. In Border Cave in South Africa, a baboon's leg bone about forty-three thousand years old carries twenty-nine notches, cut with four different edges at different times.

He asked if it was a lunar calendar. The machine said it had been interpreted as a possible lunar tally.

The boy in him wanted it to be, and he wrote that down, because it was what he wanted, and under it: *any number near thirty is near a month. Open.*

He showed nobody the page, because a man who shows his whole hand before he's checked it is only guessing out loud.

That night

He went to bed late. Sleep took him straight to the library: shelves going up out of sight, a smell of dust and salt, and somewhere among the stacks the Ledger, keeping its quiet. A page turned by itself, with the number from the kitchen at the top of it. Sand hit him in the face

before he saw anything. Then there was wind, and a girl on the roofs of a village half sunk in its own midden, watching her brother walk into a storm. He was standing outside the wall in a boy's body with his eyes screwed up against the grit, and nobody marked him.

• • •

The storm came in on the shortest day, which everybody said was a bad sign, and Tam went out into it anyway.

Gwen watched him go from the top of the midden, where the roofs of the houses humped out of the heap like the backs of seals. The wind was coming straight off the sea, tearing the tops off the dunes. The whole bay behind her was white. Tam did not look back. He went bent into the wind with his pack on his shoulders and his staff in his fist, a dark shape against the pale grass of the links, and then nothing.

"He's away," she said, when she came back in.

Her grandmother did not look up. She was in her usual place on the stone bench by the hearth, with her bad leg stretched out to the warmth and something in her lap.

"He was always going to be away," she said. "Shut the door. You're letting the sea in."

Gwen dragged the door slab across and pulled the hide over it. The storm dropped to a long low grumble that she felt in the floor more than heard. The house closed round her. There was the square hearth in the middle with the dried dung and heather glowing in it, and the two beds on either side, built of upright slabs like boxes and heaped with dry bent grass and skins. The little cells let into the thickness of the walls. And straight in front of her, facing the door, so that it was the first thing anyone saw, the dresser. It had two shelves of flat stone on stone legs, with the best pots on them and a whalebone dish for the red paint. There were the big pots with the grooved patterns her grandmother had pressed into the clay when she was young. There was a stone ball carved all over with knobs that everybody touched.

Outside, all round, was the midden, ash and shells and bones and old turf packed against the walls up to the roofs. The village sat inside its own rubbish like an egg in a nest. Gwen had been born in the bed

on the left, and so had Tam. She thought now that it was the best place in the world, and that her brother had just walked out of it into a storm.

"Why couldn't he wait?" she said. "One more day."

"Because they said the shortest day. They're setting the corner stones over by the lochs. They want the young men there when they do it, so the young men will remember."

"He'll not see anything in this."

"No," her grandmother agreed. "But he'll have been there."

• • •

The thing in her grandmother's lap was a cord.

It was not much yet. It was a length as long as her forearm, twisted hard and tight out of plant fibre, and near the end of it there was one knot.

"What's that for?"

"That's tonight," said her grandmother.

"A knot's not a night."

"It is if you tie it right."

Her knuckles were swollen like barnacles, but her fingers still knew the work.

"Every night from now," she said, "I'll tie a knot. One for each sleep. When the moon's full I'll tie a fat one, so I don't lose my place. And I'll keep tying until the sun comes back round to tonight. Then we'll see how many there are." She turned the cord over. "They're building a house for the dead by the lochs, the biggest there's ever been, and they're setting the door so the sun shines in when it goes down on the shortest day. Down the passage and onto the back wall. But to set the stones right they need to know the day. Not near enough the day. The day. So the old men watch the sun, and the old women count." She smiled a little at the fire. "And some years the old men are wrong, and some years the old women are right."

"That's a lot of knots," said Gwen.

"It's one at a time," said her grandmother. "Like everything."

• • •

Her grandmother tied a knot every night. After a while Gwen began to wait for it the way you wait for the last thing of the day: the fire banked, the door slab dragged across, and her grandmother feeling back along the cord with her thumb, counting under her breath, and tying one more.

When it was a moon old she made Gwen tie one.

"Not like that. You'll have it out in a week. Over, and under, and through, and pull. There. Feel it."

"It's only a knot."

"It's tonight," her grandmother said. "Tonight's never coming again. You'd want it tied right."

• • •

In the spring, when the days had stretched and the lambs were coming, Gwen walked to the lochs with two of the women from the next house, carrying dried fish and barley meal on their backs. They came down at midday to the place where the two lochs lay side by side, one fresh and one half salt from the sea. A thin neck of land ran out between them.

Gwen stopped at the top of the slope and stared.

On the near shore there was a ring of stones, thin and tall and flat-faced, like great grey blades set on end, taller than three men. In the middle of them was a big square hearth of stone like the one at home, as though the ring were a house and the sky its roof. Out on the neck of land there were buildings bigger than any house she had imagined, and more people than she had known there were in the world. And on the shore, close by the ring, there was a village of houses like her own with smoke going up from its roofs, and a great hall among them. She felt she could have walked into any one of them and known where to sit.

They found Tam at the house of the dead.

It was not a house yet, only a broad oval platform that the men had built up out of clay carried from the edge of the loch. A wide ditch was being dug round it. In the middle of the platform, alone, in a square, like four people waiting for a fifth, there stood four great stones.

They were the corners of the room that would be, Tam said, when he had stopped hugging her. The walls would go up between them, slab on slab, leaning in until they met at the top. And from the front of it, there, towards the south-west, where the hills of the southern island stood up beyond the water. The passage would go out there, long and low, built of single slabs longer than a boat.

From the place between the four stones two cords were pegged out across the clay, side by side, running towards the south-west, like the edges of a path not built yet.

"That's where the light will come," he said. "You couldn't see a thing on the shortest day, it was blowing like the end of the world. But the next clear evening the sun went down there, between the hills. Near enough, they said. Now they want it to a hair."

"Grandmother's counting."

"I know. There's women counting in every village from here to the north end." Some of the grin went out of his face. "How is she?"

That evening a stranger sat by their fire, a long-faced man who had come over the sea from the south with a boat full of axes to trade. When he heard them talking about the passage, he spoke of the sea to the south-west. Further than he had been himself, he said, there was a great river with a hill built by hands beside it, and a little window cut over its door. On the shortest day, he said, the sun went in at the window at its rising and ran down the floor to the back. His grandfather had told him so.

"At its rising?" said Tam. "Here it goes in at its going down."

The stranger shrugged: people did things differently in different places, and that it was all the same sun.

Gwen thought about that, walking home under the long light. The same sun went in at a door at one end of the world in the morning and at another door at the other end at night. In between, in a house dug into a midden by the sea, an old woman tied a knot.

• • •

The summer was short and bright: the boats out after the fish, the children down at low tide for limpets, the barley ripening behind the dunes. And every night the cord grew, until her grandmother had to wind it round a flat stick. Every night before she tied the knot she felt back along the turns as far as the last fat knot of the full moon, and counted.

"Why do you count back?" Gwen asked her once. "You know how many there are."

"I know how many I think there are. That's not the same. Every night you check the ones you've done. If there's a mistake, you want to find it while it's small."

"And if there is one?"

"Then you say so, and you put it right, and you don't pretend it wasn't there." Her fingers were slower than they had been in the winter, and sometimes she had to start again. "A count you can't trust is worse than no count at all. Because people will build on it."

In the autumn her fingers got bad, and some nights she could not pull the knot tight. On those nights Gwen did it for her. She sat on the edge of the bed with the cord across both their knees, while her grandmother counted back with her thumb and said yes, and yes, and there, and Gwen tied the knot and pulled it.

"You've the hands for it," her grandmother said one night, with her eyes closed.

"It's only knots."

"No," said her grandmother. "It's the year."

• • •

By the next winter the cord took two sticks to hold it. The storms were back. Each evening the sun went down a little further south behind the hills of the southern island, slowing, as it always did when the shortest day was near.

Her grandmother could not walk to the lochs, and they both knew it without saying.

"But they'll want to know if we agree," she said. "So you'll take it. Go the day before, and sleep there. On the day the knots run out, you stand in that passage, wherever they've got to with it, and you watch. And then you come home and tell me."

Gwen looked at the cord in her lap, and it was heavy. She had not expected it to be heavy.

"What if the knots don't come out right?"

"Then you tell me that," said her grandmother. "That's what it's for."

• • •

She crossed the island in a grey wind with the cord wrapped in a skin inside her cloak. In the morning she went up to the house of the dead and saw what a year had done. The walls were rising between the four stones, a little above a man's height now and beginning to lean in. The room was already a room, square and dark, with the great stones at its corners. The passage had been laid along the line of the cords. The mound was beginning to rise round it all. And the line from the back wall to the door was there, open and straight, pointing south-west across the water at the hills.

Tam took her to the old women, six of them from six villages, wrapped against the wind on the bank of the ditch, and each had a cord. They laid theirs out side by side on the grass. The oldest, a tiny woman with a voice like a gull, went along them all with her thumb, counting the fat knots and then the thin ones, while Gwen knelt beside her grandmother's cord and held her breath.

Four of them came out the same, her grandmother's among them. The last knot was tomorrow.

"Tell your grandmother," said the gull-voiced woman. "Tell her old Fetha says she's still a fool about the full moons, but her count's good."

• • •

The next day was the shortest day, and it was clear.

All afternoon people came up and stood on the banks and the half-built mound. The old men stood at the back of the room between the great stones. Tam stood with the young men along the top of the passage slabs. And because she had brought the cord, Gwen was let into the passage itself, near its inner end. She crouched against the cold stone with the cord in her arms and looked down the long slot of it towards the bright south-west, and waited.

The sun came down the sky red and swollen and slid along above the hills with the slow tiredness it had at the end of the year.

Then the sun came to the end of the passage.

First there was a glow on the stones at the outer end, red on grey. Then it was a line of light lying along the floor, coming towards her the way the tide comes in over flat sand. The walls went red and gold and every crack in them stood out sharp. It came up the passage and over her feet and her knees and the cord in her arms. Then it was past her and across the floor of the room. It climbed the back wall between the two great stones and lay there, a long bar of red-gold light on bare stone, in a room that had no roof yet and did not need one.

Nobody made a sound, and Gwen heard the wind, and a curlew out on the loch, and her own heart.

Then the sun touched the hills and went down behind them, and the light on the back wall drew in and went out. The people on the banks let out their breath all together, a long sigh that went round the platform like the sea going out over shingle.

• • •

Her grandmother was in the bed when Gwen got home, propped up on the skins with the lamp lit, looking like she had been waiting there since Gwen left. She did not ask anything, only held out her hands. Gwen put the cord into them. Her grandmother felt along it from the first knot, the one she had tied on the night of the storm, to the last, the one Gwen had tied in the house by the loch. Her lips moved, and it took a long time.

"Well?" she said at last.

"It came. On the day the knots ran out. Straight down the passage and onto the back wall between the stones, and it stayed till the sun went behind the hills. Four cords were the same as ours. Old Fetha says you're a fool about the full moons, but your count's good."

Her grandmother laughed, a small laugh, and it hurt her, Gwen could see. But she laughed anyway, and lay back with her eyes closed and the cord in her hands.

"Then that's the year," she said.

"What will you do with it?"

Her grandmother opened her eyes and looked at the cord, and then at Gwen.

"Nothing. It's done. It's only a cord now. You can use it for the nets." She held it out. "But you'll start another tonight. And next year it'll be yours they lay out on the grass by the loch."

Gwen took it. After a while she went to the wall cell where her grandmother kept her fibre and twisted a new length between her palms the way she had been shown, until it was hard and tight. Then, as the lamp burned down and the wind went over the village, she tied one knot in the end of it, over and under and through, and pulled it, and felt it.

"That's tonight," she said.

"That's tonight," said her grandmother, and slept.

• • •

The old cord went into the nets, and the nets wore out. Gwen tied the cords for her grandmother as long as she lived, which was longer than anyone expected. Then she tied them for her daughter, who had better hands than either of them. None of the cords lasted, because cords don't. The counting lasted.

And the house of the dead by the lochs went up, year after year, until the mound covered it and the grass grew over it. Every winter, for weeks either side of the shortest day, whenever the sky was clear, the setting sun came down the long passage. It lay on the back wall between the great stones, whether there was anybody there to see it or not. It was still coming in at that door long after the village by the bay was empty. The sand had blown over it and buried it.

The master page

The page did not close at once. He was in the library again with it open in front of him and the Ledger at his elbow, the way a foreman stands by you while you sign for a load.

He laid the year on the shelf beside the others. The salmon run on the Bann; the equinox stones on the Alentejo plain; the rows at Carnac; the roof-box at the Boyne, where the midwinter sun goes in at its rising. The stranger by Gwen's fire had told of that one. The Ledger let the line stand in ink as far as the published evidence went, from tomb plan to tomb plan, and wrote the rest in pencil: nobody can show that a story crossed the sea. Ahead, on shelves he had not reached, there were pages waiting with the same knot in them: a Gaulish count a boy is forbidden to write down, and an earl's men sheltering in Maeshowe from a midwinter storm four thousand years on, cutting runes on its walls.

Every page in this book was building one part of a single machine, and Orkney's part was the count. Days kept by hand, checked back every night, agreed between six villages. Without a count, nothing on a later page could be laid against anything else.

In the morning he would take the rhythm to the Core, the framework he had taught himself to code from nothing. In plain words its master equation says three things: the world is a stack of rhythms; something steadies them so they don't run away into noise; and the past comes round again into the present, late, and goes on shaping it. It could tell him whether a rhythm was really in the numbers. It could not tell him that a grandmother tied a knot, or that anyone in Orkney had heard of the Boyne. That was the bones' work, and the dates', and they would say what they said whether he liked it or not.

Nor could it answer the sixth word, the one he had left off the margin. Gwen had asked it the small way, *what's that for?*, and been told *tonight*, and a year later *the year*: so you'd know when the light was coming back. The dream had shown him the rest. Six villages had to agree on one count, and people were going to build on it, stone on stone, for years. On a stock-take you would sooner have no figure than a wrong one, because a wrong one gets ordered against, and Orkney had known the same thing five thousand years before anybody had a handset. You stood on the same day as your neighbours or you were lying to them.

A knot. *Not yet*, every night, all winter, *not yet*, until the sun came down the passage and said *now*.

On the next page, four hundred years on and a long way south, a long pale-green stone would come down out of the north to a chalk plain, where people gathered at midwinter with their pigs.

The lamp was still on in the kitchen when he came down, and before he made the tea he opened the Core's catalogue at *the 18.6-year lunar cycle tracked by five stone monument sites* and changed Orkney's line: Brodgar contested; Maeshowe the sun, not the moon. He wrote the reason beside it and saved the file, and the machine stamped the date on it, which was its own kind of knot.

CHAPTER 6

The Last Stone from the North

The River Avon and Salisbury Plain: the found year, 2500 BCE



Illustration 7. *The Altar Stone on its sledge, hauled towards the Plain.*

Belfast, by day

SIX TONNES, AND GARY KNEW what six tonnes was, for he had slung loads like it at Kilroot. He had done it with a radio on his chest and a ticket in his wallet that said he was fit to. He had watched a load that size turn slowly on its chains in a wind off the lough while every man on the ground went quiet. You did not forget the quiet.

So he read the paper the way he would read a lift plan.

The Altar Stone at Stonehenge: about five metres long, a metre wide, half a metre thick. It is pale green sandstone glittering with mica, lying flat at the foot of the tallest trilithon.

He asked the machine where it was from. It said the Altar Stone was generally thought to come from south Wales, like the bluestones.

"Was thought, maybe. Now look at 2024," he said.

In 2024 a team led from Curtin University dated tiny grains of zircon, apatite and rutile in fragments of it. They matched them to the Old Red Sandstone of the Orcadian Basin in the far north-east of Scotland, at least seven hundred and fifty kilometres away, by sea, likely.

"Orkney, then, surely," Gary said, because he wanted it to be, for the last page's sake.

A second paper that year ruled out Orkney itself. Not the islands, then, but the mainland coast below them. He crossed Orkney out, and it stung a wee bit, and that was how he knew it was the right thing to do.

"Every stage of the monument, every date range, side by side," Gary told the machines, and they had it in a minute. Then he read the papers himself, because a table can't tell you which of two experts is guessing.

For *when*, the first Stonehenge, around 3000 BCE, was a ditch, a bank and a ring of pits, with bluestones from the Preseli Hills, two hundred and twenty-five kilometres off, set among the cremated dead. The sarsen circle and the great trilithons, from West Woods a day's walk north, went up in the second stage, in the centuries either side of 2500.

He asked when the Altar Stone went in, and the machine said it was installed in Stage 2, around 2500 BCE.

"The paper says *may*, so put the *may* back," he said. The *Nature* paper said it may have gone in during that stage. Mike Parker Pearson and his colleagues, writing the same winter, said it could not be older than the trilithon it lies beneath and might be any time after. But its long sides lie near the midwinter sunrise, so neatly that it was most likely laid flat on purpose, the way recumbent stones lie in the circles of north-east Scotland.

So he followed the midwinter to Durrington Walls, the great village a short walk up the Avon. There most of the pigs had been killed at about nine months old. Born in spring, they were killed at midwinter, and isotopes in their teeth show some raised as far off as Scotland,

north-east England and west Wales and walked to the Plain. The village and its feasts are dated around 2500 BCE. Its house floors are laid out like Orkney's, and then there was *who*.

"Who was standing round the fire?" he asked.

"In 2019 seventy-three ancient British genomes showed that the farmers who came about 4000 BCE, by way of the Mediterranean and closest to the first farmers of Iberia, had almost entirely replaced the hunters before them."

He asked what came after, and found the rest himself. From about 2450, newcomers with steppe ancestry and Beaker pots arrived, and the old population's share of the island's gene pool fell by around nine-tenths within a few centuries.

He laid the three windows over each other: the stage the stone could first belong to, the midwinter feasts, and the last generation before the newcomers. They met at one place, *2500 BCE*, and he wrote it, and the question mark, and under it the doubt, that *if the stone came in a later stage, the dream has the wrong crowd at the fire*.

The moon he kept in its own column, because the four Station Stones make a rectangle whose long sides point to the most southerly moonrise and most northerly moonset of the 18.6-year cycle.

He asked if it was deliberate.

"Historic England said in 2024 that they believe so."

"Believe. Fair enough, then," Gary said.

A team watched the moon there through the standstill of 2024 and 2025. At Chaco Canyon, three slabs throw a dagger of light on a spiral at the solstice, and some argued it marked the moon's swing too. A 1985 re-examination doubted it worked as a calendar at all. His old notes had called it Mogollon, and Peru, and he corrected both: Ancestral Pueblo, with the Mogollon their southern neighbours, and Peru another continent. He kept only the question: *what if* people with no contact anyone can show watched the same moon swing? It would not be a network, but something stranger.

Then a Greek: Diodorus, quoting the older Hecataeus, tells of an island beyond the Celts, no smaller than Sicily, with a precinct of Apollo and a temple "spherical in shape". The god comes back every nineteen years, the year of Meton, and the kings, the Boreadae, keep the succession in the family.

He asked if that was Stonehenge, and the machine said some writers had suggested so.

"Some writers have suggested everything," he said.

The 1935 editor thought of Stonehenge; John Toland, in 1726, of Callanish. Gary tried to hear it as a Greek would. The god's own people lived at the back of the north wind, praising him every day, a picture of the right life at the edge of the world. Read in its own tongue it was a hymn, not a survey. He wrote it down as what it was, and kept the rest of his thinking off the page.

That night

The library smelled of chalk that night, and of woodsmoke. The Ledger was further along the stacks than usual. The page it held open for him was heavy, the way a lift is heavy before it leaves the ground. When it turned, the number at the top was his. Then the page was dusk on the shortest day, forty people on the ropes, and a young woman seeing the ring with the sun going down through the middle of it. He was a boy at the back of the line. His hands were on the rope too.

• • •

They came up out of the river valley onto the Plain at dusk on the shortest day, forty of them on the ropes and the stone behind them on its sledge. Kerra saw the ring for the first time with the sun going down through the middle of it.

She had heard about it all her life, but hearing was not the same. It stood on the long bare rise of the downs ahead of them, black against a sky the colour of a forge. It was a great round of stones taller than any house, with flat stones laid across their tops so that the whole ring

was joined into a single thing, like people holding hands. Inside it, higher still, she could see the tops of the great arches, two uprights and a stone across. The tallest of them stood at the far side, facing her. And the sun was going down exactly behind it. It sat in the gap between the two tallest stones like an ember in a doorway, red and swollen and low. The light came out through the gap and down the slope towards them, and lay along the grass, and touched the ropes in their hands.

The man in front of her on the rope stopped hauling, and so did the man in front of him. All along the line the ropes went slack. Nobody said anything, and behind them the sledge creaked and settled on its rails.

"Well," said Hodh, beside her, and began to cough.

• • •

He had coughed all the way from the coast. The cough was dry and came up from somewhere deep in his chest. It doubled him over when it came. He was a small dark man from the far west, from a country of high bare hills across a wide estuary. He spoke the language of the Plain with a lilt that made the young men laugh. He had joined the haulage party at the river mouth because, he said, somebody had to look after the stone.

"Look after it?" Kerra had said. "It's a stone."

"It's a stone that's come further than you'll ever go," Hodh had said. "Somebody should be civil to it."

He was a stone-dresser, and his people had been coming to the Plain since long before anybody could count, bringing the blue stones from their own hills. Some of the blue stones he had dressed himself, when they were being moved into the new arrangement inside the great ring. In his hills, he told her, the stones came out of the crags ready-made, in pillars, like somebody had stood them up inside the rock and only left them for people to find. You put in your wedges at the right place and the pillar came away whole. That was why his

people's stones were the ones the Plain had wanted first, long ago, before the great grey ones: because they were already standing up when you found them.

Kerra was seventeen, and she came from a settlement at the mouth of the river where it ran out into the sea, a place of fish weirs and salt and boats. She had never been more than three days' walk from it in her life until this autumn, when the boat came.

• • •

It had come out of the east along the coast in the last good weather, two boats lashed together with timbers across them, long and low and very heavy in the water. The men in them were thin and burnt brown and so tired they could hardly stand. They spoke a language nobody at the river mouth knew. Between the two hulls, lashed to the timbers, lay the stone.

It was not like any stone Kerra had seen. It was long, as long as three men lying head to foot, and flat, and squared on its sides, and it was not grey. It was a pale soft green, the green of a duck's egg. When the sun came out on it the whole face of it glittered, as though somebody had scattered it with tiny flakes of silver. It was heavy beyond believing, and it took every man from three settlements, and three days, to get it off the boats and onto a sledge.

The boatmen would not say much, because they could not, but they could point. They pointed back the way they had come, east, along the coast. And then, when the old men of the river mouth kept asking, they pointed north. And when the old men still did not understand, the eldest of the boatmen took a stick and drew in the wet sand a long line for the coast. Then he turned it, and went on drawing, up and up, north and north, until he came to the rocks at the end of it and could go no further. He stood there with the stick in his hand and looked at them, and they understood.

"Further than the blue stones," Hodh had said, when he heard. "Much further."

"How much? Where's it from, though?" Kerra said. "What's up there?"

"I don't know," said Hodh. "I don't think anybody here knows. I think that's the point of it."

The boatmen rested at the river mouth for half a moon, and then, before the weather broke, they went home. Before they went the eldest of them came up the beach to where the stone lay on its sledge. He put both his hands flat on it, and stood with his head down for as long as it took. Then he picked up a stone from the beach, a fist-sized lump of the hard grey stone of the downs that the rivers brought down to the sea, plain as a loaf. He held it up to the old men with a question in his face, and they nodded. He put it inside his clothes against his skin and went down to the boat.

"That's all?" Kerra had said. "They bring that, and all they take back is a lump of grey?"

"They'll have been told to bring something back," said Hodh. "Something from here. Something to show." He looked after the boats. "Somebody at the other end will want to know what we're made of."

• • •

So they hauled it, forty of them on the ropes in two teams turn and turn about, up the river valley towards the Plain. They laid the timber rails in front of the sledge and greased them with fat, and hauled the sledge along them. They picked up the rails from behind and laid them again in front, up through the woods along the river. She had never worked so hard, and her hands bled and hardened and bled again. And she learned that Hodh could not haul, because the cough would not let him, but that he walked beside the stone all day with his hand on it. Every evening when they stopped he went over it with his fingers inch by inch, checking it for cracks, the way her mother checked a child for fever.

"Why do you keep doing that?" she asked him one night.

"Because if it's cracked, I want to know before it breaks. Not after." He coughed, and wiped his mouth, and looked at his hand, and wiped it on the grass. "Same with anything."

She did not ask him about the cough, and nobody did. Once, when it was very bad, she brought him water and sat beside him till it passed. He drank, and looked at her over the rim, and said, "It's the dust of my trade, girl. Every stone I ever dressed, a little of it went in. That's all. That's all it is." And she understood from the way he said it that it was not all, and that he knew it was not all, and that he did not want to be asked again.

• • •

Now they stood on the Plain in the last of the light with the ropes slack in their hands, and watched the sun go down between the two tallest stones.

It went down slowly, and the gap filled with red. The red narrowed to a thread along the ground and went out. The sky over the ring went from red to rose to the clear cold green of a winter evening, with one star in it.

"That's the door," said Hodh softly. "That's the door for the shortest day. The sun goes out through there, into the dark, and the year turns." He looked at the stone on its sledge. "And they want this put at the foot of it."

• • •

In the morning, before the work began, Hodh took her into the ring.

It was bigger inside than it had looked from the slope, and quieter, the stones keeping the wind out the way the walls of a house do. The great grey uprights stood round them in their circle, each one twice the height of a man and more, smoothed and squared by years of pounding with hammerstones. The lintels lay across their tops in a continuous ring, joined end to end. Hodh showed her how each lintel had a hollow cut in its underside that sat over a knob left standing on the top of the upright, and each one was cut at its ends with a tongue and a groove. That way it locked into the next.

"Like a carpenter's joint," he said. "Like you'd join two timbers. They built it like a house of wood, only out of stone, so it would stand for ever. Whoever thought of that was either very clever or very frightened of rot."

Inside the grey circle stood the five great arches in a horseshoe, opening towards the north-east, the tallest at the closed end facing the way they had come. And among them, inside and between, set in their own rows, were the blue stones. They were much smaller than the grey ones, and not really blue. They were a dark grey-green speckled with white, like a starling's breast in winter, and when Hodh knocked on one with his knuckle it rang.

He went to one near the middle and put his hand on it.

"I dressed this," he said. "When I was younger than you are now. My father brought me. They were moving them all round that year, into the new rows, and he let me take the face off this one, and I made a mess of the corner, there, see, and he made me leave it so I'd remember." He laughed, and the laugh turned into the cough. He leaned on the stone until it passed. "I remember."

"How long ago did your people bring the first ones?"

"Before my father's father's father. Before anyone's. They were the first stones here. Before the grey ones, before the arches. There was only a bank and a ditch then, and a ring of holes, and the blue stones standing in the holes, and the dead put in beside them." He looked round at the great grey circle. "Then they built all this round them. Like a house built round a hearth."

Kerra looked at the speckled stone under his hand, and at the great arch beyond it, where the sun had gone down the night before. She looked at the empty place at its foot where the green stone would go.

"And now one from the north," she said.

"And now one from the north." He took his hand off the blue stone. "West, and here, and north. If anyone ever comes from the south, and the east, they'll have the whole world in one place." The cough took him again. "That might be what they're after. I don't know. I only dress them."

There were people everywhere on the Plain, more than Kerra had ever seen. They had come for the midwinter, as they came every year, from every part of the country and beyond. They were camped along the river below the ring and up at the great settlement on the slope above the river to the north-east, where the houses stood in their hundreds. The smoke of their fires went up in a grey haze over the whole valley. The smell of roasting pig lay over everything, for they had brought pigs from everywhere, Hodh said, from the west and the east and the far north. They were driven on foot for days and weeks, fattened for this, killed and eaten here at the turn of the year, so that everybody who came ate something from everybody else's country.

On the second night the old men sent for her.

She did not know why. She went up through the camp with Hodh beside her, coughing, to one of the big houses in the settlement, where the fire was high and the floor was hard white chalk. The old men of the Plain were sitting round it in their cloaks, a dozen of them, grey and lined and very still. They had all been to see the stone. She had watched them walk round it that morning, slowly, and bend to look at the glitter on it, and put their hands on it, the way the boatman had.

"You were at the river mouth," said one of them, "when it came?"

"Yes."

"You saw the boatmen."

"Yes."

"Then tell us," said the old man, "where it came from."

Kerra looked at Hodh, but he gave her nothing, only a small nod that meant: you saw it, you say it.

She thought about lying. She thought about saying it came from the west, like the blue stones. That was a thing these men knew and could put a shape to, and it would have pleased them. She thought about saying she did not know, which was true, in a way. Then she thought about Hodh going over the stone in the dark every night with his fingers, looking for cracks, and saying he wanted to know before it broke and not after. She said what she had seen.

"From the north," she said. "Further north than any of the others. The boatman drew the coast in the sand, and he went north with it until he came to the end of the beach and couldn't go any further, and he pointed past that. From a country I've never seen." She hesitated. "I don't think anybody at the river mouth has seen it. I don't think anybody here has."

Nobody said anything.

It was not the silence she had expected. She had thought they would argue, or laugh, or ask her what kind of fool believed a boatman with a stick. Instead the old men sat round the fire without moving, and one or two of them looked at each other. One of them, the oldest, put his hand over his eyes and sat like that. The fire cracked, and outside somebody was singing.

"Thank you," said the first old man at last. "Go and eat."

• • •

"Why did they go so quiet?" she asked Hodh, outside, in the cold.

He walked a while before he answered.

"The blue stones came from the west," he said. "From my hills. That's where the sun goes down at the end of the day, so it's the direction of the dead, some say, and that's fitting for a stone. The great grey ones come from the downs a day's walk to the north, from here, from home. That's fitting too. A house should be made of its own country." He hacked into his fist. "But north's the dark quarter. North's where the sun never goes. North's the way you'd go if you wanted to get to the edge of everything, and there are stories, old ones, older than the ring, that say there's an edge up there, and people at the edge, who aren't quite people. Who see you coming before you see them." He shrugged. "They're old men. They've heard those stories since they were children. And then a girl from the river mouth comes in and tells them, plain as you like, that a stone has come to them from past the end of the beach."

"Do you believe the stories?"

"I believe in the stone," said Hodh. "I've had my hands on it. The rest I'll leave to them." He smiled at her, in the dark. "You did right. You told them what you saw, not what they wanted. That's rarer than you'd think."

• • •

It took eight days to dress the stone and bring it into the ring.

Hodh dressed it and would not let anyone else. He lay on his side on a hide beside it to work, with a hammerstone in one hand and a smaller one in the other. He kept a bowl of water by him to keep the dust down, because, he said, there was no sense making a bad thing worse. One end of the stone he squared off clean, at right angles to its sides. The other end he cut away on a slant, a long clean bevel, the way the feet of the big grey uprights were cut. When Kerra asked him whether that meant it was to stand or to lie, he said the old men had not told him. He was not sure they had decided. He coughed while he worked, and stopped when he had to, and lay with his eyes closed, and then started again. Kerra brought him water, and sat with him, and handed him what he needed. She watched the green stone come up out of the green stone under his hands, like it had been in there all along waiting to be let out.

"There," he said, on the last evening, and lay back on his hide and looked at it. "That's as good as I can make it. It's better than it was. It's not as good as it'll be in a hundred years, when the weather's had it."

They laid it on the ninth day, at the foot of the tallest arch, across the line of the midwinter sunset. It went in a bed cut for it in the chalk with antler picks. Kerra was on the ropes as it went in slowly, levered off its sledge and down its ramp and into its place, with a hundred people pulling and old men shouting and Hodh lying on his side at the edge of the bed with his face nearly touching the stone, watching it settle, and saying, softly, under all the shouting, "There. There. Now. There," until it was down.

That evening, when the sky was clear, they stood in the ring and watched the sun go down through the great arch again. The last of the light came through the gap and fell across the new stone at its foot.

The green of it caught fire, and the flakes of silver in it shone all over its face like the frost on a field. Then the sun went down and the light went out of it. It was a long pale stone lying in the dusk at the foot of two grey ones, and that was all.

• • •

Hodh went home in the spring.

He went down to the river mouth with the people from the coast. Kerra walked with him the whole way, slowly, because he could not walk fast any more, and they did not talk much. At the river mouth there was a boat going west along the coast, towards the wide estuary and the high bare hills beyond it, and he got into it.

"You'll come back next winter," she said. "For the turn of the year."

"Maybe." He sat in the bottom of the boat with his back against the side and his hands in his lap, the cracked, square, dusty hands of his trade. "I'd like to see my hills again. The pillars, standing up in the rock. I'd like to sit up there one more time and look out west." He looked at her. "You'll look after it. The green one."

"It's a stone," she said, and found she could not say anything else. He laughed, and coughed, and the boat pushed off.

He did not come back the next winter. Kerra did not hear when, or how, and in the end she decided she did not need to. She told herself he had got to sit on his hills and look out west. She went on believing it for the rest of her life, which was a long one. And every midwinter that she could, she walked up from the river mouth to the Plain. She stood in the ring and watched the sun go down through the great arch and lie for a moment on the green stone at its foot.

The master page

When the boat had gone, the page stayed open, and the Ledger laid it on the shelf next to Orkney's.

This lock he could make in ink. A stone from the far north-east lying at the foot of the arch like the recumbent stones of its own country; house floors at Durrington laid out like Orkney's; pigs walked

from Scotland to a midwinter feast. The scientists who traced the stone wrote only that moving it implied a high level of organisation among people who left no writing; Parker Pearson's team went a step further and called the monument a work of island-wide unification. That was the system this page built. Orkney's page had made the count. This one made the gathering: stones from west and home and north brought to one ring, and a whole island told, once and for ever, *when the sun goes out through the door on the shortest day, be here.*

Other lines went in pencil. Diodorus' round temple pointed forward to Apollo's island on the next page, and the kings who kept it in the family pointed back to the man laid in the heart of Newgrange whose parents were close kin. The moon's 18.6 years at the Station Stones he set beside the nineteen he would meet in Athens, with a gap between, because they are not the same number.

He would run it through the Core next. Of its master equation's three parts he liked best the memory term: the past reaching forward into the present and steering it. Stonehenge looked like that from a height: bluestones set up first, then lifted and set again inside the sarsens centuries on, a house built round its own old hearth. The Core could tell him whether the sun's year and the moon's eighteen-and-a-bit rang together in the numbers. It could not tell him who hauled the stone, or what they hoped for. And it would not let him off the other thing the bones said: within a few hundred years of the last great stone going up, the people who raised it had mostly gone from the island. A model can carry an echo. It can't grieve, and it had no business pretending to.

When he was a boy he had thought it was the *alter* stone, the one that changes things. It isn't spelled that way. But it had altered the map, right enough, in one afternoon in 2024.

The sky was the one invitation everybody could read; that was the size of it. Kerra had asked the other question on the beach, *where's it from, though?*, and the honest answer then was Hodh's: nobody here knows. The honest answer now was a basin of old red rock in the far north-east, seven hundred and fifty kilometres off, and not Orkney. And when they came, somebody had to stand by the fire and tell the

old men the truth about where the big stone came from, though a lie would have pleased them more. People had walked for weeks on the strength of that count. At Kilroot you looked a sling over for cuts before every lift, and the one time a man didn't was the one he'd be explaining for the rest of his life.

Further along the shelf, on a bright island in a southern sea, girls were laying locks of their hair on a tomb for two maidens from the north, and a gift wrapped in straw was coming down the roads to meet them, from hand to hand.

In the morning, with the wind off the lough rattling the back door, he typed one line into the method notes at the top of the Core's entry for Stonehenge, where he would have to read it every time he opened it: *Check the stone for cracks before it breaks, not after.*

PART II

Transmission

Seven Roads to the Edge

CHAPTER 7

Straw and Gifts for the Island of Light

Delos, in the Cyclades: the found year, 430 BCE



Illustration 8. *The offering from Tenos arrives at the quay on Delos.*

Belfast, by day

THE CRANES HAD TAUGHT HIM to look for cracks, and for a while Gary had worked in freight as well, and freight teaches you one thing above everything else. A load is only as good as its paperwork. Every hand that touches it signs for it, and if one signature is missing you do not know what you have.

So he read Herodotus like a waybill.

"Who carries the gifts to Delos, according to Herodotus?" he asked.

The machine gave him a tidy list, and Andros was on it.

"Andros is skipped, and that's the whole point of Andros. Read it again," he said.

It apologised, and he went and read Book Four himself.

There the Delians say that sacred things wrapped in wheat straw come from the Hyperboreans, the people beyond the north wind, to the Scythians, then from people to people as far west as the Adriatic. They go south to Dodona, the first Greeks to receive them, to the Malian Gulf, across to Euboea, and city to city as far as Carystus. Then, leaving Andros out, the Carystians carry them to Tenos and the Tenians to Delos. Every hand is named, and Andros is skipped with no reason given, and he liked Herodotus for not making one up.

But there were three waybills for the one load. Callimachus, two centuries later, starts the sheaves at Dodona, with no Scythians at all. Pausanias, six centuries later, sends them through the Arimaspi and the Issedones, the Scythians, Sinope on the Black Sea, Prasiae in Attica and the Athenians. Same load, three routes: either the road changed, or the story did, or both.

Then came the check that mattered. Herodotus says the two northern girls, Hyperoche and Laodice, lie at the foot of an olive tree on the left of the entrance to the temple of Artemis, and the earlier pair, Arge and Opis, behind the temple, facing east. In the 1920s French archaeologists found two old monuments in very nearly those places. There were spindle whorls at the one where girls laid their hair wound on spindles, and at the other, human remains, a tomb of Mycenaean type with a long entrance passage, and pottery a thousand years older than Herodotus. The written source and the ground agreed, and that doesn't happen as often as you'd think.

He made himself check the next bit just as hard, because it was his own. His *Full Story* said that in Herodotus Apollo visits the Hyperboreans every nineteen years. He read the five chapters again, line by line, and it isn't there. He had asked the machine first, and it had agreed with him at once and given him a chapter number.

"You only told me what I'd already said, and that's worse than being wrong," he told it. The nineteen years come from Diodorus, three and a half centuries later, quoting Hecataeus: the god returns

every nineteen years, “the period the Greeks call the year of Meton”. He corrected his own paper, felt the small sting, and wrote *good* in the margin anyway.

That was the thread he followed to the year. He had his machines line up every ancient date that touched the nineteen, then checked each against its source. Meton of Athens set out his cycle in the archonship of Apseudes: 235 months of the moon fitted into 19 years of the sun, so the festivals stop drifting through the seasons.

He asked what year, and the machine said Meton introduced the cycle at the summer solstice of 432 BCE.

“Usually given as. There’s a 2021 paper argues for a year earlier,” Gary said.

He noted it, and that Babylon came to the same count and made it fixed rule by about 380. Two or three centuries later a craftsman built it into bronze: the Antikythera mechanism carries a dial of 235 months on a five-turn spiral.

Herodotus was writing in those same years, and two summers after Meton’s solstice, in 430, the plague came into Athens, and Thucydides, who caught it and lived, describes it. Four years after that, in 426, the Athenians purified Delos and carried away the graves on the island. So if he wanted the maidens’ tomb still heaped with hair, Meton’s count newly made, and the plague on the boats, there was one year that held all three. *430 BCE.* He wrote it, and the question mark. Under it: *nobody says Herodotus stood on Delos that year. That’s mine.*

“Who carried it?” he asked.

“Genomes from Bronze Age Greece, read in 2017, show the Mycenaeans carried a share of ancestry, a few per cent up to around a sixth, from a northern source related to the hunters of eastern Europe and Siberia. The Minoans did not.”

He checked the paper: something had come down from the north into Greece, long before Herodotus, and it was people, not straw. He wrote both on the page and left a gap between.

And one more, under *what if*: in the fourth century BCE, within a lifetime or two of Meton, people on the coast of Peru built thirteen towers along a ridge at Chankillo, a toothed horizon that the sun walks along through the year. No ship went between Athens and Peru, so what if nobody needed one, and the sky was the only messenger?

He asked what was in the straw, and the machine said Herodotus does not say.

"Aye, well, neither would I," he said.

Last, he read the straw with a Greek's eyes. Herodotus's word for what came wrapped in it was *hira*: holy things, the god's things. A Greek did not unwrap what belonged to a god, any more than a postman opens the post. And Herodotus, to make sense of it, reached for a custom he had seen himself: the women of Thrace and Paeonia keep wheat straw by them when they sacrifice to Artemis the Queen. Not cargo, then, but an offering, wrapped in the harvest, carried by many hands to a god's house.

He wrote *offering?* in the margin, small, and went to find out whether anybody better qualified had thought of it first.

That night

The library was warm that night, and bright at the far end, the way a room is when a door stands open on the sea. The Ledger had the page ready. He did not see it turn; he smelled it, salt and pine resin and something sweet burning, and then it was the grey before dawn, a hollowed step under an olive tree, and a girl kneeling at a tomb with a bronze knife shaking in her hand. He was sitting on the temple steps with a boy's skinned knees, and the sparrows took no notice of him.

• • •

Euphrosyne cut her hair on the morning the boat came from Tenos. She did it badly, because her hands were shaking.

Her mother had given her the little bronze knife the night before, and the spindle, a plain one of olive wood that had been her grandmother's. She had sat up half the night turning them over in her lap.

In the morning she went down through the town before it was properly light, along the edge of the sacred precinct where the temples stood white and grey in the dawn. She went to the place every girl on the island knew: the old tomb at the foot of the olive tree, on the left hand as you went in at the gate of the temple of Artemis.

There was nobody else there, which was how she had wanted it. She knelt in front of the tomb on the worn stones, where so many girls had knelt before her that the stones were hollowed. She took a lock of hair from behind her left ear, where it would not show too much, and cut it, and it took three goes. When it was done she sat back on her heels and looked at the dark tress lying across her palm and felt, absurdly, like crying.

Then she wound it round the spindle, carefully, turn on turn, the way her mother had shown her, as if it were thread and she were about to spin it. She tied it off and laid it on the tomb with the others.

There were a great many others, and some were fresh and glossy. Most were old and grey, the spindles cracked, lying in heaps among the roots of the olive tree. Among them, here and there, were the offerings of the boys, a few hairs wound round a green shoot, gone brown now and dry. Every girl on Delos laid her hair here before her wedding, and every boy before he came of age, and had done for longer than anybody could say.

"For Hyperoche," Euphrosyne said, very quietly, "and for Laodice. Who came from the north and didn't go home."

She did not know what else to say, and nobody had told her. So she stayed a while on her knees in the cool of the morning, with the light coming up gold over the hill behind the town. She thought about the two girls in the tomb, who had been her age, or near it, and had come all the way from the end of the world carrying gifts for the god. They had died here, a long way from their mothers, and been buried under an olive tree by strangers. And she thought that in four days she

would be married, and would go to live in another house on the far side of the town, which was not very far, and was very far. She put out her hand and touched the spindle once more before she got up.

• • •

The boat from Tenos came in at midday.

She heard the shouting from her mother's courtyard and ran down to the harbour with half the town. The boat from Tenos only came once a year, at this season, before the feast, and everybody knew what it carried. It was a small boat with six oarsmen and an old man in a white cloak sitting very upright in the stern with something in his arms. It came in round the point under oars, slowly, while the priests of the god stood waiting on the quay in their robes and the crowd pressed in behind them.

The thing in the old man's arms was a bundle of straw.

That was all it was, to look at: a bundle of wheat straw, golden and dry, about the size of a sleeping child. It was wrapped round and round and tied with cords of the same straw twisted together. When the boat bumped against the quay the old man stood up with it, and two of the oarsmen helped him ashore. He walked up the quay to where the chief priest of the god was waiting, and stopped, and held it out.

"From the Carystians," he said, loudly, so that everyone could hear, "who had it from the Euboeans, who had it from the people of the Malian gulf, who had it from Dodona, who had it from the peoples of the Adriatic sea, who had it from those beyond, who had it from the Scythians, who had it from the Hyperboreans, who live beyond the north wind. The people of Tenos bring it to the god."

"The god receives it," said the chief priest, and took the bundle into his arms.

The crowd let out its breath.

"What's in it?" a small boy said beside her, too loud, and his mother hushed him.

"Nobody knows," Euphrosyne whispered to him. "Nobody opens it. That's why it gets here." Somebody began to sing, and others took it up, an old slow hymn that the women sang at the feast. The chief priest carried the straw bundle up from the harbour towards the sanctuary, past the marble lions on their terrace above the sacred lake. The priests and the people of Tenos and the whole crowd followed. Euphrosyne went with them, and found that she was singing too.

• • •

The priest who kept the old tombs was called Kleandros. He was very old, older than anyone on the island, and he had been Euphrosyne's grandfather's friend. He lived in a small house at the edge of the precinct and spent his days sweeping the steps of the temples and talking to anybody who would stop. She found him that afternoon sitting on a block of marble in the shade near the temple of Artemis, with his stick across his knees, watching the sparrows.

"I saw your hair on the tomb," he said, before she could speak. "You cut it too close to the ear. Your mother will scold."

"I couldn't get the knife to go."

"Nobody can, the first time." He moved along the marble to make room for her. "Sit. You've a question. You've had it since you were six."

She sat, and out of a great many questions she picked one.

"Where does it come from, really? The straw."

Kleandros looked at her for a long moment. Then he looked away, north, over the roofs of the treasuries and the white heads of the temples and the blue strait beyond, towards Tenos and Euboea and the mainland and everything that lay past them.

"From the Hyperboreans," he said. "From the people beyond the north wind. You heard them say it on the quay."

"But where is that?"

"Nobody knows," said Kleandros. "That's the honest answer, and I'll give you the honest answer, because you're nearly a married woman and you can bear it."

Nobody on the island had ever been there, he said. Nobody on Tenos had, or in Carystus, or anywhere along the road that he had ever heard of. The people of Tenos got it from the Carystians, and the Carystians from the next city along on Euboea, and they from the people of the gulf. Those had it from Dodona, in the mountains, where the oak tree speaks. Dodona had it from the peoples to the north-west, along the Adriatic sea, and they from peoples whose names he did not know. So it went all the way up into the north, to the Scythians, on the great grass plains where the rivers froze in winter so hard you could drive a cart over them. The Scythians got it from the Hyperboreans, who brought it to their border.

"And nobody, in all that long road, opens it. Each people takes it from its neighbour and carries it to the next and hands it on. That's the custom."

"Why don't they bring it themselves?"

The old man smiled.

"They did, once. That's the story. The first time, they sent it with two girls. Hyperoche and Laodice."

The girls were sent with gifts for Eileithyia, who helps women in childbirth, as thanks for an easy birth, so the Delians say. Five men of their people went with them to keep them safe on the road. They came all that way and brought it here. Then he lifted his hand and let it fall. They never went home. The two girls died here, and were buried under the olive tree where she had laid her hair that morning. The five men stayed, and were honoured, and their families after them, and the Delians called them the Bearers still.

"And the Hyperboreans, so the story goes, were angry. Not with us. With the road. They didn't want to go on sending their daughters and their sons away and never having them back. So after that they carried the gifts only as far as their own border, wrapped in straw, and asked their neighbours to send them on. And the neighbours did. And here it is."

Euphrosyne was quiet for a while.

"And before them?" she said. "Before Hyperoche and Laodice?"

Before them, he said, there were two others, Arge and Opis, who came long before with the gods themselves, when the island was new. The women sang to them at the feast in the old hymn that Olen made, the man from Lycia who made the first hymns of Delos. They collected gifts for them. When the thigh bones were burned on the altar, the ashes were carried round and scattered on their grave, behind the temple of Artemis, facing the sunrise. He nodded towards it. "You've seen it."

"I thought that was just a pile of ash."

"Everything's a pile of ash, if you don't know the story," said Kleandros.

• • •

Once, everybody knew, the island had floated loose on the sea. Then Leto, carrying the children of Zeus and driven out of every land by Hera's jealousy, came to it, and it took her in. It was fixed then to the bottom of the sea with pillars. Leto knelt under a palm tree by the round lake and clung to it, and gave birth to Artemis first and then to Apollo. And the island, which had been grey and nameless, became bright: Delos, the clear one, the one you can see, the island of light. There was a palm tree still growing by the lake, and people said it was the same one, though Kleandros said that was nonsense.

"Then what is it?" Euphrosyne said. "The gift. What's in the straw?"

"I don't know."

"You must know. You've seen it every year."

"I've seen straw every year." He rubbed his knees. When he was a boy he had asked the old priest who kept the tombs before him, who was older then than Kleandros was now. He had not known either, nor had the priest before him. "It's carried in, and laid before the god, and it's not opened. That's the custom." He looked at her sidelong. "There are people who'll tell you it's first fruits. Barley. Wheat. Because of the straw. That's a reasonable guess. Wheat comes wrapped in straw, so they say, and what else would you wrap in it?"

"But you don't think it is."

"I don't think anything. I keep the tombs." He smiled.

There had been a man here, a few summers back, he said, from across the water in Caria, over towards the sunrise. He was a talker. He had been everywhere, Egypt, Babylon, the Scythians, and he wrote down everything anybody would tell him, every story, and then argued with it. He had sat where she was sitting, and Kleandros had told him all of this, the road and the girls and the Bearers, and he had written it down. Then he had said that he knew himself that the women of Thrace and Paeonia, up in the north, have straw with them when they sacrifice to Artemis the Queen. He had seen it.

"So he believed that much. That the straw came from somewhere real." Kleandros chuckled. "And then I told him about Abaris."

"Who's Abaris?"

"Another one from the north. A Hyperborean, so they say, who went round the whole world carrying an arrow, and ate nothing the whole way. Some say he flew on it."

The man from Caria had put down his pen and laughed, and said he was not going to write that one down. If there were men beyond the north wind, he had said, there must be men beyond the south wind too, and nobody ever came here bringing straw from them. Kleandros laughed again, and then stopped, and was quiet. "He was a clever man. I liked him. But he only wanted to write down what he could believe. I've kept these tombs for sixty years, and I'll tell you, child, the things you can believe are the smallest part of it."

• • •

On the first day of the feast Euphrosyne sang with the chorus of girls.

It was the last time she would. The old blind poet's hymn, the one every child learned, said the girls of Delos knew how to sound like every people in the world, so that each man listening thought it was his own tongue he heard. She stood in the second row in her white dress with the other girls, under the sun, before the altar, and sang. When they came to the verse for the girls from the north she found she was singing it louder than the rest. Old Kleandros, on his block of marble at the edge of the crowd, had closed his eyes.

The feast went on for three days. On the third, the last, before the great sacrifice, the straw bundle was to be carried from the house where it had been kept, under guard, into the temple of the god and laid before him.

It had always been carried by the chief priest. But this year the chief priest was ill, with a fever that had come over on a boat from Athens, where, people said, a sickness was killing people in the streets. He could not stand. And the second priest had gone to Athens on the god's business and not come back. And so on the morning of the third day there was a short, sharp argument among the priests in the porch of the temple, which Euphrosyne could see but not hear. At the end of it old Kleandros came limping down the steps on his stick, looked through the crowd, found her, and beckoned.

Her mother pushed her forward, and she did not want to go, and she went.

"They want me to carry it," he said, very low, "because I'm the oldest, and the oldest is next after the priests. And I can't. Look at me. If I carry it up those steps I'll drop it, or I'll drop dead, and either one will be an omen they talk about for a hundred years." He took her wrist in his dry, light hand. "So I told them the girls from the north came here as maidens, and a maiden should carry it in. And they argued, and I'm old and I'm stubborn, and I won. You laid your hair on the tomb four days ago. You'll be a wife tomorrow. Today you're still one of them."

"I don't know what to do."

"You hold it, you walk, you lay it down. That's all anybody's ever done. Don't drop it. Don't open it." He let go of her wrist. "And don't be frightened. They were girls like you."

• • •

They put it into her arms in the porch of the treasury where it had been kept, and it was heavy.

That was the first thing: she had thought it would be light. She had seen the old man from Tenos carry it, and the chief priest. They had held it easily, the way you hold a bundle of straw. She had expected to

feel nothing in her arms but dry stalks and air. It was not like that. The straw was dry and light and pricked her through her dress. But in the middle of the bundle there was something that was not straw. It was heavy, much heavier than it had any right to be, so that she had to shift her arms and hold it close.

It was hard, and she could feel it through the straw against her ribs. It was hard and round and smooth, smaller than she would have guessed, about the size of a ripe fig, perhaps a little bigger, and it was cold. Through the dress and the straw, she could feel that it was cold, the way the water of the sacred lake was cold in the mornings.

She walked, and the musicians began, and the crowd parted in front of her and closed behind. She went along the paved way between the treasuries with the bundle in her arms. The priests came behind her, and the people sang the hymn of Olen that called the girls from the north by their names. She climbed the temple steps one at a time, because she could not see her feet. She passed between the columns into the cool and the dimness, where the god stood.

At the top of the steps she stumbled, only a little: her foot caught the edge of the last step and she lurched and caught herself. The bundle shifted in her arms, and at the bottom of it, where the straw had been tied off, the straw parted a little, the width of a finger.

She looked down, because she could not help it.

Through the gap in the straw she saw something smooth and grey, not gold. Not the gleam of a jewel, or the brown of grain, or the colour of anything that was precious, that she had ever been told was precious. Only grey, the grey of rain coming in off the sea, of a stone on a beach, of nothing at all. And as she looked at it, it seemed to her that it had stopped being cold and was warm, warm as her own skin, from her holding it.

She pressed the straw closed with her fingers and did not look again.

She walked the last few paces across the temple floor to where the priests had laid a white cloth on the stone before the god. She knelt, as she had been told, and laid the bundle down on the cloth, gently, with both hands, like you would lay down a sleeping child. Nobody

had told her what to say, and it seemed wrong to say nothing, so she leaned over it, so close that the straw touched her lips. She whispered to the girls from the north, to Hyperoche and Laodice under their olive tree, to whoever had wrapped it at the end of the world and sent it on its long road from hand to hand.

"What did you bring? And what did you want to take home?"

The straw said nothing. Behind her the music rose and the crowd sang the names of the northern girls. The priests came forward with the knife and the fire for the sacrifice. She got up off her knees and went back out between the columns into the hard white light.

• • •

Kleandros was waiting at the bottom of the steps, leaning on his stick. He looked at her face. Then he looked away, out over the sacred lake, where the palm tree that was not the same palm tree stood in the sun.

"Well," he said. "You didn't drop it."

"No."

"Heavy, was it?"

For a moment she did not answer. She thought about lying, and could not see why she should.

"Yes," she said. "Heavier than it should be. It's not wheat."

The old man nodded slowly, like a man hearing something he had always known and never been allowed to say.

"No," he said. "I didn't think it was."

"Do you know what it is?"

"No. I told you the truth, child. I don't know." He was quiet for a while. "But I'll tell you what I think, since you've carried it. I think something comes down that road from the end of the world every year, from people we've never seen, and passes through every hand between there and here without anybody opening it, and gets laid in front of the god, and I think that's the point of it. Not what's inside. That it comes. That it's still coming. That every people on that road, every year, picks it up and carries it and hands it on, and none of

them knows why, and none of them stops." He looked at her. "That's a kind of thread, isn't it? Like the hair you wound on your grandmother's spindle. It holds the year together."

Euphrosyne looked at the temple, white in the sun, and the crowd, and the smoke of the sacrifice beginning to go up into the blue.

"What happens to it after?" she said.

"That is a question for another day," Kleandros said. He took her arm, to help himself down the last step, though he did not need help. They went together through the crowd towards her mother. "Tomorrow you'll be married. Go and let them fuss over your hair."

• • •

She was married the next day, in the house on the far side of the town, and had children, and grew old on the island of light. In time her daughters went down to the olive tree in the mornings with spindles of their own. They laid their hair on the tomb of the girls from the north, as she had done. Every year, at the same season, she went down to the harbour with the rest of the town to watch the boat from Tenos come in, with its straw bundle in the arms of an old man in a white cloak. She sang the hymn of Olen with the other women, and called the northern girls by their names.

She never told anybody what she had seen through the straw. Not her husband, not her daughters, not the priests. It did not seem to be hers to tell. But on the morning before each of her daughters was married, when she gave her the little bronze knife and the olive-wood spindle that had been her grandmother's, she would take the girl's face in her hands and speak, as if it were part of the custom, which it was not. "Whatever they tell you is in the straw, it's heavier than that. Carry it carefully."

And far to the north, on roads that no one on Delos had ever seen, at the edge of some country without a name, somebody was already wrapping next year's gift in straw.

The master page

Delos, the clear one, the island you can see. Gary had laughed the first time he read that, because nothing about it was clear. Not the road, not the gift, not the girls. The only clear thing was that it kept coming.

The Ledger laid the page beside the two before it, and he saw what this one built. Orkney's page had made the count; the Plain's had made the gathering. This page made the relay: a thing handed on from people to people who would never meet, each one signing for it with nothing but their hands, and arriving on time because a count of nineteen years kept the feast where the season was. Count, gathering, relay. He could feel the machine taking shape and did not say so out loud.

The locks he made in ink were few. Meton's arithmetic: 235 moons come to 6,939.69 days, nineteen years of the sun to 6,939.60, a gap of about two hours. The Core could hold a rhythm like that in its sleep. So could a pocket calculator. He would type it in plainly in the morning, so nobody, including him, would mistake it for a discovery. And the ground at the olive tree, which agreed with the book.

The rest went in pencil. Diodorus' northern temple, and Stonehenge, and the Hyperboreans as his own Atlantic stone-builders: three true things, and however he stacked them he could not get a fourth out of them. The 18.6 at the Station Stones and the 19 at Athens, side by side with a gap between. Ahead, on shelves he had not reached, there was a clerk counting days north of Britain for Pytheas, a school at Croton where number and calendar were one craft, and an Egyptian page that would send him backwards a thousand years. The Ledger did not mind which way a road ran, only that it was signed for.

Down the side of the Core's page for Delos he would set the three things his framework was built to look for: a rhythm, a steadying hand, a memory with a long fuse. The yearly gift was the rhythm. The custom that nobody opened it was the steadying hand. And the memory was the girls, dead perhaps a thousand years by Herodotus's day and still sung by name. He could map that. Where the straw came from, the map would never know.

A small boy on the quay had asked the only question anybody ever asks about it: *what's in it?* Nobody on Delos knew, and nobody was meant to; that was the custom doing its job. What stayed with him was the choice in the story. The northerners sent their daughters once, and lost them, and would not do it again. They did not stop giving. They changed how they gave, and trusted every stranger on the road to carry it the rest of the way and not open it. Belfast had its own version: the parcel left in with the neighbour, the key under the mat for whoever was coming. You don't send your children into the dark for a god. You find another way, and you hand what matters over the hedge.

The nineteen years Diodorus gave were for the feast. Everybody on the road, from the Scythians to the boat from Tenos, had to be ready at the same season without a letter between them, and all Meton's arithmetic came down to was turning up when you said you would.

And the next page was already turning towards a hot room cut into a cliff in Egypt, where a scribe was waiting for a lamp.

Euphrosyne's words were the last thing on the page. He had made them up himself, and he still could not get rid of them: *Whatever they tell you is in the straw, it's heavier than that.*

CHAPTER 8

The Road Through the Duat

Thebes, and the Great Place in the western cliffs, in the seventy days after the death of Menkheperre Thutmose: the found year, 1425 BCE; and a temple in the Fayum, under the Romans

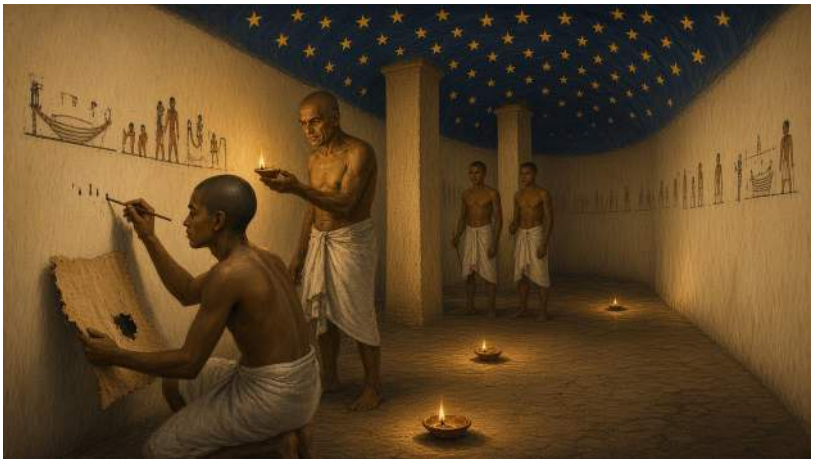


Illustration 9. *Pa-iry at the wall of the burial chamber, making two small strokes in a gap.*

Belfast, by day

AT SHANNONBRIDGE, GARY'S JOB HAD been to stand where the crane driver could not see and tell him what was happening at the other end of the hook. You said exactly what you saw. You did not guess, because a guess under a crane is how somebody gets hurt.

"When did Thutmose the Third die?" he typed.

"Thutmose III reigned from 1479 to 1425 BCE."

"How do you know? Somebody had to write it down at the time."

That was the right question, and the machine had no answer of its own, so he went and found it: the king's throne name was Menkheperre. His tomb, KV34, is cut high in a cleft at the end of the Valley of

the Kings, across the river from Thebes. On the walls of its burial chamber, shaped like the ring round a king's name, is the earliest complete copy of the Amduat, *what is in the Duat*, the Book of the Hidden Chamber. It is the sun's journey through the twelve hours of the night, painted on a yellow ground to look like an unrolled papyrus. His map said *c. 1450*, a round number, and he didn't trust round numbers, so he followed the moon.

He had his machines gather every dated line about the king, and checked them by hand. A machine gives you a date with great confidence and no shame, aye. The carved record was steadier: in the king's annals at Karnak, the battle of Megiddo falls in his twenty-third year, on a day of the new moon. The Egyptians kept a civil year of exactly 365 days, and a Demotic papyrus of the Roman period, Carlsberg 9, sets twenty-five of those years against 309 months of the moon.

"Twenty-five, not nineteen," he said.

"Correct. The lunar phases recur on the same civil date every twenty-five years."

And nearly every eleven, it added, so a new moon on a known day gives a short list of years.

That is why the books give three starts for the reign: 1504, 1490 or 1479 BCE. Most scholars now take 1479. *The method travels further than the number*, he wrote, and marked it *open*.

Then the end: an officer called Amenemheb wrote in his own tomb that the king died in his fifty-fourth year, in the third month of winter, on the thirtieth day. Count forward from the accession in 1479 and that gives 1425 BCE.

Now the walls.

"When was the burial chamber painted?" he asked.

"The Amduat in KV34 was painted during the reign of Thutmose III."

He asked it who said so, and it said most guidebooks did.

"Aye, well. The guidebooks weren't down there with a lamp."

The guides do say the chamber was painted while the king lived. But in 1975 John Romer saw that the decoration had been done after the four side rooms were sealed. He argued it was finished after the king's death, under his son Amenhotep II. And Herodotus, a thousand years later, says the embalming took seventy days.

He laid it out like a coroner. Body first: a king was dead in 1425, with seventy days before the burial. The chamber had its side rooms walled up and its walls bare, and a book had to be painted on them before the door was shut. He wrote *1425 BCE*, and the question mark, and under it: *Romer's reading, not everyone's. If the walls were painted in his lifetime, the dream is twenty years late and has no deadline.*

Who. A family: Thutmose III was the son of Thutmose II by a lesser wife, Iset, and for twenty years or so he shared the throne with Hatshepsut, his father's chief wife and half-sister. It was a dynasty that married inside itself, like Diodorus' northern kings. He asked if the genetic tree was solid.

"In 2010 scientists drew a five-generation tree from eleven royal mummies of the later dynasty," the machine said.

He pressed it on what the other geneticists had made of the tree.

It said other geneticists had questioned it, because DNA that old in that heat degrades and is easily contaminated.

"Fair play to them," Gary said. He wrote down the tree and the argument and did not choose. And much older, one man: in 2025 a team read the genome of a man buried in a pot at Nuwayrat between about 2855 and 2570 BCE. He was about four-fifths North African Neolithic and a fifth from the eastern Fertile Crescent. One man, the researchers said, not a nation. But enough to show the road from the Tigris to the Nile had been walked.

Where it was kept. The House of Life, the temple workshop where sacred books were copied. And from across the desert, the old story in his notes: *Inana and Enki*, where Enki keeps the *me*, the arts of civilised life, in his temple at Eridu. He gives them away in drink, and Inana sails them home to Uruk. Two houses, one question: where do you keep what you know, and who may carry it out? He wrote that under *interpretation*, because no text joined them.

He read the book the way its scribes had, and to them the Duat was not a hell. It was the hidden country the sun crosses every night to be made young again. The book was a working guide to it, whose captions promise that knowing it is of use to a man on earth as well as below. To a scribe the words were part of the road.

And the practice he loved most: Egyptian copyists, when their source was damaged, did not fill the hole with a guess. They wrote two signs that meant *found destroyed*, and went on.

He asked if it was real, because he wanted it too much.

"Yes. The phrase *gm wš*, 'found destroyed', is attested in Egyptian manuscripts."

He checked it anyway, against the published papyri, and it was. He didn't say the year out loud all day. Seventy days is a short time to paint a night, and he felt every one of them.

That night

He fell asleep reading the First Hour, and the library went dark the way a room goes dark when the lamp is carried out. The Ledger left it unlit and turned the page in the dark. He could not see the number at the top, only feel the page under his fingers, rough as papyrus. Then the dark thinned into the last of an evening on a flat roof, cooking smoke going straight up, and a young woman twisting old temple linen into wicks for her brother. He was sitting on the next roof with his chin on his knees, and the smoke went past him.

• • •

Meret twisted the wicks.

It was not her work. Her work was the fine linen, the royal linen, so thin you could read a scribe's hand through it. The women of the temple workshop wove it on the long looms in the shade of the colonnade from first light until the heat drove them in. But at the end of each day there were the scraps, the cut ends and the frayed selvages

and the old worn cloths sent back from the god's house. The overseer gave them to whoever was quickest. Meret was always quickest, because she had a brother who needed light.

She sat on the roof of the house in the last of the evening with the scraps in her lap and a pot of fat beside her. She twisted them one by one, rolling the linen between her palm and her thigh until it was tight and round, then dipping it and rolling it again. Below her the city was going quiet, and the smoke of the cooking fires went straight up into the still air. Across the roofs she could see the great gates of the god's house standing up gold in the last sun. Beyond them, on the far bank, were the cliffs of the west, where the sun went down and the dead went with it.

Pa-iry came up the ladder as she was finishing. He was seventeen and thin and inky to the elbows. He had the look he always had now, coming home, of a man who has been somewhere very dark and has not yet got used to the light.

"Twenty," she said, and held up the bundle. "That's tomorrow and the next day."

He took them and turned them over in his fingers. "These are better than the ones the stores give out."

"Of course they are. I made them." She wiped her hands. "Sit. Eat. Tell me what it's like down there."

He sat, and ate, and did not tell her, because he could not.

• • •

He had been chosen on the day the king flew to heaven. The House of Life was the school and the scriptorium at the god's house, where the sacred books were kept and copied. Of all the boys in it, he had been the one the old chief of the scribes had picked. His hand was steady, the old man said, and his eye was clean. Most of all, he did not guess: when Pa-iry did not know a sign he said so. When a copy was damaged he did not fill the gap with what he thought ought to be there. That was rarer than a steady hand, he said, and more use.

So Pa-iry had gone across the river to the west, to the Great Place, the hidden valley in the cliffs where the king's house of eternity stood cut and waiting with its walls still bare. He had gone down into it with the master draughtsman and the painters, carrying the book.

And before he went he had sworn the oath.

Meret had been there, at the edge of the courtyard of the House of Life, with the other families. She had watched her brother kneel before the old chief of the scribes and put his hands between the old man's hands and swear. She had not heard the words; nobody outside the House of Life ever heard them. But she knew what they meant, because everybody knew. What he learned in there, and what he saw in the king's house, he would not speak of outside it, not to his mother, not to his sister, not to any woman he might marry, not to anyone, as long as he lived.

"It's the book of what's in the Duat," he had said, the night before, the last thing he could tell her. "Of the hidden chamber. The road the sun takes through the night. They're putting it on the walls, every hour of it, before the seventy days are out, so that when they carry the king into the dark he'll know the way." He had looked at her with his serious inky face. "And after tomorrow I can't tell you anything else about it. Ever. You understand?"

"I understand."

"I mean it, Meret."

"I know you do," she had said. "That's why they picked you."

• • •

Every morning the way to the king's house went across the river by boat before dawn, and up through the fields and the desert edge on foot. Then they went into the valley, and along its floor between the high cliffs to the furthest of its branches. There was no shade and the heat came off the rock like the breath of an oven. The door was high up in the cliff, in a cleft where the rain came down in a waterfall once in a lifetime. You went up to it on a ladder, past the scratched names of the workmen on the rock.

Inside it was dark at once.

You went down, steeply, by stairs cut in the rock and corridors that bent away from the light. The air got hot and dead and still. There was a deep shaft halfway, square, cut straight down into the rock. You crossed it on a plank with your lamp held high and your eyes on the far side, and you did not look down. Then there was a hall with walls covered from end to end with names, hundreds of them. They were the names of every god and power that lived in the twelve hours of the night, set out in rows like the names of soldiers on a muster. And at the bottom of a last stair, at the very heart of the rock, there was the chamber.

Pa-iry could have described it to Meret, if he had been allowed to. He thought about it at night, lying on his mat on the roof, describing it to her in his head.

It was shaped like a king's name; that was the first thing. Instead of square like a room it was long, with the ends rounded, like the ring that is drawn round the king's name when you write it. To be in it was to be inside the king's name itself. The boy at the edge had a word for that shape, cartouche, from a book at home; nobody in the room needed one. Two square pillars held up the roof, and the four small side rooms were already stacked with the king's things and walled shut. The ceiling was painted dark blue and covered in yellow stars. And the walls, all of them, from the floor to above his head, had been washed with a pale yellowish ground, the colour of an old papyrus that has been unrolled many times. The whole room looked like a scroll, a great book unrolled round the walls of a cave, and you were standing inside it, reading.

That was what they were writing on. They were writing it the way the book itself was written, the way he had copied it in the House of Life. It was not in the great carved and coloured signs of the temple walls, but in the quick sure hand of a book, black and red ink. The fig-

ures of the gods were drawn as simple lines, like the drawings in the margin of a papyrus. As if the king were not being given a temple. As if he were being given a book to take with him.

• • •

The master draughtsman was called Amenemhat. He was fifty and bald and very fierce and had been painting the walls of kings' houses since before Pa-iry was born. He did not like boys from the House of Life, who knew the signs, he said, but not the walls.

"Read," he said, on the first day.

Pa-iry unrolled the book, which had come across the river in a sealed chest. It was old, older than the House of Life's oldest scribe. It was a long roll of papyrus gone the colour of honey, with the twelve hours of the night written out on it in columns. Each hour had its three bands, the boat of the sun in the middle band and the gods and powers of the hour above it and below. He held it up to the lamp, which was one of Meret's wicks floating in a dish of oil, and he read.

The book was written backwards, and that was one of its secrets, one of the things only the House of Life knew. The columns were set down in the reverse of the usual order. The hidden chamber was a reversed world, where the sun went backwards through time from old age to birth, and the writing went backwards with it. So Pa-iry read, and Amenemhat drew the guidelines on the wall in red, and then the painters came after with the black. The figures went up one by one in the lamplight, and the columns of words went up beside them, backwards.

The First Hour: the sun's boat entering at the western gate of the horizon, where Set stands against the riverbank. Pa-iry read that and stopped. He had always been told that Set was the enemy, and here he was in the king's own book standing at the gate. Amenemhat said, without looking round, "Keep reading, boy. The book knows its business better than you do." Then the baboons who sing as the god goes by, and open the doors for him. *Open your doors to me*, the god says to

the gods of the hour. Pa-iry read it out, and Amenemhat painted it. The baboons went up on the wall in a row with their arms raised in welcome, simple and black.

That was the first day, and there were not many days to do it in.

There was no ten-day week for them now, not with the seventy days running. Amenemhat kept the count on a sherd of broken pot by the shaft, one stroke a day. They went down every day, and every day began the same way: the plank over the shaft, the hall of names, the last stair, and the heat closing round them like a hand. The smoke of the lamps hung under the starry ceiling in a grey film, and the painters worked stripped to their short kilts. Pa-iry did not faint. He stood with the roll held open to the light and read, column after column, backwards. The words went on in his head when he lay down at night: the names of the hours, and of the gates, and of the powers who guarded them, the ones who were honoured and the ones who were destroyed.

In his head, on the roof, he went on describing it to Meret.

He told her about the Fourth Hour and the Fifth, the dry land of the hidden god, where the boat had to become a serpent and slide. The creatures there only heard the sun's voice go by in the dark. He told her about the Sixth Hour, the deepest, where the sun's soul went down to the bottom of the night and found its own body lying there, and joined with it. The whole of the dark held its breath.

He told her all of it, every night, in his head, and none of it out loud. And sometimes, when she looked up from her wicks and caught his eye, he had the strange feeling that she had heard it anyway.

• • •

"Is there a boat?" Meret asked him, one evening, on the roof, without looking up from her wicks.

For a while he kept quiet.

"Everyone knows there's a boat," he said at last, carefully. "The sun goes in a boat by day across the sky. Everyone knows that."

"And by night, when it's gone down behind the cliffs?"

"Everyone knows the sun goes under the earth at night, and comes up in the morning."

"In a boat? Under the ground, all night, in a boat?"

"I can't tell you, Meret."

She nodded, and went on rolling. After a while she said, "It's a long way, for a boat. All night. All the way from the west to the east, under everything." She held up a wick and looked at it in the dusk. "I'd want a good lamp."

He laughed despite himself, and then he did not, because it was too close.

"Is it dark?" she said.

"Meret."

"I'm not asking about the book. I'm asking about the king's house. You go in there every day. Is it dark?"

"Yes," he said. "It's very dark."

"Then that's all right," said Meret. "That's all I wanted to know." And she gave him the next day's wicks.

• • •

In the third week of the seventy days, the overseer of the weaving came down the colonnade with a scribe at her elbow and stopped at Meret's loom.

Meret did not stop, because you did not stop for the overseer; you went on, and let her look. The linen on the loom was the finest she had ever made. It was so fine that when she held a finished length up to the sky she could see the shape of the sun through it. It had taken her since the flood went down to weave a span of it the length of a man.

"This one," said the overseer to the scribe. "And the next two lengths she makes. Mark them."

The scribe made a mark on his board. Meret went on throwing the shuttle, over and under and back, and did not ask, because she knew better. But the overseer, who had known her mother, bent down as she went past and said quietly, by her ear, "For the House of Life. They'll

write on it, his son's words for him. It's to go round the king when they carry him into the dark." And she straightened up and went on down the colonnade.

That night on the roof Meret told her brother. She told him the way she might have told him that the price of onions had gone up, rolling her wicks, not looking at him. When she had finished he sat very still.

"I can't tell you what they'll write," he said at last.

"I didn't ask you."

"I know. I'm only saying." He picked up one of her wicks, and put it down again. "But it'll be beautiful, what they write. And it'll be on your linen. And it'll be with him. All the way."

Meret pulled the wick she was making tight and laid it with the others.

"Then that's two of us on the road," she said. "You on the walls, and me round him. That's all right." She looked out across the river at the black line of the western cliffs. "I'd want somebody I knew, going all that way in the dark."

On the fortieth day the book failed them.

It happened in the Seventh Hour, when they had come round the chamber wall by then, hour by hour. Now they were in the hour where the great serpent of chaos lies across the river to swallow the water and stop the boat. There it is cut and bound so that the boat can pass. Pa-iry had the roll open under the lamp, reading, and he came to a place where he could not read, and stopped.

The papyrus was damaged: some small creature, long ago, had eaten a hole the size of his thumbnail through two columns.

"Well?" said Amenemhat.

"It's gone, master. Two columns. Here, and here."

The old draughtsman came and looked, holding the lamp close, and scowled at the hole like it had insulted him personally.

"You know this hour," he said. "You've copied it. What goes there?"

"I think I know," said Pa-iry. "I think it's the name of one of the ones who binds the serpent. But I'm not sure. There are two it could be."

"Then pick one. Pick the likelier and we'll paint it and be done."

"I can't pick one, master."

"It's the king's road!" Amenemhat said. "You want to leave a hole in the king's road?"

"I don't want to put a wrong name in it," said Pa-iry. His voice shook, but he kept on. "If I put a wrong name in, it's there for ever. Every scribe who copies from this wall after us will copy the wrong name. Every king after this one. And nobody will ever know it was a guess." He swallowed. "There's a way of writing it. The House of Life uses it. When a copy's damaged, you don't fill it. You write *found destroyed*. So whoever comes after knows there was something there, and that we didn't have it, and that we didn't pretend."

Amenemhat looked at him a good while in the lamplight, while the painters stopped work and watched.

"*Found destroyed*," the old man said at last. "In the king's own chamber."

"Yes, master."

"They'll say we were careless."

"They'll say we were honest," said Pa-iry. "Or they'll say nothing, because they won't notice. But it'll be true."

There was a long silence in the hot dark. Then the old draughtsman snorted, and turned away, and picked up his brush.

"Write it, then," he said. "Small. And if the king's soul gets lost in the Seventh Hour because some beetle ate his road, I'll tell the gods whose fault it was."

So Pa-iry wrote it, small, in black, in the gap where the names should have been: two signs that meant *found destroyed*. And the painters went on past it, and the boat of the sun went on through the hour. The serpent of chaos was bound and cut on the wall, as it was every night under the earth, gap or no gap.

He did not tell Meret about that either. But that night, on the roof, he took one of her wicks and turned it over and over in his fingers. When she asked him what was the matter he said, "Nothing. I did something right today, I think. I'm not sure." And she said, "Then it was probably right," and went on rolling.

• • •

They finished on the sixty-sixth day, with the Twelfth Hour.

It was painted on the curved end of the chamber, at the far end of the king's name. That was where the great stone coffin of yellow quartzite would lie, shaped like the king's name too. Pa-iry read it out by the light of the last of Meret's wicks. His voice was not steady, and Amenemhat did not shout at him for it.

The Twelfth Hour was the end of the night, and the end of the road. In it, the boat of the sun was taken by the gods and drawn on a rope through the body of a great serpent. It went in at its tail and out at its mouth, the whole length of it. The sun that had gone into the west at evening as an old man, tired and ram-headed, came out of the serpent's mouth at the other end as a child. It came out as the beetle that rolls the morning up over the eastern horizon, new. And the old body, the night body, was left behind, at peace, lying at the edge of the world. And the doors of the east were opened, and the god came out into the light.

They painted the beetle last, and Amenemhat did it himself, at the very end of the wall, small, at the top of the curve. It was a few quick black lines for the body and the legs and the round of the sun pushed up before it. Then he stood back and looked at it with the lamp in his hand, for a good while, without a word.

Then he said, "There. He has his road."

• • •

Pa-iry came home very late that night, across the dark river, and Meret was waiting on the roof.

She did not ask him anything; she had a lamp lit, one of her own wicks in a dish beside her, and a bowl of bread and onions and beer. She watched him come up the ladder slowly, like an old man, and sit down beside her, and put his face in his hands.

"It's done," he said through his fingers. "It's finished. All twelve. He has it."

She put her arm round him and did not say anything for a while. Across the river the cliffs of the west were black against the stars. Somewhere in them, deep in the rock, in the dark, there was a room shaped like the king's name with a whole night written round its walls.

"Can you tell me one thing?" she said at last. "Only one. Not the names. Not the secrets. Just one thing, so I know."

He took his hands from his face and looked at the lamp, at the little flame on the twisted linen she had made, burning steady in the still night air.

He thought of the oath. He thought of the old man's hands round his hands. He thought of the hole in the papyrus and the two small signs he had written in it. And then he thought of the Twelfth Hour, and the beetle at the top of the curve. He found that there was one thing he could say without breaking anything at all, because everybody in the world already knew it.

"In the end," he said, "it's morning."

Meret was quiet, and then she laughed softly and leaned her head against his.

"Well," she said. "I could have told you that."

"I know."

"I'd have liked to see it, though. The beetle."

He looked at her. "How do you know there's a beetle?"

"Because I'm not a fool, and I've been to the god's house, and the sun comes up every morning like a beetle pushing a ball of dung, and every child in Thebes knows it." She took the bowl and put it in his hands. "Eat. You can't take an oath against bread."

He ate, and the lamp burned down, and the wick she had made from the god's old linen glowed at the end and went out. In the east, beyond the god's house and the city and the fields and the desert, the sky over the hills began to lighten, and a line of gold came up behind them. Gold had come up there every morning since the world began, whether there was a book written about it or not.

• • •

Fifteen hundred years later the king of Egypt was a Roman who had never seen the river. In a temple by the great lake in the Fayum, a young priest sat in the House of Life with a reed pen and a pot of black ink and copied a book that was already old.

It was written in the people's script, the quick joined-up hand that had grown out of the book hand over the centuries, and it was a conversation. In it a young man, whom the book called *the one who loves knowledge*, asked questions of a master. The master might have been Thoth himself, the ibis-headed, the baboon, the lord of writing. He asked about how to hold the pen and what the palette meant, and about the House of Life and the chamber of darkness where knowledge was kept. He asked about the sacred places of the land and a journey across the waters of the underworld. The master answered. The young man asked again. That was the whole of it, page after page, a seeker and a teacher going on talking, as if the talking itself were the thing being taught.

The young priest had never been to Thebes, never seen the valley in the western cliffs, or the room built in a cartouche's shape. He could not have read the old book hand on its walls without a great deal of trouble. But he knew, because his teacher had told him, and his teacher's teacher had told him, that the book he was copying was one of the ways a man was let into the House of Life. It was done by asking, and by being answered, and by writing it down without changing it.

Halfway down the column his copy failed him: the roll he was copying from was torn, and a line was missing.

He sat for a while with the pen in his hand. He thought he knew what the line said; he was nearly sure.

Then he dipped his pen and wrote, small, in the gap, the two signs that every scribe of the House of Life had learned to write since before anyone could remember: *found destroyed*.

And went on copying.

The master page

The Ledger did not shelve this page with the others. It laid all four open at once, the way the House of Life laid out an hour of the night, in three bands.

In the top band, the sky: the sun's year at Maeshowe and on the Plain, the moon's 18.6 years at the Station Stones, Meton's nineteen, Egypt's twenty-five, the twelve hours of the Amduat, and a new moon at Megiddo that fixed a king's years. In the bottom band, the families: Orkney's fathers keeping their line while wives came over the water, the man in the heart of Newgrange, the northern kings who kept it in the family, the Eighteenth Dynasty marrying inside itself. In the middle band, where the boat goes, the things that moved: grooved pots, a vole from the Belgian coast, pigs walked to a feast, a green stone from the far north-east of Scotland, straw from beyond the north wind, a man at Nuwayrat with Mesopotamia in his blood.

And he saw what this page added. Orkney had made the count, the Plain the gathering, Delos the relay. Egypt made the record: a count written down so that a moon seen at a battle can still date a king three and a half thousand years later, and written honestly, with the holes left showing. Count, gathering, relay, record. Four parts of one machine, and the book not half done. Ahead were an Irish monk told to leave a passage out of a tale, and the last priest at Philae carving the old script for nobody to read.

In the morning he would lay the bands in the Core, his reading glass over the master page: it shows where the lines rhyme, never who wrote them. Its three parts were the three bands, more or less: rhythms piled on rhythms, a steadying hand, a memory that returns late. His own audit reported its hits and misses at equal weight, and

his journal said it straight: *a compelling pattern is not automatically a mechanism*. The pass marks set for the Grail work went further: no figure from the Core counts as evidence for the history. The Core does not get a vote.

So he did what Pa-iry did. Wherever his old notes had joined two things with a line and no source, the Ledger wrote, small: *found destroyed*. Between the Altar Stone and the Hyperboreans. Between Meton and Maeshowe. Between Enki's house and the House of Life. Between Chaco and the Plain. It was the most honest map he had ever made, and it had more holes in it than any he had seen.

What was left was not a network. It was a habit. On an island at the top of the world and a chalk plain and a bright island in the Aegean and a hot room cut into a cliff, people had looked up at the same sky and counted, and checked their count, and handed it on. Ask a boy on a roof in Thebes and that was the whole of it: they tracked the stars so they would know the way in the dark, and be ready when the light came back.

Meret had kept a rule that nobody wrote down. She lit her brother's way and never asked him to break his oath. She asked him once, on the roof, what it was like down there, and then had the grace never to ask again.

The dedication at the front of his papers still carried the line he had put there for Tina. *If you don't quit, you can't lose.* Pa-iry would have understood it, and Meret better.

On the next shelf, a long way east, a boy who carried his verses by heart was setting out for a mountain, and his sister was following him against orders, with an ear for any word he might drop.

Over the roofs the sky had begun to go grey, and then pale, and the stone on the windowsill was the same colour. In the end, it's morning. The Amduat takes twelve hours of the night to say so, and he had sat up in the chair for every one of them.

CHAPTER 9

Sixty Days to the Mountain

The Kuru country between the two rivers, and the road north to the snow mountains, in the late Vedic centuries: the found year, 1000 BCE



Illustration 10. *Devadatta, Ila and the teacher reciting by the fire at the ford.*

Belfast, by day

HE WAS STILL STIFF FROM the chair when he thought of RGIS, where the rule on a stock-take was two counts of every shelf by two different people. If the two counts did not agree, a third person counted the shelf again while the first two stood and watched. Nobody trusted one count of anything, however good the counter was. A tired man in a cold warehouse will see what he expects to see and write it down with a clear conscience.

"How were the Vedas kept?" he asked the screen.

"The Vedas were passed down orally, word for word, for more than three thousand years before they were written down."

"Word for word. Says who?"

He sent it off to fetch the sources and then read every one of them himself. A machine will hand you a wonder with the same flat confidence it hands you a bus timetable. What he found, once he had read it all, was better than the wonder the machine had sold him. The hymns were learned from a teacher's mouth, and they were not only learned straight through. There was a way of saying them word by word, with every joint pulled apart and every word set down by itself. There was another way of saying them in overlapping pairs, one-two, two-three, three-four, so that a word dropped anywhere would break the pattern and be heard. The later manuals set out even more tangled weavings, forwards and backwards. In 2003 UNESCO listed the chanting as a heritage of humanity. The oldest written copies anybody has are the best part of two thousand years younger than the verses. It was a stock-take with the counts built in, and the counters were children. He liked that very much.

The map said Kailash. He went looking for the mountain in the old texts and it was not there. The hymns know the snowy mountains, himavant, and they know one mountain by name, Mujavant, where the best of the sacred plant was said to grow. The fever charm in the Atharvaveda sends the fever away to the Mujavants and the Gandharis, the far peoples of the north-west, like a parcel to an address nobody wants. Kailash itself turns up in the great epics, centuries later. So the dream would get a mountain, but not a name. He wrote *Kailash* in pencil and drew a box round it.

Then he went after the year a layer at a time, like taking up a floor board by board to see what is under it.

The first sheet was the land, and the land had moved. The later hymns, and the prose books that came after them, had moved east from the seven rivers of the Punjab. They had come into the country between the Yamuna and the Ganga, the Kuru country. In that same country the archaeologists find a fine grey pottery painted with black lines and dots, Painted Grey Ware, from around 1200 to 600 BCE. With it, from about 1000 BCE, comes iron, the dark metal, as the Atharvaveda calls it, to set beside the copper.

The second sheet was the sky, which is harder to argue with and easier to get wrong. The oldest calendar manual, the Vedanga Jyotisha, puts the winter turning of the sun at a particular star group, Dhanishtha. If you run the sky backwards, that was true somewhere around 1400 to 1150 BCE, and Pingree worked it out at about 1180. But the text itself is in late language, and Witzel and others put it in the last centuries BCE. The line may describe an old sky, or it may have been worked out on purpose later. The hymns had their own count already, a wheel with twelve rims and three hundred and sixty spokes. And there was Varuna, who knows the twelve months and knows the extra month that is born after them.

The third sheet was the road itself, and the one word that sat under every road in the old books. The word for a holy bathing place, tirtha, meant first of all a ford, a place where you could get across. The pilgrim's road to the holy waters and back is told at length in the Mahabharata, later again. The fords are older than the stories about them.

He laid the sheets one over the other and looked at what showed through. There were grey pots and the first iron in the Kuru country, an oral school with its counts built in, and a hymn-book that already knew the thirteenth month. Where they overlapped, he wrote it at the top of the page: *1000 BCE*, and gave it its question mark, which on this page was doing a lot of work.

Under that, the strongest doubt: *If the solstice line is a later sum and not an old sighting, the sky gives me nothing, and the year stands on pots and iron. And no text of the period names the mountain.*

"So the mountain's a guess," he said.

"The identification of the pilgrimage mountain with Kailash is traditional rather than Vedic."

"Aye, dead on. That's what I said, only longer."

He sat a while with the stone off the windowsill turning in his hand. It was the way he used to turn a shackle pin in a gloved hand at Kilroot, waiting for the driver to take the strain. He did not write down

what he thought the mountain was. He wrote the word *tirtha*, a ford, on a yellow sticky note and put it on the edge of the screen where he could see it.

That night

The word ford was the last thing in his head. When the library came, it was warm and smelled of woodsmoke and cow dung, and the Ledger was already turning a page, slowly, the way you turn a page somebody else is still reading. At the top stood the number he had written, 1000. Below it lay a yard of beaten earth under a thatched roof, with boys' voices saying the same words twice. He was a boy himself at the edge of the yard, behind a wattle wall, and a girl was there before him.

• • •

The boys sat in a row on the swept earth under the eaves of the teacher's house. The teacher said the verse, and the boys said it back, twice, and the teacher said it again.

Devadatta was the oldest of them, fourteen at the turn of the rains, and he had the longest memory in the yard. He had been in the teacher's house since he was eight, fetching firewood and water and cattle-fodder in the mornings. In the afternoons he learned the verses, the hymns of the teacher's family. They had come down from the teacher's teacher and from his teacher before him, mouth to ear, back past anybody's counting. He could say forty of them straight through without a stop. He could say them word by word, with each word lifted out and set down by itself, so that you heard where every word began and ended. And he could say them in the pairs, which was the hard way: one word and the next, then that word and the next again. It went all the way down the verse like a man crossing a river on stepping stones, putting each foot down twice.

Behind the wattle wall at the side of the yard, where the women of the house ground the barley, his sister Ila sat with the quern and ground, and listened.

She was eleven, and she was not supposed to listen. Nobody had ever said so, exactly, but the verses were for the boys in the yard and the quern was for her, and that was the way of it. She ground slowly when they were saying something she liked. The verse about the dawn she liked best. The dawn was a young woman who came each morning in her red and white and woke every creature that walks, and went on ahead of the sun, and never grew old.

That evening, by the cooking fire, Devadatta said the dawn hymn over under his breath to set it for the night. Ila said, without looking up from the pot, "It's *shining* in the fourth verse. Not *shimmering*. You said *shimmering*."

He stopped. He went back to the fourth verse and said it in the pairs, slowly, putting each foot down twice. The pairs broke where she said they would, like a step that was not there.

"How did you know that?" he said.

"I heard it." She stirred the pot. "Every day for three years, through a wall."

"You're not supposed to."

"I know." She tasted the barley and added salt. "How do you know you've got it right, Devadatta? When you're on your own, with nobody to hear you, and the teacher's asleep. How do you know?"

"You say it in the pairs," he said. "If a word's wrong, the pairs won't hold. You feel it go, like a stone turning under your foot."

"And if you've learned it wrong in the first place, both ways wrong, the same wrong?"

He opened his mouth and shut it again. "Then somebody else hears it," he said at last. "The teacher. One of the others. That's why there's more than one of us."

"Or a girl behind a wall," said Ila, and gave him his bowl.

• • •

The news came at the dark of the moon, with a cart from the east.

The teacher's eldest pupil, a grown man with a house of his own now, came to sit with the teacher and talk late. Devadatta, lying awake on his mat under the eaves, heard most of it. The sacred plant that the

priests pressed for the great rites, the plant of the hymns, had become hard to get. What came down the roads from the north now was dry and old and mixed with other stalks. The priests of the king's household had begun using stand-ins, and saying the stand-in was as good. The teacher said it was not. He said the hymns named the mountain where the true plant grew, and somebody ought to go north while they still knew the road. They should bring some back and press it once properly, so that the young ones would know the taste of it and could tell the difference ever after.

"You're too old for the mountains," the eldest pupil said.

"Then I'll take a young back to carry it," said the teacher. "And a young mouth to carry the verses, in case mine fails on the road."

In the morning he chose Devadatta.

Ila heard it from behind the wall, and kept it in until night. Then she came and sat at the end of his mat in the dark.

"How far?"

"Sixty days to the mountain, the teacher says. If the rivers are down."

"And back."

"And back." He could hear her breathing. "You can't come, Ila. It's not for you. Father would never let you. The teacher would never let you. You know that."

"I know that," said Ila. She was quiet for so long that he thought she had gone to sleep sitting up. Then she said, "You'll count the days in verses. I know you will. One hymn every night, so the hymn tells you the day. Forty hymns, forty days."

"There are sixty-two in the teacher's book."

"Then you'll get there with two to spare," said his sister, and went back to her own mat.

• • •

They went north at first light, three days after the new moon. The teacher rode on the ox-cart with the provisions and the pressing stones wrapped in cloth. Devadatta walked beside the ox with a goad, and a carter who knew the first part of the road would leave them at the hills.

The country went by slowly, the way country does behind an ox. Barley fields, and then rice in the low wet places, and the villages of thatch and wattle with their cattle pens of thorn. In every village there was the same grey pottery that his mother kept for best, thin and fine and the colour of a pigeon's back, painted with black lines and rows of dots round the rim. The women brought them water in it, and curds, and once a little honey. The smiths in the bigger villages had the new metal, the dark metal. Devadatta stopped to look at a sickle of it hanging over a doorway, black and plain and harder than the copper his father kept, until the teacher called him back to the cart.

Each night, by whatever fire they had, he said one hymn of the teacher's book from first verse to last. The next night he said the next one, so that the hymn was the day. The teacher listened with his eyes shut and corrected him twice in the first ten days and not at all after that. The carter, who had never heard the verses said so close, sat on the far side of the fire with his mouth a little open.

On the seventeenth night the teacher did not listen. He lay under the cart wrapped in his blanket, shaking, with his teeth knocking together. In the small hours he was burning hot and talking to people who were not there.

The carter looked at him and said it was the fever of the hills, the one that came every third day. He said it the way people say the name of a dog that bites.

• • •

In the third week they came to the big river. The road went down a cut bank to a ford, and the brown water spread out wide and shallow over a bed of pale stones.

Devadatta was at the water's edge with the ox, letting it drink, when he heard someone shout his name from the road behind.

He turned round and it was Ila, coming down the bank at a run, barefoot and filthy to the knees, with a bundle on her back and her hair full of dust. She stopped in front of him and stood there breathing. She did not say anything, because there was nothing she could say that he would not already know.

"How long?" he said at last.

"Since the morning you left. I stayed a day behind. I slept in the villages." She swallowed. "The women give you a place by the fire if you say you're following your brother. They all think it's lovely."

"It is not lovely." He was so angry that his voice cracked in the middle, the way it had started to do that year. "You'll go back. Today. The carter goes back from the hills; you'll go with him. Father will be out of his mind. The teacher…" He stopped.

Ila had seen the teacher, lying on the cart under his blanket, grey in the face, with his eyes closed.

"What's wrong with him?"

"The fever. Every third day."

She went past Devadatta to the cart without a word. She put her hand on the teacher's forehead and then on his neck, the way their grandmother did, and turned round.

"Has he had water, or the kushtha root to chew, or the charm said over him?"

"I don't know the charm," said Devadatta. "It's not in our book."

"It's not in yours," said Ila, and began to say it under her breath, there by the cart. They were the words every house knew half of. The fever was sent away to the north, to the Mujavants, and to the far Balhikas, its own people, its kin, with its sister the cough and its cousin the itch, and never to come back. Devadatta, listening, realised that they were walking towards the fever's home, and did not say so.

That was when the old woman spoke. She was sitting on a flat stone at the edge of the ford, where she must have been sitting all along, though he had not seen her. She wore a faded cloth, and she had a grey painted bowl of the good kind in her lap and a stick beside her.

"Ila," she said. "You took your time."

Ila looked round, and so did Devadatta.

"How do you know my name, grandmother?"

"I heard it," said the old woman, comfortably, as if that settled it, and held out the grey bowl. It was full to the brim with water from the ford, clear water, the silt gone to the bottom. "Here. Take it up the mountain for me. Carry it carefully; it's a long way. And when you come back down past this ford, you'll tell me. What did you carry out, and what do you bring back?"

Ila took the bowl in both hands. She looked at it, at the old woman, at her brother.

Devadatta stood at the edge of the river with the goad in his hand and the whole of his duty in front of him like a wall. A girl did not go to the mountain, and a girl did not hear the verses. His father would beat him for letting her come a step further, and the teacher, if he lived, would never forgive him. All of that was true. And on the cart the teacher was burning, and Ila was the only one of the three of them who knew what to do about it.

"She comes," he said to the carter. "As far as the mountain. And back with me."

The carter only shrugged; it was not his sister.

On the cart, the teacher opened his eyes. He looked at Ila steadily, and at the bowl, and he said nothing at all, but he did not say no.

"The river hymn, boy," he said then, quite clearly. "Before we cross."

So Devadatta stood in the shallows with the ox and said the hymn where the poet asks the rivers to bend down and let the carts across. The rivers answer him and bend, the way a mother bends to her child. Then they went over the stones of the ford with the water at their knees, Ila in the middle with the bowl held up against her chest, not spilling a drop.

When he looked back from the far bank, the flat stone was empty.

• • •

It took them another twenty-six days to reach the snow.

The fever came and went on the third day, and Ila nursed the teacher through it with the kushtha root and broth and the charm. On the good days he walked a little, and on the bad days he rode. The carter left them where the hills began. They took two porters from a hill village, short broad men who spoke a language Devadatta could not follow and laughed at the ox. Then they left the ox too, and went up on foot, with the pressing stones on a porter's back and the grey bowl in Ila's hands.

Above the last village there were no fields, only pines, and then no pines, and then stone, and the cold got into everything, the blankets and the bones and the sound of their voices. The nights were very clear, and Devadatta had never seen so many stars; there were stars in places where at home there was only dark. He said his hymn to them every night, shivering, with his hands under his arms.

On the fifty-ninth night, high on the mountain, in a hollow under a cliff of snow, his memory broke.

He was in the middle of the fifty-ninth hymn, with the teacher wrapped up beside the fire with his eyes shut, listening, and the porters asleep. He came to the seventh verse and it was not there. It was simply not there, like a rung missing from a ladder, so that his foot came down and went through. He went back to the start of the verse and came at it again and it went again. He tried the pairs, and the pairs ran out in the same place. He sat there with his mouth open and his eyes stinging and the cold in his throat. The teacher did not open his eyes, and the silence went on and on.

Then, very softly, from the other side of the fire, Ila said the verse.

She said it straight through, the whole of it, in her plain grinding-stone voice. She said it exactly as the teacher said it, with the voice rising and falling in the right places. Then she stopped and looked at the fire, and he could see that she was frightened.

Without looking at the teacher, Devadatta said the verse after her, word by word, lifting each word out and setting it down. Then he said it in the pairs, one foot down twice on every stone, and the pairs held, all the way to the end. He said it again, and they held.

"Again," the teacher said from under the blanket, without opening his eyes. "And then the eighth."

So he said it again, and after it the eighth verse and the rest of the hymn. Nobody said a word about where the seventh had come from, then or ever.

• • •

In the morning, where the snow ended at a hollow of dark stones, there was a spring, the coldest water Devadatta had ever touched. Among the stones above it grew a low plant with jointed stalks. The teacher knelt, holding on to his pupil's arm, and said it was the one. They cut it carefully and wrapped it in wet cloth, enough for one pressing. Then the teacher sat on a rock in the thin sun and said the first verse of the hymn of the pressing over it, and could not go on for a while.

Ila went down to the spring with the grey bowl.

She tipped the water from the ford into the spring, slowly, all of it, so that the river water and the mountain water ran together over the stones. Then she knelt and filled the bowl again from the spring, clear and freezing. She stood up with it held against her chest.

"That's what you carry out," she said to her brother. "And that's what you bring back. The old woman will want to see it."

"She might not be there."

"I think she'll know," said Ila.

• • •

They came down past the ford on the hundred and third day, by the teacher's count and Devadatta's hymns, which agreed. The flat stone at the edge of the water was empty, and they waited by it for half a morning, and nobody came. A man fishing downstream said there was no old woman at this ford that he knew of; there was an old woman at the next village who sold curds. Ila went and asked her. It was a different old woman, with a different voice.

So Ila stood at the ford with the bowl of mountain water and said it to the river instead, because somebody had to be told. "I carried out the river," she said. "I bring back the mountain. And a verse."

"You can't say that last part to anyone," said Devadatta. "Not at home. Not ever. You know that."

"I know." She looked at him, and for a moment she was not eleven, she was a great deal older. "But you'll say it. The seventh verse. Every time you say that hymn, all your life, you'll be saying it the way I said it on the mountain. That's enough."

He knew that it was. He also knew that she would go back behind the wall, to the quern, and he would go back to the yard. Nobody would ever write down which of them had held the verse on the mountain when it fell. He took the bowl from her, carefully, for the last part of the road, so that she could walk with her hands free.

The master page

The Ledger laid the page on the shelf beside the others and did not need to hold it down. It lay flat by itself, like a thing that has been carried a long way.

In ink: the hymns were kept by mouth, and kept with checks built in. Word by word, in pairs, in the later weavings, by more than one voice, until the written copies came the best part of two thousand years after. The grey painted pots and the first dark iron in the Kuru country. The extra month that is born after the twelve. A tirtha was a

ford before it was a holy place. In pencil: that the mountain was Kailash, which no text of the period names; and that the plant on the mountain was the plant of the hymns, which nobody now can identify for certain. The old woman at the ford went in pencil too, in the margin where the book keeps the ones who greet travellers by name, because no source has her, and the book needs her.

And he saw what this page added. The House of Life had made the record: the count written down, with the holes left showing. The mountain road's part was the recitation: the count kept in the mouth instead of on the wall, and checked by its own pattern, so that a word gone missing breaks the pairs and is heard. It is a record that walks. It can cross a river and climb above the snow and come back down. It can also fail at night on a mountain with nobody to catch it but whoever happens to be listening through the wall. A pencil line ran forward from it, a long way, to a teacher at Nadia handing the same verses to a clerk who wanted them on paper.

Next day he would open the Core and lay the rhythms side by side: the twelve rims and the three hundred and sixty spokes, the extra month, the five-year round of the calendar manual, the nineteen, the twenty-five, the eighteen point six that is not nineteen. The Core can tell him whether the numbers rhyme. It cannot tell him whether a girl ever said a verse on a mountain, or whether a girl's ear was ever the third counter at a stock-take. Only the girl could have told him that.

How do you know you've got it right? You say it in the pairs. You let more than one person hear it. And you do not mind, much, who the other person is.

Ila would have said the count was for the walking: how many days out, which moon to turn for home on, when the extra month was due before the rains came wrong. What the dream showed him was the people holding it, and that some of them were behind a wall. The stock-takes had been the same. The figure that saved the night was usually got by somebody who was never on the sheet, and the best you could do was say their name when you handed the count in.

On the other side of the world and more than two thousand years on, in a dry country of flat-topped rock, a girl was carrying seed corn up a path to a village on a mesa, and waiting to be asked what she had brought.

Down in the margin, under the old woman at the ford, the Ledger had left room for one more name. He did not know whose it was going to be, and he did not fill it in.

CHAPTER 10

Footprints on the Mesa

Orayvi, on the Third Mesa, in the years of the great drought, with the family come down from the country of the cliffs: the found year, 1290 CE



Illustration 11. *Dawn from the mesa edge, with footprints across the rock.*

Belfast, by day

AT SHANNONBRIDGE HE WAS THE spotter, the one man on the ground whose whole job was to watch the load and tell the crane driver what it was going to do before it did it. The driver could not see round the side of the boiler house, and the lads on the ground could only see their own wee corner. Everything went through the spotter's eyes and hands. The thing the old hands drummed into him was that you call it early. If you wait until the load has swung before you say so, you have already lost it. You learn where it will be in a few seconds' time from where it is now. You call that, and you are nearly always right, and when you are wrong you say so at once.

The spotter's job was still in his head when the answer about the Hopi landed on the screen.

"The Hopi are descended from the Anasazi of Chaco Canyon," it said. "Their sun priests used the Sun Dagger on Fajada Butte as a precise solar calendar, and still use it."

He sent the machines off for the papers and read what they brought home by hand, the beach stone pinning the corner, and he went slower than usual, because this one was not only his to read.

The first thing he did was say plainly to himself what he would leave out. Some of the old journals from the end of the nineteenth century printed things about Hopi ceremonies that were never meant for strangers. The Hopi Tribe has its own preservation office and has asked, more than once, that such things be kept out of books. So he kept to what the Hopi have published themselves, and to what the astronomers measured standing at the edge: where the sun came up, and when the planting was called. Nothing from inside the ceremonies went onto the page, and nothing would.

Then he put a line through *the Anasazi*, because it is not a Hopi word, and the Hopi do not use it for their ancestors. Leigh Kuwanwisiwma, who ran the tribe's preservation office for many years, wrote that the Hopi name for Chaco is Yupköyvi. It was one of the places, he wrote, where clans lived for a while on the long migrations before they came to the mesas. The Hopi call the old ruins footprints, and they mean it. He wrote *one of the footprints* beside the line.

Next went *a precise solar calendar*, with a note under it. In 1977 three researchers found that three slabs on Fajada Butte throw a dagger of light across a spiral at midsummer. An astronomer called Zeilik went back in 1985. He found that it would not work as an accurate calendar, and that it was more likely a shrine to the sun. That argument is still going on. *Still use it* went the same way. Nobody at Hopi watched the sun from Chaco; they watched it from home.

That part is solid ink, and it is the spotter's job exactly. Alexander Stephen, living at First Mesa in the 1890s, wrote down that a sun watcher stood at a fixed place every morning. He watched where the sun broke the line of the eastern hills, which had names. McCluskey,

working through Stephen's journal and later reports, showed that the watcher did not wait for the sun to reach a mark. He called it days ahead, from how far it had still to go, so that people had time to get ready, and he checked himself as he went. The watching is in the record for the 1890s. For 1290 it is a pattern laid back over the years, nothing more.

He started with the trees, the nearest thing to a witness that century had left, and the bottom of the pile was the great drought in their rings, from 1276 to 1299. Over it went the country of the cliffs to the north and east, emptied by about 1300. On top lay Wesley Bernardini's reading, built with Hopi knowledge, of clans coming in to the mesas one by one from 1275 onwards, each on its own road. The year all three would allow went at the head of the page, *1290 CE*, and after it the question mark, with a bit of space left beside it, like a seat kept for somebody who hadn't arrived yet.

The doubt went in below: *Orayvi was already old by 1290, but nobody can name a clan that came that year, and the Hopi tell their own histories their own way. The sun watching is from the 1890s and laid back by pattern. The girl is mine, and so is her family. I have given her no Hopi name, because it is not mine to give*.

From the map he took away the four roads, which were somebody else's book and not the Hopi's. He gave the girl a mesa, a drought, and an uncle who could watch the sun. A spotter, aye, calling it before it happens. That part stayed in his head and off the page, and he moved the stone aside so he could see the hills drawn there.

That night

The photograph of the mesa edge slid down to his knee as he went, the line of the eastern hills ruled across it in pencil. Dust and juniper smoke rose round him, and the dry smell of corn stored too long. The

Ledger spread the page on sun-warmed sandstone. It moved it along the edge, from one notch in the skyline to the next, the way you follow a line with your finger so as not to lose it.

• • •

The girl came up onto the mesa at the end of a long walk, behind her grandmother. She had a bundle of seed corn on her back and her feet bleeding a little inside her sandals.

She was ten, and she had walked for most of a year. Her family had come down out of the country of the cliffs to the north, where the springs had gone to mud and then to nothing. Up there the corn had stood up green in the spring and died yellow before it could set. They had stopped in other places on the way, and stayed a season, and gone on. Now there was a village on the mesa top in front of them, grey stone on grey rock, with smoke going straight up into a white sky.

They did not go in. Her mother stopped them on the path below the houses. They put their bundles down and sat on them, all eleven of them, and waited. The girl did not know what they were waiting for.

"Why don't we go up?" she asked her mother.

"Because it's their house," her mother said. "You don't walk into somebody's house. You wait at the door until they ask you what you've brought."

The girl looked down at her bundle. It was corn, the last of the seed, which nobody had been allowed to eat even on the worst days.

"Is that what we've brought?"

"That's part of it."

She thought about the long walk behind them, and the places they had stopped and left. "Where do we stop?" she said. "In the end. Where do we stop?"

Her mother was quiet for a while, looking up at the houses on the edge of the rock. “Where they let us,” she said. “If they let us.”

• • •

The men from the village came down in the afternoon, three of them, older men with their hair tied back. They sat down on the rock facing her family, and nobody said anything while the shadows shortened.

Then the oldest of them asked who they were and where they had come from. Her grandmother answered, because she was the eldest of their clan. She told the long way they had come, place by place, from the country of the cliffs. She named the springs they had drunk at and the places they had stayed, and the girl was surprised at how many there were. Her grandmother said the names slowly, so that the men could know them. The men listened hard, every name a thing put down on the rock in front of them.

When she had finished, the old man asked what they could do.

Her grandmother looked at the girl’s uncle, her mother’s brother, who was sitting at the end of the row with his hands on his knees.

“He watches the sun,” her grandmother said. “Where it comes up. He can tell you the day before it comes, so you know when to plant, and when the cold is turning. He learned it from my brother, who learned it from our grandfather. We carried it with us.”

The old man looked her uncle up and down, slowly.

“We have a man who does that,” he said at last. “We’ve always had one.”

“Then he can show yours what he sees,” her grandmother said, “and yours can show him what he sees, and you can see whether they agree.”

The old men talked among themselves in low voices. Then the oldest one stood up and told them they could stay below the village for now, by the spring at the foot of the rock, and plant the strip of sand he would show them, and they would see.

• • •

That night they lay on the sand by the spring under more stars than the girl had ever seen. Her grandmother told the story she told at every new stopping place, so that nobody would forget it.

A long way back, she said, before her own grandmother's time, there had been great houses in a canyon far to the south and east, bigger than anything the girl had ever seen. They had walls four storeys high, and hundreds of rooms, and round rooms dug into the ground. They were built in the shape of a half moon. Roads ran out from them in straight lines across the country. They went up cliffs and over hills, as if the people who made them did not care about going round anything. Some of the walls were set to the sun, her grandmother said, so that on certain days the light came through the gaps in a certain way. The people knew where they were in the year.

"Who lived there?"

"Our people lived there for a while," her grandmother said, "and other people's people. Lots of clans. Then they left, a long time ago, before any of us, and went north to the cliffs, and some went other ways. Nobody lives in the great houses now. There's only wind."

"Why did they leave?"

"Because it was time to go on." Her grandmother pulled the blanket up round the girl's shoulders. "That's what the old people said. You stay somewhere until it's time to go on, and you leave your footprints there, so the ones who come after know you passed. The great houses are footprints. The cliff houses are footprints. Every place we stopped this year, we left a mark on the rock."

The girl had seen her uncle do it, at every stopping place, scratching the sign of the clan on a rock face with a hard stone before they left, so that it would be there after them.

"Will we leave footprints here?"

"If they let us stay," her grandmother said, "this is where the footprints stop for a while."

• • •

Before dawn the next morning her uncle shook her awake and put his finger to his lips. She went with him up a narrow path in the dark to a ledge on the east side of the mesa, below the houses, where a flat rock stuck out over the drop.

He sat down on the rock with his back very straight. She sat beside him with her knees pulled up under her blanket, and they waited for the sun. Below them the land went away east for ever, grey and then blue as the light came. At the far edge of it was the line of hills, flat in some places and broken in others, with notches and humps and one high peak away to the north.

"Watch the line," her uncle said. "Not the sky. The line."

The sun lifted out of a low place between two humps, a little to the south of east, so bright at once that she had to look away.

"Now," her uncle said. "Where it came up. Remember that place."

He showed her how to remember it. You did not look at the sun, which burned. You looked at the hill beside it. You fixed it with your body, sitting in exactly the same place on exactly the same rock, so that the hills did not move and only the sun did. Every morning, he told her, the sun comes up a little further along the line. In the winter it walks south, and then it stops, and stays in its house for a few days, and turns round and walks back north. In the summer it walks north and stops in its other house, and turns again.

"How do you know where it'll be tomorrow?"

"Because I know where it was yesterday," her uncle said, "and the day before, and the day before that. It walks the same road every year. Not fast, not slow. If you know how far it's gone, you know how far it's got to go."

She asked him how he knew the names of the hills here, when they had never been here before. He laughed quietly and said that he didn't. He would have to learn it before anything else. Their hills at home had had names. These hills were strangers.

• • •

The village's own sun watcher came down to the ledge three days later. He was an old man with a bad hip who walked with a stick, and stood looking at her uncle on the flat rock a good while, saying nothing.

Then he sat down beside them, which the girl had not expected, and began, very slowly, to name the hills.

He named them from south to north, every hump and notch and gap along the line, the way her grandmother named the springs on their road. Each one had a name, and some had stories that he told in a few words and some he did not tell at all. Her uncle listened with his eyes half shut, and said each name back after him. The girl said them too, under her breath, so that she would have them.

When he had finished, the old watcher pointed. It was a notch a good way north of where the sun was rising now.

"When it comes up there," he said, "we plant the early corn. How many days?"

Her uncle studied the line, at the place where the sun had come up that morning and the notch the old man was pointing at, and at the hills between.

"Twenty-two days," he said. "Twenty-three, maybe. The road's longer there, where it bends."

The old man let it be. He got up on his stick and went back up the path to the village. Her uncle sat on, and the girl did not know whether that had been the right answer.

• • •

Every morning after that they went to the ledge before dawn, her uncle and the girl. Every few mornings the old watcher came down and sat with them.

The two men did not talk much. Her uncle said where he thought the sun would be in five days. The old watcher said where he thought it would be, and then they waited and saw. Most days they agreed. On the days they did not, the sun showed which of them was right, and the one who was wrong said so, and nobody minded. The girl liked those mornings best. She liked the way the two old heads turned together when the light came, like birds on a wall.

She learned the names of all the hills, the village names. She taught them to her little cousins on the sand by the spring, along with the names of the springs on their own road from the cliffs. Some nights she lay awake and said them both, the village's hills and her family's springs, one after the other. She found that she could hold the two of them at once without dropping either.

The drought did not break. Clouds came up in the afternoons over the hills, high and white, and went away again without rain. The corn in the village fields stood short and pale. Her mother carried water up from the spring to the little plants in their strip of sand, one jar at a time. The girl carried the small jar beside her. Some of the other families in the village had gone on, people said, to the south, where there was a river.

The girl asked her grandmother whether they would have to go on too, and her grandmother said that it was not time yet. She said the sun had not told anybody to go.

• • •

On the twenty-second morning the sun rose just short of the notch, so close that its edge seemed to touch the rock, and the girl held her breath.

On the twenty-third morning it came up in the notch itself, round and red and filling the gap in the hills exactly, like a lid set down on its pot. The old watcher was there, and her uncle, and the girl between them, and none of them said a word until it had cleared the line.

Then the old watcher laughed, a short dry laugh, and put his hand on her uncle's shoulder.

"The road's longer there, where it bends," he said. "I've been telling them that for forty years, and they never believe me."

He went up the path to the village faster than the girl had ever seen him go, leaning on his stick. Not long after a man stood on the highest roof and called out across the houses that it was time to plant the early corn. The girl heard it from the ledge, thin on the wind. She knew without anybody telling her that the call had been made a long way back, on the first morning, when her uncle had said twenty-two days or twenty-three.

• • •

That evening the old men came down to the spring again and sat on the rock facing her family, as they had on the first day.

The oldest said that they could build against the north wall of the village, where there were rooms nobody had lived in since the last family went on. He said that her uncle could sit on the ledge with their own watcher, while the old man's hip still let him climb. After that he could sit by himself, until he had taught somebody. He said that the seed corn should go in the ground with theirs, in the village fields, in the morning. There was no point in keeping it in a bundle any longer.

Her grandmother thanked them, and the girl's mother cried a little, very quietly. Her uncle sat with his hands on his knees and looked east at the line of hills going dark.

The next day, before they moved their bundles up, the girl went with her uncle to a smooth rock face at the foot of the path. She watched him scratch the sign of their clan on it with a hard stone, as he had done at every place they had stopped. This time, when he had finished, he gave her the stone. He let her do the last line herself.

"Is it still a footprint," she asked him, "if we're not going on?"

Her uncle thought about it, the way he thought about where the sun would be.

"It's a footprint for whoever comes next," he said. "So they know we stopped."

• • •

On the first night in the new room, the girl climbed out onto the roof after everyone was asleep. She sat with her knees under her blanket and looked east.

There was no sun, only stars, and the line of hills was a black edge against them. But she knew where the notch was without seeing it. She knew where the sun would come up tomorrow, a little further along than today, and the day after that a little further still, all the way to its summer house in the north, and then back. She said the names of the hills along the edge, from south to north, the village's names, and then the names of the springs on the road from the cliffs, her family's names, the way she had every night.

"Where do we stop?" she said to herself, very quietly.

Here, she thought, until the sun says otherwise. And when it did, they would leave a footprint, and go on.

The master page

Along the mesa edge the Ledger set this page on warm rock, and ruled one line across it, the line of the eastern hills, with the notches marked and every mark named in a hand the Ledger did not write.

In ink: the great drought in the tree rings from 1276 to 1299, and the country of the cliffs emptied by about 1300; clans coming in to the mesas over generations, each on its own road, as Hopi tradition tells and Bernardini's archaeology bears out; Chaco named by a Hopi writer as one of the places clans stayed and left; the Fajada Butte daggers found in 1977 and doubted as a calendar in 1985; and the horizon watching recorded at First Mesa in the 1890s, with the call made days ahead. In pencil: a sun watcher at Orayvi in 1290 doing what the record shows a century and more ago; the family from the cliffs; the twenty-three days. *Found destroyed*: whatever the people of the great

houses knew about the light in their walls, which went north with them and was not written; and everything from inside the ceremonies, which is not lost but is not the Ledger's.

The new page sat beside the old ones. Egypt had written its count down and the mountain road had sung its own. The mesa's part was the watch: the year read off one line of hills from one place, every morning, by one person who calls the day before it comes and says at once when he is wrong. It is the spotter at Shannonbridge, calling the swing before it swings. Behind it, already in ink, stands the mountain road, where the count lived in the mouth and the order was the check. And back in a different hand is the line his old notes drew between Chaco and the Plain, which stays *found destroyed*; Chaco does not need the Plain to be worth a page. Still in pencil ahead lies an ocean with no hills on it at all, where the watchers would have to read the line of the sea instead.

Come morning it would go to the Core, which can run the sunrise along any horizon for any year he likes and tell him to the day when the sun would sit in a notch. What it has no way of knowing is the names of the hills, or which of them have stories, or whether he is allowed to know. Those are not questions for a machine.

Where do we stop? Where they let us, her mother said. Here, until the sun says otherwise, the girl said. Both were true, and the second one was hers.

The girl had needed it for the early corn, and to know the sun was not lost when it went south in the winter, only walking its road. What the dream added was the old man on the ledge letting a stranger's child sit beside him, because two people watching the same line can find out whether they agree. Gary knew that one from every new site he had ever started on: the first week you learn the names of things and keep your hands in your pockets, and the men worth listening to are the ones who say straight out when they've called it wrong.

A long way south and west, and more than six hundred years before, on an ocean with no stones in it at all, an apprentice of seventeen was being sent up to the bow of a canoe for the night watch, and would have to come home and say what he had seen.

For weeks after, he would stand at the bedroom window in the dark, with the photograph of the mesa propped on the sill, until the light came grey over the roofs, and catch himself waiting for it to come up in one particular place, between two chimneys across the street.

CHAPTER 11

The Ice at the Edge of the Sea

‘Avaiki, the old islands of the west, and the sea far to the south of them, in the time of Ui-te-rangiora as the genealogies count it: the found year, 650 CE



Illustration 12. *Teariki in the bow, pointing at the pale shapes on the southern sea.*

Belfast, by day

AT TOTAL CARGO THE PAPERWORK had a phrase for a box nobody had opened, and the phrase was *said to contain*. The shipper said there were four hundred cartons of tiles in the container, and the carrier wrote *said to contain* in front of it. That meant the carrier was passing on what he had been told, and was not swearing to it. Gary had always liked the phrase, because it was the most honest three words in freight.

“Who was Hui Te Rangiora?” he typed into the box.

"Hui Te Rangiora was a Polynesian navigator who sailed to the southern ice around 650 AD. A 2021 study confirmed that Māori and Polynesian voyagers were the first to reach those waters."

"Confirmed," he said. "Get me the study. And get me whoever answered it."

He had the machines pull everything, the old book and the new papers and the replies to the papers. Then he read them himself, line by line, with the kettle going cold. A machine will tell you a thing is confirmed when what it means is that somebody wrote it in a journal once.

The old book was S. Percy Smith's *Hawaiki*, printed in 1899. Smith had the story from Rarotongan genealogies and chants, and he counted the generations back to Ui-te-rangiora, as he spelled it, and found forty-eight of them. At twenty-five years a generation that came to about 650. In that age, Smith wrote, the people's head-quarters were in Fiji, with colonies in Tonga and Samoa. Their great canoe was called Te Ivi o Atea, and the story said its timbers were men's bones. The wonders came in a list, like a manifest. There were rocks that grow out of the sea, and monstrous seas, and a woman who lives in the mountainous waves whose hair waves about on the water. There was *the frozen sea of pia*, with a deceitful animal that dives to great depths, and a foggy, misty, dark place not seen by the sun.

The new papers were harder going. In 2021 Priscilla Wehi and her colleagues gathered the Māori and Polynesian stories of the far south and argued that the journeys deserved a place in the history of the southern ice. Atholl Anderson and his colleagues answered the same year, and their answer was a proper checker's answer. Forty-eight generations at a realistic length, they said, puts the man in the fifth century, not the seventh. The Rarotongan language had no word for ice before the Europeans came. In the old phrase, *'uka* was foam, and *pia* was the white arrowroot whose starch the islanders beat into a fine white flour. So the old words might mean a sea of foam as white as arrowroot. They said Te Rangi Hīroa had warned in 1926 that the Rarotongan traditions carried a great deal of information picked up

after the Europeans arrived. And Ngāi Tahu scholars, in their own reply, said the story was an origin story about the beginnings of things, not a ship's log.

Then the sheet that settled where, if not when. Wilmshurst and others in 2011 dated the first settlement of eastern Polynesia by hundreds of radiocarbon samples. The Society Islands came first, around 1025 to 1120, and the remote islands, Rarotonga among them, about 1190 to 1290. In 650 there was nobody living on Rarotonga at all, and nobody there to tell a story. That was not a problem for the story, which had never claimed otherwise. It put its people in the west, in the old islands, Fiji and Tonga and Samoa. Rarotonga was where the story was kept and written down, and not where it happened. The map had it the wrong way round, so he moved the dream west, and noted the change.

He dealt it out like a hand of cards, and the first card was Smith's arithmetic, a single year from a long count of names. The second card was the same count at a fairer length, which slid the man back two centuries. The third was the islands, which said that whenever he lived, he lived in the west. The fourth card was the words themselves, the oldest thing in the story. Whatever he saw, the people who kept the story called it by the name of something white and soft that you could hold in your hand. The overlap was not a found year but a received one, and he wrote it at the top all the same, the way the tradition gave it: *650 CE*. He put the question mark after it in pencil, which meant he could rub it out, and he knew he never would.

And under that the doubt, in the same plain hand: *If the generations are thirty years long, he is two hundred years older. If pia is arrowroot and 'uka is foam, he may never have seen ice in his life*.

He turned back to the screen. "So which is it, foam or ice?"

"The sources disagree. Anderson and colleagues argue the phrase describes sea foam; Wehi and colleagues read it as frozen sea."

"Aye. They do."

He had worked enough days under cranes to know that wind decides more than men do. A story about a lift always grows a wee bit in the telling down the pub. He had stood on enough shores, fishing,

to know what foam looks like when a big sea has been beating on the rocks all night. It lies thick and white and standing, like something you could walk on until you tried. He did not write down which he thought it was. He wrote nothing down about it, the way a carrier signs for a parcel and writes *said to contain*.

That night

The window was open and the lough was loud, and somewhere in the noise of it the library's shelves ran down to a beach. The Ledger left the page on the sand and let the wind turn it. Up in the corner was 650, the number he had written, with its mark of doubt curled beside it like a wave. Below lay a warm night and surf on a reef, and a man and a youth on the wet sand, pointing up. He was a boy at the water's edge, and the stars were all wrong.

• • •

"There," said Ui-te-rangiora. "Where the black of the sea meets the black of the sky, just left of the dark headland. Watch."

Teariki was seventeen and had been watching for three years, on this beach and on others. He knew what was coming, but he watched anyway, because the master would know if he did not. Out on the edge of the world a star came up, small and red, and climbed a hand's breadth while they stood there in the warm water to their ankles.

"Name it."

Teariki named it, the way he had named it a hundred times before.

"Where does it rise?"

He told him, by the headland and by the place on the horizon where it always came up, the same place every night of a man's life.

"And when it's too high to steer by?"

"Then the next one, in the same house of the sky. And then the next. All night, one after the other, coming up out of the same place in the sea like birds out of one hole in a cliff." Teariki said them, the whole path, the stars of the south road in their order, the way the master had made him say them a hundred nights on this beach. "And

when there are clouds, the swells. The long swell from the south-east that never stops, and the short one that comes and goes with the wind."

"And when there is fog, and no swell, and no star?"

Teariki was quiet. "Then I don't know, master."

"Good," said Ui-te-rangiora. "That's the right answer. Remember you said it."

They walked back up the beach to the canoe shed. She lay there under the thatch on her rollers. She was longer than any canoe on the island, her twin hulls bound together under the deck, the great sails of plaited pandanus rolled along their booms: Te Ivi o Atea, the Bones of Atea. The children of the village said that her timbers were made of men's bones, the bones of the master's enemies. The master never told them otherwise, because it kept them out of the shed.

"Are we really going south?" Teariki said. "Further than anybody?"

"Further than anybody in this generation. Not further than anybody." Ui-te-rangiora put his hand flat on the hull, the way you touch the back of a sleeping child. "The old men say there are things down there nobody living has seen. Rocks that grow up out of the sea. A woman with hair as long as a canoe. A sea that is white."

"How will they know we went?"

The master turned and looked at him in the dark.

"The chiefs, you mean. The ones who say I've grown old, and should leave the sea to younger men." He laughed, but not as if it were funny. "They'll know because I'll tell them, and you'll stand beside me while I tell them. And because I'll bring something back that they can hold in their hands."

• • •

They went out through the gap in the reef at the turn of the season, when the wind came round.

There were forty of them, men from the master's own household and from two others. The canoe sat low with water in bamboo and in coconut shells, and fermented breadfruit sealed in leaves, and dried fish, and coconuts. There were cakes of pia that Teariki's mother had

made for him. She had beaten the white root and washed it and dried the starch on mats in the sun until it was a flour as white and fine as anything in the world. She had baked the cakes in leaves in the earth oven and wrapped them, and given them to him at the canoe shed. She did not cry until the canoe was through the reef. He had seen her, small on the beach, with her hands over her face.

For many days the sea was the sea he knew, blue and warm and full of fish. The birds went out from the islands in the morning and came back at night, so that you could tell where land was by the way they flew. Then the birds stopped coming, and after that the stars began to change.

He had known they would. The master had told him so, on the beach. But it was one thing to be told, and another to watch it happen. The stars he knew sank lower each night behind them in the north. New ones came up out of the south that had no names in the path he had learned, or only the names of old men's stories. The master named them anyway, standing at the steering paddle with his face turned up. Teariki said the names back and kept them, and the master checked him each night as he had on the beach.

Then it grew cold. That was the strangest thing of all.

Teariki had never been cold in his life except for a little before dawn after rain. Now it came every day a little more, a cold that got into the plaited sail and into his hands and into the cakes of pia in their leaves. The men wrapped themselves in bark cloth and then in mats and then in everything they had, and huddled in the lee of the deck house, and still shook. The sea went from blue to grey and from grey to the colour of slate, and it got up into swells such as none of them had ever seen. They were not waves but moving hills, slow and huge. They lifted the whole canoe up and up until you could see the next three hills marching after it, and then let it down into a valley where the sky was a strip of grey overhead.

On one of those days, Teariki saw the woman's hair.

It was lying on the face of the swell, long and brown and gleaming. There were dozens of strands of it, as thick as his wrist and longer than the canoe, waving slowly as the sea went under it. He shouted

and pointed, and the men crowded to the side and stared. The master leaned over the side with a paddle and lifted a strand of it out of the water. It was a plant, a great rubbery weed with a root like a fist. The men laughed at Teariki, but not for long, because none of them had ever seen it either.

On another day something came up beside the hull, black and sleek with a face like a dog and great round eyes, and looked at them, and dived. They saw it go down a long way into the grey water, deeper than any fish would go, and it did not come up again. A man in the bow made the sign against bad spirits. Teariki thought it had been laughing at him.

• • •

It was then that the men began to say they should turn back.

They said it at first among themselves, in the lee of the deck house, where they thought the master could not hear. Then one of the older men, from another household, said it out loud at the steering paddle. They had seen the woman's hair and the deceitful animal, he said. They had seen stars nobody living had a name for. That, he said, was enough wonders for any telling a man could want. The water was running low and the cold was killing them, and there was nothing to the south but more grey.

Ui-te-rangiora heard him out to the end, leaning on the steering paddle. Then he said, "Three more days. And then we turn."

Teariki, crouched at the foot of the mast with the sail rope in his frozen hands, understood then that the master was not going on for the wonders. He was going on for the chiefs at home, the ones who said he had grown old. He was going on because he had said he would bring back something they could hold. And Teariki, who loved him, wanted to stand up and say *master, the men are right*, and wanted just as much to follow him to the end of the world. He said nothing, and hated himself for both.

On the second of the three days the fog came down.

• • •

That night the master sent him to the bow.

"You've the youngest eyes," Ui-te-rangiora said. "Sit there and watch the water, and tell me what you see. Not what you think it is. What you see."

So Teariki sat in the bow of the left-hand hull, with a mat over his shoulders that was wet through and frozen stiff at the edges, and watched. There were no stars. For the first time in many days there was no swell to speak of, only a long slow heave under the hull, and a silence that pressed on his ears. The fog was so thick he could barely see the end of the hull in front of him. The canoe went on through it very slowly, under a scrap of sail, and the water talked along her sides in a low voice.

Some time in the middle of the night, the voice changed. Teariki held his breath.

It went from a hiss to a whisper, and then to something softer, a rustle, as if the canoe were going through dry leaves. Teariki leaned out over the bow and looked down at the water.

The sea was white.

Not white with foam, the way the reef was white where the waves broke. It was flat, and still, and white all over, as far as he could see into the fog, which was not far. It looked thick, like the starch in his mother's bowls before she poured it out on the mats. It lay on the black water like a skin. Where the bow pushed through, it parted, in plates and pieces that turned and touched with a small sound, and closed again behind.

He opened his mouth to shout. Then he remembered what the master had said, and made himself look again, and look longer.

"Master," he called softly, back along the hull. "The sea's gone white. Like pia. Like my mother's pia, before it dries. It's flat. It's moving in pieces. I don't know what it is."

The master came forward along the deck, slowly, stooping, and crouched beside him in the bow, and looked.

He looked until his neck was stiff, and he did not say anything at all. Teariki could hear his breathing, and the small sounds of the white pieces touching along the hull, and nothing else in the world. Then the master put one hand on Teariki's shoulder, and left it there, and went back to the steering paddle without a word.

Teariki watched the white sea until the fog began to go grey with the morning. His eyes ached, and he was so cold that he could not feel his hands on the hull.

• • •

In the morning the canoe was not moving, and the men woke and stood up and stared.

The fog had lifted a little, enough to see a stone's throw, and all around the hulls the white had set and lay hard against the wood. A man put his foot on it, gingerly, and it held his foot. Then it cracked under his weight with a sound like a breaking branch, and he snatched his foot back, and the water underneath was black and very cold. Further off, through the fog, there were shapes that Teariki thought might be the rocks that grow out of the sea, pale and tall, with their tops lost in the mist.

For a while nobody spoke, and then somebody began to pray, and the others joined in.

Ui-te-rangiora was sitting by the steering paddle with a gourd in his lap, and the stopper in the gourd, and his hands round it. Teariki saw a broken place in the white crust along the side of the steering hull. Someone had leaned down in the night and pulled a piece away.

"We turn now," the master said, in an ordinary voice. "Paddles. Break it away from the hulls. North."

It took them all morning to get clear, breaking the white with paddles and the butts of spears, until the canoe was in black water again and the sail could be set. Nobody looked back except Teariki, who could not help it. Behind them the fog closed over the white sea and left no sign it had ever been there.

At midday, when they were well clear, the master called Teariki to him, and took the stopper out of the gourd, and showed him.

It was water, only water, cold and clear, with a little grit in the bottom.

They looked down at it together for a long moment.

"It was hard," Teariki said. "In the night, it was hard. I saw it. That man stood on it."

"I know."

"And now it's water."

"Now it's water," said Ui-te-rangiora. He put the stopper back in, carefully, as if the gourd still held what he had put in it. "Say nothing to the men."

• • •

They came home at the end of the season of rains, with eleven fewer than had gone out. The rest of them were so thin that the women on the beach did not know their own husbands until they were close.

There was a gathering in the chief's house, three nights later, when the men had eaten and slept. The master stood up in the middle of it and told the whole voyage, from the gap in the reef to the gap in the reef, in the old way. He chanted the names of the stars they had followed and the days they had sailed. Then came the wonders in their order: the rocks that grow out of the sea; the monstrous seas; the woman in the waves with her long hair; the deceitful animal that dives deep; the fog, and the dark place not seen by the sun. Last came the white sea, the frozen sea of pia, that held a man's foot and closed round the hulls in the night.

Then, at the end, he took out the gourd and held it up.

The chief, an old man who had been young with Ui-te-rangiora, held out his hand for it, and the master gave it to him. The chief took out the stopper and looked in, and tipped it, and a little cold water ran out onto the mat, with a little grit.

There was a silence in the house.

"Water," said the chief.

Ui-te-rangiora did not answer. He stood very straight, and Teariki saw, as he never had before, how old he was.

The chief looked round the house and found Teariki, sitting by the door with his mother.

"You were there," he said. "You're the one who sat in the bow, they tell me. Your master says the sea was hard. Was it hard?"

Teariki stood up, and everybody in the house looked at him. He could feel his mother's hand on his leg and the master's eyes on his face. He knew exactly what the master needed him to say, and how easy it would be to say it, and that the master would never ask.

He thought of the beach, and the stars, and *tell me what you see, not what you think it is*.

"It was hard in the night," he said. "A man stood on it. I heard it break. In the morning it was round the hulls like a crust on a pot. By midday, in the gourd, it was water. I saw both." He swallowed. "I don't know what it was. I only know what I saw."

The chief weighed him with a long look, and then at the master.

Ui-te-rangiora was smiling, a small smile, and tired, and the first one Teariki had seen on his face since the fog.

"That's my apprentice," he said to the house. "You'll have noticed he doesn't lie. Put what he said in the telling, beside mine. Both of it. That's how it goes into the chant."

So it went into the chant: the white sea, hard in the night and water by noon, and the boy in the bow who said he did not know. Afterwards, on the beach, the master gave the gourd to Teariki to keep. It was empty now, and it smelled of nothing at all.

"How will they know we went?" Teariki said.

"Because you'll tell it," said the master. "Right."

The master page

The Ledger laid the page down on the shelf beside the mountain road, and it lay on its side a little, like a canoe pulled up on sand.

In ink: the tradition as Smith printed it in 1899, with its forty-eight generations and its list of wonders; the dates for the settlement of the eastern islands, which leave Rarotonga empty in 650; and the honest argument of 2021, both halves of it, Wehi and Anderson, frozen sea and arrowroot foam. In pencil: the year, which is a count of names

and not a find, and slides two centuries with the length of a generation; and the ice itself. *Found destroyed*: whatever the first tellers said, in the first voice, before anybody wrote anything down. What reached the paper in 1899 had been told and retold for a thousand years and had picked up what later sailors saw.

He saw what this page added. The mountain road had made the recitation: the count kept in the mouth and checked by its own pattern. The mesa had made the watch, one line of hills read every morning and called ahead. To that the canoe added the survey. You go out to look, and you come back, and you tell exactly what you saw, star by star and day by day, and the telling is checked by the men who were there. The proof you bring home may melt in the gourd by noon. The telling does not, if it is told right. Behind it stood the season held in an old man's head on the Bann, and ahead of it, in pencil, the Greek who wrote of a sea you could neither walk on nor sail. Further ahead was an Irish monk in a boat of hide.

He meant to feed the Core the rhythms at first light: the star path rising out of one place in the sea night after night, the long swell that never stops, the turn of the season. The Core can say whether those rhythms are in the numbers. It cannot say whether the white sea was ice or foam, and it cannot say whether the man was real. He typed that in as well, so nobody could say he hadn't.

How will they know we went? Not by the gourd. By the telling, if the boy in the bow tells it right, and by the men beside him who can say he did.

The boy in the bow had wanted the stars to come up out of the same place in the sea every night, so he could find his way home in the dark. What he got was harder: a morning when he had to say which white thing he had seen and which he had only hoped for, with the people at home waiting to build on it. *I don't know* is a terrible thing to have to say in front of a chief. Gary had said it to site managers with a programme to meet, and it had never once been the wrong answer, only the unwelcome one.

A long way north and west, a scribe in a cold scriptorium was being told to leave a woman with a silver branch out of the tale he was copying.

In the kitchen the window had been left open, and the lough was loud in the dark. Rain had come in on the sill, and it stood in beads on the stone there like a swell with nowhere to go.

CHAPTER 12

The Silver Branch

Druim Snechta, the ridge of snows, in the north of Ireland, one winter in the house where the little book was made: the found year, 720 CE



Illustration 13. *Dearbhla brings the hoarfrost branch to the scriptorium door.*

Belfast, by day

THE FIRST THING THEY TAUGHT him at the solicitors' office in London, in his first weeks there, was that you never tidied a witness statement. If the witness said *he come in the door*, the copy said *he come in the door*. If you could not bear it, you put *sic* after it in square brackets. That small word meant *thus*, and it meant *I know, and I left it*. The partner who taught him said that a copy which improved the original was not a copy any more. It was evidence that somebody had been at it.

"When was the Voyage of Bran written?" he typed.

"Immram Brain was written down by Irish monks in the eighth century, preserving an ancient pre-Christian myth unchanged."

"Unchanged," he said. "Who checked?"

He set the machines fetching the texts and the editions and the arguments. Then he went through them himself with a pen, like he had gone through bundles in London. A machine will tell you a story came down unchanged for the same reason a bad clerk tidies a statement. It sounds better, and nobody has asked it to show its working.

He took the story first, the way the machine had given it to him and the way the translation had it. Bran son of Febal hears music behind him, sweet enough to send him to sleep, and wakes with a branch of silver beside him, white with blossom. That night a woman in strange clothes stands on the floor of his royal house. She sings to him of a branch of the apple tree from Emain, and of islands in the western sea where there is no grief and no death. Then she goes, and the branch leaps out of Bran's hand into hers, and he cannot hold it. He sails with three companies of nine men. On the sea he meets Manannán son of Ler driving a chariot over the waves, who sings that what Bran sees as sea is a flowering plain. Manannán's song, in the middle of it, foretells the birth of Christ. Bran comes at last to the Island of Joy and the Land of Women, and stays what seems a year. When he comes home, one of his men, Nechtan son of Collbran, leaps ashore. The moment he touches the earth of Ireland he is a heap of ashes, as if he had lain in the ground for hundreds of years. Bran tells his wanderings from the boat, writes the verses in ogam, and sails away, and his wanderings are not known from that hour.

Then the word count, which was the first thing he checked, because it was the kind of thing a clerk checks. The story says the woman sang fifty quatrains and Manannán thirty, and every copy anybody has gives twenty-eight of each.

He had three things to go on, and he stacked them the way a scribe stacks quires, the oldest at the bottom.

The first sheet was the language, and what the scholars made of it. The scholars who know Old Irish put the tale's forms in the late seventh or early eighth century, round about 700, older than the copies by a long way. The oldest manuscripts that carry it were written centuries later, in other hands, and every one of them is a copy of a copy.

The second sheet was a book that does not exist. Later scribes, copying Bran and other tales, wrote in the margin that they had them out of the *Cín Dromma Snechtai*, the little book of Druim Snechta. That is Drumsnat in County Monaghan, an hour and a bit down the road from Belfast, and the little book itself is lost. Thurneysen put it in the first half of the eighth century. Mac Mathúna, who edited Bran, argued the book could be as late as the end of the ninth century, or even the tenth.

The third sheet was the monks themselves, and what they did with it. A Christian prophecy sits in the middle of a song about an otherworld of women, and somebody put it there on purpose. Whoever wrote this down was a churchman who loved the old story enough to keep it and was worried enough about it to fit Christ into it.

Where the three overlapped he wrote it at the top, *720 CE*, and beside it the question mark, which the scholars who put the book two hundred years later would have wanted bigger.

Under that went the doubt, in his plain hand: *If Mac Mathúna is right about the little book, my scriptorium is a hundred and fifty years too early. The abbot I'm about to dream never sat in that chair*. And under that, a second line, because he owed it: *No source says any abbot told any scribe to leave anything out. That part is mine and nobody else's*.

The map had the copying in a midlands monastery. He moved it to Druim Snechta, because the book had a name and a place and he would rather stand where the evidence stood. The map also had a Sister Dearbhla bringing the light. A nun would not be carrying lights into a monks' scriptorium at night. So he made her a girl of the monastery's own tenant families, the people who farmed its land and owed it work, and he wrote the change in the notes.

"So nobody knows if the verses that went missing were cut or just lost," he said.

"That is correct. The discrepancy between the stated and transmitted numbers of quatrains has no agreed explanation."

"Aye, grand. Neither do I."

He sat a while with the stone off the sill in one hand and the pen in the other. He did not write down whether he would have left the woman out, if it had been his job and his abbot. There was nobody in the kitchen to tell the joke to, so he put a small *sic* in the margin of his own notes, beside the word *unchanged*.

That night

His notes were under his cheek when he went. Cold met him in the library, and the smell of smoke and wet wool. On a slanted board the Ledger lifted a page rather than turning it, like a leaf of calfskin still stiff with frost, and in one corner, small, was the number he had written, 720. Beneath it was snow on a ridge, a round bank of earth and a light in a doorway. He stood in the snow as a boy, and a girl was coming up the path with the light.

• • •

The light came at the same time every evening in winter, just as the bell for vespers stopped and the dark began to come up out of the valley.

Eochaid heard her before he saw her. Her feet came on the frozen path inside the bank, and then her breath at the door of the writing-house, and then she was in. Dearbhla was twelve, and red in the face from the cold. She had a bundle of rushlights under her arm wrapped in a scrap of sacking, and a little pot of embers in a sling. She set the rushlights in their iron holders along the edge of his board, three of them, and lit them from the pot. The writing-house came up out of the dark around them: the slanted board and the stool, and the satchel of books hanging on its peg. There was the cracked bowl of pumice, and the cow's horn of ink fixed in its hole at the side of the board.

"You're late," said Eochaid, without looking up.

"I'm not. The bell's only just stopped." She blew on her fingers. "You're early."

That was true enough. He had been at the board since terce, and his hand was so cold he had to hold it under his arm every few lines to make it work. He was twenty-three, and he had been a monk of Druim Snechta for nine years and a scribe for four, and he was copying out the tales.

The abbot wanted a book of them, and not a great book like the gospels in the church with their painted beasts. He wanted a little book, a cín, of the stories the people told at their fires and the poets told at the king's. There was the conception of Cú Chulainn, and the prophecy of Conn, and the adventure of Connla who went away with a woman from the sea, and the voyage of Bran son of Febal. A man had lent the abbot an old copy from another house, worn soft at the edges. The abbot wanted the tales set down fair and kept at Druim Snechta, because the old copy had to go back. Besides, he said, a thing that was only in one book was a thing that was halfway to being lost.

Eochaid read aloud as he copied, as everybody did; you could not copy a line in silence, it would not go into your hand. So each evening Dearbhla sat on the floor by the door with her back to the wall and her knees up under her cloak, keeping the rushlights trimmed. She heard the tales come out of the old book a line at a time, in his voice, and go down onto the new calfskin.

She could not read a letter, and she had never wanted to, she said, until this winter.

• • •

Bran's voyage he had started at Martinmas, when the first frosts came.

He read the opening to her that first night: the music, and the sleep, and the silver branch. Dearbhla listened with her mouth a little open. When he got to the woman in strange clothes standing on the floor of the king's house, she sat up.

"My grandmother tells that," she said. "Only the woman doesn't sing. She comes out of the sea with the branch, and she just looks at him, and he goes."

"Here she sings fifty verses," said Eochaid. "It says so. *Fifty quatrains the woman from unknown lands sang on the floor of the house to Bran son of Febal.*"

"Fifty!" She was delighted. "Read them."

He read her the first. *A branch of the apple tree from Emain I bring, like those one knows; twigs of white silver are on it, crystal brows with blossoms*. And the next, about the island far off with the sea horses glittering round it. Then he stopped, because the bell was going for the first office of the night. He laid the old book face down on the board, gently, the way you would lay down a sleeping child.

"Who decides what goes in?" Dearbhla said, getting up. "Into your book. Out of all the stories. Who decides?"

"The abbot."

"And what goes out?"

"The abbot," said Eochaid, and she laughed and went home with the pot of embers through the snow.

• • •

The abbot came the next morning, and he was not laughing.

He was an old man, and kind, and very tired. He stood at Eochaid's elbow with his hands in his sleeves and read what had been copied so far, his lips moving. Then he turned the old book over and read the woman's verses that were still to come.

"Leave these," he said.

Eochaid put the quill down.

"The woman's song," said the abbot. "Leave it out. Put in the music and the branch and the sleep, and the woman comes, and she goes, and he sails. Then go on to Manannán, all of it, every verse, because Manannán foretells the Son of God, and that is the heart of the thing. But not the woman's song."

"Father, it's half the tale."

"It's half the tale," the abbot agreed. "And it's the half that sends young men to sea. There's a boy gone from the house at Cluain this autumn, did you hear? Took a boat from the coast and went west, looking for an island with no death in it. He'd been reading Bran." The

old man looked at the page, at the first two quatrains in Eochaid's careful hand. Something in his face was sorry. "I'm not asking you to burn anything. The old book goes back where it came from, whole. I'm asking you not to make another."

"Yes, Father," said Eochaid, because there was nothing else a monk could say.

That evening, when Dearbhla came with the light, he copied the sleep and the music and the branch again onto a new leaf. He wrote that the woman came, and that she went. He did not read the verses aloud, and Dearbhla sat by the door and listened to him not reading, and said nothing until the bell.

Then she said, "He told you to leave her out."

"Yes."

"Are you going to?"

He did not answer that, and she did not ask it again.

• • •

The tale went on, and the winter with it.

On the page, Bran went out over the sea with his three companies of nine. After two days and two nights he saw a man coming towards him over the water in a chariot, and it was Manannán son of Ler. Manannán sang to him that the sea was not the sea. *What is a clear sea for the prowed skiff in which Bran is*, he sang, *that is a happy plain with profusion of flowers to me from the chariot*. Speckled salmon leap from the womb of the white sea, he sang, which to him were calves and lambs.

In the writing-house the snow came down for three days without stopping, and the rushlights guttered in the draught from under the door. Dearbhla brought turf for the fire in the guest-house and porridge for the brothers. For Eochaid she brought a bowl of hot milk that she was not supposed to bring. His ink froze in the horn on the coldest morning, and he had to hold the horn in his armpit to thaw it before he could write. In the margin, very small, where nobody would look, he wrote in Irish that his hand was cold and the light was poor.

On the page, Manannán foretold a great birth after ages, the son of a woman whom mate would not know, who would take the rule of the many thousands. Eochaid copied every verse of it, as the abbot had asked, and found that he was glad to.

Every night when Dearbhla had gone, and the brothers were back on their pallets after the night office, he went back to the writing-house. He went alone, and lit one rushlight from the fire in the guest-house, and copied the woman's song.

He did it on spare leaves, offcuts, the ends of skins that were too small for the book. He did it without saying the words aloud, so that it went into his hand slowly and badly, like a man walking in the dark. There were twenty-eight quatrains. He counted them twice, in the old book and in his copy, the way the old men had taught him to check a psalm, and they agreed. Twenty-eight, where the heading said fifty.

He went through the old book leaf by leaf by his one rushlight, and there were no more, and there was nowhere they could have been. Whoever had made the old book had had twenty-eight, or had written twenty-eight. Somewhere before that, in some other cold room, the other twenty-two had gone, or had never been, and nobody had written down which.

So at the end of the woman's song, small, in the margin, he wrote: *The heading says fifty. Twenty-eight are here. The rest are not in the old book.*

• • •

On the ninth night, Dearbhla came back.

He heard her feet on the frozen path and looked up with his heart going, because nobody came out after the night office. At the door it was her, with no rushlights and no embers, wrapped to the eyes, with snow on her shoulders.

"You'll be killed," he said. "Your father…"

"My father's asleep. You've only the one light. You'll ruin your eyes." She held up what she had brought, a branch off one of the old apple trees in the abbot's garden, cut short and bare of leaves. In the cold every twig of it had grown a thick fur of white frost that caught

the rushlight and glittered. Snow lay along the top of it in little white knots like blossom. "I can't bring you more lights; they're counted. So I brought this. It's silver, look. It catches what there is."

She stood it in the corner by his board, where it caught the one flame and threw it back, and the writing-house was a little brighter. Then she sat down in her place by the door.

"Read it to me," she said. "The woman's song. Just this once. I won't tell."

He looked at her as if she were a page he could not read. Then he picked up the spare leaves and read it to her, the whole of it, the twenty-eight verses, in a low voice. The frost on the apple branch shone in the corner while he read. He read the island with the sea horses round it, and the plain where the games are played, and the music without grief, without sorrow, without death. At the end, the woman's last command: *Do not fall on a bed of sloth. Let not thy intoxication overcome thee. Begin a voyage across the clear sea, if perchance thou mayst reach the land of women.*

For a while Dearbhla was quiet.

"She doesn't say *stay*," she said at last. "Your abbot's afraid she says *stay*. She doesn't. She says *go*." She pulled her cloak tighter round her. "It's the going she wants."

• • •

He was reading the verse again when the door opened and the abbot came in out of the snow.

He had a rushlight of his own cupped in his hand, and he stood in the doorway and looked. He saw the boy at the board with the spare leaves, and the girl on the floor by the door. In the corner the frosted branch was catching both lights now and throwing them back.

Nobody said anything. Dearbhla got slowly to her feet, but Eochaid did not; his legs would not do it.

The abbot came to the board, and set his light down beside Eochaid's, and held out his hand, and Eochaid gave him the leaves. The old man read them standing, all of them, his lips moving, to the end, to the small note in the margin. He read that twice.

"Twenty-eight," he said.

"That's all there are, Father. In the old book. I counted twice. I didn't leave out any, and I didn't make any up. I didn't want to put in what I thought ought to be there. I wrote that the rest were gone, so whoever comes after knows there were more, and that we didn't have them."

The abbot read the note in the margin twice, then once more.

"There's a boy at Cluain," he said at last, "who went to sea after a song. And here is a boy at Druim Snechta who sat up all night in the cold to copy it right, and says so, and says what he hasn't got." He laid the leaves down on the board, square, very gently. "God knows which of them I'm more afraid for."

He turned then to Dearbhla.

"And you. Out after the office, in the snow."

"I brought him the light, Father. He only had the one."

The abbot looked at the apple branch in the corner, with its frost and its snow, catching the two flames.

"So I see," he said. Then he did something that Eochaid remembered all his life. He reached out and touched the branch, very lightly, and a little of the frost came away on his fingers. He looked at it on his fingertips, glittering, and then it was water.

"Put it in the book," he said to Eochaid. "The woman's song. In its place, after the branch, before Manannán, so the child comes at the end of it and not at the start. Twenty-eight. And your note, just as it is." He picked up his light from the board. "Then copy the voyage to its end, the ashes and all, so they know what happens to the man who jumps ashore."

He turned in the doorway and looked at Dearbhla. "And you, home. Now. And I never saw you."

He went out into the snow, and they heard his feet go away along the path, slow and careful, the feet of an old man on ice.

• • •

They finished Bran on the eve of the Nativity, in daylight, with the door open on a white world.

The last part of the tale was short. Bran came home after what seemed a year, and his people on the shore did not know him; the voyage of Bran was among their old stories, they said. Nechtan son of Collbran leapt from the boat. As soon as he touched the earth of Ireland he was a heap of ashes, as if he had been in the ground for many hundred years. Then Bran told his wanderings from the boat to the people on the shore, from the beginning to that hour. He wrote the verses in ogam, and bade them farewell, and from that hour his wanderings are not known.

Eochaid wrote the last line and sat back, and Dearbhla, by the door, had her chin on her knees.

"He wrote them down," she said. "Bran. Before he went."

"In ogam. The old letters, the strokes on the edge of a stone."

"Could you write my name in that?"

He took a peeled hazel stick from the woodpile by the door, and his knife. He cut her name along the edge of it in the strokes and notches of ogam, the way names were cut on the old stones by the roads, and gave it to her. She turned it in her hands, and ran her thumb along the notches, and could not read a stroke of it.

"It says Dearbhla," he said.

"I know," she said. "I believe you." And she put it inside her cloak, and went home through the snow with the empty pot of embers swinging on its sling. The apple branch stayed in the corner of the writing-house until the frost went off it, and then it was only a branch.

The master page

The Ledger did not lay this page flat. It stood it in the corner, like a branch, where it could catch what light there was.

In ink: a tale whose language belongs round about 700; a little book of Druim Snechta that later scribes say they copied from, which no one has seen for centuries; a woman who, by the heading, sang fifty quatrains and, in every copy, sang twenty-eight; and a prophecy of Christ set in the middle of an otherworld song by a churchman who would not throw the song away. In pencil: that the book was made in the first half of the eighth century, against the scholars who would put

it later. In pencil too, in the margin where the book keeps its greeters at the edge, the woman with the silver branch, who stands on the floor and names the land and tells a man to go. *Found destroyed*: the other twenty-two quatrains, and the hand that lost them, and whether it was an order or an accident.

He saw what the page added. Egypt made the record, with its holes left showing, and the mountain road kept the recitation in the mouth. The canoe brought back a survey that melted in the gourd. Druim Snechta's part was the copy: the thing carried forward by a hand that is not the first hand, faithful to what is there and honest about what is not, including the parts its own masters would rather leave out. It is the House of Life's *found destroyed* again, two thousand years later and a sea away, with no line between them but the same small honest habit. Ahead, in ink, was a boy on Iona with a white pebble, and in pencil, a long way ahead, a woman on a rock at the edge of the fog with a branch in her hand.

For the Core he noted three things to try: the rhythm of the hours, the bell every three hours through the night, the count of quatrains against the heading. The Core can tell him whether twenty-eight and fifty make any pattern in the numbers. It cannot tell him who lost the twenty-two, or whether any abbot ever stood in that doorway. The doorway was the scene's, and the scene was his.

Who decides what goes in? The abbot, the story says, and the abbot changed his mind. The scribe decides what he will not pretend. And sometimes a girl decides, by bringing a light.

The boy had wanted the bell to go at the right hour in the dark of the year, so the feast fell on the right night and the brothers rose when they should. The dream had shown him that copying was its own kind of sky-watch: you keep what was there, you mark what is gone, and somebody after you builds on it. Gary had learned that in a solicitors' file room, though nobody there had called it holy. If a page was missing from a bundle you wrote *page missing* on a slip and put the slip where the page should have been. You did not write the page.

A thousand years earlier and a long way south, a clerk of fifteen was writing *six* in a ship's log, and crossing it out, and writing it again.

He woke with his cheek on his notes and the word *sic* printed backwards on his skin. Outside it had begun to snow, a little, over Belfast.

CHAPTER 13

The Man Who Went to Thule

Massalia, and the long way north round Prettanike to the island six days beyond it, in the generation of Alexander: the found year, 325 BCE



Illustration 14. *Pytheas with his rod under the low northern sun.*

Belfast, by day

AT ORBIT SECURITY IN LETTERKENNY, the year he ran the office there, the incident book had one rule written inside the cover in somebody's capitals: NEVER TIPP-EX. If a guard wrote the wrong time on a callout, he put a single line through it, so you could still read what was under it. Then he wrote the right time beside it with his initials. Gary had liked that rule more than he could have explained to anybody at the time. A book you could not tidy was a book you could trust, because every mistake in it was still there, struck through and owned.

He thought of it when the machine came back about Pytheas.

"Pytheas of Massalia sailed to Iceland in 325 BCE and measured the latitude of Thule with a gnomon," it said.

That was three claims in one sentence, and he did not believe any of them on sight. He set the machines fetching the fragments and the editions and the arguments. Then he went through them himself, by hand, with the tea going cold at his elbow.

The first thing he checked was the man's own book, and it does not exist. Pytheas wrote an account of the Ocean, and not a page of it survives; everything we have is other men quoting him, and most of them did not like him. Polybius sneered that a private man, and a poor one, could never have made such a voyage. Strabo, writing three hundred years later, called him the arch-falsifier, and then went on quoting him without seeming to notice. Thule, Strabo reports him as saying, is a six days' sail north of Britain, and is near the frozen sea. Pliny adds that one day's sail beyond Thule is the frozen sea itself. A Greek teacher of astronomy called Geminus quotes him plainly: the barbarians showed us the place where the sun goes to rest. In those parts, Geminus says, the night was very short, two hours in some places and three in others.

"Iceland" was the machine's word, not the fragments'. Scholars have argued for Iceland, for the coast of Norway and for Shetland, and the honest answer is that nobody knows. The one man who could have told us is in pieces in other men's books. He put a single line through *Iceland* on the printout, wrote *Thule* beside it, and Tipp-Exed nothing.

The second claim was the gnomon, and it was better than the machine knew. Hipparchus, who did not trust much, wrote that at Byzantium the ratio of a gnomon to its noon shadow at midsummer is the same as the one Pytheas gives for Massalia. That figure is a hundred and twenty to forty-two less a fifth. Somebody had stood a pole in the ground at home and measured its shadow, and the figure was good enough for the best astronomer of the next two hundred years to use. Hipparchus also passes on Pytheas's note that there is no star at the pole. The pole is an empty place, and three stars stand near it and make, with that point, a rough four-cornered shape. That was a man who looked, Gary thought, because nobody makes up an empty sky. He liked him for it.

"So did he measure Thule with it, or is that you filling in a gap for me?" he asked.

The machine said yes. He went looking and found no fragment that says so. The latitudes the later astronomers worked out for the far north may have come from Pytheas's figures for the length of the day, and they may not. He wrote *may* in the margin twice and boxed it.

The third claim was the year, and he took it in three goes. First came the window, which is all anybody really has. Aristotle, who wrote a good deal about the north and died in 322, never mentions him, and Dicaearchus, who was dead by about 285, had read him and disbelieved him. Then came the scholars: Roller puts the voyage in the 320s or shortly after, Cunliffe a little before 320, and an older reckoning puts it as late as 310 to 306. Last were the Carthaginians, who held the straits at the Pillars of Heracles, and the argument that a Greek of Massalia went overland across Gaul to the Atlantic rather than try to sail past them. Where the three agreed he put the year in the margin, *325 BCE*, and a big question mark after it, because twenty years is a wide window to be standing in.

Under that went the doubt, in his plain hand: *It's a window and not a date, and the window is twenty years wide. I don't know where Thule is, and nobody does*. And under that, because he owed it, a second line: *The clerk is mine. Nothing says Pytheas had a boy keeping his days, or which boats took him north*.

The dream had been going to put him in a Greek ship sailing out past the Pillars. He changed it, because a Massalian ship would have had to slip past Carthage, and the likelier road was overland to the Atlantic and then north in the boats of the people who lived there. Pliny, quoting an older Greek, says the Britons crossed the sea in boats of wicker sewn round with hide. He gave the dream one of those and put the change in the notes.

The last thing he noticed was the unit, and it was the thing he liked best. Pytheas seems to have given his distances in days, six to Thule and one more to the frozen sea. Some scholars think it was the men who came after him who turned the days into stadia, at a thousand or

so to a day's sail, and then laughed at how big the numbers came out. If that is right, Pytheas wrote down what he could count, and the men at home wrote down what they wanted it to mean.

He sat with that a wee while and thought of the incident book in Letterkenny, and the single line through the wrong time. Dead on, he thought, and it was the incident book's rule all over again. He did not write that thought down. He set the stone from the windowsill on the printouts to hold them flat, and went to bed.

That night

The printouts were on his knees in the chair, and the room began to tilt and kept tilting, the slow lift and drop of a boat on a long swell. A leather case smelling of fish oil and salt lay open, and the Ledger let the wind unroll the page from it. Somewhere a boy said a number out loud, and then, checking, said it again.

• • •

The tin came across the sand at low tide in carts, just as the merchants in Massalia had said it would, and Nikon wrote that down first, because it was the thing he had been sent to write.

He was fifteen, and he was a clerk, and he had never been further from Massalia than the salt pans before this spring. Three men at home had put up the silver for the voyage. They wanted the harbours, the prices, the names of the headmen who sold tin and amber, and the days between one place and the next. They had sent Nikon to keep their tablets and to keep his mouth shut. They had sent Pytheas, as far as Nikon could tell, because nobody else would go.

On the beach at Belerion the tide was a long way out. The people of the headland were easy with strangers and used to traders. They beat the tin into blocks the shape of knucklebones and carried them across the wet sand to the island while the sea was away. Nikon counted the carts and wrote the number, and asked a man the price of a block through the trader who spoke for them. He wrote that too, very small, in the wax tablet the merchants had given him.

Pytheas was not looking at the tin at all. He was looking at the sea, a long way out, where it had gone.

"It goes further out here than at home," Nikon said.

Pytheas pointed with his stick. "At home it hardly goes at all. Here it goes out half a morning's walk, and comes back as high as a house. And it's biggest when the moon's full. Write that."

"It isn't for the merchants."

"No. Write it in mine."

Pytheas kept his own roll in a leather case, and Nikon kept that for him as well. Pytheas's hands were always busy with a stick, or a pole, or the line of the sea. The roll was not for selling; it was for everybody, Pytheas said, and it had to be right.

That night, in the trader's house, with the tin stacked against the wall like grey bread, Nikon wrote the tide into it under Pytheas's eye. Then he sat back and looked at the two things he had: the merchants' tablet with its prices, and the roll with the moon and the sea.

"Who's going to believe any of this?" he said.

"Nobody, at first," Pytheas said, and laughed. "Write it anyway."

• • •

They went north up the coast of Prettanike in the boats of the coast, because no ship from home would have lasted the weather. The boats were frames of wicker and light wood with oxhides stitched over them, greased against the sea. They had oars, and a square sail of leather that cracked like a whip when the wind took it, and they rode over the swells like gulls. Nikon was sick for six days and then, quite suddenly, was not.

At every long stop Pytheas set up his pole. It was a plain straight stick of ash, smoothed and marked, and he stood it upright in the ground with a plumb line beside it and waited for noon. At noon the shadow was shortest, and he laid his marked cord along it and called the length, and Nikon wrote it down.

"At home," Pytheas said, the first time, "on the longest day, the pole is a hundred and twenty and the shadow is forty-two less a fifth. Say it."

Nikon said it.

"Now look at this one."

The shadow was longer, and it was longer every time they stopped, as if the sun were tired up here and could not climb so high. Nikon worked it out on his fingers, a little longer and then a little longer again. Then he saw it suddenly: the shadow was telling them how far they had come. It did not need a coast or a harbour or a man who spoke for the headmen. It needed a stick and a cord and the sun.

"The merchants want it in stadia," he said. "How far from Belerion to here. How far to the amber."

"I know what they want."

"I don't know what to put, then."

Pytheas squinted along the cord while he answered. "Days. How many days we sailed, and what the shadow was. Up here I don't know how far a day is, and neither do they. Let the men at home turn days into stadia if they like. I won't pretend to know a thing I can't count."

Nikon wrote the days in the merchants' tablet, and in the roll he wrote the shadow and the days, and he found that he wanted to write them neatly.

• • •

The nights got shorter as they went. In the south of Prettanike the dark had come down at a decent hour, the way it did at home. Up the long west side of the island it came later each week and went away earlier, until there was hardly any of it at all. It was only a deep blue hour and then the light coming back. Nikon could not sleep in it, and Pytheas did not try.

One of those nights they sat on a headland with their backs against a rock and their cloaks round them. Pytheas showed him the sky.

"At home," he said, "they'll tell you there's a star at the pole. The one the whole sky turns round. Find it."

Nikon looked and kept looking, following the Bear round with his eyes the way the sailors at home had taught him. The sky turned slowly over them, and there was no star where the turning was.

"There's nothing there."

"Good. What is there?"

Nikon looked again, harder. "Three stars. Near it. Not on it." He drew them on his knee with his finger. "And the empty place makes four. Like a field with a corner missing."

"Then write that the pole is empty, and three stars stand by it, and the empty place makes the fourth corner with them. Everybody at home who says there's a star there has said it because somebody told them. Write what you see when you look," Pytheas said, and leaned his head back against the rock.

Nikon wrote it in the roll by the light that never quite went away. He wrote *empty*, and looked up again to make sure, and it was still empty.

• • •

At the northern end of Prettanike, at the headland the coast people called Orkas, the land stopped. Beyond it there were islands, and beyond the islands there was the sea, grey and very wide. The headman there had a boat of hide bigger than any they had seen, and men who had been north.

Pytheas asked him, through two speakers, what was north.

"Thule," the headman said, or something that came into Greek as Thule. "Six days' sail. Then the sea is thick."

"Six," Pytheas said to Nikon. "Write that he said it."

So they went. Pytheas paid with something Nikon did not see; he suspected it was some of the merchants' silver, and he did not write that down anywhere. The boat had eight oars and the big leather sail, and a skipper who never seemed to sleep. Nikon sat in the bottom with the roll in its case inside his tunic, against his chest, where it would stay dry.

On the first day they lost sight of the islands. On the second the birds were different, and on the third the fog came down, white and wet and close, and the skipper had them take in the sail and ship the oars. They lay on the sea all that day and all the short night, rising and

falling in the white, hearing nothing but the water on the hides. Nobody sailed anywhere, and Pytheas sat with his hands round his knees and said nothing, and Nikon could not tell if he was afraid.

On the next morning the fog lifted, and the skipper looked at the sky and the swell and said something short, and they went on.

On the seventh dawn since Orkas they saw Thule.

• • •

It was a low, wet country with hills behind it, and people who came down to the shore to look at them. They were not unfriendly, but they had little grain. They lived on a kind of millet and on roots and wild greens, and they made a drink of grain and honey that Nikon drank too much of and was sorry. They threshed their grain indoors, in a long barn, because the sun was never clean or long enough to dry it outside, and it rained most days.

On the longest day Pytheas set up his pole on the beach, and the shadow lay out along the sand much longer than Nikon had ever seen it. Then that night the people took them up the hill behind the houses and showed them where the sun went down. It went into the sea in the north-west, slowly and slantwise, and then it did not go far. Nikon lay on the grass and watched the glow move along under the edge of the sea towards the north. Before he could fall asleep, it came up again in the north-east.

"Two hours," Pytheas said. "Maybe three. Write that they showed us where the sun goes to rest."

Nikon wrote it, and then he turned back in the roll to write the passage, and that was where it went wrong.

He wrote *six days' sail from Orkas*, because that was what the headman had said, and what Pytheas had told him to write. Then he stopped and counted the dawns on his fingers, sitting there on the hill in the half dark: the first day, the birds, the fog, and on. He had seen seven dawns, not six. He drew a line through *six*, the way the merchants had taught him to correct a tablet. He had the stylus on the leaf to write *seven* when Pytheas put a hand over his.

"What are you doing?"

"I counted. It was seven dawns. They'll check it at home."

"How many days did we sail?"

Nikon opened his mouth, and shut it again, and thought of the fog, and the sail in, and the oars shipped. They had lain on the sea a whole day and gone nowhere.

"Six," he said slowly. "And one lying in the fog."

Pytheas took his hand away. "Then write six. And write the fog beside it, so they know. Write what we sailed, Nikon. Not what we sat through."

Nikon wrote *six* again, beside the six he had struck through. He did not scrape the first one out, though somebody at home might see the line through it and think he had been careless, but he wanted them to see it. He wanted them to know he had thought about it and been told, and then he wrote the fog in small letters underneath.

• • •

One more day north, Pytheas said. The headman's men would not take them further than that.

On the next day the sea changed. It did not freeze the way Nikon had half expected, into something you could walk on. It only thickened, slowly. The air was full of a wet grey that was not quite fog, and the water under it was not quite water. There were soft, heaving masses on it, lighter than the sea, and between them the sea itself seemed to be breathing. The oars caught in it, and the boat slowed and slowed, and the skipper's face went still.

"Like a sea lung," Pytheas said quietly. He had seen the soft grey jellyfish that washed up at home, the ones the fishermen called sea lungs. This was what it looked like: everything joined, land and sea and air, and none of them itself. "You couldn't walk on that. You couldn't sail through it."

The skipper said a word and turned the boat.

That evening, on the shore of Thule again, Pytheas sat beside Nikon while he wrote it.

"Careful now," he said. "This part matters more than the rest. Write that I saw this, the thing like a sea lung, with my own eyes. Then write what they told us is beyond it, the frozen sea, the place where nothing goes. But write that they told us. Put the seen and the told in different places, so nobody mixes them."

"Why does it matter, if it's all true?"

Pytheas tapped the roll. "Because I don't know that it's all true. I know what I saw. When somebody comes after me with a better boat, he'll need to know which bits to trust. Give him that."

• • •

They came south again in the autumn, with the nights getting longer every week and the stars coming back. In the tin country the merchants' agent took the tablets and was pleased with the prices and puzzled by the days, and Nikon kept the roll.

On the last night before they went inland towards home, he sat with Pytheas on the sand at Belerion. The sea was coming in again, the great Belerion tide, under a full moon.

"Who's going to believe us?" Nikon said. He had asked it at the start, and meant it as a joke. He did not mean it as a joke now.

Pytheas thought about it for a while. "Some won't. They'll say a poor man couldn't have gone so far, and that the sea doesn't thicken, and that the night is never two hours long. They'll call me a liar."

"That must bother you."

Pytheas smiled into the dark. "A little. But a man who wants to call you a liar has to write down what you said first, so people know what he's arguing with. And then it's written. And one day somebody reads it and goes." He picked up the roll in its case and put it in Nikon's hands. "So make it right. Six, and the fog. Empty, and the three stars. Seen, and told."

Nikon held it on his knees. Over the sea the stars went round the empty place in the north, and he found it without looking for it, and it was still empty.

The master page

Down went this page, flat on a wet rock, with a stone on each corner to keep the wind off it.

In ink: a man of Massalia who went north round Prettanike and wrote it down; six days' sail to Thule and one day more to a frozen sea, in Strabo and Pliny; the barbarians who showed him where the sun goes to rest, in Geminus; the empty pole and its three stars, in Hipparchus; the shadow at home, a hundred and twenty to forty-two less a fifth; the tin carried over the sand at low tide, in Diodorus. In pencil: the year, somewhere in a window twenty years wide; where Thule was; the boat of hide, which is the pattern of that coast and not a record of his voyage; and the boy with the roll. *Found destroyed*: the book itself, every page of it, kept now only in the books of the men who called him a liar.

He saw what the page added. Egypt made the record, and the mountain road kept the recitation, and the canoe brought back its survey. Druim Snechta handed on the copy. Massalia's part was the measure: a pole and a shadow, a count of days, a sky looked at instead of believed, and a line drawn between what was seen and what was told. The struck-through six is Druim Snechta's honest note turned into a number. Ahead, in pencil, was an Irish monk in his own boat of hide, going north into the same fog.

The Core's share was the arithmetic: the shadow ratio, the day lengths, the six and the one. The Core can turn a shadow into a latitude faster than he can read the figure out. It cannot tell him where Thule was, or whether a boy ever sat in the bottom of a hide boat with a roll against his chest, and it does not pretend to.

Who's going to believe us? Not Polybius, and not Strabo. But Strabo copied him out to call him a liar, and that is why there is anything left of him at all.

Nikon had needed it to know how far north they were when there was no shore he recognised, and how many days to count home. The dream showed Gary something else: the sky is the one measure the merchants cannot buy, the same for anybody who looks up. And the

six was still there under its line, where anybody could read what had first been written and what it was changed to. Gary had once seen a stock sheet go up to the office with a figure scraped out and another written in over the top, and he had seen the face on the man who got it. Pytheas let them call him a liar for two thousand years, so long as the number was right.

On the next shelf, two hundred years earlier and back in the warm sea, a boy of Sybaris was being sent to live among his city's rivals at Croton, to find out what a number is for.

When he looked again the stones were off the corners and the page had gone with the wind, north, the way everything on it had gone.

PART III

Integration

Whole Cloth

CHAPTER 14

The School at Croton

Croton, on the instep of Italy, and the coast south from Sybaris, in the years after the teacher came from Samos: the found year, 530 BCE



Illustration 15. *Theano plays the single string beside Philon.*

Belfast, by day

ON THE KITCHEN WALL AT the Lifehouse, the year he did the admin there, the house rules were pinned up in a plastic sleeve, and most of them explained themselves. A few looked daft until somebody told you what they were for. Then they did not look daft at all; they looked like something that had happened once, to somebody, and was not going to be let happen again. He had been taught on the business course in Donegal that a rule without its reason is the first one to get broken. At the Lifehouse he learned it properly, from the other side, watching new staff decide which rules mattered by whether they could see the point of them.

That sheet was on his mind when the search on Pythagoras came back.

"Pythagoras founded his school at Croton in 530 BCE, proved the Pythagorean theorem, discovered the musical ratios by listening to blacksmiths' hammers, and was married to Theano," it said.

"Four," he said to the screen. "That's four."

That was four claims in one breath. "Who says each of those, and how long after him did they say it?" he asked, and he sent the machines after every one of them. Then he went through what they brought him himself, by hand, with a pencil behind his ear.

The first thing he checked was who said any of it, and when. Nothing Pythagoras wrote survives, and he may never have written anything. The full lives of him were written by Diogenes Laertius, Porphyry and Iamblichus, seven and eight hundred years after he died, and they are full of wonders. The early lines about him are few and short. A poet of his own time mocks him for hearing a dead friend's voice in the yelp of a beaten puppy. Heraclitus, a generation later, sneers that learning a great many things did not make Pythagoras wise. That is nearly all there is from anyone who could have met him.

He took the theorem first, because it was the one everybody knew. The relation between the sides of a right-angled triangle was known in Babylon more than a thousand years before him, worked in clay tablets with the numbers set out in rows. No early source gives Pythagoras a proof of it. The story of him sacrificing an ox for joy at finding it is late. He put a single line through *proved* and wrote *no* beside it.

The hammers went next, and they went quickly. The story of the master passing a smithy and hearing the hammers ring out the octave, the fifth and the fourth comes from a writer of the second century after Christ. It does not work, because a hammer's weight does not set the note the way the story says. What does work is a plain string under tension. Stop it at half its length and it sounds the same note higher, the octave, two to one. At two-thirds it gives the fifth, three to two, and at three-quarters the fourth, four to three. An early follower of the school, Hippasus, is said to have shown the same with

bronze discs of different thicknesses. The ratios are old and true; the smithy is a story. He crossed out *hammers* and wrote *string* in the margin.

Theano went third, and she was harder. Diogenes makes her the master's wife, Iamblichus lists her as the wife of another man in the school, and some later writers make her a daughter. He gave the dream the plainest of the choices, a girl of the school who was his pupil, and put the change in the notes.

He built the year from the bottom up, with Aristoxenus underneath, a writer of the fourth century before Christ who knew the last of the Pythagoreans. Through Porphyry he says that Pythagoras left Samos at forty, when the tyranny of Polycrates grew too harsh to live under, and went to Croton. Above him he put Polycrates, who took Samos round about 535. Last, over both, came the chronographers, who put the master's prime in 532 or 531 and liked to make a man's prime his fortieth year. Where all three let the light through, he put *530 BCE* at the top of the page. Its question mark leaned on a man's fortieth birthday that nobody had seen, so he made it a big one.

Then, in his plain hand, the doubt: *The date leans on a habit of setting a man's prime at forty, and the war is twenty years off.* The war came in 510, when a demagogue at Sybaris drove out five hundred of its richest men, they fled to Croton, and Croton, at the master's urging, would not hand them back. The wrestler Milo led Croton out against Sybaris, Sybaris was destroyed, and they turned the river over the ruins. In 530 the two cities were only rivals and neighbours, rich and hungry side by side. The map had the boy's father warning that the war was coming soon. Gary changed it to a father's fear and not a forecast, because a man in 530 could be afraid but could not know.

The last thing he checked was the rules, because they were the part that interested him most. The lives are full of strange ones: do not eat beans, do not stir the fire with a knife, do not pick up what falls from the table. Some of them are old enough that Aristotle collected them, which makes them older than the wonders. Nobody now is sure what

most of them were for. He looked at that a wee while and thought of the plastic sleeve on the kitchen wall. Then he wrote the second line under the doubt: *Philon is mine, and so is the calendar lesson*.

Aye, he thought, the rules were the bit he believed. That one didn't go in the notes either, not even in pencil. He set the stone on the printouts and put the light out.

That night

Rain on the window, the printout of the ratios on his chest, and then under the rain a single string was plucked. Something stopped it halfway along its length and plucked it again, the same note higher. On a wooden board with a little bridge under it lay the Ledger's page, and the Ledger tuned it more than turned it, sliding the bridge along until the note came true.

• • •

His father put him on the boat at Sybaris with a cloak too good for the weather. He gave him a purse too heavy for a boy of thirteen.

"Croton has the best physicians in Greece," his father said, "and its wrestlers win at Olympia, and it has a man from Samos who teaches them to count. Learn what he teaches." He looked up the coast towards the south, where Croton was, as if he could see it. "We're rich and they're hungry, Philon. One day that will come to blows, the way it does between neighbours. When it does, I want one of mine who knows how they think."

Philon did not believe in the blows, because Sybaris was the richest city he had ever heard of, with its broad streets and its feasts and its fields running down to two rivers. Nobody would fight Sybaris, he said, and his father laughed and kissed him and let him go.

They went down the coast for two days, past the long brown beaches and the dark hills behind them. On the second evening the boat came in past a long beach and a river mouth, and there was Croton, its walls and its harbour and its smoke. Beyond it, on the cape to the south, stood the sanctuary of Hera. On the beach below the town a

man as big as two men was wrestling three youths at once, and winning, and laughing. That was Milo, the sailors said, who had won at Olympia more than once. Philon stared at him all the way into the harbour.

At the house of the school a girl met him at the gate. She was perhaps fifteen, in a plain white dress, with her hair tied back and ink on her fingers. She looked at his good cloak and his heavy purse and did not seem much impressed by either of them.

"You're the one from Sybaris," she said. "I'm Theano. I'm to tell you the rules."

• • •

The rules were the strangest things he had ever heard, and she said them in a perfectly ordinary voice.

"Don't stir the fire with a knife. Don't eat beans. Don't pick up anything that falls from the table; leave it. If a man is loading a burden, help him; if he's putting one down, don't. Don't step over the beam of a balance." She counted them off on her inky fingers as she went. "And the first one for anybody new: you listen. In the circle, when the teacher speaks, you don't speak. You listen."

He asked her how long for. She said some of the older ones had listened for five years, and he said *five years* so loudly that a man across the courtyard looked round.

"You're not staying five years. But you listen." She saw his face, and something in her own face nearly laughed. "Go on. Ask."

"What's it for? Any of it. The beans."

Theano thought about it honestly, which he had not expected. "Some of them I know," she said. "The knife and the fire, I think, is about not stirring up a man who's already angry. The beam of the balance is about not stepping over what's fair. Some of them nobody has told me. You keep them first, and then you find out. That's the rule under the rules." She turned towards the house.

He kept them, because he did not know what else to do. He did not eat beans, though he had liked them at home. When a cup fell from the table at supper on his first night he bent for it. Twenty people looked at him, and he left it where it was.

On the second day the steward of the house came for his purse. Philon held on to it out of habit, and the man waited without any hurry until he let go. Everything a pupil brought went into the common keeping, the steward said, and came back out if he left; among friends, things were held in common. Philon had grown up in a house where every jar had a seal on it and every slave knew which seal was whose. For the next hour his hand kept going to his belt for a strap that was not there; the first time, his stomach turned over, and by the fourth he was nearly laughing. He had never known the purse was heavy till it was gone. That evening he ate the same barley and greens and fish as everyone else, off the same plain dishes, and nobody asked what his father was worth.

• • •

In the mornings Theano taught the younger ones on a board with a string along it.

It was only a plank, a plain board longer than his arm with a gut string pegged tight along the top and a small wooden bridge underneath that she could push up and down. She plucked the whole length. The note came out low and round, and he could feel it buzzing in the wood where his knee touched the board. Then she set the bridge at the exact halfway mark and plucked one half.

"Same note," Philon said, before he could stop himself. Then he remembered he was supposed to be listening, and went red.

"Same note, higher," Theano said. "That's allowed. You're not in the circle." She marked the place on the plank with her thumbnail. "Half the string. Two to one."

She slid the bridge to two-thirds along, and the new note was different but sat down beside the first one without any argument. "Three to two." At three-quarters the third note fitted as well, though it took a breath longer to settle. "Four to three."

Philon sat very still. His father's stewards weighed things all day in the market at Sybaris, oil against bronze, wool against grain, and he had watched them since he was small enough to stand under the scale. He had seen the beam come level a thousand times, and this was the beam coming level. The sound came out right because the numbers were right, and if you got the numbers wrong, the sound told on you.

"One, two, three, four," Theano said, and laid her finger on the plank four times. "Add them."

"Ten."

"That's the whole of it," she said. Everything that sounded right was in those four, and the teacher said the body was the same. The physicians here kept a man well by keeping him in tune, hot against cold, wet against dry, not too much of any, and sickness was a string pulled too tight.

He went to bed that night with the three notes going round in his head. When the house was dark, he did the other thing she had told him about. Every pupil did it last thing, lying down, going back over the day: where he had gone wrong, and what he had done, and what he had left undone. He had gone wrong at supper, with the cup, and he had done the string. He left undone a letter to his father, and he lay awake half the night composing it.

In the middle of the summer he took a fever, the kind that came up off the marshes at the river mouths every year, and for four days the room went hot and cold around him. A physician came up from the town, a short man who smelled of vinegar and crushed herbs. He did not shout at the fever or hang anything round Philon's neck. He looked at his tongue and felt the heat of his forehead and the cold of his feet, and asked what he had been eating and how he had slept. Then he took away the rich things that had come in a parcel from Sybaris, and gave him barley water and cool cloths. "Too much heat," he said to Theano, as if the boy were a pot left too long on the fire. "Bring him down slowly. Don't drown him."

At night, when the fever was worst, Theano sat by the door with the practice board across her knees and played the three notes over softly. She played the whole string and the half and the two-thirds, again and again, until he slept. On the fifth morning he woke clear-headed and starving, and she was asleep against the doorpost with the board still in her lap. He lay and looked at her and at the string for a long while. He thought that the physician and the string were the same craft, and that nobody at home would believe him if he said so.

• • •

At the end of his first month the teacher spoke in the circle about the moon.

Philon had seen him before, across the courtyard, a tall man of about forty in white, with a quiet voice that carried further than it should. In the circle the pupils sat on the ground under the plane trees, the older ones at the front and the new ones at the back. Philon sat at the very back, with his mouth shut, which was the hardest thing he had done in his life.

The teacher drew in the dust with a stick. He made a big circle for the year, and twelve small ones round it for the months of the moon. "Twelve moons," he said, "fall short of the year by eleven days. Every farmer knows it. The festival creeps backwards through the seasons until the feast of the harvest falls in spring. So the cities put in a month, now and then, when somebody notices." He drew a thirteenth small circle, outside the ring. "Now and then is not a rule."

Then he drew eight big circles in a row beside the first. "Eight years. Ninety-nine months. Three of them put in, at the third year, the fifth and the eighth. Then the moon and the sun come round together, near enough, and the festival is back where it began. Near enough is a phrase I'll come back to," he said, and laid the stick down.

Philon's hand had gone into his tunic for a stylus he had not brought. He did the sum instead, in his head, in his father's market way, the way you work out who owes what at the end of a season. It came out: eight years, ninety-nine moons. He wanted to shout it across the circle.

He did not, though it nearly choked him. He listened, which was the rule.

Afterwards, in the shade by the well, he caught Theano by the sleeve. “It comes out,” he said. “Ninety-nine. I did it.”

“It nearly comes out.” She drew it in the wet mud by the well with her finger. “It slips. A day and a half in every eight years. Not much. In a man’s life, ten days. In a city’s life, the festival walks off again, slower.” She wiped it out with her palm. “He knows. That’s what he meant by near enough. Somebody will find a better count one day.”

“So what’s it for? If it’s wrong.”

Theano looked at him as if he had finally asked a proper question. “It’s for being wrong by a day and a half instead of by eleven days. And for knowing that it’s a day and a half, and saying so.” She stood up and wiped her hands on her dress. “Anybody can be wrong. The number tells you how much.”

• • •

The letter from his father came at the end of summer, carried up the coast from Sybaris.

It was short, in a steward’s hand. The talk in the agora at Sybaris was ugly, his father wrote. Croton’s men had jeered at Sybaris’s at the games, and Sybaris’s men had jeered back. The old men were saying things they ought to have kept for the wine. It was nothing, very likely, but Philon was to come home before the winter storms, and he was to bring what he had learned.

He read it three times and then went and sat on the wall above the harbour, where nobody would see his face.

He did not want to go. He had not known that he loved anything until he had to leave it. He loved the string and the circle, and the strange rules and the plain white clothes, and the sums in the dust. He loved the evening going-back-over, and Theano’s inky fingers, and the way nobody here pretended to be right when they were only nearly right. At home they would ask what he had learned and he would say *ninety-nine*, and they would laugh at him.

On his last day the teacher called the circle, and Philon sat at the back, and the teacher looked straight at him.

"The boy from Sybaris is going home," he said. "He has listened all summer. Let him speak once." He waited until the circle was quiet. "What's the number for?"

Everybody turned round, and Philon's mouth was completely dry. He thought of the string pulled too tight, and the cup on the floor. He thought of his father's stewards in the market, and the festival walking off through the seasons a day and a half at a time.

"For finding out when you're wrong," he said. "And by how much. And then saying so."

The teacher looked at him for a moment and nodded once. Then he went on to something else, like a man moving past an ordinary answer from an ordinary boy.

Theano walked him down to the harbour in the morning. At the boat she gave him something wrapped in a scrap of linen. It was the gut string off the practice board, coiled small.

"We've another," she said. "Put it on a plank at home. Stop it at half and see if they laugh then."

Then the boat put out, and he held the coil in his fist all the way up the coast, until Croton was gone behind the headland. He could still feel the shape of it in his palm when the walls of Sybaris came up out of the evening on the second day.

The master page

The Ledger stretched this page along a board like a string, and when he touched it, it hummed.

In ink: Aristoxenus, through Porphyry, on the master leaving Samos at forty under Polycrates; Croton's physicians reckoned the first in Greece, and Milo of Croton winning at Olympia; the ratios of the octave, the fifth and the fourth, two to one, three to two, four to three; the strange rules, some of them old enough for Aristotle to collect; the eight-year cycle of ninety-nine months that the Greek cities used to keep their festivals in season. In pencil: the year, which leans on a prime set at forty; Theano as a girl of the school and not a wife;

the board with its string; the evening going-back-over, which comes to us in verses written down much later; and any lesson on the calendar at Croton at all. *Found destroyed*: every word the master said in his own voice, and the reasons for most of the rules. Struck through in the margin are the smithy's hammers and the proof that was never his.

He turned the new page over against the others. The scribes of Egypt had written their count, the mountain road had sung it and the canoe had brought home a survey. Druim Snechta handed on the copy, and Massalia took the measure with a pole and a shadow. Croton's part was the school: the first time the pieces were brought into one room and taught as one craft, number and music and the body and the moon. It had rules to hold them together and a silence to keep them in. Behind, inked already, were Delos and its nineteen years, which Meton would fix a century after this, the better count Theano promised. In pencil, further on, a school that keeps its secrets can lose them in a night, and there would be a fire on the hill at Croton.

He would give it to the Core first thing: two to one, three to two, four to three; ninety-nine months in eight years, and the day and a half of slip. It would run the ratios and the slip faster than he could type them. Of what the master said under the plane trees, or what any of the rules were for, it knows nothing, and the hum of the laptop fan was the nearest it would ever come to music.

What's the number for? The boy's answer is still the best one the book has found. It is for finding out when you are wrong, and by how much, and then saying so.

Philon wanted the harvest feast to fall at harvest and the farmer to sow in the right month, so the whole city kept the same day. Under that, the dream had been harder on him. The sky is where you find out that near enough and right are two different things, and that the gap between them can be measured: a day and a half of slip in eight years. Gary had watched a crane driver set a load down a finger's width off its mark and lift it again without being asked, and as far as he was concerned that man had been to Croton.

Nearly five hundred years on, over the mountains in Gaul, a boy was being taught a count of months from a knotted cord, and was not allowed to write it down.

The note went on after the page had gone quiet, the string's note and then the same note higher, two to one, and it was still in his head at the kettle the next morning. He hummed it, flat.

CHAPTER 15

Twenty Years of Verses

The country of the Sequani, between the Saône and the Jura, and a grove above the Doubs, in the year Caesar wrote down what he knew of the druids: the found year, 53 BCE



Illustration 16. *Ategnatos runs the knotted cord through his fingers while Marcus listens.*

Belfast, by day

WHEN HE WAS BANKSMAN AT Shannonbridge, there was a laminated card of crane signals cable-tied to the rail by the cabin steps. Nobody ever looked at it with a load in the air. You did not stop to read how to say *stop*. The signals were in your arms and your shoulders before they were anywhere else: the flat palm, the slow circle, the crossed arms that meant everything halts now. The card was there for the inspector and the new start. The real record was in the bodies of the men on the ground, and it was checked every lift, by whether the load came down where it should.

The card was still in his head when the druids came back on the screen.

"The druids memorised a thirty-year lunar calendar over twenty years of training and were forbidden to write it down," it said. "The Coligny calendar of about 55 BCE is their calendar, recorded in bronze."

"Half of that's from Caesar," he said, "and the other half's from somewhere else entirely. Show me which is which."

He put the machines to gathering passages, editions and arguments. Then he sat down and read them, every one. He went through them himself, by hand, with the stone holding the printouts flat against the draught.

Caesar first, because nearly everything comes back to Caesar. In the sixth book of his war in Gaul he stops the fighting for a few pages and describes the druids. Young men come to them in great numbers to be taught, he says, and some stay twenty years. They learn by heart a great number of verses. They do not think it right to commit the teaching to writing, although for almost everything else, public and private accounts, they use Greek letters. He gives two reasons, as a Roman would see them. The first is that the teaching should not spread among the common people, and the second that the pupils, trusting to the written word, should not let their memories go slack. He adds that the Gauls count time by nights and not by days, so that the night comes first and the day after it.

That was ink, but it was a Roman's ink. Caesar was the man at war with these people, writing for readers in Rome. Some of what he says about the druids may come from an older Greek writer, Posidonius, a generation before him. He wrote *outsider* in the margin and underlined it, and moved on to the next line of the machine's summary.

The thirty years were not Caesar's at all. They came from Pliny, a century later, who says the druids begin their months and their years on the sixth day of the moon, and their age after thirty years. That was a different source, a different century, and the machine had stitched the two together into one sentence. He separated the two of them on the printout with a pencil line.

The calendar was the third claim, and the worst. The Coligny calendar is real: a bronze plate found in 1897 in a field near Coligny, in the Ain, close to the Roman road from Lyon to Vesontio. It had been broken into nearly a hundred and fifty pieces. It sets out five years of lunar months, sixty-two of them, with an extra month put in every thirty months to keep the moon in step with the sun. The months are marked good or not good, and each one is split at the middle by a word that seems to mean the returning night. But it is in Roman letters, and it was made in the first or second century after Christ, the experts disagree which, a century or two after Caesar, when Gaul was Roman. Nothing on it says *druid*. The machine had moved it back a century or more to make it fit. He put a single line through *55 BCE* beside it and wrote *first or second century* above it.

The year went down in layers, like a road being laid, with Caesar's description at the bottom, which sits in his account of the year 53, the summer he crossed the Rhine for the second time. Over that he put the Sequani, whose country ran from the Saône to the Jura. Their chief town was at Vesontio on its loop of the Doubs, which Caesar describes from having marched into it. The top layer was the Roman wine traders, who were already in the towns of the Saône before the war and whose amphorae turn up all over that country. Where they agreed, he headed the page *53 BCE* in ink. He left the question mark off at first, because Caesar had been there; then he remembered who had written Caesar's book, and put it on.

Below, the doubt, plainly put: *Caesar is an enemy describing what he wanted Rome to know, and Coligny is a century or two later. Nobody can show the druids' count was Coligny's count*. And under that, because he owed it: *The knotted cord is mine. No source puts a cord in a druid's hand.*

The map had the year as 55, and he moved it to 53, because that was where Caesar had put his pages on the druids, and he wrote the change in the notes.

He sat a wee while with the bit about memory. Caesar had meant it as a criticism, or as a curiosity, and maybe it was. But he had seen men on the ground at Shannonbridge who could not have told you the page number of a single signal and who never once got one wrong.

Aye, he thought, dead on: a thing you carry in your body is checked every time you use it, and a thing on a card is checked when the inspector comes. He left that unwritten and went up to bed.

That night

His arms were folded across his chest, the way you hold them for the last signal of a shift, and the room went dark and green round them. Water ran fast over stones somewhere close. The Ledger's page came to him tied up with a cord, and the Ledger undid it knot by knot, counting under its breath, before a single word could be read.

• • •

The old man made the cord in front of him on the first morning, so that Ategnatos would see it made.

It was a long cord of twisted wool, dyed dark. The old man sat on a stone at the edge of the clearing with the cord across his knees and tied the knots one after another without looking at his hands. Thirty knots he tied, each a finger's width from the last, and then a bigger knot, a fist of a knot, and then he began again. Below them, through the oaks, the river ran round its great loop under the town, brown and fast with the snow off the mountains.

"A knot is a month," the old man said. "Thirty knots are thirty months. After thirty months, the moon has run ahead of the sun by a whole month, and so we give the sun a month back. That's the big knot."

Ategnatos was twelve. He had been brought up the hill from the town by his mother's brother three days before, with a cloak and a pair of shoes and a bag of barley. His uncle said he was lucky, and that he would learn for twenty years and come out of the trees knowing more than a king. He did not feel lucky; he felt cold, and he could not stop looking at the cord.

"Is that the count?" he said. "Is it all on there?"

"It's on there this morning." The old man held up the end of it. "Tonight I'll burn it."

Ategnatos stared at him. “Why burn it? Is it a sin to keep it?”

“Ask me that again when you’ve had it a year,” the druid said.

“The cord is for your eyes for one day,” he said. “After that it’s for your head. Watch the knots. Then shut your eyes and see them. Then say them.”

• • •

So he learned the count the way he would learn everything in the grove, by saying it.

The months went in pairs and threes in the verses, each with its name and its number of nights. Some had thirty nights and were good, and some had twenty-nine and were not good. On those you did not start a journey or a marriage or a war if you could help it. Every month was cut in two at the middle by the returning night, when the moon turned back from full and began to go. You counted the nights first and the days after them, because the dark came first, the old man said, and everything was born out of the dark. You began the month on the sixth night of the moon, when she was strong enough to throw a shadow. And after thirty years, which was an age, you began the whole wheel again.

The older pupils said the verses in the evening round the fire under the oaks. There were boys of fifteen and young men of twenty-five. There was one man with grey in his beard who had been in the grove for nineteen years and had one year left. When he said his verses the others went quiet, because he never stumbled, and he never had to go back. Ategnatos listened to him the way he had once listened to his father’s old dog breathe at night, without knowing he was doing it.

He stumbled all the time, and lost the eleventh month, and the big knot, and the order of the good and the not good. The old man did not shout; he said, “Again,” and Ategnatos said it again, and then again, until his tongue knew the way before his head did. It was like walking a path in the dark that you had walked in the day.

On the evening of that first day the old man fed the cord into the fire a knot at a time. Ategnatos watched the wool catch and curl and go black, and he said the knots under his breath as they burned, to see if he still had them.

He had most of them.

That night he saw why the month began when it did. The old man took him out of the trees to the bare shoulder of the hill, where the sky was open, on the sixth night after the new moon. The moon was a bright bow low in the west, and the old man stood him with his back to it and made him look at the ground. There, faint on the pale grass, was his own shadow, thin as a stroke of charcoal, but there. "That's the start of the month," the old man told him. "When she's strong enough to cut you out of the dark. Not before." Then he turned the boy round and named the stars for him as they came out over the mountains, the ones that went round and the ones that wandered. He said that the young men learned how big the world was from them, and how far the stars ran in a year. Ategnatos lay awake for hours afterwards on his bracken, with the shadow still on the backs of his eyes. He found that the first five knots were there without his calling them.

• • •

The Roman boy came up the river in the spring with his father's wine.

There were Romans in the town now. There had been traders before the war, men from the south with their big clay jars of wine who came up the Saône and stayed in the market and paid well. Now there were soldiers too, passing through towards the north and the Rhine. The town had been told it was a friend of Rome and was expected to act like one. The grove was quiet about the soldiers, and the old man said nothing about them at all, which Ategnatos had learned was how the old man said the most.

Marcus was thirteen, a year older than him. He had come with his father to sell wine to the headmen and to buy salt meat and hides to send back down the river. Ategnatos met him because he was sent

down to the town for barley. The Roman boy was sitting on the quay with his feet in the water and a flat wooden thing on his knees, scratching at it with a pointed stick.

"What's that?"

Marcus held it up for him to see. It was two little boards hinged together, and the inside of each was filled with dark wax, and the wax was covered in small marks. "Tablets. My father's accounts. Who owes us for the wine, and how many jars, and when."

Ategnatos knew the marks. They were Greek letters, the same ones the headmen in the town used to keep their own counts of cattle and grain and debts; even the old man could read them. He did not know what they said here. He looked at them, and kept looking, and then he looked at the river, so as not to look at them any more.

"You can read them?"

"Of course," Marcus said, surprised. "Can't you?"

"We're not supposed to. Not for what we learn."

Marcus did not understand that at all, and Ategnatos did not try to explain it. But the next day, when he came down again with a reason he had made up, the Roman boy was on the quay again, and on the next day too.

In the middle of the summer the soldiers came through the town in earnest, a long column on the road north. With them came a Roman with a box of tablets who sat at a table in the square and wrote. Ategnatos was down for barley that day and stood at the back of the crowd and watched. The Roman wrote the names of the headmen, and of their sons, and how many cattle each man had and how much grain was in each store. The headmen stood in front of him one at a time and answered. When he had finished he closed the tablets and tied them with a cord and put them in the box. The box went north on a mule with the column, and the square was very quiet after it had gone. Ategnatos walked back up the hill slowly. He understood something then about the old man's quiet that he had not been able to put

into words before, and he did not try to put it into words now. He said the count all the way up the hill instead, under his breath, and it was all still there, where nobody could put it in a box.

• • •

They were friends by midsummer in the way boys are who have only half a language between them and do not need the other half.

Marcus taught him Roman words for the things on the quay, *amphora* and *salt* and *river*, and laughed at his accent. Ategnatos taught him the names of the fish and showed him how to take a trout with his hands under the bank. Marcus told him about the south, where it was hot all year and the sea was blue, and where his father had a house with a red roof and a garden. Ategnatos told him about the grove, not the verses, but the oaks and the fire and the man with grey in his beard who never stumbled.

"Twenty years," Marcus said. "Twenty years of saying things over. Doesn't your head get full?"

Ategnatos thought about it before he answered. "No. It gets bigger."

On the night of the full moon they sat on the quay after the traders had gone in, and Ategnatos said the count for him, quietly, in his own tongue. The months and their nights, the good and the not good, the returning night in the middle of each, and the big knot every thirty months. Marcus did not understand the words, but he heard the shape of it, and when Ategnatos stopped he was quiet for a while.

"My father would write that down," he said. "In about a morning. Then he'd never have to say it again."

"Then he'd never have it."

Marcus looked at him in the moonlight, puzzled by him and a little impressed. Then he held out the tablets and the stylus.

"Go on," he said. "Write something. Anything. I'll show you the letters. Nobody will know."

Ategnatos looked at the smooth dark wax and the bronze point of the stylus. He felt, very clearly, the cord burning knot by knot in the fire, and the old man's voice saying *again*. He thought of the soldiers on the road north, and of everything they took and wrote down and carried away.

"Is it a sin?" he said. "If I write?"

"It's a ledger," Marcus said, and shrugged, the way he did about everything. "That's all writing is. It's how my father knows who owes him."

• • •

He went back up to the grove in the dark with the question going round in his head like a knot that would not untie.

The old man was still awake by the embers. He did not ask where Ategnatos had been; he knew, Ategnatos thought, the way he knew which nights were good. He only moved over on his stone to make room.

"The Roman boy writes everything," Ategnatos said at last. "His father writes who owes him. He says I could write the count in a morning."

"He could," came the answer. "He'd have it on a board. And when the board's in a soldier's bag on the road north, he'd have it there too. A count you carry is checked every time you say it. Say it wrong and the month's wrong, and the farmer sows on the wrong night, and you know it by harvest. A count on a board is checked when somebody remembers to look."

"So it's a sin."

The old man let the silence run. The fire ticked, and somewhere below them in the dark the river went round its loop.

"It's a rule," he said. "We keep it because of what's been taken. And because a head that leans on a board stops working. But I've never once said writing was a sin. Only that this isn't for writing. Not yet. Not while they can take it."

• • •

At the end of the summer Marcus's father had sold all his wine, and they were going back down the river.

Ategnatos came down to the quay at first light to see them go. The boat was loaded with hides and salt meat. Marcus was sitting on the bales with his tablets on his knees, and when he saw Ategnatos he jumped down and held them out without a word, with the stylus.

Ategnatos took them. The wax had been smoothed flat and ready, and he did not write the count. He had decided that on the hill, in the dark, with the old man beside him. He held the stylus the way he had watched Marcus hold it all summer. Very slowly, in the Greek letters the headmen used for cattle, he wrote the only thing he knew how to spell, which Marcus had taught him on the second day for a joke: MARCUS. Then, under it, slower, he wrote his own name, and gave the tablets back. Marcus looked at the two names, crooked and too big, and for once he did not laugh.

"That's not a ledger," Marcus said.

"No."

"It's not a sin either."

Ategnatos found he was smiling. "No. It's so you know who you met."

Then the boat pushed off into the current, and Marcus held the tablets up over his head until the bend in the river took him. Ategnatos stood on the quay and said the count under his breath, all of it, every knot and the big one, to see if he still had it, and he did.

The master page

The Ledger did not lay this page flat. It coiled it like a cord on a stone, and left the end hanging, so that it could be untied.

In ink: Caesar's druids, with their twenty years, their verses learned by heart and their rule against writing the teaching down, and their Greek letters for everything else; the Gauls who count nights before days; Pliny's months and years begun on the sixth day of the moon, and his age of thirty years; the bronze plate from Coligny, with its five years of sixty-two months, its extra month every thirty months, and its good and not good, made a century or two later in Roman letters. In pencil: the year, set where Caesar set his pages; the old man and the grove above the Doubs; Coligny's count as the druids' count; and the cord, which is mine. *Found destroyed*: every one of the verses, by the rule that kept them, and the bronze plate itself, which was found in nearly a hundred and fifty pieces.

Set beside the rest, the page made its own claim. Egypt wrote, the mountain road recited and Croton built a school with a silence at its heart. The grove's part was the memory: a count carried in the body for twenty years, checked every time it was said, kept off the page not out of ignorance but out of fear of who would carry the page away. It is the mountain road's recitation with a war around it. In ink, looking back, it answers the rule at Croton that a newcomer listens first. Pencilled in ahead are the black-robed women on the shore at Mona, and a Roman cavalryman swimming the strait.

What the Core could chew on was the numbers: thirty months to the extra month, sixty-two months in five years, thirty years to an age. It can show how near the bronze plate comes to keeping the moon and the sun together. Not one verse can it give him, though, or whether any druid ever knotted a cord. The cord was his, and the notes said so.

Is it a sin? Not the writing. The old man never said so. The sin would have been to put the count where a soldier could carry it off, and the boy wrote a friend's name instead.

Ategnatos had it for the farmer: sow on a good night and not a bad one, and begin the month when the moon is strong enough to throw a shadow. What the dream added was plainer, and Gary knew it from the job. A count you carry is checked every time you use it, the way a signal is checked every time a load comes down. Nobody on a lift

keeps the signals on a card in his pocket; he'd be dead before he found it. You say them until your hands know the way, and when somebody comes looking for the page, you write a friend's name on it.

Six hundred years later, across the sea on Iona, a boy on kitchen duty was cutting bread for a sailor who had just come back from the north with salt in his beard.

His arms were still crossed over his chest when the grove let him go, the way you cross them for the last signal of a shift. Somewhere a car went by on a wet road, and he gave it the stop signal without moving a finger, and it did not stop, and that was all right too.

CHAPTER 16

The Boy and the White Pebble

Iona, the island of the monastery, one summer in the time of Columba, with the open sea to the north of it: the found year, 580 CE



Illustration 17. *Lugaid in the kitchen doorway on Iona with the white pebble.*

Belfast, by day

THE FIRST JOB HE HAD after school was kitchen porter in a restaurant in Belfast, and the thing he remembered best about it was that the porter heard everything. Nobody talked to the boy at the sink, so everybody talked in front of him. The chefs argued about the pass, and the waiting staff came back through the swing doors with the night's stories still on their faces. The owner did his sums out loud by the walk-in fridge. He had stood there at nineteen with his arms red to the elbow and the whole place going past him. He had kept most of it, because nobody had told him not to.

The sink was on his mind when the search on Iona turned up its answer.

"Cormac Ua Liatháin sailed fourteen days north from Iona in about 580 and reached Iceland," it said, "where he met the creatures of the Arctic sea."

"Where does it say Iceland?" he asked. "Give me the line, not the gist."

The machines went off for the text, the translations and the arguments. Then he read every page of it himself, his stone from the beach on the corner of the printout.

The source is a single book, the Life of Columba that Adomnán, the ninth abbot of Iona, wrote round about the year 700, a century after the saint died. The oldest copy of it was written on Iona itself by a monk called Dorbbéne, who died in 713. That copy left the island long ago and is in a library in Schaffhausen, in Switzerland, which is about as good as early books get. Adomnán was writing about his own house, from what the old monks had told the younger ones, and he says so.

Cormac is in it three times, each time at sea. The first time he goes looking for what Adomnán calls a desert in the ocean, an empty place where a hermit could live alone with God, and does not find it. The second time he is the one in the boat, and the saint says he will not find it this time either. The third time is the long one, and it is the one the machine meant. For fourteen summer days and as many nights, Adomnán says, his boat ran under full sail before a south wind, straight to the north, beyond the bounds of human wandering, to where return seemed impossible. Then the sea filled with small loathsome creatures about the size of frogs. They stung, and they struck at the keel and the sides and the prow as if they would pierce the hides the boat was covered with. On Iona, at the same hour, the saint saw it in his mind, rang the bell, and set the brothers praying. The wind went round into the north, and Cormac came home.

That was ink, in the sense that the book says it. The book does not say Iceland. Nor does it say how far fourteen days was, or what the creatures were, and people have guessed jellyfish and worse. He put a single line through *Iceland* on the printout and wrote *north* beside it. Then, because he was honest with himself, he wrote one more line

under that. An Irish geographer called Dicuil, writing in 825, says that clerics had lived on an island he called Thule about thirty years before, where the sun at midsummer hardly set at all. That was two hundred years after Cormac, and it was not about him. But it meant that boats out of these islands did go that far, and came back and said so.

He built the year up from Columba, who came to Iona in 563 and died there in 597, by the Irish annals. Next over it came the king of the Picts whom the saint visited on the river Nesa, where he blessed a white pebble for the king's druid, and the annals have that king dead by 584. Uppermost was the book itself, which gives no year for any voyage at all. The overlap gave him *580 CE*, which he wrote large, and after it a question mark about the size of a wee white pebble.

His doubt, written plainly below: *Adomnán dates nothing, and he was writing a saint's life, where the point is the saint. The year is mine, inside a window of thirty-four years*. And under that, because he owed it: *Lugaid is mine, and so is the kitchen*.

He looked at the frogs for a wee while, and at the bit where the saint rang the bell. Whatever it was, the book was clear about one thing. The man came back, and told it, and somebody in the house kept it for a hundred years until Adomnán wrote it down. Aye, that was the sink all over, and no mistake. He went to bed grinning about it, and he was still grinning in the dark.

That night

With his sleeves still pushed up, the way you push them for the sink, he went under into peat smoke and wet wool and the smell of bread in an iron pan. By a fire, on a wooden board, the Ledger flipped its page over, quick and careful, like a flat loaf done on one side. Somewhere down at the shore a bell was ringing for the boats.

• • •

The bell went in the middle of the afternoon, which was the wrong time for a bell, and Lugaid was elbow-deep in the bread trough when it did.

He was twelve, and he had been on Iona since he was seven, fostered to the house by a family who wanted a son in the church and could spare one. He did kitchen duty every third week with Brother Diarmait, who was old and deaf and kind. Diarmait made the bread for the whole island in a stone oven built into the bank. Lugaid liked the kitchen better than anywhere on the island. It was warm, and it was the one place where everybody came, sooner or later, and talked in front of a boy with flour on his arms.

Every boat that came in to the island came past the kitchen door in the end, and Lugaid had the same question for every one of them. "What did you find?" The brothers back from Mull with a load of timber laughed and said timber, and the ones back from Ireland said rain and cousins and a new psalter to copy. He kept all of it, the way you keep crumbs in a fold of cloth. He had been asking since he was seven, and Brother Diarmait said he would grow up to be an abbot or a nuisance, one or the other, and that the island had room for both.

The bell kept going, on and on over the noise of the wind. Brother Diarmait could not hear it, so Lugaid pulled his sleeve and pointed, and they went out together into the wind.

Everybody was coming in from the fields and the shore towards the church. There were monks with their hoods back and the lay brothers from the barley and the two boys who minded the cows. The

wind was strong from the south, as it had been for days, and it pushed at their backs all the way up the path. At the church door the old abbot was standing in the wind with his hand on the bell rope. He was very old by then, and thin, and his eyes were on the north.

"Pray for Cormac," he said, when they were all there. "He's gone further than a man should go, and he's in danger now. Pray that the wind turns."

They knelt in the church, and Lugaid knelt at the back between Brother Diarmait and the door. He did not know how to pray for a wind, so he said the psalms the way he had been taught and thought very hard about a boat. He had seen Cormac's boat on the beach in the spring before it went. It was a big frame of wicker and light wood with ox hides stitched over it, greased black. He thought about the hides, and the men in it, and the open sea he had never seen the end of.

How long they knelt he never knew. At some point the noise at the door changed. The wind that had been leaning on the door all afternoon from the south let it go. There was a quiet, and then the door rattled again from the other side.

Brother Diarmait could not hear it, but Lugaid could. He put his hand flat on the cold earth floor, as if he could feel it there too, and he counted the days since the boat had gone, and it came to fourteen and more.

• • •

The boat came in eleven days later, at dusk, on a north wind.

Lugaid saw it from the kitchen door, a dark shape coming round the end of the island under a ragged sail. Then he saw the brothers running down to the beach to drag it up. He was told to stay where he was, and so he stayed, and kept the fire up, and set out the bread. He was still there an hour after dark when the door opened and Cormac came in on his own, grey with salt from his eyebrows to his boots.

He was not old, but he looked old that night. He sat down on the bench by the fire without saying anything and held his hands out to it, and they shook. Lugaid had seen him before the voyage, a big brown man who laughed on the beach, and this was not the same man.

"Is there bread?" Cormac said.

Lugaid gave him the end of the loaf and a bowl of milk. Then he gave him more bread, and the heel of the cheese that Brother Diarmait kept for the sick. Cormac ate it all without a word, slowly, like a man remembering how. When he had finished he sat looking into the fire. Lugaid sat on the other end of the bench and waited, because nobody had told him to go.

"What did you find?" he said at last, when it seemed nobody else was going to ask.

"Nothing," Cormac said. "Sea."

• • •

He told it to the fire more than to the boy, and Lugaid kept still and let him.

They had gone looking for an empty place, the way Cormac always did. They wanted a rock in the ocean with nobody on it, where a man could live alone with God and the birds. The south wind had taken them, and they had let it, because that was the point; you went where you were sent. It had blown for fourteen days and fourteen nights without a break, straight north, so steady that they hardly touched the sail. The islands had gone first, then the birds they knew, and then the dark at night had shrunk until there was hardly any of it at all. There was only a grey hour, Cormac said, and then the sun coming back up on the wrong side of the sky.

"And no land," he said. "Not a rock. Not a bird that knew land. The sea just went on, the same colour, and it went on, and there was nothing in it at all, and then there was too much."

Then there were the creatures, which had come up out of the water in their thousands one day near the end. They were small soft horrible things about the size of a frog, and they stung if they touched you. They had beaten against the boat from every side, the keel and

the sides and the prow. The men had seen the hides bulge, Cormac said, and they had thought the skins would go. They thought they would drown out there in the middle of nothing where nobody would ever know.

"And then the wind turned," Lugaid said.

Cormac looked at him properly then. "How do you know that?"

"We prayed for it. He rang the bell, in the afternoon, and we all prayed. I felt it go round at the door." He counted on his floury fingers. "It was the fourteenth day. Or the fifteenth."

Cormac looked at him a while, and then he put his face in his salty hands. Lugaid did not know whether he was laughing or crying, and he did not think Cormac knew either.

• • •

Cormac stayed on the island all that summer, mending. He came to the kitchen most evenings, because it was warm, and because the boy did not talk unless he was talked to.

Lugaid showed him the white pebble on the third evening. He had found it two summers before. He had gone with the abbot's party up the long lake and the river to the king of the Picts, to carry the bag of books. The king had a druid called Broichan, who was keeping an Irish girl as a slave and would not let her go. The abbot had stooped at the river Nesa and picked a white stone out of the water and blessed it. He said it would heal the druid if he let the girl go, and not if he did not. Lugaid had not seen the healing, but he had stooped at the river beside the abbot and picked up a white pebble too, the same as the saint's. He had kept it ever since in a fold of his tunic.

"They say his floated," he told Cormac. "In the water, like an apple. And the druid drank and he was well, and they let the girl go."

Cormac turned the pebble over in his big fingers. It was milky white and smooth, with one grey line through it like a hair.

"Does yours float?"

"No," Lugaid said. "I tried. It's just a stone."

Cormac laughed for the first time since he had come home, a short surprised sound. He gave it back.

"Keep it," he said. "A stone that does what it's told is a fine thing to have. But a stone that's just a stone is a fine thing too. It's the one you can trust."

• • •

On a clear night near the end of the summer Cormac took him up the hill behind the house, where the brothers went to look for weather. They sat on the wet grass in the half dark that was all the night there was in that season.

"You asked me how I knew it was north," Cormac said, though Lugaid had not asked him anything. "When there was nothing to see."

He showed him the star that does not move, low over the sea, and the others that turn round it all night like cows round a stake. He showed him how the sun went down in the north-west in summer and came up again in the north-east, a little way along, as if it had only gone behind a wall for a while. Out there, he said, it had hardly gone down at all, and that was how he knew how far they had come before he knew anything else. The sky had told him before the sea did.

"I cut a notch on the gunwale every time the sun came round to the same place," he said. "Fourteen notches. You can't keep count in your head out there, it goes soft. You keep it in the wood."

Lugaid lay back on the grass and looked for the star until he had it. He thought about the notches, and the bell, and his own counting with his fingers on the church floor. Two counts had been running at the same time, one out there and one in here. Neither of them had known about the other until a man came in and asked for bread.

"Who keeps the count when you're gone?" he said.

"You do," said Cormac. "You've been doing it all summer."

• • •

Before the summer ended he was making ready to go again. Lugaid knew it before anybody said it. Cormac stopped coming to the kitchen in the evenings and started going down to the boat on the beach and standing beside it.

"You found nothing," Lugaid said to him, on the last evening, when he did come in. "You said so. Why go back?"

Cormac sat on the bench and was quiet for a while. "I didn't say it right," he said. "That first night. I was cold." He held his hands out to the fire, steady now. "I found where it stops. Where there's no more rock, and no more birds, and the sea is nobody's. Nobody had told it before, so nobody knew it was there. Now they do. The house knows it." He looked across the fire at the boy. "You know it."

"So you found something."

"I found the edge," Cormac said. "That's a thing. You can't go past what nobody's told you is there."

Lugaid looked into the fire a while. "Take me next time."

"No." Cormac said it gently. "Somebody has to be in the kitchen when we come back. Somebody who heard it the first time, and kept it, and can tell the abbot after us, and the one after him." He stood up and laid a hand on the boy's head. "Keep your stone. Keep the bread hot."

• • •

The boat went out on the next morning's tide, on a fair wind from the east, and the whole house went down to the beach to bless it. Lugaid stood at the back with Brother Diarmait, and when the sail was a grey mark against the grey sea, he took the white pebble out of his tunic. He held it in his fist until it was warm. Then he went up the path to the kitchen and stoked the oven. He said the fourteen days over to himself while the fire caught: the south wind, the islands going, the grey hour, the sea that went on, the creatures, the bell, the turn.

He would say it to the cook who came after Brother Diarmait, he thought, and that cook would say it to the boy after him, because that was how the house kept things.

The master page

Here the Ledger chose a board by the oven for the page, where it would stay warm and nobody would forget it.

In ink: Adomnán's Life of Columba, written on Iona round about 700, with its oldest copy, Dorbbéne's, kept now at Schaffhausen; Cormac's search for a desert in the ocean; the fourteen days and nights north before a south wind, beyond the bounds of human wandering; the stinging creatures the size of frogs against the hides; the saint at home ringing the bell and the wind turning; the white pebble from the river Nesa and the druid Broichan. In pencil: the year, set inside a window of thirty-four years; what the creatures were; how far north fourteen days could take a boat; and Dicuil's clerics in Thule two hundred years later, a different voyage that shows the same road. *Found destroyed*: whatever Cormac said himself, before it was told and retold for a hundred years by men who had it from the kitchen.

The page found its place among the others. Egypt kept the record and the mountain road the recitation, while Druim Snechta copied what was there and marked what was not. Croton kept its silence, and the grove kept a count in the body. Iona supplied the harbour: the house the boats come back to, where a man who went to the edge can sit by the fire and say *nothing, sea*, and a boy can keep it until somebody writes it down. It is Druim Snechta's little book seen from the other end, the stories before they reach the vellum. Already inked, behind, is Thule, the island six days north of Prettanike, and Pytheas's sea that you could neither walk on nor sail. Further on, in pencil, wait the Easter quarrel that Iona's monks would carry to Whitby, and the statutes that would one day be signed on this same island.

The Core would have it by breakfast: fourteen days, a south wind, the sun that hardly sets. It can work out how far a hide boat might run in fourteen days before a steady wind. It is blind to what Cormac saw, or whether a boy on kitchen duty ever felt the wind turn at the church door. The kitchen knew more than the Core did, that summer.

What did you find? Nothing, sea. It turned out to be the most useful answer anybody gave in that house all summer, because it marked the edge.

Lugaid had wanted to know where north was when there was nothing else, and how many days to count before he could start to worry. The dream answered him from the other end: the sky over the edge is

the same sky as over the kitchen, and somebody has to stay under it and keep the count for the ones who went. Gary had been the one at the sink often enough to know the job. The porter hears everything, keeps the bread hot, and is there when they come in off the boats wanting to tell it, even when what they have to tell is *nothing, sea*.

Fifty years earlier, in another boat of hide, the youngest monk was on the stern oar when the green island came up out of the fog, and Brendan was asleep in the bow.

In Belfast the rain came round to the north in the night and started hitting the other side of the house, the way a wind will when somebody on an island is ringing a bell for it.

CHAPTER 17

The Boat of Hide

The western ocean, out of Kerry, in a boat of ox hides with fourteen monks and three who came late, in the years the story gives as seven: the found year, 530 CE



Illustration 18. *Brendan hails the hermit from the bow of the hide boat.*

Belfast, by day

THE RAIN WAS STILL ON the back windows in the morning. At Vodafone, in the call centre in Belfast, the rule for the notes was that you wrote what the customer said, not what you thought of it. Every note began the same way, with *customer states*: customer states the bill is wrong; customer states the phone was never delivered; customer states he rang last month and was promised a call back. You did not write *customer is lying* or *customer is confused*, even when he was one or the other. The next agent to open the account had to see what was

said and judge it fresh. He had liked that rule more than he let on. It was the first job that told him the record and the opinion go in different boxes.

The machine had been sure of itself about Brendan.

"Saint Brendan sailed from Ireland to North America in the sixth century," it said. "Tim Severin proved it in 1976 by sailing a leather boat to Newfoundland."

He wrote *customer states* at the top of the page, for his own amusement, and then he sent the machines out. He checked what came back himself, line by line, with the windowsill stone holding the corner down.

The story is called the Voyage of Saint Brendan the Abbot. It is in Latin, and it survives in more than a hundred manuscripts, which tells you how much it was loved. The oldest of them are from the tenth century. The text itself was written round about 800, or a bit later, and nobody agrees where. Brendan was a real man, and the Irish annals have him founding the monastery at Clonfert round about 558, and they have him dead in 577. So the book that made him famous was written two hundred years and more after he was buried, by someone who never met him.

What the book says is that he heard of a land promised to the saints, out in the ocean to the west. He chose fourteen monks and fasted forty days. Then he built a boat with a frame of wood covered in ox hides tanned in oak bark, the joints smeared with fat, with spare hides carried in the bottom. Three more monks ran down to the shore at the last and begged to come. Then the book does a strange thing, because it does not send them in a line across the sea at all; it sends them round. Every year for seven years they keep Maundy Thursday on an island of sheep. They keep Easter on the back of a great fish called Jasconius, and the weeks until Pentecost on an island of white birds that sing the hours. They keep Christmas with the silent monks of Ailbe. In between they meet a column of crystal standing in the sea, an island of smiths who throw burning slag at them, and a land in a fog. At the end of the seventh year, a youth meets them on the shore of

the promised land and greets every one of them by name. He tells them to take as much fruit and as many precious stones as the boat will carry.

He read that twice, and wrote in the margin that the book is a calendar before it is a map.

Severin was ink, of a different kind, and it was the part the machine had got half right. In May 1976 he and a small crew sailed a boat of ox hides on an ash frame from Brandon Creek in Kerry, by the Hebrides and the Faroes to Iceland. They wintered there, and in June 1977 they came ashore on an island off Newfoundland. He put a single line through *proved it* and wrote *proved it could be done* beside it. That is a real thing to have shown, and it is not the same thing at all. Someone else had a citation in the pile from 2014 that the machine had leaned on. He could not find the paper, so he left it out.

He laid it out in the order the evidence came: first Brendan's life, from the annals, with his death in 577. Over it went the old tradition that has him at sea as a younger man, before Clonfert. On top lay the book, written two centuries after and dated by nothing. Through all three showed *530 CE*, and he wrote it across the top. The question mark after it was the tradition's as much as his, and he said so in the margin.

The doubt followed: *The year is the tradition's, and the tradition was set down long after. The book even has him die soon after he gets home, which the annals will not allow if he sailed young. Nobody can say there was a voyage at all, only that there was a man, and a book, and that the boat could have been built*. And under that, on its own line: *Dubán and the stern oar are mine; the seven years are the book's*.

On the map he moved the chapter from *Brendan reaches America* to *Brendan goes round*. The customer states, aye, and the agent writes it down. He put the kettle on and sat with the stone in his hand, thinking about the note that says what was said.

That night

The headset marks were still pressed into his hair, he could have sworn, when his head went down on the table. Tallow, wet leather and the sea were in the air. Against the spray the Ledger had wrapped its page in an oiled hide, and it opened it on its knee one fold at a time, the way you open a thing that has been a long while at sea and might be wet in the middle.

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Dubán was sixteen and the youngest in the boat, and the stern oar was his because nobody else wanted it.

It was a long heavy blade lashed to the right side of the stern, and it steered, and it did not let you sleep. You stood to it with your feet braced on the ribs and your whole weight on the shaft. You felt every wave come up under the hides like a hand under a blanket, and the hides moved under it. He had learned that on the first day out of the creek in Kerry, and the thing that frightened him most. The boat was not stiff like a wooden boat. It gave, and breathed, and came back. After a week he stopped being afraid of that and started to trust it.

"How will we know when we're there?" he asked the abbot, on the second night, when the coast had gone and the sea was the only thing left.

Brendan was sitting in the bow with his back to the mast, wrapped in his cloak. He was not a young man even then, and his face in the last of the light was mostly lines.

"Somebody will tell us," he said.

"Who?"

"The one who's there," said the abbot, and he pulled the cloak up over his head and went to sleep, which was what he did when he had decided a question was finished.

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The island of sheep came up out of the sea on the Thursday before Easter. Dubán saw it first, because the man on the stern oar sees everything that comes up over the bow.

It was green, with streams running off it. It was covered in sheep, great white sheep bigger than any in Kerry, who looked at the monks as they came ashore and did not move out of their way. There was a man there who met them, and nobody ever learned his name; the brothers called him the steward, because he looked after them. He brought them bread and fish and a lamb for Easter and asked for nothing. On the Thursday the abbot washed the brothers' feet, as the Lord did. Dubán sat with his wet feet on the grass and could not think of a single thing to say.

On the Saturday the steward told them where to keep Easter. They rowed to a small dark island nearby with no grass on it at all. They went ashore and set up the pot and lit a fire for the meal, and then the island moved. Dubán felt it before anyone, through his feet, the same way he felt the waves through the stern oar, a long slow shift like a sleeper turning over. They ran for the boat and threw themselves in. From the boat they watched their fire go away across the water on the back of the island, still burning.

"It's a fish," said Brendan, quite calmly, sitting in the bow. "The biggest in the ocean. He's called Jasconius. He's always trying to put his tail in his mouth and he never can."

The brothers laughed, the way you laugh when you have nearly died, too loud and too long. Dubán did not laugh, because he was looking at the fire on the fish's back, getting smaller, and he was thinking that they had not found the land at all. They had found a fish, and a man who washed their feet, and a great many sheep.

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The island of birds was close by. They stayed on it from Easter until Pentecost, fifty days, which was the longest Dubán had ever stayed anywhere that was not Kerry.

There was a great tree there by a spring, and the tree was white with birds, so many that you could not see the leaves. At each of the hours, when the brothers sang, the birds sang with them. At vespers they sang the psalm with the monks, beating their wings in time, and Dubán stood under the tree with his mouth open and forgot his part. One of the birds came down onto the prow of the boat and spoke. It told them they had seven years of this ahead. Dubán's heart went down into his stomach, because he had thought the land was a summer away.

"Every year you'll keep the Lord's Supper with the steward," the bird said, "and Easter on the back of the fish. From Easter until Pentecost you'll be here with us, and Christmas you'll keep with the community of Ailbe. And at the end of the seventh year you'll find what you came for."

That night on the stern oar, with the boat tied up in the lee of the island, Dubán counted it on his fingers. Thursday, the fish, the birds, the long empty summer, Ailbe at Christmas, then round again, seven times round. He had thought they were going somewhere, and they were going round and round.

He did not know yet if that was a punishment or a lesson. He turned it over for days, and could not decide.

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The first winter they came to the island of Ailbe, where the monks did not speak.

There were twenty-four of them, and they had lived there for eighty years without a word, and nobody in all that time had grown old or fallen ill. They spoke with their hands and their eyes. The abbot of the place took Brendan by the hand and led them into a church where the lamps hung unlit. At the hour of vespers an arrow of fire came in through the window, lit every lamp, and went out again. The lamps burned all night and in the morning they had used no oil.

Dubán did not see the arrow come in. He had been looking at the old abbot's face, which was turned up to the lamps, and which was the happiest face he had ever seen on a man. When the brothers asked him afterwards what the arrow was like he told them the truth: that he had been looking at the wrong thing. Brendan heard it and laughed at him, kindly, in the bow.

"No," he said. "You were looking at the right thing. Write that down in your head."

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In the third spring Dubán asked Brendan how the steward always knew when they were coming.

They were becalmed a day short of the island of sheep, and the sea was flat and white under a thin moon. The brothers were asleep on the hides in a heap like puppies. The old man was awake in the bow, as he usually was at that hour. Dubán left the oar lashed and went forward along the ribs. He sat down beside him, which he had never done before.

"He knows the feast," Brendan said. "Everybody knows the feast."

"But it moves," Dubán said. "Last year we came in the first warm days, and the year before it was cold, and the lambs were small. It's never the same day."

The abbot was pleased with him; Dubán could tell from the way he shifted under the cloak. He pointed up at the moon, low over the water, a little past half.

"Easter comes after the first full moon of the spring," he said, "once the day has grown as long as the night. The sun says when spring is, and the moon says which night. So the feast walks about inside the

spring, and every house that keeps it has to watch both of them and do the reckoning." He held up two fingers, one for the sun and one for the moon, and put them side by side. "The day moves. The round stays. The steward watches the same sky we do, so he's on the shore when we come."

Dubán looked at the moon for a long while. He had kept Easter every year of his life and never once thought about who had counted it out.

"Do they all count it the same?" he asked. "At home, and here, and everywhere?"

"No," said Brendan, and he laughed quietly into his cloak. "And that will be a fight one day, when you and I are long gone. But tonight we only have to agree with the steward."

The wind came up from the west before morning, and the island of sheep was green on the bow by noon. The steward was on the shore with the bread.

• • •

In the years that followed the round went the same way, and the boat went from feast to feast like a bead going round a string.

They saw the column of crystal one calm summer, standing straight up out of the sea and so high that they could not see the top of it. It was wrapped in a net with meshes big enough for the boat to pass through. They rowed in through the net and measured one side of the column against the oars. On a ledge inside they found a chalice and a paten of the same stuff as the net, and Brendan took them, so that the brothers at home would believe. They saw the island of the smiths, black and smoking. Huge men threw lumps of burning slag at them from the shore, and the sea hissed where the lumps came down. Dubán rowed so hard that his hands bled, and he did not care. When they were clear the old abbot said a prayer for the smiths, which nobody else had thought to do.

Two of the three who had come late were gone by then. The book would say how, a long time after. Dubán kept that part quiet in his own head and would not say it to anybody.

He grew, and his arms got thick on the oar, and his voice dropped. He learned the stars over the stern the way he had once known the turns of the lane at home. He learned which way the wind would come round after a still day, and how the hides felt before a squall. Every year at the island of sheep the steward brought the bread, and every year the steward said, "You've grown, Dubán," and every year Dubán was surprised that the man knew his name.

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At the end of the seventh year, after Pentecost, the steward came with them in the boat.

He sat in the bow beside Brendan and pointed the way. They sailed for forty days to the east, though the sea felt like the west, and Dubán stopped trying to know. At the end of the forty days there was a fog so thick he could not see the bow from the stern oar. For a day and a night they rowed through it without seeing anything at all. Dubán steered by the feel of the water and the voice of the steward calling back, a little to the right, a little more, and then nothing.

They came out of the fog into light. There was a shore, and trees heavy with autumn fruit, though it was not autumn. The light did not come from anywhere in particular. A river ran through the land, wide and quiet, and on the near bank a young man was standing waiting for them, as if he had known the hour.

He came down to the water and embraced each of them as they stepped ashore. To each of them he said his name. "Brendan." "Dubán." He named every one, the old and the young, the ones who rowed and the one who cooked, and he did not get one wrong.

Dubán stood on the shore with his bleeding hands, and something in his chest let go that had been held tight since Kerry.

"This is the land you've been looking for all this time," the young man said. "You couldn't find it at once, because God wanted to show you his secrets in the great ocean first. Take as much of the fruit as your boat will carry, and as many of the stones, and go home. Your days are nearly done."

Dubán looked across at Brendan. The old man's face was turned to the river, and it was the face he had seen once before, on the abbot of Ailbe under the lamps.

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They loaded the boat in the strange light. Dubán carried fruit down to the shore in his arms until the boat sat low. He put one red stone in the fold of his habit for himself, and nobody told him not to.

On the last evening he sat on the stern with his feet in the water and the youth came and sat beside him.

"You asked how you'd know," the young man said. Dubán had never told him that.

"Somebody would tell us," Dubán said. "That's what he said."

"And did they?"

Dubán thought of the steward every Thursday, and the bird on the prow, and the old abbot laughing in the bow. He thought of the seven times round. He thought of his name, said on a strange shore by someone he had never met, and how it had sounded like his own name in his mother's mouth.

"Everybody told us," he said. "Every year. We just had to go round enough times to hear it."

They sailed for home in the morning, and Brendan went back to his own monks, and Dubán with him. He kept the red stone all his life. When the young ones asked him in the evenings what the land was like, he did not tell them about the fruit. He told them about the round.

The master page

For this page the Ledger used a plain board, and pinned the corners down with four small stones, one for each quarter of the year.

In ink: Brendan of Clonfert, a real abbot, who founded his house round about 558 and died in 577, by the annals; the Voyage, written in Latin round about 800 or later and loved enough to survive in more than a hundred manuscripts, the oldest of them from the tenth cen-

tury; the boat of ox hides on a wooden frame, tanned in oak bark, its joints smeared with fat; the fourteen monks and three latecomers; the round of feasts kept seven years over; and the youth on the far shore who greets each of them by name. Severin's boat of 1976 and 1977, which reached Newfoundland and proved it could be done. In pencil: the year of the voyage, which is the tradition's; whether anybody sailed at all; and every face in the boat but the abbot's. *Found destroyed*: whatever Brendan himself said about the sea, if he said anything, before the book was written over it.

Laid by the others, this page showed something new. Egypt put the count in writing, the road put it in the mouth and the grove kept it in the body. Iona was the harbour where the boats come back and the stories are kept. Brendan's part was the round: the year walked as a circle of feasts, the same island at the same season every time, until the travellers know the sea by its calendar and not by its distance. The book is a map drawn as a year. Inked already, astern, is the other boat of hide out of Iona, fourteen days north before a south wind and home to a boy with bread; and back further are the navigators of the wide ocean, who also steered by what came round again. Ahead of it, in pencil, is the quarrel over reckoning the feast that Iona would carry to Whitby, and further on the last dream of the book, a woman at the edge of a fog who will greet a traveller by name.

Given the figures, the Core could reckon how long a hide boat might last in salt water and how far it could go in forty days. It could not say what the youth meant by his secrets in the ocean, or whether the boy on the stern oar heard his name. Some things you only find out in the boat.

How will we know when we're there? Somebody will tell us. The book's answer and the boy's turned out to be the same, and it took seven years.

The boy on the stern oar had wanted to know which island was next, and when to get the lamb ready, and how many more times round. The dream's answer was slower than that. A year kept carefully is itself a way of travelling, seven years of the same feasts on the same islands until the sea is a calendar and not a distance, and the

ones who keep the round are known by name when they arrive. The boat had brought its own small rules home with it: look at the face and not the arrow, pray for the smiths, and write down what was said in a different place from what you thought about it.

Across the ocean and three hundred years on, a girl was climbing a stone platform before dawn behind an old man, to watch for the morning star over a temple that no longer lined up with anything.

On the far shore the youth was still going down the line of them, saying each name. Gary waited to hear his own, and the page closed before it came.

CHAPTER 18

The Venus Table

Uaxactún, in the forest of the Petén, at the end of the stone-dating, and the long road north from there to Chichén Itzá: the found year, 889 CE



Illustration 19. *Watching the morning star from the platform at Uaxactún.*

Belfast, by day

AT RIVERRIDGE, ON THE SERVICE desk, half the calls in any week were about the bin calendar. People rang because the brown bin had not been lifted, or because they had put the blue one out on the wrong week. Some rang because Christmas had come and moved every collection along by a day and nobody had read the leaflet. He had learned to keep the calendar open on the second screen and talk them through it. The round was fixed, and it was the people who drifted, and when the round itself moved, for a bank holiday or a broken lorry, that was when the phones lit up. A calendar is only any use if everybody keeps the same one.

He had that in his head when he went to the Maya.

"Group E at Uaxactún was built to mark the solstices and the equinoxes," the machine said, "and was in use as a sun-watching station by around 400 CE."

He loosed the machines on it and then sat with their haul, checking by hand, with the stone on the corner of the site plan.

The first part is ink, as far as it goes. Uaxactún is a Maya city in the forest of northern Guatemala, a day's walk or so north of Tikal. In the 1920s the archaeologists working there, Frans Blom first and then Oliver Ricketson, noticed something about one group of buildings. If you stood on the pyramid on the west side, the one the diggers call E-VII-sub, and looked east, there were three small temples on a long platform opposite. The sun came up behind the northern one at midsummer, behind the southern one at midwinter, and behind the middle one at the equinoxes. People have found the same pattern, a pyramid facing a long platform, at dozens of Maya sites since, and they call them all E Groups after this one.

Then he got to the second part, and it was in pencil at best. When Aveni and Hartung measured the lines in the 1980s, the solstice sightlines worked, more or less. The equinox one did not, because the middle temple stands in the way of a clean view. Later digging found that the pyramid and the temples were built and rebuilt over centuries. The tidy lines in the textbooks join buildings from different periods. There may never have been one moment when you could stand up there and watch it work the way the drawing shows. What was built first, the pyramid with its plaster masks, goes back to the last centuries before Christ.

He struck out *400 CE* and sat back. The machine had picked a year from the middle of the city's life, and the city's life was a thousand years long.

So he went looking for a different year, one with a date carved on it. The Maya cut dates into stone in a count of days from a fixed start, and Uaxactún has its own first and last of them on its monuments. The last is Stela 12, and it marks the end of a period in the count that falls in 889 CE, and after that there is nothing. Tikal's last carved date was twenty years before that. All over the southern lowlands, the

stone-dating stops within a generation, and the cities empty. In the same years, far to the north in Yucatán, Chichén Itzá was busy putting up dated inscriptions, most of them between about 870 and 890. A carved band from its round tower, the Caracol, has been read as 889 or close to it.

He put the evidence in order the way you'd stack slabs, heaviest first. The last stone at Uaxactún went at the bottom; the dates rising at Chichén went over it; and on top went the Venus table in the Dresden Codex, one of only four Maya books that survived the Spanish. It runs from a starting day in 934 and may well have been worked out at Chichén itself. The one year they all let through was *889 CE*, so that went at the head, with its question mark riding on it like a passenger. On the map he struck *c. 400* and wrote *889: the road from Uaxactún to Chichén Itzá*, because that is where the evidence pointed, and he wanted the chapter to say so.

Below came the doubt: *The Dresden book was painted, most think, a good few centuries after 889, and the Venus table in it starts in 934. Nothing proves a family walked from Uaxactún to Chichén, and the drought is argued over. Ixkan is mine, and so is her grandfather, and so is the road.*

And under that, a line he had been thinking about since the Pacific: *This is not one of the Ledger's streams*. Nobody in the Petén had heard of Iona, or Croton, or the grove. They had their own sky and their own keepers, and they worked it out on their own. It goes on the shelf beside the others, not behind them.

The Venus count he did by hand, on the back of an envelope. The morning star is gone and back again every five hundred and eighty-four days, near enough. Five of those come to two thousand nine hundred and twenty days, which is eight years of three hundred and sixty-five. Eight years again, he thought, the same eight as Croton. Aye, dead on. He let that go and did the next sum, which was the one that made him sit up. Sixty-five of them make a hundred and four years, exactly twice another number he would need tomorrow.

That night

Under his hand the envelope's numbers were smudging when the heat came, and wet leaves, and smoke from a resin that burns sweet. This time the page was a folding book of bark, painted on both sides. The Ledger opened it a panel at a time, so that each new panel came round like a day, and the colours on it were still wet.

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Ixkan's grandfather woke her in the dark by tapping on the doorpost twice, which meant come quietly, and she was out from under the blanket before her mother could stir.

She was eleven, and she had been going up the pyramid with him before dawn since she was six. Her brothers had been too restless for it and her sisters too sleepy, so it was her. They went across the plaza in the dark. The forest was loud all round them, the frogs and the insects and something heavy moving far off, and the white plaster of the old pyramid came up out of the black in front of them like a held breath.

Her grandfather was a daykeeper, which meant that he kept the count of days for anyone who needed it: for planting, for naming a child, for a marriage, for the day of a king. People called him Ah K'in, the one of the sun, as they had called his father. His knees were bad, and he went up the steep stairs of the pyramid on all fours, slowly. Ixkan went up behind him with her hand flat on his back so he would not tip.

At the top they sat on the cold plaster facing east. Across the dark plaza was the long platform with its three small temples, the north one and the middle one and the south one. Over them, very bright and low, was the morning star.

"There it is," she said.

"There it is," her grandfather said. "Count it."

She knew what that meant, because every morning it was up she put a grain of maize in the gourd at home. When it was gone, she put a grain in a different gourd. The first gourd had a great many grains in it by now.

"Grandfather," she said, "where does it go? When it's gone?"

He did not answer that morning. He was watching the sky go grey behind the temples, and his lips were moving, and she knew not to talk while he counted.

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That day the sun rose well to the left of the north temple, nowhere near any of them, and her grandfather laughed.

Ixkan had heard the story all her life. The old builders were supposed to have had made the platform so that the sun would come up over the north temple on the longest day and over the south temple on the shortest. She had believed it the way you believe the shape of your own house. Now her grandfather pointed with his chin at the long platform and the three temples and the sun coming up in the gap.

"The north temple has been built over three times since the first one," he said. "Each king wanted it bigger, and the south one too. And the middle one is in the way of everything." He shrugged his thin shoulders. "The old builders made a line, and the kings who came after made it beautiful, and between them they bent it. The stone tells you a story about the sky. It isn't the sky."

"Then how do you know the day?"

He held up his hands, both of them, and wiggled the fingers. "With these. And the gourds. And the book." He tapped her forehead with one dry finger. "And this, if you look after it."

She looked at the three temples and felt something tip over inside her, gently, like a pot that has been leaning for a long time and finally goes. The story was not true, and her grandfather did not mind at all.

• • •

The book was kept in a niche in the wall of his house, wrapped in cloth. He took it out that afternoon when the heat was too much to do anything else.

It was made of bark, beaten flat and coated in white lime. It was folded like a screen, so that you could open one page or a whole wall of them. The pages were painted red and black, with rows of dots and bars and the faces of the days. Ixkan could read the numbers, because he had taught her. A dot was one and a bar was five, and the shell was nothing.

Her grandfather opened it to the Venus pages, where the four parts of the star's year were set out as he had taught her. There were two hundred and thirty-six days when it was the morning star, then ninety days gone, then two hundred and fifty days as the evening star, then eight days gone under the sun, and round again. It came to five hundred and eighty-four, nearly, but not quite, because the star was not as tidy as the book.

"So it goes under the sun," Ixkan said. "That's where it goes."

"That's where it goes for the eight days," her grandfather said. "The ninety, I don't know. Nobody knows. It goes into the underworld, the old men say, and walks about down there, and comes back up the other side as the evening star. It's late when it comes up, or early, by a day or two, and we fix the count." He showed her a place at the bottom of a page where a line of numbers had been painted over and done again. "My grandfather fixed that one. And I'll fix one. And you'll fix one."

She looked at the painted-over numbers until her eyes stung. She had thought the book was the sky's own. It was more like her mother's weaving, which was always being taken back a row and done again.

"Five of them," her grandfather said, holding up his hand, "and the star comes up on the same day of the year again. Eight years. When you're nineteen, it'll be up on your birthday, the same as the year you were eleven."

• • •

In the month of the first planting a woman came to the house with a new baby wrapped in a cloth against her chest, and asked the day-keeper for a day.

Giving days was the thing he did most often, more than the star. Every child needed to know its day, the day of the two hundred and sixty that it was born on, because the day would follow it all its life like a dog. The woman sat on the floor by the door with the baby asleep and waited, as people always did, as if she were at the healer's. Her grandfather took the bag of red seeds down from its peg and the little crystals he kept with them, and cleared a space on the mat.

"You do it," he said to Ixkan.

She had watched him do it a hundred times. She knew the twenty day names in their order, from Imix to Ahau, and the thirteen numbers that turned against them, so that a day came round with the same name and number only once in two hundred and sixty. She counted back from the day the count was at that morning to the night the baby had come, moving the seeds in their rows with one finger and saying the names under her breath. The woman's eyes went from her to the grandfather and back to her again, with her lips pressed together.

Ixkan said the day aloud. It was a good one, a day for a child who would keep things safe.

Her grandfather counted it again himself, quickly, with his eyes half shut, and nodded once. The woman let out her breath and smiled and touched the baby's head, and went away down the path with the day in her keeping, and Ixkan sat on the mat with her hands shaking a little and did not know why.

"You got it right," her grandfather said.

"What if I hadn't?"

"Then I'd have fixed it," he said, "and she'd never have known, and you'd have known for the rest of your life." He tipped the seeds back into their bag. "That's why I counted it after you. Not because I thought you were wrong. Because two is better than one."

• • •

The rains failed again that year, for the third time Ixkan could remember.

The maize came up thin and yellow and stood in the fields with its leaves rolled tight against the sun. The big water tanks in the plaza went down until you could see the plaster on their floors, cracked like old skin. Her mother began rationing the maize cakes, a few at a time, with her lips tight. Families began to go. They went quietly, before dawn, with their bundles, the way you leave a feast when you don't want to be seen to leave. Some went south and some went east. The ones with cousins in the north went north, up the long road to the cities by the northern sea where there was said to be water in great wells in the rock.

"Where do they go?" Ixkan asked her grandfather, watching another family go past with a dog trotting behind them.

"North," he said. "Most of them." He was quiet for a while, watching the dog. "It's the same question as the star."

The lords of the city held on. That year was the end of a katun, a round of twenty of the count's years. The count said a stone should be raised for it as the kings had always done. So the carvers came to the plaza in the heat and squared a block of limestone and began to cut. Ixkan sat in the shade of the wall every afternoon and watched them. One of the carvers was young and let her hold the sketch while he worked. She watched the dots and the bars come out of the stone, the day and the period and the name of the lord. The carver blew the white dust away and frowned. Her grandfather came to check the date, and nodded, and went away.

"Will there be another one?" she asked the carver. "In twenty years?"

He looked at the stone and then out at the fields, where the maize was dying.

"Maybe," he said. "Somebody will have to be here to cut it."

• • •

They left all the same, after the next harvest, which was worse than the last.

Her mother's brother was in the north, in a city called Chichén, by a great well in the rock. He had sent word with a trader that there was work for anyone who could carve, or plaster, or keep the count. Her grandfather said that was him. Her father said that was all of them. They packed in three days. That was what it took to wrap the book properly and dig up the jar of cacao beans under the floor and decide what to leave.

Ixkan's last morning in the city, her grandfather did not go up the pyramid. He could not do the stairs any more, so she went alone in the dark, on all fours as he had done. At the top she sat facing east and looked at the morning star over the three small temples. They were dark and quiet and lined up with nothing in particular, and she counted.

She put the grain of maize in her mouth instead of the gourd, since the gourd was packed, and held it there under her tongue all the way down.

• • •

The road north was long, and the forest gave way to low scrub, and the scrub to flat white country with no rivers at all.

They walked for many days. Her grandfather rode part of it on her father's back and part of it on his own feet. Every morning before dawn, wherever they had slept, he woke Ixkan by tapping twice on whatever was nearest, and they looked east together. She kept the count on a knotted string now, since the gourds were packed, a knot for every morning it was up. Then, one morning, it was not there.

She looked and looked. The sky went grey, and pink, and then the sun was there, and there was nothing, and she felt a cold drop go down inside her, the kind you feel when somebody has died.

"It's gone under," her grandfather said. He did not sound worried at all. "Now we count the gone days."

So she counted the gone days, a knot a morning in a new string, all the way north. In the towns they passed through, people asked where they were going, and her father said Chichén, and the people nodded as if everybody was going to Chichén now. On the road they met traders coming south with salt and cloth. One of them was a talkative man with a scar through one eyebrow, and he said that at Chichén the lords were building a round tower for watching the sky, with windows set to look where the evening star went down at the furthest north it ever went. Her grandfather asked a great many questions about the windows. The trader did not know the answers, but he was happy to guess.

• • •

They reached the city of the great well on a hot evening, dusty to the knees. Her uncle met them at the edge of it with water in a gourd.

Ixkan had not known there were so many people in the world. There were new buildings everywhere, half-finished, with scaffolds of poles lashed together and men climbing on them like ants. The ground was white rock, and in the middle of it, the trader had been right, a round tower was going up on a square platform. It was still only half its height, wrapped in poles.

She left her family arguing about where to sleep and climbed the nearest platform she could find, which was not very high. She sat on the edge of it and looked west, where the sun had just gone.

The sky over the scrub was the colour of the inside of a shell. And there, low and bright, just above where the sun had gone down, was the evening star.

She counted the knots on her gone-days string with her thumb. It was close enough, and her grandfather would fix the rest.

"Where does it go?" her little brother asked, behind her. He had climbed up after her and was standing with his thumb in his mouth.

"It goes under," Ixkan said, "for a while. And then it comes up on the other side." She took his hand and pointed it at the star. "And then you count it."

The master page

Pleated into panels, this page lay red on one side and black on the other, and it did not sit on the same shelf as the others. It stood beside them.

In ink: Uaxactún in the Petén and its Group E, the pyramid facing three temples on a long platform; Blom's and Ricketson's sightlines of the 1920s and Aveni and Hartung's measurements of the 1980s, which kept the solstices and lost the equinox; the building and rebuilding of the group over centuries; Stela 12, the city's last carved date, in 889; the run of dated inscriptions at Chichén Itzá, most of them between about 870 and 890; and the Venus table of the Dresden book, with its four parts of two hundred and thirty-six, ninety, two hundred and fifty and eight days, its corrections, and its starting day in 934. In pencil: the drought, which is argued over; whether anybody walked from Uaxactún to Chichén; what the windows of the Caracol were for. *Found destroyed*: nearly every Maya book, burned by the Spanish in the sixteenth century, so that four survive, and every count kept on maize grains and knotted strings.

He looked at what the page added, and at where it stood. Egypt made the record, and the grove kept the memory, and Iona was the harbour. This page is not part of that line. It is the same work done on the other side of the world by people who never heard of any of it. Uaxactún's part was the city: the place where the stone said one thing about the sky and the daykeeper's hands said another, and where the city stopped cutting dates and walked north with the count in a bundle. Behind, in ink, is the eight-year round that the Pythagoreans read in the same star at Croton, a parallel, not a link. A city in a lake waits ahead, in pencil, in the highlands, where the count of fifty-two years was about to come round.

He meant to put it to the Core in the morning. It would confirm that five rounds of the star make eight years and sixty-five make a hundred and four, and that the windows of the Caracol line up with something or other within a degree or two. Whether a girl climbed that tower it cannot say, nor what was in a daykeeper's head when he fixed a row of numbers. The daykeeper's hands had known things the stone never did.

Where does it go? Under, for a while, and then up on the other side. The girl's answer was the book's answer, and the city's.

The girl had needed it because the morning star had to be counted, and somebody had to count it, and when it went you counted the gone days instead. The dream's answer was the bundle on her back. A count you can carry is worth more than a temple that lines up, because cities empty and the count goes on. Her family had believed their own eyes over the stone, fixed the numbers when they were wrong and said that they had, and taken the book with them when they left.

Six hundred years on, in a city built on a lake, a boy was grinding red paint for his master on the morning a runner came in from the coast with news of strangers.

Before it was light he went out the back in his socks to look. Over the roofs to the east, low and very bright, was the morning star, and he stood on the cold step and counted it, for no reason he could have given a daykeeper.

CHAPTER 19

The Fifty-Two Years

Tenochtitlan, the city in the lake, and the market at Tlatelolco, in the year One Reed, twelve years after the last New Fire: the found year, 1519 CE



Illustration 20. *Grinding the red at the Tlatelolco market.*

Belfast, by day

HE HAD WORKED ON THE lifting crews at Kilroot, the power station on the north shore of Belfast Lough. He slung loads and led a small team there, in the years when the coal was coming to an end. Everybody on site knew the coal units were going. The talk in the cabins was of the gas and the batteries and what would be built next. The old men who had worked there half their lives talked about the station the way you talk about a relative in a home. He was not there the night it finished, but he read about it after, like everybody else. It went off at midnight

at the end of September 2023, after more than forty years of burning coal on the shore of the lough. One fire out, and the next thing not quite lit.

The cabins at Kilroot were in his mind when the Aztecs came up on the screen.

"The Aztecs held the New Fire every fifty-two years to stop the world ending," it said. "In 1519 they were preparing it. Then Cortés arrived, and they took him for the god Quetzalcoatl, come home."

"Which year?" he asked it, and it did not know.

Fetching was the machines' job; the reading he did himself, by hand, with the stone weighing down the page.

The fifty-two years are ink, as far as anything here is. The Mexica, like their neighbours, ran two counts together, a year of three hundred and sixty-five days and a sacred count of two hundred and sixty. The two came back into step every fifty-two years. They named the years with four signs and thirteen numbers, so that a year called One Reed came round once in fifty-two. When the round closed they bound up the years like a bundle of sticks and made a new fire. Sahagún, the Franciscan who sat with Nahua elders and their painters in the decades after the conquest, sets it down in his great book. Every fire in the valley was put out, and the hearthstones and the household gods were thrown into the water. Pregnant women were masked with maguey leaves and shut up in the granaries, in case they turned into wild beasts. Children were kept awake so they would not turn into mice. On a hill called Huixachtlan the priests watched the stars they called the marketplace, which we call the Pleiades, until they passed the top of the sky. Then a fire was drilled on a captive's chest, the man was killed, and the new fire was carried by runners to every temple and every house.

Measured against that round of fifty-two, the machine had the date wrong, as it turned out. The last New Fire was held in 1507, in the year Two Reed, so the next was due in 1559. In 1519, when the Spanish came ashore on the Gulf coast, the ceremony was twelve years behind them, not in front. He put a line through *preparing* and wrote *remembering* above it.

The god was pencil, and pale pencil at that. The story that the Mexica took Cortés for Quetzalcoatl coming home is in the books written after the fall. Historians now think it was mostly made afterwards, by Spaniards and by Nahua writers trying to explain what had happened. The omens are the same, the fire in the sky and the temple that burned itself and the lake that boiled, all written down decades after the city was taken. Instead of striking those through, he wrote *after* beside them in the margin.

He set the sources one over another, and the first layer was the year One Reed, 1519, when the strangers landed. Then the New Fire of 1507, near enough in living memory that a woman of thirty would have stood through it. Over both went the things that survive. There was the great Sun Stone, carved most likely in Moctezuma's reign and dug up in the main square of Mexico City in 1790. There was the Codex Mendoza, painted round about 1541 for the Spanish viceroy by native painters, and the Templo Mayor, uncovered from 1978. He headed the page with what they shared, *1519 CE*, and then the question mark, which for once he hardly needed, because the strangers had written the year down themselves.

Then came the doubt, put as plainly as he could put it. *Most of what we know was written after the fall, by the people who won or by people who had to live with them. The omens and the god are later, and Tlalli is mine, and so is Citlali, and so is the night she spent in the granary*.

The map had said that the boy watches the priests prepare the New Fire, and he changed it to *remembered, and planned for*. Aye, that's the bit that hurts, because they were counting towards 1559 like anybody would. He turned the stone over with that in mind, and thought about the cabins at Kilroot, and the old men talking about what came next.

That night

The line through *preparing* was still under his hand as he went down into smoke and wet stone, with something sweet and bitter in the air like chocolate. Across the Ledger's knees lay a long folded strip painted in red and black. It opened the strip out the way a painter opens bark when there is a year to add at the end.

• • •

Tlalli ground red for the master every morning, and on the morning the runner came he was grinding it.

He was twelve, and he had been apprenticed to the painter for two years, and so far he had done almost nothing but grind. The red came from the little insects that the farmers dried on the cactus. You ground it with a stone on a flat stone and mixed it with a gum, and it had to be fine as smoke or the master sent it back. The black came from soot, which was easier. The master said that a painter was a man who held the black and the red, which was a way of saying that he held the writing, and the knowledge, and what was true. Tlalli liked the saying, though he did not yet hold anything but the grinding stone.

Beside him, the master was painting the new year in its square that morning. He did it at the start of every year, on the long strip of bark paper that held the city's years, one after another, each in its little square. He had drawn the reed in its square and was painting the single round dot beside it that made it One Reed.

"What comes after?" Tlalli asked, because he had been wanting to ask for days.

"After One Reed? Two Flint. Then Three House, then Four Rabbit, then Five Reed." The master did not look up. "You know this."

"No. After all of them. When it gets to the end."

"Then we bind the years," the master said, without looking up. "One Rabbit, and then the binding at Two Reed, and the New Fire, and it starts again." He dipped the brush in the red. "You were born in a New Fire year. Ask your mother about it, she'll tell you more than I can."

Tlalli went on grinding the red and counted on his fingers. He had been born in Two Reed, at the start of the bundle. The next binding would come when he was fifty-two, which was so old that he could not picture it at all.

• • •

His mother sold cloth in the great market at Tlatelolco, at the north end of the island, and that evening he went to help her pack up.

It was the biggest market in the world, everybody said, and Tlalli believed it. There were rows for everything. There was a row for maize and a row for beans, a row for turkeys and a row for little dogs, a row for feathers and a row for the gold workers, a row for the stone blades and a row for the herbs. There were rows of pots and rows of mats and rows of rope. There were judges who sat in a house in the middle and settled quarrels about short measure. His mother's place was in the cloth row, under an awning, where the plain white cotton cloaks were piled up and folded in twenties. A small white cloak was worth so many cacao beans, and a big one was worth more. People paid with either, and his mother kept it all in her head and never once got it wrong.

"The master says you were in the granary," Tlalli said, helping her fold. "When I was born. At the New Fire."

His mother stopped folding and looked at him.

"Did he," she said. "Well, I was. Before you were born. You were in me, as big as a melon." She went back to folding the cloaks. "I'll tell you on the way home."

• • •

She told him in the canoe going home along the canal, with the city going dark round them and the fires coming up one by one on the roofs.

"Every fire was put out," she said. "Every one, in every house in the valley, all at once. My mother put ours out with water and threw the hearthstones into the canal, and the old god stones too. We swept the house and threw out the old pots. It was so dark, Tlalli. You can't think how dark, a whole city with no fire."

The women who were carrying were masked with a maguey leaf and put into the granaries, his mother said. That was in case the night went wrong and they turned into beasts and ate the people. She had sat in the dark in the maize bin with the leaf over her face and her hands on him, inside her, and listened. Outside, all over the city, the children were being kept awake. Her small brothers were pinched and slapped and shaken all night so they would not fall asleep, because a child who slept that night would turn into a mouse.

"Did they?" Tlalli asked.

"Did any turn into mice? No. Nobody slept." She laughed out loud. "Your uncle cried all night because your grandmother kept pouring water on his head."

Everybody had gone up on the roofs and the walls, she said, and looked south-east to the hill of the star, where the priests were. They were watching the marketplace, the little bunch of stars, to see whether it would pass over the top of the sky. If it did not pass over, the sun would not come back, and the night would last for ever, and the star demons would come down and eat the people.

"And it passed," Tlalli said.

"It passed. And there was a fire on the hill." She was quiet for a while, paddling in the dark. "You know how they made it. On a man's chest. I didn't see that. I was in the granary. I heard the whole city shout."

The runners had come down from the hill with the fire on pine torches, she said. They took it to the great temple first, and then to every temple in every quarter, and then the people had run with it from the temples to their own houses. Her mother had lit the new hearth with it before dawn, and put new hearthstones down, and everybody had put on new clothes. Then they had let the women out of the granaries.

"And then you," she said. "Two months after. In the new fire year."

• • •

The runner came from the coast in the spring, and Tlalli saw him because he was grinding red on the step of the master's house, which looked out over the causeway.

He came at a run, with his hair in a knot and his cloak tied back, and he did not stop at the market or at the temples. He went straight up the causeway towards the palace, and behind him came another runner, carrying something rolled up under his arm.

By the afternoon the whole city knew that strangers had come ashore on the coast to the east, on the edge of the sea. They had come in houses that floated, big as hills, with cloth wings. They had pale faces and beards, and they wore metal on their bodies, and some of them rode on deer as tall as a roof. They had a weapon that made thunder, and the smoke from it stank.

The master was called to the palace that evening, and he came back very late, and very quiet. He sat down on his mat in the dark and did not take off his cloak.

"They want it painted," he said. "Everything the runners have seen. The houses on the water, the deer, the men, the dogs, what they eat, what they say, what they carry. They want copies, in case the first ones are lost." He looked across at Tlalli. "We'll need a great deal of red."

• • •

A merchant came through the cloth row a few days later with a pack on his back and dust on his legs to the knee, and he stopped at Citlali's awning because he had bought from her before.

He had been on the coast, he said, trading for feathers and salt, and he had seen the strangers' things with his own eyes. He untied a corner of his cloak and showed them, there under the awning, a string of beads. They were green and blue and clear, harder than any stone Tlalli had ever touched, and the light went right through them the way it goes through water in a bowl. The strangers had given strings like it to the coast people the year before, the merchant said, when the first of their floating houses came, and taken gold and cloth for them.

"How many cloaks will you give me for it," he said to Tlalli's mother, holding the string up so that it caught the sun.

Tlalli wanted it more than he had ever wanted anything. He looked at his mother, and she looked at the beads for a long moment, turning them a little in the merchant's fingers without touching them.

"None," she said.

The merchant laughed, a little put out. "Nobody in the valley has seen a thing like this."

"That's why," his mother said. "I know what a cloak is worth, because I know how long it takes to spin and weave it, and I know what the woman next to me will take for hers. I don't know what that is worth. Neither do you. Nobody does yet." She folded another cloak and put it on the pile. "Come back when somebody knows, and I'll give you the right price."

The merchant went off up the row, grumbling, and sold the string by the evening to a feather worker for more than it was worth. Tlalli watched the beads go and felt sore about it all day. That night, in the canoe, his mother told him he could be cross with her as long as he liked, so long as he remembered why.

"You count what you know," she said. "The rest, you wait."

• • •

For a month after that, Tlalli did nothing but grind.

The runners kept coming, every few days, with reports and with painted sheets from the painters on the coast. The master copied them onto clean strips and added what the runners said. Tlalli ground red and ground black and watched the strangers come out onto the paper under the master's brush. There were the floating houses with their tall poles, the long beards, the deer with the men on their backs, the great dogs with their tongues out. The master painted them in the same black and red as the year signs, in the same careful outline. It made them look like they had always belonged on a page.

Tlalli asked him why he painted them like that, the same as people.

The master did not stop working. "Because that's how we paint," he said. "What's in front of you is what you paint. I don't know what they are. Nobody does. If I paint them as gods, and they're men, then the page is a lie. If I paint them as men, and they're gods, then the gods will forgive me for being careful." He finished an outline and sat back to look. "You write what you were told. You don't write what you think of it."

Tlalli ground the red and thought about that until her arms ached.

In the market, people talked, and Tlalli heard them in the cloth row as he folded. Some said the strangers were gods. Some said they were men from beyond the sea, the way the merchants came from beyond the mountains. Some said they would go away again, and some said they would not. His mother sold cloth to all of them. When the talk got

loud she said, quite calmly, that people had been saying things at Tlatelolco since the market was built, and that the price of a white cloak was still the price of a white cloak.

• • •

On the night of the full moon Tlalli climbed up on the roof of the master's house with his mother, who had brought the supper. They looked south-east across the dark water towards the hill of the star.

"That's where it was," she said. "Where they made the fire."

He looked at the hill, black against the sky, and then he looked up for the marketplace, the little bunch of stars, and could not find it. His mother said it was hiding near the sun at this time of year. In the cold months it would be back, and climbing. On the night of the binding it would cross the top of the sky at midnight, the way it always had. That was what the priests on the hill would be watching for.

He counted it out on his fingers again, as he had on the first morning, and this time he counted all of it. Two Flint, Three House, Four Rabbit, all the way round, four signs and thirteen numbers, to One Rabbit, and then the binding, forty years from now. He would be fifty-two, and he would be a master painter by then, with an apprentice of his own grinding red on the step. He would paint the binding of the years in its square on the long strip, and the fire on the hill. He would put the old pots out in the canal like everybody else.

"What comes after?" he said. "After the next one?"

His mother laughed and put her arm round him. "Another bundle," she said. "That's what always comes after. Another bundle, and another fire, and people folding cloth in the market." She handed him a tamale out of the basket. "Eat. You'll be fifty-two soon enough."

So he ate the tamale and looked at the dark hill. He thought about the binding of the years, and the strip of paper growing a square at a time, and the fire that would come down the causeway on the torches when he was old.

He did not know, and nobody on that roof knew, that there would be no next binding on that hill, nor any square for it on the master's strip.

The master page

This page the Ledger unfolded as a long strip, and at the far end of it, where the next square should have been, it left the bark bare.

In ink: the fifty-two-year round of the two counts, the naming of the years with four signs and thirteen numbers, and the binding of the years with its new fire, as Sahagún's Nahua elders and painters described it after the conquest; the last New Fire in 1507, in the year Two Reed, on the hill of Huixachtlan; the year One Reed, 1519, when the strangers came ashore on the Gulf coast; the market at Tlatelolco and its cloaks and cacao; the painters who held the black and the red; the Sun Stone, the Codex Mendoza, and the Templo Mayor. In pencil: the omens, and the returning god, both set down after the fall; and every face in the canoe. *Found destroyed*: the painted books of Tenochtitlan, almost all of them, burned in the conquest and the years after, so that what we have of the fifty-two years we have mostly from the painters' grandchildren, telling it to friars.

He weighed what the page added, and where it fitted. Egypt made the record, and the grove kept the memory, and the boat of hide went round. Like the Maya page, this one stands beside the line, not in it. From Tenochtitlan came the renewal: the round that is closed on purpose, every fire put out and every pot thrown away, so that the count can start again clean. It is the only page on the shelf where the people chose the dark, to see the fire come back. Back, beside it, is the girl from Uaxactún counting the gone days of the morning star, and the hundred and four years of the Venus table, which are two of these bundles laid end to end. Ahead lies the fire at Croton, in pencil, where a house of counters burned and the counting scattered.

The Core can show that three hundred and sixty-five and two hundred and sixty come back into step every eighteen thousand nine hundred and eighty days, and where the Pleiades stood over the valley on the night of the binding in 1507. It has no reach into what a woman thought in a granary with a leaf over her face and a child inside her. For that there were only the grandchildren, telling the friars.

What comes after? Another bundle, his mother said, and she was right about everything except the year.

The boy grinding paint had wanted to know when to put the fire out, and that the stars would pass, and that it would come back. The dream gave him the harder half: a people who keep the round are promising their own grandchildren that somebody will be here to light it. The painters had kept that promise the only way left to them. They painted what they were told and not what they thought of it, kept the price of a cloak the price of a cloak, and told the friars the rest.

Two thousand years back and across the world, a grown man was running down a hillside in the dark towards the house of the counters at Croton, which was burning.

The next binding of the years was due in 1559. Nobody lit it.

PART IV

Fragmentation

The Taking Apart

CHAPTER 20

The Burning at Croton

Croton, on the instep of Italy, in the summer after Sybaris fell, and the road and the sea north to Metapontum: the found year, 510 BCE



Illustration 21. *Myrto listens as her father plucks the single string.*

Belfast, by day

FOR TWO YEARS HE WAS lead administrator for six small businesses that shared one office, one accountant and, on a bad week, one temper. The year one of the six folded, the owner went off to England and the files went with nobody. What was left of the firm was in other people's heads and inboxes. The bookkeeper had the invoices, a lad in the yard had the delivery notes in his van, and the woman on the phones remembered which customers paid late and why. He spent a fortnight going round them with a notebook and putting the business back together, a piece from each, and it was never the whole of it. It was enough to close the books honestly, and he had been proud of that, in a tired sort of way.

The six firms came back to him when the screen gave its answer on Croton.

"In 510 BCE an enemy called Cylon burned the Pythagorean school at Croton," it said. "Pythagoras was killed, and all his writings were destroyed in the fire."

Once the machines had found the sources, he checked every one himself, the old beach stone weighting the page.

The first thing to go was the line about his writings. Every old source that talks about it agrees that Pythagoras wrote nothing down, and that the school's teaching was kept in the members' heads and passed on by mouth, so there was nothing on paper to burn. He crossed out *all his writings* and wrote *what they carried in their heads* beside it.

The death went next, in a different way, because the stories do not agree. In one, told by Diogenes, the master escaped the fire and was caught at the edge of a bean field that he would not cross. In another, from Dicaearchus, he got away to Metapontum up the coast and died there in a temple of the Muses after starving for forty days. In others he was not in Croton at all that night. Cicero, more than four hundred years later, went to Metapontum and was shown the seat where he used to sit. He wrote *the stories disagree* and left it at that.

The fire itself is in pencil, but a firm pencil. Aristoxenus, writing about two hundred years later from men who knew the survivors, says that a rich man of Croton called Cylon, turned away by the school, turned the town against it. After Croton destroyed Sybaris in 510 there was a quarrel over the conquered land, and his party set fire to a house where the members were meeting, the house of Milo the wrestler. Polybius, much later, describes meeting-houses burning all over the south of Italy. That is the next generation's trouble, fifty years on. Porphyry says that the knowledge, kept unwritten in their breasts, perished along with those who carried it. Only a few sparks survived, among those who happened to be away.

He layered the year, the war of 510 at the bottom, when Croton took in the exiles of Sybaris and then destroyed the city. Over it lay the quarrel over the land, and on top the fire at Milo's house, which Aris-

toxenus puts soon after. All three met at *510 BCE*, so that headed the page, and its question mark went in small and worried, because Aristoxenus's *soon after* was doing most of the lifting.

The doubt went in after: *Aristoxenus is two centuries late and had his own side. Some put the fire later, and the great burnings Polybius describes are fifty years later still. Philon is mine; he would be thirty-three that summer. Myrto is mine, and so is the farm.*

In the chapter map he changed *c. 510 to 450* to *510, with the later burnings heard of only as an aftermath*. Aye, he thought, the six firms all over again, and every one of them with a wee bit. He sat on a while with the stone heavy in his palm, and didn't write that bit down.

That night

Across his chest lay the notebook, open at a list of names, each with a line after it saying what that person had kept. Dry grass, goats and woodsmoke filled the room, and after them a smoke that was not woodsmoke. One edge of the Ledger's page was charred, and it turned the page very gently from the unburnt side, so that it would not crack.

• • •

Myrto's father kept a string in a box, and on the evening before the fire he took it out to show her.

It was an old gut string, dark with handling, coiled small and tied with a thread. He fixed it to a plank in the courtyard of the farm between two pegs, as he had done once or twice before. He set a little wooden bridge under it. He plucked the whole string, and it hummed in the plank. Then he moved the bridge to the middle and plucked the half, and the same note came back higher.

"Two to one," he said. "That's the first thing they taught me at Croton, when I was a boy not much older than you. A girl taught me. She gave me this string when I went home."

Myrto was ten. She had heard about Croton all her life, and about Sybaris, which was where her father and grandfather came from, and which was gone, though she had never seen it. The family had come

to Croton when the men of Sybaris drove them out, five hundred rich houses at once. The men of Croton had taken them in, because the teacher on the hill said they must. Then there had been the war, and the river turned over the ruins. Now there was this farm in the hills above Croton, and her grandfather sitting all day by the door saying nothing.

"What are the other notes?" she asked. "You said there were more."

"Three to two, and four to three." Her father moved the bridge and plucked, and moved it again. "There was a whole teaching after that. The numbers in the sky, the months, the body. I only had a summer of it." He coiled the string back into its box and tied the thread. "I know the beginning. Somebody else knows the rest."

"Who remembers the rest?" Myrto said.

He laughed and tugged her hair, as he always did when she asked too much. "The ones on the hill. They have it all in their heads, every word, and they don't write any of it down. That's the rule. Go and get your supper."

• • •

There were men on the road that evening, more than usual, walking down towards the town in twos and threes, not talking much.

Myrto saw them from the wall where she sat with the goats. Her father saw them too, and he stood by the gate looking after them. The quarrel in the town was about the land of Sybaris, the land that had been won in the war. The people had wanted it shared out, and the council had not shared it, and the council was full of the teacher's men. Myrto did not understand the half of it. She understood that her grandfather had stopped eating his supper, and that her father had begun locking the gate at night, which he had never done.

"They meet at Milo's house tonight," her father said to her mother, by the fire. "All of them, the old ones and the young ones."

"Then they'll be safe enough," her mother said. "Who'd touch Milo?"

Myrto went to bed in the loft and lay looking at the stars through a chink between the roof tiles. She found the bright one sitting low over the west, the evening star, which her father said the teacher reckoned was the morning star too, only seen at dusk instead of dawn. She liked that. One star with two names, depending on when you looked.

• • •

She woke to her father's voice, and to a light on the wall of the loft that was the wrong colour for morning.

When she put her eye to the gap in the tiles she saw the glow. It was down the hill and across the valley, over the roofs of Croton, a red light coming up out of the town and lighting the bellies of the smoke. Her father was already in the courtyard with his cloak half on. She heard the gate bang. By the time she had her sandals on and was down the ladder, he was a dark shape running down the hillside in the dark towards the town, and her mother was calling after him from the gate.

Myrto went after him without deciding to; her legs did it, as they did when the goats got out.

It was a long way down, through the olive terraces and the vines and then the lane between the walls, and the closer she got the louder it was. There were people in the lane, some running towards the fire and some away from it. There was a smell like burnt hair, and it got into her throat. She lost her father and found him again by the shape of his shoulders against the light. He was standing at the corner of a street where the light was brightest, with his hands open at his sides.

The house was a big one, with a high wall and a gate, and the gate was shut. Men were standing in front of it with sticks, not doing anything, only standing. Behind them the roof of the house was burning. The fire made a noise like wind in the olive trees.

• • •

Myrto never knew how her father got round to the back of the house. She remembered running behind him along a wall, and a ditch, and a small door in the back wall with a beam across it on the outside. She

remembered him pulling at the beam until his hands bled. She remembered it coming away all at once and the door swinging open, and smoke rolling out of it like a fleece.

People came out through the smoke, bent double, coughing, holding each other. Some were old men in white, and some were boys hardly older than Myrto. Her father stood at the door and pulled them through, one after another, and pointed them to the ditch and the dark. Myrto stood in the ditch and held their hands as they came down into it, because nobody had told her not to and there was nobody else to do it.

The last one out of the door was a woman carrying a small boy on her hip. Her white dress was black down one side, and her hair was singed, and she was not crying. She set the boy down in the ditch beside Myrto and turned to go back.

Her father caught her arm. "Theano," he said. "There's nobody else. The roof's going."

The woman looked at him in the light of the fire, and Myrto saw her know him.

"The boy from Sybaris," she said.

"You can't go back in."

"There's nothing to go back for," Theano said, and she stopped pulling against him. "That's the thing. There's nothing in there to carry. There never was. It was all in us." She looked at the people in the ditch, and at the small boy, and at Myrto. "What we've got out is what there is."

Not everyone came out, and Myrto knew that, though she did not see it. She knew it from the grown-ups in the ditch, who counted each other in whispers and broke off and never said the last number.

• • •

They went up the hill to the farm before dawn, all of them, keeping off the road.

There were eleven of them, and the small boy, and her mother met them at the gate with water and asked nothing. They sat in the barn on the straw with their burnt clothes and their red eyes, and for a long

time nobody spoke at all. The old men sat very straight. One of the boys was shaking, and Myrto's grandfather, who had not spoken to anybody in a month, went and sat beside him and put a blanket round him and kept his arm there.

When the light came, Theano made them speak.

"We say what we have," she said. "Now, while we're all together, before we go. Each one. What you know, and how you know it. The rest of us listen."

So they did, one at a time. An old man with a burnt beard said the rules, all of them, from the beans to the balance beam, slowly, as if he were reading them off the inside of his eyelids. A young man said the numbers that make shapes, the three and the six and the ten laid out as pebbles in a triangle. He drew them in the dust of the barn floor. Another said the months, and the eight years of ninety-nine moons, and stumbled, and Theano gave him the next part and he went on. A thin boy said what the physicians said about hot and cold and the body in tune. Somebody said the evening questions, where did I go wrong, what did I do, what did I leave undone. Each one stopped where their part stopped, and there were long silences between them, and in the silences you could hear the goats.

Myrto sat in the corner and listened, and understood that nobody had all of it. They had a piece each, like the people in the barn who each had a burnt patch on a different part of their clothes.

When it came round to her father, he got up without a word and went into the house and came back with the box. He took out the coiled string and fixed it to the plank in front of them all. He played the whole and the half, two to one, and the old men nodded the way you nod at a name you have not heard in years.

"That's mine," he said. "It's only the beginning. It's all I had time for."

"The beginning is the part people lose first," Theano said. "Because everybody thinks somebody else has it."

• • •

They left the barn the next night, in twos and threes, the way the men had come down the road the evening before.

Some went south to Locri, and some north along the coast to Tarentum. The rest went to Metapontum, where there was word that the teacher himself had gone, or was going, or had been seen; nobody could say which. Theano was going to Metapontum with the small boy. Myrto's father carried the boy down to the boat on the beach below the farm, after dark, with the water black and quiet and the town still smoking behind the headland.

At the boat Theano crouched down in front of Myrto.

"You held their hands in the ditch," she said. "I saw you."

Myrto did not know what to say to that, so she kept her mouth shut.

"Do you know the string? The half?"

Myrto nodded, because her mouth was dry. "Two to one. Same note, higher."

"Then you've a piece too." Theano took Myrto's two hands in hers, which were blistered across the palms. "You asked who remembers the rest. Your father told me, last night, in the barn. It's all of us. A piece each. So you say yours, every night, before you sleep, so you don't lose it. And one day somebody will go round all of us and write it down, and it won't be the whole of it, but it'll be honest." She stood up and brushed the sand off her knees. "That's what we're for now."

Theano climbed in after the boy, and the boat slid off into the dark. Myrto stayed on the beach, her father's hand on her shoulder, and watched it go out across the black water towards the north. Out over the sea in the west the evening star was up, the one with two names.

"Same star as the morning," her father said, quietly, like a man saying over a piece of his own. "The teacher said so. Only seen when the day is going."

Myrto said the half-string to herself under her breath, two to one, and held on to it.

The master page

The Ledger put this page on a board scorched along one edge, and set the coiled string in the middle of it, where the fire had not reached.

In ink: that Pythagoras wrote nothing and the school kept its teaching unwritten, by mouth; Croton's taking in of the five hundred exiles of Sybaris, and the war of 510 in which Croton destroyed Sybaris and turned the river over it; Porphyry's report that the knowledge died with the men who held it in their breasts, and that a few sparks were kept by the ones who were away; and Philolaus, two generations later, the first Pythagorean to put the teaching into a book. In pencil: Cylon, the quarrel over the land and the fire at Milo's house, from Aristoxenus two centuries on; how the master died, which the stories tell four ways; and the eleven in the barn. *Found destroyed*: everything the members carried in their heads and did not get out of the house, which is most of it, and which nobody can now list.

This page went on the pile with the others. Egypt's scribes wrote, the mountain road recited, and the grove carried the count in bodies that were burned rather than written. Croton built the school. Croton's part was the scattering: the first time the whole was broken into pieces because it lived only in people, and the people ran in different directions, each with a part. It is the six firms, and the notebook going round. Inked in, behind, is the boy from Sybaris at the practice board, and the coiled string Theano gave him at the harbour. Ahead, pencilled, is another burning, by a strait in the far west, where the keepers would again have nothing written to save.

The Core can lay out the pieces the barn remembered, the rules and the triangle numbers and the ninety-nine months, and show where they join and where they don't. Who was in the house it will never know, or what was lost with them. Numbers don't burn, and they don't grieve either.

Who remembers the rest? All of them, a piece each. The girl heard that in the barn, and the book has no better answer for Croton.

Myrto had wanted to know the evening star when she saw it, and that it was the same star she had seen in the morning. What the dream added was that the sky does not burn. A count kept up there can be found again by people who have run in six directions. Gary thought of the six firms sharing one notebook between them, and of the beginning of everything, which nobody ever writes down because each of them is sure somebody else has it.

Five hundred and seventy years on, and at the far western edge of the empire, a girl stood among women in black with torches on the shore of a strait, and watched the soldiers swim.

In the ditch below the burning house the grown-ups were still counting each other in whispers, and stopping, and starting again from one.

CHAPTER 21

Mona

Mona, the island across the strait from the mountains, and the lake in its western marsh, in the spring the governor brought his army to the shore: the found year, 61 CE



Illustration 22. *Cunovinda looks across the strait at the Roman fires.*

Belfast, by day

THE YEAR HE WORKED AS a junior in a solicitors' office in London, the thing he learned first was what a statement was. It was one person's account, signed and dated, and it was only ever one side. A good file had the other side's statement too, or at least a note of what the other side would say. A thin file had one statement and a great deal of confidence. The partners were patient with him about most things, but on that one they were sharp. Never read a statement as if it were the event, one of them told him. Read it as what that person wanted the court to believe happened.

That was in his head, with the burning at Croton not long behind him, when the answer on Anglesey arrived.

"In 61 CE the Romans massacred the druids on Anglesey," it said. "They cut down the sacred groves and ended druidism in Britain."

The machines brought back a pile, and he read down through it by hand, the stone squatting on the corner of the page.

There is one statement, and it is Roman. Tacitus, in the fourteenth book of his Annals, written something like fifty-five years after, says that Suetonius Paulinus, the governor of Britain, prepared to attack the island of Mona. It was strong in people and a refuge for fugitives. He built flat-bottomed boats for the infantry, because the strait was shallow and uncertain. The cavalry crossed by the fords, or swam beside their horses where the water was deeper. On the shore stood the enemy's line, dense with arms and men, and women running through it in funeral black with their hair loose, carrying torches, like Furies. The druids stood round them raising their hands to the sky and pouring out curses. The soldiers froze at the sight as if their limbs were paralysed. Then, urged on by their general and each other, they went forward, struck down everyone in front of them, and wrapped them in their own fire. A garrison was put on the island, and the groves were cut down. Then news came that the province had risen behind them under Boudica, and the army turned and marched away.

It is a vivid statement, and it is one side. Tacitus's father-in-law, Agricola, served on the governor's staff that year, so the account may come from somebody who stood on the beach. But it is an enemy describing, the way Caesar was, and when Tacitus says the groves were given over to savage rites and the altars drenched in the blood of captives, that is the prosecution's case.

He put a line through *ended druidism in Britain*. Tacitus himself has Agricola attacking Mona again, in 77 or 78, so the first conquest did not hold. Druids turn up in Gaul in Tacitus's own Histories eight years after the strait, prophesying from the burning of the Capitol.

The other side's statement, if there is one, is in a lake. In 1942, when the RAF were building the airfield at Valley, the workmen dug peat out of a bog called Llyn Cerrig Bach to spread on the sand. In it

they found swords and spearheads, chariot fittings, cauldrons, a bronze plaque, and a gang-chain for leading captives by the neck. It had all been thrown into the water as offerings over a long time, from before the Romans came into Britain until round about the time of the governor's army. Whatever the people of Mona thought, they put it in the lake, and the lake kept it.

He put the evidence down like a solicitor would, the statement first. Tacitus's year, the consuls of 61, went at the bottom, and over it the rising under Boudica, which some historians would rather put in 60. The lake and what was in it went on top, and through all three showed *61 CE*, which he printed above everything else, question mark and all, because the ones who put it in 60 were not daft.

Below that, the doubt: *Tacitus is the only witness, he is the enemy's witness, and he wrote half a century after. Some put the attack in 60. Cunovinda is mine, and so are her mother, the children she takes to the lake, and the boy who swam*.

The map changed as well: he moved the girl from a young woman to a girl of twelve, and her brother off the cart and into the marsh. The other side's statement, aye, at the bottom of a bog. He never wrote the comparison down, not in any of the notebooks. He turned the stone in his hand a wee while, the way he did when a file was thin.

That night

Tacitus was open on his chest at the page about the women in black, and the air turned to salt and bog water and burning pitch. Dripping like something just lifted out of a lake, the page lay in the Ledger's hands. It laid the page flat to drain before it would let him read it, one corner at a time.

• • •

The Romans had been on the far shore of the strait for three days, and Cunovinda had been watching them from the rocks every one of them.

She was twelve, and her mother was one of the women who served the grove, and so she was allowed on the rocks when other children were sent inland. Across the water, under the mountains, the soldiers were building boats. They were not boats like the island's, but flat and wide and ugly, like doors laid on the water. There were hundreds of men over there, and smoke from their cooking fires. She saw the flash of metal when they moved, and at night their fires made a line along the shore like another coast.

She came down off the rocks at dusk to find her mother sitting outside the house with black cloth across her knees, sewing.

"What's that for?" Cunovinda asked.

"For standing on the shore," her mother said, and bit off the thread. "So they can see us."

"And then what?"

Her mother did not answer that. She held up the cloth in the last of the light and looked it over, like she would look over a fleece to see if it was clean.

"What will be left?" Cunovinda said. "After. If they come over."

"The island," her mother said. "The lake. You." She folded the black cloth over her arm. "Go and find your brother. He's down at the boats again."

• • •

Old Vercassos, who kept the count for the grove, took her down to the strait that night to look at the water.

He was the oldest of the men in the grove, so old his beard had gone yellow at the ends, and he walked with a stick of blackthorn. Cunovinda had known him all her life. He had taught her the nights of the moon on her fingers when she was small, because the grove counted in nights, not days, and the month began when the moon did. He sat down on a flat rock with a grunt and pointed with the stick at the strait. It was empty now, a wide mouth of black mud with the sea gone out of it and a thin channel of silver down the middle.

"Look at that," he said. "Two nights after the full moon, the sea goes out further than at any other time of the month. It won't go out so far again until the dark of the moon." He poked at the mud with the end of his stick. "They know it, over there. They've a man who watches it, same as me. They're waiting for this. Tomorrow, when the light comes, the water will be lowest. That's when they'll walk."

"Can't we stop the sea going out?"

He laughed, a dry cough of a laugh, and shook his head. "Nobody stops the sea. You count it. That's all anybody ever does with it."

He made her count the nights of the moon back to the full, and forward to the dark. Then he made her say the times of the low water for each, the way she had learned them, one long breath of numbers.

"Good," he said. "Now you know when they'll come. That's worth more than a spear." He leaned on his stick and looked across at the fires. "Your mother will be on the shore. You won't. You'll take the little ones inland, to the lake, before the light. You know the way in the dark."

"I want to stand with her."

"I know," said Vercassos. "And somebody has to know the way to the lake."

• • •

Her mother came to find her on the rocks late that night, when the fires across the water had burned down to a row of red points.

She sat down beside Cunovinda without a word and began to unplait her hair, the way she had done every night when Cunovinda was small. Her fingers were quick and cold. When the plait was out she combed the hair loose with her fingers until it hung down Cunovinda's back. Then she did the same with her own, so that the two of them sat side by side with their hair down like the women of the grove on a feast night.

"Tomorrow we stand on the shore in the black with our hair like this," her mother said, "and the torches in our hands. We run up and down the line and we shout. Do you know why?"

Cunovinda shook her head.

"So that when they come up out of the water, the first thing they see is the dead," her mother said. "The black is for the dead. The fire is the fire of the other world. They are frightened of the dead, the soldiers, more than of spears. If they stop, even for a breath, the men have a chance." She was quiet for a while. "It is the oldest thing we know how to do."

Then she took something out of the fold of her dress and put it into Cunovinda's hand. It was the little bronze disc from the house. It had a pattern of curling lines that went round and round and never ended, and it had hung by the door for as long as Cunovinda could remember.

"Carry that to the lake for me," her mother said. "You'll know what to do with it when you get there. Don't give it to anyone else."

She kissed the top of Cunovinda's head, and stood up, and went back to the house with her loose hair moving behind her in the dark.

• • •

They went before dawn, eleven small children and Cunovinda, with her brother Taso holding the end of her cloak.

The night was grey and very still, and they went along the paths between the fields with the dew soaking their legs. The little ones did not cry, because they had been told not to, and they had felt the grown-ups' fear all week like you feel weather coming. Behind them, towards the strait, there were torches moving, many of them, gathering on the shore in a long line.

At the top of the rise Cunovinda stopped and looked back, though she had been told not to.

The sky over the mountains was going pale, and the sea had drawn right back. The strait was a great sheet of grey mud and shining channels, and across it, from the far shore, the soldiers were coming. The flat boats were being dragged and poled through the channels, packed with men. Horses were coming down into the water with their riders beside them. Where the channel was deep the horses swam, heads up, and the men swam with them, holding the manes. She had never seen a horse swim, and had not known they could.

On the near shore, the island's line stood waiting. She could see the warriors in front, and the druids behind them in their white with their arms raised to the sky. Among them were the women in black with their hair loose and their torches, running up and down the line. One of them was her mother, but she could not tell which. They all looked the same from up here, black and fire, and that was the point.

All along the line the soldiers stopped in the water, stopped dead, pinned where they stood by the sight of it.

Then somebody among them shouted, and they came on.

Cunovinda turned round, and picked up Taso, though he was too big to be carried, and took the children down the other side of the rise where they could not see.

• • •

The lake was in the west of the island, among the marshes, a long shallow water with reeds round it and a rock above it like a shelf.

She had been there before with her mother, at the feasts, when the grove gave things to the water. She knew the path in over the tussocks where the ground was firm. She brought the children up onto the rock shelf, where there was a hollow out of the wind. She sat them down in a row and gave them the bread from her bag, and counted them, eleven, and Taso, and herself.

They waited there all day while smoke went up in the east, first thin and then thick, a long dark smear blowing out over the sea. The little ones asked when they could go home, and Cunovinda said soon. She taught them the nights of the moon on their fingers to pass the time, the way Vercassos had taught her. They said them over and over, the full and the waning and the dark. By the middle of the afternoon even the smallest one could say the nights to the full moon without help.

In the evening a woman came along the path from the east. She was one of the grove women, though not her mother, and her black dress was torn and her torch was gone and her hands were empty.

She came up onto the rock and looked the children over, one by one, without saying anything. Then she sat down among them, and the smallest one climbed into her lap, and she let him.

"Is my mother coming?" Cunovinda said.

The woman looked at her, and then out at the smoke.

"The groves are burning," she said at last. "They're cutting the oaks." That was all she said, and Cunovinda did not ask again. She sat with her arm round Taso and watched the smoke until it was dark.

• • •

On the third day a soldier came to the lake.

He came alone, along the path from the east, on foot, leading a horse. He was young, not much more than a boy, with yellow hair cut short and a face burnt red by the sun. He did not have his helmet on, and he saw the children on the rock and stopped dead, and for a long moment nobody moved at all.

Cunovinda stood up without knowing why. She had the bronze from her mother's house in her hand, the little disc with the curling pattern on it that her mother had given her to carry the night before. She had been holding it in her fist for three days.

The soldier came no closer. He looked at the children, and at her, and at the lake. Then he said something in a language she did not know, not the soldiers' Latin she had heard shouted across the strait, but something lower and rounder. He pointed at himself and said a word that might have been his name. He pointed at the horse, and made a swimming motion with his arms, and laughed a little, and stopped, remembering where he was.

Cunovinda walked down off the rock to the edge of the water, past him, close enough to see that his hands were shaking. She held up the bronze disc so that he could see it. Then she threw it, as hard as she could, out over the reeds. It went into the lake with a small sound, and the rings went out from it across the water.

The soldier watched it go and did not try to stop her. When the rings had gone, he nodded once, as if somebody had said a thing in a court and he had heard it. Then he turned his horse round and led it back the way he had come.

• • •

On the fifth day word came along the paths that the soldiers were leaving.

Nobody knew why they were taking down their camp by the strait and going back across the water as fast as they had come, and leaving only a few men behind. A rider had come for the governor, people said. Something had happened far away in the east of the province, a great rising. The soldiers were needed there, and the governor was taking them.

Cunovinda brought the children back from the lake on the sixth morning. Some of their mothers were there to meet them, and some were not. The grown-ups who were there took the ones whose mothers had not come, without being asked. Her own mother was not there, but old Vercassos was, with his arm bound up and his yellow beard burnt short on one side. He took Taso by the hand and said nothing, and Cunovinda understood, and she did not cry until a long time after.

That night he took her down to the strait again. The tide was coming in, filling the channels, covering the mud where the soldiers had walked.

"You asked your mother what would be left," Vercassos said. He had heard that somehow, as he always heard things. "She told you. The island. The lake. You." He leaned on his blackthorn stick. "And the count. You took eleven children to the lake and brought eleven back, and every one of them can say the nights to the full moon now. That's not nothing. That's the grove."

Cunovinda watched the water come in over the mud, all of it, steady, like nothing had happened there at all.

"And what's in the lake," she said.

"And what's in the lake," Vercassos said. "The water keeps what it's given. Longer than us."

The master page

The page came out wet, and the Ledger did not dry it. It let it lie on the board with the water standing in the creases, the way things come up out of a bog.

In ink: Tacitus, *Annals* 14.29 to 30, on Suetonius Paulinus at the strait, the boats and the swimming horses, the women in black with torches, the druids' curses, the groves cut down, and Boudica's rising pulling the army away; Agricola on the staff that year, and back in 77 or 78; and the hoard of Llyn Cerrig Bach, dug out of the peat at Valley in 1942. In pencil: the year, 61 or 60; the prosecution's charge of blood on the altars; how many died; and every face on the shore. *Found destroyed*: the groves themselves, and whatever the druids of Mona knew and had never written down, which the soldiers wrapped in their own fire.

He set this page with the others. Egypt had trusted ink and the grove memory, and Croton had broken when its keepers scattered. Mona's share was the felling: the first time an empire came for the keepers on purpose, with boats and a governor, because it could not stand a memory it did not hold. And yet the felling could not reach the count in a child's fingers, or the lake. In ink, behind it, is the old druid of the Sequani burning his cord sooner than let it be taken. Pencilled further on is a priest far up the Nile, cutting the old signs into a gate for the last time, on the birthday of a god.

He would give it to the Core in the morning, and it would give him the lowest tides in the strait. What the women in black were thinking is past its reach, or who the boy with the horse was. Neither does Tacitus.

What will be left? *The island*, her mother had said, and *the lake*, and then, folding the black cloth over her arm, *you*. She was right in every part, and the lake held up its end for the best part of nineteen hundred years.

Cunovinda had needed to know which morning the sea would go out furthest, and so which morning they would come. Under that lay the dream's answer. An army can take the grove, but it can't take the moon or the tide off you, or a count already in the fingers of eleven children. The other lesson Gary had learned years before in a solicitors' office, and the shore had learned it the hard way: you read the other side's statement as a statement, and never as the whole of what happened.

The smell of the bog was still in his nose when he woke, and the stone cold in his hand, the way a thing is cold that has been a long time under water. Outside, the tide would be coming in over the mudflats on the lough, and he lay and listened for the birds on it.

CHAPTER 22

The Last Priests of Philae

Philae, the island of Isis at the First Cataract, on the edge of Egypt and Nubia, on the first of the five days upon the year: the found year, 394 CE

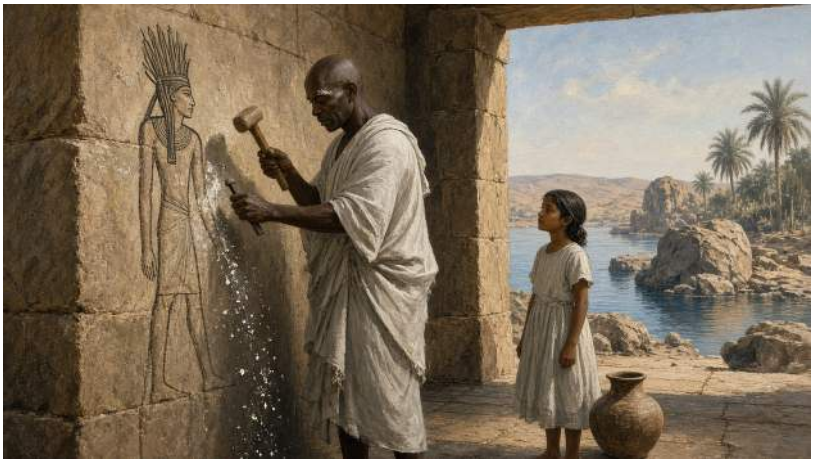


Illustration 23. *Nesmeterakhem cuts the figure of Mandulis into the gate at Philae.*

Belfast, by day

IN THE SUMMER OF 2016 he moved on to a technical support desk in Belfast. One of the first things he learned there was that every big network had an old box still running in a back room somewhere. Nobody liked to touch it. The manuals had gone, the company that made it had been bought twice, and its logs came out in a format that nobody under fifty could read at a glance. There was usually one engineer near retirement who could. When that man was off, the tickets about the old box just sat there, and everybody hoped it would keep running until he was back. He used to wonder what would happen the week after the man retired for good.

The old box was on his mind when the screen came back on Philae.

"The Theodosian decrees of 391 closed the temples of Egypt," it said, "and the last hieroglyphs were carved at Philae in 394 by the last priest of Isis, before the temple closed."

Everything the machines turned up he read himself, the stone holding the corner of a photograph of a gateway.

The inscription is ink, as good as ink gets. On a gateway that the emperor Hadrian added to the temple of Isis at Philae, there is a rough figure of the Nubian god Mandulis, a sun god, with a tall crown. Beside his head are two short columns of hieroglyphs. Before Mandulis, son of Horus, they say, by the hand of Nesmeterakhem, son of Nesmeter, the Second Priest of Isis, for all time and eternity. Under the god's hand a longer note in demotic, the everyday script, gives the date: the birthday of Osiris, his dedication feast, year 110. The years were counted from the emperor Diocletian, and that day was the twenty-fourth of August 394. Nobody has found a later hieroglyphic inscription anywhere in the world.

The rest of the claim came apart in his hands. The decrees of 391 are real enough, and they forbade sacrifice and closed the temples to visitors, and the great Serapeum at Alexandria was pulled down about the same time. But Philae was on the frontier, and the Nubian peoples to the south kept coming to worship Isis there. Nesmeterakhem was no last priest, because his family served the temple for at least three generations, and demotic notes were still being scratched on its walls in 452. The temple was only shut for good in the reign of Justinian, round about 535, when a general called Narses arrested the priests and sent the statues to Constantinople. He scored out *before the temple closed* and wrote *a hundred and forty years before* above it.

Set he had to come at sideways, through the old kings. They had honoured Set as a strong god of the desert and the foreign lands, and Seti, the father of Ramesses the Great, was named for him. Over the next thousand years he became the murderer of Osiris and the enemy of every god. In the late temples he is speared on the walls as a hippopotamus. By 394 that change was a thousand years old, and nobody at Philae could have remembered it. They could only have seen the old names and wondered.

He worked from the bottom of the pile up, with the decrees of 391 against the temples underneath and the gateway and its date over them. On top went the long tail of the temple, Nubian pilgrims and demotic scratches, out to Justinian. The year that showed through all of them was *394 CE*, and he headed the page with it. The question mark was hardly needed, since the priest had dated his own gate to the day, but he put it in anyway, small, out of habit.

The doubt came next: *The day is exact, but who stood at the gate is not. The Nesmeter family may have been Nubian, and the hand on the stone is rough. Tabes is mine, and so is her grandfather, and so is her mother's church across the water*.

He went back to the map and moved the old priest off the chisel and gave it back to the man who signed it. Aye, he thought, the old box in the back room, and one man left who could read the logs. He left it unsaid, and the stone stayed on the photograph a good while.

That night

Heat came first, with the photograph of the gate under his hand, then river smell and the sound of water going fast over rocks somewhere close. On warm stone the Ledger turned its page slowly, the way an old man turns the leaves of a book he can still read and knows nobody else can. Out on the water a boat was being poled upstream against the flood.

• • •

Tabes carried water up from the river to the gateway every morning, and on the morning the priest began on the god, she was carrying it.

She was eleven. She lived with her grandfather in two rooms against the outer wall of the temple, where he had been a doorkeeper since before her mother was born. He could not read the signs on the walls. He could say what most of them meant, because he had heard them read aloud all his life, and that was nearly as good, he said. The priest who was carving was Nesmeterakhem, the scribe of the house of writings, whose father was the Second Priest of Isis and whose

grandfather had served here before that. He was a quiet broad man with a Nubian face and dust in his eyebrows. He had drawn the outline of the god on the stone of Hadrian's gate in charcoal. It was a tall figure with a crown like a bundle of reeds, and was beginning to cut it.

Tabes set down the water jar and watched the chisel go in, in short taps, with the white dust falling. It was not beautiful work, and even she could see that the lines wandered.

"Who will read it?" she asked, without meaning to say it aloud.

Nesmeterakhem did not stop tapping. "Mandulis," he said. "He's the one it's for. And whoever comes after."

"Who comes after?"

He did stop then, and looked at her over his shoulder, and she thought he was going to be cross. He only wiped his forehead with the back of his wrist and left a white streak on it.

"That," he said finally, "is a very good question for a water carrier."

• • •

A few mornings later Nesmeterakhem took her up onto the temple roof before dawn. She had asked him where the dog star was, and he said it was easier to show her than to say.

They went up a narrow stair inside the wall, in the dark, with his hand on her shoulder to steer her. The roof was wide and flat and still warm from the day before. From up there she could see the whole island lying in the river like a boat, and the black rocks of the cataract, and the desert hills on both sides. The river was loud below them, and the east was only just going grey.

"There," he said, and pointed low over the eastern hills. "Watch the place where the sky is lightest. Not above it. Just beside it."

She watched until her eyes watered and saw nothing. Then she saw it, a single white spark sitting on the edge of the hills, very bright and trembling, with the grey coming up behind it.

"That's Sopdet," Nesmeterakhem said. "Isis in the sky. For a long while in the spring you can't see her at all, because she's too near the sun. Then one morning she comes up out of the east just before the sun does, for a moment, and the river begins to rise. The priests used

to watch for that morning from this roof and from every roof down the river, and when they saw her they knew the year was turning." He folded his arms against the cold. "We still watch. There are fewer of us."

She asked him whether the river knew about her, and he laughed, very quietly so as not to wake anybody below. Nobody knew what the river knew, he said, but the two of them kept to the same time, and had done for longer than anyone had been writing it down.

Then he crouched down and drew three quick signs of the everyday writing in the dust with one finger. He told her what they were: the star, the year, and the word for opening, which was what the old priests had called the first day. She drew them after him, badly. He rubbed hers out with his palm and made her draw them again until they were right.

Once the sun was up the star was gone, washed out into the light, and Tabes found she had been holding her breath.

• • •

Her grandfather explained the day to her that evening, sitting on the step with his feet in the dust and the river loud below them.

"In five days the year ends," he said. "Not the soldiers' year, ours. Twelve months of thirty days each, and then five days that don't belong to any month, the days upon the year. On the first of those Osiris was born, and that's the day Nesmeterakhem wants his god finished. On the second Horus, on the third Set, on the fourth Isis, on the fifth Nephthys. Then the year turns over and begins again."

"Set too?" said Tabes. "He gets a birthday?"

"Everybody gets a birthday." Her grandfather grinned at her in the dark of the doorway. "Even him."

He showed her where the dog star had come up a month before, just before the sun, low in the east over the desert. The river had begun to rise the same week, as it always did, and it was rising now. You could hear it in the cataract below the island, louder every night, the water piling over the rocks.

"The star tells the river," her grandfather said, "and the river tells the fields, and the priests tell the year. It's all one thing." He was quiet for a while. "Or it was. Now there's the soldiers' year, and your mother's year, and ours. Three years in one house." He shook his white head slowly. "It's a wonder anybody knows when to eat."

• • •

Her mother's year was the thing Tabes did not talk about.

Her mother lived across the water at Syene, in the town, and came to the island once a month with bread and to look at her father's knees. Two winters ago she had taken Tabes to the church at Syene, where the priest in the long black robe had put her into the water and said words over her. He had given her another name that Tabes was not to use on the island. She had a small wooden cross on a string under her dress. Her grandfather had never seen it, or he had seen it and never said, which was more like him.

That same priest at Syene said the temple on the island was a house of demons. Tabes had thought about that a great deal. She thought about it carrying water up past the carvings of the gods on the walls, Isis with her wings and Horus with his falcon's head and the little cows. They did not look like demons; they looked like her grandfather's friends.

One morning she asked him about the man with the spear on the inner wall, the one stabbing a little hippopotamus under his feet.

"That's Horus," her grandfather said, "killing Set. Set killed his father Osiris and cut him in pieces, and Isis gathered the pieces, and Horus grew up and took his revenge. That's the story on the walls."

"So Set is the bad one."

"He is now." Her grandfather scratched his chin. "My grandfather told me there was a time when the kings were named after him. Big kings, the ones who built the great temples down the river. Seti, that means Set's man. Set was the strong god of the desert, the one who stood in the prow of the sun's boat and speared the serpent every night. And then, somehow, he was the enemy." He shrugged his thin

shoulders under the linen. “Nobody remembers why. It was too long ago. The gods change sides, child. Or we change them for them, and forget we did.”

Tabes thought about the cross under her dress, and the name she was not to use on the island, and held her tongue.

• • •

On the first of the five days, the birthday of Osiris, Nesmeterakhem finished the god.

He had been working at it every morning. Now Mandulis stood on the stone of the gate, rough and tall, holding his staff, with the crown of reeds on his head. In the evening, when it was cooler, Nesmeterakhem came back with a smaller chisel, and Tabes’s grandfather came with a lamp, and Tabes came because nobody sent her away. She held the dish with the chisels in it.

Nesmeterakhem stood looking at the blank stone beside the god’s head. Then he began to cut the signs.

He did it slowly, because he had to stop and think between them, his lips moving. Once he cut a sign and stood back and looked at it and shook his head and went on. Tabes watched the little pictures come out of the stone in the lamplight: the water, the bread, the seated man, the owl, the reed. Her grandfather read them aloud as they came, under his breath, not reading them really but knowing them.

“Before Mandulis, son of Horus,” her grandfather said. “By the hand of Nesmeterakhem. Son of Nesmeter. The Second Priest of Isis.” The chisel tapped in the lamplight. “For all time and eternity.”

When the hieroglyphs were done, Nesmeterakhem knelt down and cut a few lines under the god’s hand in the quick, joined-up writing of the scribes, the everyday script. That went faster. At the end of it he sat back on his heels and blew the dust away, and the lamp flickered.

“Read that one,” he said to Tabes, and pointed at the end of the line.

She looked at it hard, because he had shown her some of the everyday signs that summer, in the dust, while she waited with the water. She found the ones she knew.

"Today," she said slowly. "The birthday of Osiris. Year one hundred and ten."

"Good," said Nesmeterakhem, and he looked at the stone, and not at her, a good while. "Then somebody can."

• • •

The soldiers' boat came to the island at the end of the month, after the new year, with a tax man in it and a letter.

The letter was read out in Greek on the quay, in front of the priests. The Greek meant nothing to Tabes, though the faces said plenty. Her grandfather explained it afterwards. It was the emperor's law again, the one that had been read three years before and then half forgotten: no more sacrifices, the temples of Egypt shut to visitors, nobody to go into them to pray. The tax man had been told to read it at every landing, so that nobody could say they had not heard.

"And here?" Tabes said.

Her grandfather looked south, up the river, towards the cataract and Nubia beyond it, where the pilgrims came from.

"Here they'll leave us alone, for a while," he said. "They need the southern people to keep the peace, and the southern people come here for Isis. So the soldiers will look the other way." He sighed and rubbed his knee. "For a while. Everything is for a while."

Her mother came the next day and said Tabes was to come and live at Syene now. She said it quietly, kneeling by her father's chair, with her hand on his knee, and the old man looked at the wall, and went on looking, and then nodded.

• • •

On her last morning on the island Tabes carried water up to the gate one more time, though nobody needed it.

Nesmeterakhem was not there, and the gate was empty and cool in the early light. Mandulis stood on the stone with his staff and his crown of reeds, and the little columns of signs went down beside his head. She set the jar down and stood in front of them. She read the

bits she could, moving her lips the way the scribe had: before Mandulis, by the hand of, for all time and eternity. Under the god's hand she read the quick lines, today, the birthday of Osiris.

She touched the last line with one finger, very lightly, so as not to wear it.

Then she picked up the empty jar and went down to the quay, where her mother was waiting in the boat. Her grandfather was standing on the steps with his stick, and the river was high and brown and loud with the year's flood. Under her dress the wooden cross was warm against her skin. In her head the signs went round: for all time and eternity, year one hundred and ten. She did not see why she could not keep both, and nobody had ever been able to tell her.

The boat pulled away into the current. She looked back at the gate, small and pale above the water, and said it once more to herself, so that somebody would have read it.

The master page

On warm stone, with a lamp beside it, the Ledger put down this page, and the lamp was the only light in the room.

In ink: the graffito on Hadrian's gate, Mandulis and the hieroglyphs by the hand of Nesmeterakhem, for all time and eternity, with the demotic date, the birthday of Osiris, year 110, the twenty-fourth of August 394; the decrees of 391; demotic notes down to 452; Narses closing the temple round about 535; the five days upon the year; the dog star and the flood; and Set, honoured by Seti and speared on the late walls. In pencil: whether the Nesmeter family were Nubian; who held the lamp; and the girl at the gate. *Found destroyed*: the ability to read the signs at all, which went out of the world within a generation or two of that evening and stayed out for fourteen hundred years.

He put the page with the others. Egypt had kept the record and the grove the memory. Croton broke when its keepers scattered, and Mona was felled. Philae's part was the last word: the record running down not because anybody burned it, but because the last people who could read it grew old in a back room at the edge of the empire, and the young ones were taught other things. Behind, inked, is the House

of Life at Thebes, where a scribe once copied the road through the night for a king. And the script came back from this same island: a small obelisk taken from Philae, with the queen's name in Greek and in a cartouche, helped Champollion read Cleopatra in 1822. Ahead in pencil stands a cliff in Northumbria and a quarrel over which year a church should keep.

Ask the Core, and it can place the dog star in the sky of 394; it is no help with what Nesmeterakhem was thinking with the chisel in his hand, or whether he knew he was the last. Nobody knows that, and nobody should pretend to.

Who will read it? Mandulis, the scribe said, and whoever comes after. For fourteen hundred years it was only Mandulis.

Tabes had needed to know when the dog star would rise and the river come up, and which of the five days was the god's birthday. The dream gave the harder answer: the star and the flood keep their appointment whether anybody writes it down or not, and a year is only lost when the last person who counts it stops. The old man at the gate had done his work well enough, rough as it was, and he had taught the water carrier the signs. Most people get to hand on less.

Two hundred and seventy years on, and at the other end of the Christian world, a boy sat on a bench behind his bishop in an abbey on a cliff, listening to grown men argue about the date of Easter.

The lamp beside the page burned down to a bead, and nobody trimmed it. In the dark the signs on the gate went on being there, the way they would for fourteen hundred years, until somebody could read them again.

CHAPTER 23

Whitby, 664

Streanæshalch, the abbess Hild's house on the cliff above the Esk, and the coast road north to Lindisfarne, in the spring the king called the two Easters to one table: the found year, 664 CE



Illustration 24. *Ceolla by the barn fire with the Easter table; Bishop Colmán beside her.*

Belfast, by day

THE YEAR HE WENT DOWN to Dublin to sling for a crane-hire firm, he carried two cards in his wallet that said the same thing. One was the card you needed on a site in the North. The other was the SafePass you needed in the South. They both said he had done his safety training and could be trusted on a site, and the training was much the same. But a gate in Dublin wanted the southern one, and a gate in Belfast wanted the northern one. A man with the wrong card stood at the gate however good he was. Nobody on either side thought the other card was dangerous. It was only that each site had chosen whose card counted, and the choosing was the whole of it.

The two cards were in his head when Whitby came back up on the screen.

"At the Synod of Whitby in 664 King Oswiu chose the Roman date for Easter over the Celtic one," it said, "and the Celtic Church came to an end."

The machines fetched; he read, by hand, with the stone holding Bede open.

Bede is the source, and he is a good one on this, though he wrote nearly seventy years after and he was on the winning side. In the third book of his history he says that in King Oswiu's own house Easter had been kept twice in one year. The king had been brought up by the monks of Iona and kept their reckoning. The queen, Eanflæd, came from Kent and kept the Roman one, so that sometimes the king was feasting on Easter Sunday while the queen and her people were still fasting for Palm Sunday. So the king called both sides to the house of the abbess Hild at Streanæshalch, which is Whitby. Colmán, the Irish bishop of Lindisfarne, spoke for the Iona custom and said he had it from his fathers, and from Columba, and from John the Evangelist. Wilfrid, young and sharp and lately back from Rome, spoke for the Roman reckoning. He said it was kept everywhere except by the Picts and the Britons, who were fighting the whole world from their two islands at the end of the ocean. Then the king asked whether Columba had been given the keys of heaven, as Peter had. Colmán said no. The king said that he would not go against the doorkeeper, in case there was nobody to open the door to him when he got there.

He put a line through *came to an end*. There was no one Celtic Church to end, whatever the machine thought. The southern Irish had already gone over to the Roman reckoning thirty years before, and Iona itself kept its own Easter for another fifty-two years, until 716. Colmán went home to Iona with his Irish monks and about thirty of the English, carrying some of the bones of Aidan. Later he founded a house on Inishbofin off the coast of Mayo.

The argument itself was about the moon. The Iona monks used an old table that ran in a cycle of eighty-four years, and let Easter fall on the fourteenth day of the moon if that was a Sunday. The Romans

used a nineteen-year cycle and would not let it fall before the fifteenth. Most years the two tables agreed, and nobody noticed. Some years they were a week apart, and that week was the king's supper.

He stacked the year with the synod at the bottom, which Bede puts in 664. Above it lay the eclipse of the sun that Bede says came on the third of May that year, at about the tenth hour; the sky puts it on the first. On top was the plague that came in after it and carried off the new bishop of Lindisfarne. Through the three he read *664 CE* and wrote it down, and when he came to the question mark he found he was smiling, because Bede himself had the day of the eclipse out by two.

Then the doubt: *Bede is the winners' historian, writing at a distance of seventy years. The speeches are his, not the speakers' own. Some put the synod in 663, and Bede's eclipse is two days off the real one. Ceolla is mine, and so is the table he carried.*

On the map he took the speaking part off the boy and gave him a bench behind the bishop and a book of years on his knees. Two cards in the one wallet, he thought, aye. He set the stone down on Bede and looked at the date of the eclipse a while longer than he needed to.

That night

Bede lay open on his knee at the page about the doorkeeper. Sea air and sheep came in, and the smell of new vellum scraped and not yet written on. The page the Ledger held was ruled into columns, years down one side and moons across the top, and it turned it carefully, the way you turn a table when you have to keep your place in it.

• • •

Ceolla had the table in his lap for the whole journey south, wrapped in a wool bag against the spray. On the second evening on the road he asked the bishop the question he had been carrying with it.

He was twelve, and his family had given him to the monks at Lindisfarne when he was six. The Irish brothers had taught him to read, and to sing the psalms, and to keep the table. It was a long sheet

of vellum ruled into columns. It had the years going down one side and the numbers of the moon beside them, eighty-four years of them, and then back to the start. You could find any Easter on it, if you knew how. Ceolla knew how better than anybody in the house under thirty. So he was the one who carried it.

"Father," he said, when they had stopped for the night in a farmer's barn above the sea. "The queen's table and ours. Which one is right?"

Bishop Colmán was sitting on a sheaf with his back against the wall, pulling burrs out of his cloak. He was a tall grey Irishman with big quiet hands, and he did not answer quickly, ever.

"Ours is the one Columba kept," he said at last. "And Aidan after him. And the brothers on Iona keep it now."

"But is it right?"

Colmán looked up from his burrs. "Right for what?"

Ceolla did not know how to answer that, so he held the bag on his knees and looked at the fire.

"We'll see at Hild's," the bishop said, kindly enough, and went back to his burrs.

• • •

The house of the abbess Hild stood on the cliff above the mouth of the Esk. It was a long cluster of timber buildings and a church inside a bank, with the sea grey below it and the wind coming straight off it.

Ceolla had never been among so many important people at once. There were the king's men with their good cloaks and their swords left at the gate. There was the queen's priest from Kent, and the Frankish bishop Agilbert with his own clerks, and old Cedd who spoke both English and Irish and went between them all like a dog between flocks. There were Hild's own nuns and monks, and the abbess herself, a tall woman with a weathered face and a voice that carried without being raised. She gave the Lindisfarne brothers the best of the guest house and sat with Colmán by the fire the first night. Ceolla heard her say, quietly, that her own people kept the Iona Easter, and always had.

He also saw Wilfrid, who was young for a priest, about thirty, with a fine Roman haircut shaved round the crown like a ring. He walked about the abbey like he already owned it. He looked at Ceolla's wool bag once, and at the edge of the vellum sticking out of it, and smiled in a way Ceolla did not like.

• • •

On the second night Ceolla could not sleep. He took the table out to the guest house porch where a rushlight was burning, and unrolled it on his knees to look at it.

He had been there a while, running his finger down the columns, when the abbess came along the path from the church with her cloak held shut against the wind. She stopped when she saw him, and he scrambled to stand up. She told him to sit, and sat down on the bench beside him as if she were nobody in particular.

"Is that the Iona table?" she asked him, and he said it was.

She asked him to show her how it worked. He did, the years going down and the moon's days beside them, and how you found the full moon after the spring and then went on to the Lord's day after it. She followed his finger closely, the way a good reeve follows a count. When he had finished she nodded and sat back against the wall.

"I was baptised by Paulinus at York, when I was a girl," she said. "He came up from Kent with the queen of that time, and he kept the Roman Easter, the same as the queen's priest keeps it now. Then the king was killed and Paulinus went away south, and years after that Aidan taught me, and I kept it his way, the way that's on your knees." She smiled at the vellum. "So I've kept it both ways in my life, and I can't say the Lord rose any less on the one day than on the other."

Ceolla had nothing to say to an abbess, and did not try. She seemed to think that was the right answer.

"They'll decide it in there, the day after tomorrow," she said, looking towards the dark shape of the church, "and whatever they decide, I'll keep. That's what a house like this is for, to keep what it's given and keep it well. But I'll tell you something your bishop knows already, and you're young enough to be told it plainly." She put her finger on

the column of the moon. “This part isn’t theirs to decide. This part was there before Peter or Columba either, and it’ll be there after the both of them.”

She stood up and pulled her cloak round her again. She told him to go to his bed before the rushlight was out, and went off along the path towards her own house with the wind in her skirts.

He sat a little longer with the table on his knees, and the column of the moon, and then rolled it up carefully and went in.

• • •

They met in the church on the third day, with the doors shut against the wind.

The king sat at the top, with the queen’s priest on one side and Colmán on the other. The abbess sat near him, and everybody else packed in on benches round the walls. Ceolla sat on a bench just behind the bishop with the table open on his knees, in case it was wanted.

The king spoke first, and said that men who served one God and hoped for one kingdom of heaven ought to keep one rule of life, and not differ in the sacraments. He said they were here to find out which was the truer tradition, and then all follow it. Then he asked Colmán to begin.

Colmán stood up, and he did not speak for long. He said that the Easter he kept was the one his elders had kept, who sent him here as bishop. He said it was the one the blessed John the Evangelist had kept, the disciple the Lord loved, and the churches that John had looked after. Holy Columba and his successors had kept it too, he said, men whose holiness was shown by the signs God did through them. He said he did not see how he could despise it. Then he sat down, and Ceolla saw his big hands lying quiet on his knees.

The Frankish bishop Agilbert was asked to answer. He said, in his slow Frankish English, that his disciple Wilfrid could explain the matter better in the English tongue. Wilfrid stood up as if he had been waiting all his life.

He talked at length, and said that the Easter he kept was kept at Rome, where the apostles Peter and Paul had lived and taught and suffered and were buried. He said he had seen it kept the same way in Italy and in Gaul. It was kept in Africa, Asia, Egypt and Greece, everywhere in the world where the Church of Christ had spread, at one and the same time. Everywhere, he said, and he turned and looked along the benches, except these people and their accomplices in obstinacy, the Picts and the Britons. They were fighting against the whole world, he said, from two islands at the very end of the ocean, and not even the whole of those.

Ceolla felt his face go hot. He looked at the table on his knees, at the columns of years in the neat Irish hand, and at the bishop's back in front of him. The bishop in front of him did not move at all.

• • •

After that the synod went on for most of the day, and most of it was about the moon.

Wilfrid said that John had kept the fourteenth day of the moon because he was among Jews and had to go slowly. After him, he said, the Church had learned to wait for the Lord's day. He said that the Iona table let Easter fall on the fourteenth day, which was the Jewish Passover itself, and that was wrong. The Roman table, he went on, was made by Anatolius, and Colmán said his own table was Anatolius's. Wilfrid laughed, not kindly, and said that Anatolius had used a nineteen-year cycle, which the Iona monks either did not know or did not keep.

Ceolla knew the table on his knees as well as he knew the Our Father. He knew it ran for eighty-four years, not nineteen. He knew it let the fourteenth day of the moon stand. He sat very still and felt, as never before, that the thing on his knees, which he had carried and guarded and been proud of, could be wrong. It was not wrong about the moon, because the moon was the moon. It was wrong about something else, something about who got to say.

Then the king asked the question.

"Tell me, Colmán," he said. "Is it true that the Lord said to Peter, *You are Peter, and on this rock I will build my church, and I will give you the keys of the kingdom of heaven*?"

Colmán said it was true.

"Can you show me that any such power was given to your Columba?"

Colmán said no.

"Then do you both agree," the king said, "that these words were said above all to Peter, and that the Lord gave him the keys?"

They both said yes.

The king smiled like a man who has settled a dispute about a field. "Then I tell you, I'll not go against this doorkeeper," he said. "I'll obey his rules in everything, as far as I know them. Otherwise, when I come to the gates of the kingdom of heaven, there may be no one to open them, if he's turned away who holds the keys."

Everybody on the benches agreed, the high and the low, and stood up, and it was done.

Ceolla sat on with the table open on his knees, and understood that it had never been about the moon at all. It had been about whose table counted, and whose keys. The sky had not been asked.

• • •

They left Hild's house two days later, on the coast road north.

Colmán had resigned his see, because he would not keep the new Easter, and he would not stay in a place where he was the only one who did not. He was going home to Iona, and the Irish brothers were going with him, and some of the English too. At Lindisfarne he would take up some of the bones of Aidan from the church, to carry with him, and leave the rest.

Ceolla walked beside the bishop's pony with the table in its bag on his back. He had not been told whether he was going or staying. He did not know which he wanted, and nobody asked him.

On the afternoon of the first day north, out on the open moor above the sea, the light went wrong.

It went thin and grey and cold, the way it does before rain, but there was no cloud. The sheep on the moor stopped grazing and stood looking about them. The brothers stopped on the road, and somebody cried out and pointed. Ceolla looked up through his fingers and saw the sun with a bite out of it, a black curve eating into the edge of it, growing.

Some of the brothers knelt down in the road. One of the young English monks began to cry, and said it was a judgement, for Whitby, and nobody could tell which side it was a judgement on. The pony shied and tossed its head. Bishop Colmán stood with his hand on its neck and looked up through his fingers, and did not speak.

Ceolla did not kneel, because he had his hand on the bag on his back, and he was counting in his head. They had looked it up only last week, for the journey, on the table and on the queen's table too. The moon was new today; both tables said so, and agreed on that.

"It's the moon, Father," he said. "It's new today. It's gone in front of the sun. It's in the book. Isidore says it. It's the moon going in front."

The bishop looked down at him, and for the first time since the church at Whitby, his face changed. It was not quite a smile, but it was near one.

"Both tables agree on that," Ceolla said. "Ours and theirs. The sky's the same for both of them."

The black went across, and came off the other side, and the light came back warm onto the moor, and the sheep put their heads down and went on eating.

• • •

At Lindisfarne, on the morning the boat was to sail for Iona, the bishop came to find Ceolla in the scriptorium.

"You're staying," he said. "Your family are here, and your tongue is here. The new abbot will need somebody who can keep a table." He put one of his big hands on the wool bag. "Keep theirs, since it's theirs now. But keep ours too, in your head. Not to fight with. So that somebody knows what it was, and why it was, when they ask."

"Which one was right?" Ceolla asked him. He had to ask it once more, before the boat went.

Colmán thought about it, the way he always did, slowly.

"The moon was right," he said. "Both times. That's what you told me on the moor." He took his hand off the bag and stepped back. "The rest is keys."

Ceolla stood on the beach with the brothers who were staying. He watched the boat go out on the morning tide, north, towards Iona, with the bones of Aidan in it, and the grey bishop sitting in the stern with his hands on his knees.

That night, in the dormitory, he said the old table over to himself in the dark, eighty-four years of it, from beginning to end, so as not to lose it.

The master page

The Ledger ruled this page in columns, and in one column it wrote the Roman numbers, and in the next the Iona ones, and where they agreed it left them side by side.

In ink: Bede, *Ecclesiastical History* 3.25 to 27, written in 731: King Oswiu's house keeping two Easters, the king feasting while the queen still fasted; the gathering at Hild's house at Streanæshalch; Colmán speaking for the custom of Columba and of John; Agilbert handing the case to Wilfrid; Wilfrid on Rome and the whole world against the Picts and Britons; the keys of Peter and the doorkeeper; Colmán's withdrawal to Iona with the bones of Aidan, and his house on Inishbofin. The eighty-four-year table against the nineteen-year one, the fourteenth day of the moon against the fifteenth. The eclipse of the sun in 664, which Bede dates two days late, and the plague after it. Iona keeping its own Easter until 716. In pencil: whether the synod met in 663 or 664; how much of the speeches are Bede's own; and the boy on the bench behind the bishop. *Found destroyed*: the Iona Easter tables themselves, which were let go once nobody used them, so that scholars now rebuild them from fragments.

The page joined the others. From Egypt came the written record and from the grove the remembered one, and the boats had come home to Iona. Then the taking apart began: Croton scattered, Mona felled, and at Philae the last word. Whitby's contribution was the ruling. A year that had been kept two ways under one roof was made into one, not by finding out which agreed better with the sky but by deciding whose keys counted. The sky, on the moor three days later, agreed with both. Already in ink, behind, is the abbot in the boat of hide telling the boy on the stern oar that the reckoning would be a fight one day, and the harbour at Iona that Colmán sailed home to. Pencilled ahead is the same island nine hundred years later, where the chiefs of the Isles were made to sign away the bards.

Next morning the Core would run both tables for every year from 600 to 700 and print the years where they part, and put the eclipse on the moor to the minute. It knows nothing of what Colmán thought on the road north, or whether a boy kept the old table in his head. It would find the eclipse and miss the boy.

Which one is right? The moon, both times, the bishop said. The rest is keys.

Ceolla had wanted to find Easter in the table, and to know that an eclipse comes at the new moon and is not a judgement on anybody. The dream's answer was that the sky is the one witness both sides share, and it is very seldom called. Keep the table you are given and know the one you lost. A ruling is not a measurement, and Gary had sat through enough meetings to know which one usually won.

The next page was Iona again, nine hundred and forty-five years on: nine chiefs of the Isles putting their names to a set of statutes, and a bard's daughter hiding under a bench in a hall in Moidart.

Somewhere under the page, in the dormitory on the cliff, a boy was still saying the old table over in the dark, eighty-four years of it, and Gary found his own lips moving with him, a year behind.

CHAPTER 24

The Statutes of Iona

Eilean Tioram in Moidart, the castle of the Captain of Clanranald on its tidal rock, in the autumn the chiefs came home from Iona with the bishop's paper: the found year, 1609 CE



Illustration 25. *Eòghan plays the harp on the rocks below the castle as the birlinn comes in.*

Belfast, by day

WHEN HE WAS STUDYING BUSINESS in Donegal, there was a handbook for every module, and every handbook had a list of learning outcomes at the front. Everything you were assessed on was on that list. If a thing was on the list and you could write it in the right shape, you passed. If a thing was not on the list, it did not exist as far as the course was concerned, however useful it was. The best of what he learned that year was never on any list. It was in the way the older ones in the class talked about the businesses they had run and lost, at

the break, by the vending machine. Nobody assessed that, and nobody ever wrote it down. He remembered more of it than of the handbooks.

The handbooks came to mind when the search on Iona returned.

"In 1609 the Statutes of Iona banned the Gaelic language," it said. "They outlawed the bards and the harpers of the Highlands for good."

He set the machines on it, then checked what they found by hand, the stone keeping the page of statutes flat.

The statutes themselves are ink, and they are not long. In August 1609 Andrew Knox, the bishop of the Isles, sailed to Iona as the king's commissioner, and the chiefs of the Isles met him there. Most of them had spent the winter before in prison in the Lowlands, after they had been invited on board a king's ship off Mull to hear a sermon and carried off. On Iona they agreed nine articles and bound themselves to them, and among the names is Donald MacAllan, the Captain of Clanranald, of Eilean Tioram in Moidart. The articles are plain enough when you read them. The churches were to be repaired and the ministers obeyed, and inns were to be built for travellers. Each chief could keep only so many gentlemen in his household, six for Clanranald. The trade in wine and whisky was to be stopped. Every man in the Isles with sixty cattle was to send his eldest son, or failing a son his eldest daughter, to school in the Lowlands until they could speak, read and write English. No one was to carry a gun outside his house. And the chiefs were not to keep wandering bards and such vagabonds; any found were to be put in the stocks and sent out of the Isles.

He struck through *banned the Gaelic language* with one clean line. The statutes do not mention Gaelic. They make English the price of an heir's schooling, which was a slower way of doing the same work. Then out came *outlawed the bards*, because it is the wandering ones the article names. The chiefs' own household poets went on. The MacMhuirich family were poets to Clanranald for nearly a hundred and fifty years after that, and one of them wrote the history of the family into what is called the Red Book. The statutes were not well

kept, either, any more than most laws. In 1616 the Privy Council had to try again, more harshly, and made English a condition of inheriting.

There was one more thing, and it came from much later. In 1703 a Skye man, Martin Martin, described how the poets of the old schools composed. They shut themselves into a dark room for a day, lay on their backs with a stone on their bellies and their plaids over their heads, and made the poem in the dark. Only then did they bring it out to be written down in the light. He sat back and looked at the stone on the printout until the screen dimmed.

Over the next hour the year went together piece by piece, the kidnapping off Mull in 1608 first, the statutes on Iona in August 1609 over it, and over both the chiefs coming home, and the Privy Council trying again in 1616. The overlap was *1609 CE*; he wrote it large at the top, and the question mark was there for form's sake, because the chiefs had signed and the Privy Council had kept the paper.

His doubt went underneath: *Martin Martin wrote ninety-four years later, and he describes the schools when they were nearly gone. The Ledger dates the breaking of the Insular stream to 1609, which is its framing, not the historians'. Mairead is mine, and so are her father and her uncle and the song*.

The map got its correction too, in the same plain pencil. He gave the uncle a harp and a road out instead of a Lowland school, and sent the chief's own boy south in his place. Aye, the handbook and the vending machine all over again. He sat with the likeness, his hand flat on the stone a wee while.

That night

The statutes were on his chest, with the stone on top of them, the way the poets lay. Peat smoke and seaweed came in, and from somewhere below the sound of a harp being tuned, one string at a time. The Ledger's page was folded small and hidden, and it fetched it out from under a bench before it would open it.

• • •

The galley came home on a grey evening at the end of August. Mairead was on the rock at the castle door to see it, because her uncle Eòghan was sitting there with his harp and had told her to watch for the sail.

She was eleven, and her father was the chief's poet, as his father had been and his father's father, back further than anybody could count without the book. They lived in the long house below the castle wall. Her uncle was a harper and did not live anywhere. He went round the Isles from hall to hall with his harp on his back and his songs in his head, and came to Eilean Tioram every autumn for as long as the chief would have him. Mairead loved him better than anybody, which she would not have said aloud.

That evening he was playing a song she had not heard before, slow, with the tide coming in round the rock and the light going.

"What's that one?" she asked.

"It's the stars of the autumn," Eòghan said. "It's for the boats. When the seven stars come up in the east at nightfall, the long boats come home, and the singing starts in the halls. It goes through all the stars of the winter, one after another, and what each of them tells the boatmen." He played the turn of it again. "It's old. My teacher had it from his. I've never heard anybody else sing it."

"Who will sing it, when you're old?"

He laughed, and pushed her hair out of her eyes with the back of his hand. "I'll find somebody."

Then the sail came round the point, and the galley came in on the last of the tide with the chief standing in the stern. The men on the rock went quiet, because he was not smiling.

• • •

The chief had the paper read in the hall that night, after the meat.

Children were not let into the hall when the chief had business. Mairead knew that. She went in anyway, along the wall in the smoke, and got under the long bench by the fire where the dogs slept, and lay there among the dogs with her chin on her hands. Nobody saw her. Or the dogs saw her and did not care.

The paper was in the Lowland tongue, and the chief's clerk read it out slowly, a line at a time. Then her father put each line into Gaelic for the hall in his good carrying voice. Mairead watched her father's feet by the fire and listened. There was a great deal about churches and ministers that she did not understand. There was a part about inns. There was a part about drink that made the men groan. Then there was a part about the chief's household, that he was to keep six gentlemen and no more, and the hall went very quiet.

Then her father read out the part about the children.

Every man in the Isles with sixty head of cattle, he said, is to send his eldest son to a school in the Lowlands, and keep him there until he can speak, read and write English.

Mairead looked along the bench to where the chief's eldest boy was sitting. He was fourteen, and he had been fostered in her own house when he was small, so that he was the nearest thing she had to a brother. He had gone white, and his father did not look at him.

Then her father stopped, and read the next line to himself first, and his face changed. Then he read it out in Gaelic in exactly the same voice as all the rest.

The chiefs are not to keep wandering bards, or any such vagabonds, in their houses. Any that are found are to be taken, and put in the stocks, and sent out of the Isles.

Nobody in the hall looked at her uncle. Everybody in the hall did not look at him, all at once, which was worse. Eòghan sat by the far wall with his harp across his knees and his face quite still. After a while he put out one finger and touched a string so lightly that only the dogs and Mairead could have heard it.

• • •

Her father went into the dark room the next day to make the poem for the chief's homecoming.

Mairead had seen him do it all her life, and it still frightened her a little. The room was a small one at the end of the house with no window. He stopped up the door with a plaid and lay down on his back on the floor in the black. He put a flat stone from the shore on his belly. He put another plaid over his head, and stayed there from morning until dark, making the verses in his head, every line, every rhyme, every count of syllables, without a word written. Only when the poem was finished in the dark did he come out and say it aloud by the fire, and only after that did it go into the book.

That evening Mairead brought him his supper at the door of the dark room, and he came out blinking, with the stone in his hand.

"Why the stone?" she asked him. She had always wanted to know.

"To keep you here," her father said. He weighed it in his palm like a loaf. "When you're making a poem in the dark, your head goes away from you, out over the sea. The stone sits on you and reminds you that you're a man on a floor in Moidart, with a body, and a chief, and a supper getting cold." He put it down on the step. "Without the stone you'd float off. You'd make something beautiful and nobody could use it."

"Is Uncle Eòghan going away?"

Her father took his time answering. He sat down on the step beside the stone and ate a mouthful of his supper.

"The chief can't keep him now," he said. "He's signed. If the bishop's men come and find a wandering harper in the hall, the chief pays for it, and so do we." He looked at her. "My place is safe. I'm the chief's own poet, and the paper says nothing about us. Your uncle's road is the one the paper means."

"Where will he go?"

"Ireland," her father said. "To the Glens of Antrim, to our cousins there. They'll keep a harper a while yet."

• • •

Eòghan taught her the song on the shore below the castle, on the last two nights, under the stars it was about.

He did not say that was what he was doing. He only took her down onto the rocks after dark, when the tide was out and the stars were thick over the hills of Moidart, and sang it to her, verse by verse, and pointed. There were the seven stars coming up in the east, small and close together like a fist of sparks. The song said that when they stood at nightfall above the hill the boats should be drawn up for the winter. There was the Plough wheeling round the star that does not move. The song said where its handle pointed at midnight in each month of the dark half of the year. There was the red star of the hunter, and the bright one low in the south, and every one had its verse and its warning for the boatmen.

He sang a verse and stopped, and she sang it back. If she got it wrong, he sang it again. If she got it right, he went on to the next, and never said she had got it right.

On the second night they went through it from the start to the end without a stop. When she had finished the last verse he sat on the rock without speaking.

"You've a better ear than your father," he said at last. "Don't tell him I said so."

• • •

On the last evening the chief's boy came down to the long house to find her, which he had not done since he was small.

He sat on the wall outside the door beside her, in his old clothes still, with his knees drawn up, and for a while he did not speak. The seven stars were not up yet. Over the sea the sky was going green, the way it did at the end of summer. The gulls were going home along the shore below them.

"Your father has my line," he said at last. "All of it. Back to Somerled, and back past him."

Mairead said that he had, and that he said it at every feast, and she knew most of it herself from hearing it.

"Say it to me," the boy said. "The way he says it."

So she said it, sitting on the wall, in her father's carrying voice as near as she could make it, each name joined to the next one by the word for son. It was very long, and she went slowly, and twice she had to go back and get a name she had dropped. He listened with his chin on his knees and his eyes shut, and when she came to his own father's name at the end of it, and then his own, he opened his eyes and looked at the sea.

"They'll not have that in Glasgow," he said.

She told him that he could say it to himself at night, in the school, where nobody would hear. He thought about that for a while, and then asked her to say it again. She did, and this time he said it with her under his breath, a little behind, as she had once said the stars behind her uncle.

When they had finished he got down off the wall and stood in front of her, very tall all of a sudden, and fourteen. He did not thank her, because the chief's son did not thank the poet's daughter for what was his already. He only nodded, the way a man nods to another man across a hall, and went back up the path to the castle in the dark.

• • •

The chief's boy went first, the next morning, in a boat going south to the Lowlands with a man in a black coat who had come to take him to a school in Glasgow.

He was wearing new Lowland clothes that did not fit him, and he stood in the boat very straight, as the chief's son should, and did not wave. Mairead waved from the rock until her arm hurt. She thought about him in a room full of boys speaking a tongue he did not have, learning to say the names of things in it, and wondered which of his own names he would forget first.

Eòghan went on the evening tide, in a boat of the MacDonalds bound south for Islay and then across to Antrim.

He did not take long about it. He put his harp in its bag, and his bag on his back, and kissed her father on both cheeks, and put his hand on Mairead's head the way he always did. He did not say anything about the song. At the boat he turned round like he had forgotten something, and looked at her, and touched one finger to his lips, and then to his ear. Then he got in, and the boatmen pushed off, and the boat went out across the grey water of the loch towards the open sea.

Mairead stood on the rock until it was gone.

• • •

That night she could not sleep. She went out of the house and down onto the shore, where the tide was out, and lay on her back on the flat rock where her uncle had sat, and looked up.

The seven stars were up in the east, small and close. She put a flat stone from the shore on her own belly, the way her father did, so that she would not float off. Then she sang the song, under her breath, from the first verse to the last, all the stars of the winter one after another and what each of them told the boatmen. She did not get a word wrong.

When she had finished she lay still, with the cold stone rising and falling on her as she breathed. She knew that nobody else on the rock, or in the castle, or in the whole of Moidart, had the song now but her. Her uncle had it, going south over the sea. She had it, lying on a rock with a stone on her stomach, and that was all.

"Who will sing it?" she said aloud, to the stars, the way she had asked him.

The seven stars did not say anything, and they did not need to. She knew the answer as soon as she had heard herself ask it.

She took the stone off her belly and tucked it into the fold of her plaid, and went back up to the house to bed. In the morning she would sing it to her father. It would not be all of it, but just enough that he would know it was there, and keep his place in the dark room, and not ask her where she got it.

The master page

In the dark, the Ledger spread this page on a stone floor and put a stone on top of it, and only then brought a light.

In ink: the bond and statutes agreed on Iona in August 1609 under Andrew Knox, bishop of the Isles, after the chiefs' kidnapping off Mull the year before; Donald MacAllan, Captain of Clanranald, of Eilean Tioram, among the signatories; the nine articles, among them six gentlemen for Clanranald's household, the stopping of wine and whisky, the eldest sons of men with sixty cattle sent to Lowland schools to speak, read and write English, and wandering bards put in the stocks and sent from the Isles. The Privy Council's harder act of 1616. The MacMhuirich poets of Clanranald continuing for a century and a half, and the Red Book. Martin Martin's account of 1703 of poets composing in the dark with a stone on the belly. In pencil: how the statutes were read in any one hall, or whether they were; which songs went south and which stayed; and the girl under the bench. *Found destroyed*: the wandering repertoire, the songs carried round the halls by the harpers the statute named, most of which nobody ever wrote down and nobody now can sing.

The page took its place. Egypt had written, the grove had remembered, and Iona had been the harbour. Then the taking apart: Croton scattered, Mona felled, Philae's last word, Whitby's ruling. Moidart's part was the hiding. The paper made two roads out of one house, English for the heir and the stocks for the harper, and between them the song went under a bench and into a child, where no article could reach it. Inked behind it is the same island of Iona, where a boy once kept the bread hot for the boats, and where the Irish bishop

sailed home after Whitby. And on Iona the statutes were agreed. In pencil, ahead, is a cord with knots in it, carried across a dry country by people who had also been told what not to sing.

The Core could show him where the Pleiades and the Plough stood over Moidart on an August night in 1609. Of what the harper sang it can say nothing, or whether a girl learned it on a rock. The rock wasn't saying either.

Who will sing it? She will. The answer was in the asking, and the asking was hers.

The girl had needed to know when the seven stars stood at nightfall above the hill and the boats should come home. Truer still was the dream's answer: a song about the sky can be carried by one child when everything else has been counted and limited and sent away. She had put a stone on herself so she wouldn't float off, learned it verse by verse without waiting to be told she had it right, and kept her place in the dark until it was safe to sing.

Seventy-one years on, across the ocean, in a dry country under a range of mountains, a boy was untying a knot in a cord every morning as the sun came over the ridge.

He woke on his back with the stone on his chest, rising and falling, and for a moment he did not move it. Then he lifted it off and held it, and got up, and went to see what the day was like over the lough.

CHAPTER 25

The Knotted Cord

Ohkay Owingeh, which the Spanish called San Juan, on the Rio Grande below the mountains, in the first weeks of August: the found year, 1680 CE



Illustration 26. *A knotted cord on a sunlit roof edge above the river.*

Belfast, by day

IN SEPTEMBER HE PITCHED IN Dublin. For three weeks beforehand there was a sheet of paper on the wall beside the kettle with the days left written along it in a row. Every morning he put a line through one before he did anything else, before the post or the emails. Nobody else in the office paid it any heed, and it was not as if he could forget the date. But the line every morning made it real in his hand in a way the calendar on the screen never did. When the row got short he found he was getting up earlier to do it.

The sheet by the kettle was in his head when New Mexico came back on the screen.

"In 1680 Popé led the Pueblo Indians in a revolt timed by a knotted cord," it said. "They drove out the Spanish and destroyed every trace of them and their church."

The machines dug out the depositions, and he read them himself, line by line, a page of them under the stone from the sill.

The cord is in the record, and it comes from the Spanish side, out of the mouths of prisoners. The governor, Antonio de Otermín, had statements taken from Pueblo men during and after the rising, and Charles Hackett translated them and published them in 1942. Pedro Naranjo of San Felipe, questioned in December 1681, said a cord of maguey fibre was made with knots in it to show the number of days to wait, and passed through all the pueblos. Each morning a knot was untied, one day at a time. On the ninth of August two young runners from Tesuque, Nicolás Catua and Pedro Omtua, were taken by the Spanish with cords on them. The rising had been set for the eleventh, and once the runners were taken it began on the tenth.

He put a line through *Popé led*, and then thought about it. Po'pay of Ohkay Owingeh was one of the leaders, and the Spanish named him first. But the plan was carried by men in two dozen towns speaking different languages, and no one man could have ordered them. Andrew Knaut's history of the revolt makes that case, and so does Matthew Liebmann, an archaeologist who has worked with Jemez Pueblo. Then he struck out *every trace*, and pressed harder doing it. Liebmann dug the villages the people built in the mountains in the years after, and found no church in them, but the sheep kept, and the metal tools. They threw out the Spanish and their friars, but they did not throw out everything that came with them.

He also wrote down, plainly, what he would leave out. The depositions say things about what was done and said in the kivas, and some of it is the sort of thing the Pueblo peoples keep to themselves now as then. None of it went onto the page, and none of it will. Neither did any words for Po'pay, or for the two runners from Tesuque, or for anybody else who was real. They are named in the record, and they do not speak here.

The Hopi were in it too, far to the west. The friars at the Hopi mesas died in the same days. Nobody wrote down how the word reached them, or whether a cord went that far.

He went to the bottom of it first, and at the bottom was the drought and hunger of the 1670s, and the forty-seven Pueblo religious men arrested in 1675, Po'pay among them. Above it, the cord going out from town to town, and over that the tenth of August, the feast of San Lorenzo, and the twelve years the Spanish stayed away afterwards. All three agreed on *1680 CE*, which went at the head of the page with its question mark, though the Spanish had counted those days as carefully as anybody, from the wrong side.

Beneath, the doubt: *The cord story is in Spanish depositions taken under fear, some under torture, and every number in it could be bent. The names are spelt a dozen ways. Lorenzo is mine, and his grandmother, and the friar, and the cousin who runs west*.

On the map he took the runner off the road to the Hopi, which nobody recorded. He gave the cord to a boy of eleven in a house at Ohkay Owingeh who had to untie the knots. Aye, he thought, the sheet by the kettle, one line a morning. None of that went on the page. He left the stone sitting on it a long while instead.

That night

The depositions were open on his chest at a page where a man was asked what the knots meant, and the room went to dust and river mud and cedar smoke, and then to the smell of rain that had not yet fallen. This time the Ledger's page was a cord of maguey fibre. One knot at a time, very slowly, the Ledger untied it, and smoothed each length flat before it went on.

• • •

Lorenzo's grandmother woke him in the dark and put the cord into his hands before he was properly awake.

He was eleven, and he had been born on the feast of San Lorenzo, so the friar had given him that name in the church with the water. He had another name at home that his grandmother used. The cord was the thickness of his little finger, made of twisted maguey. There were knots tied along it, each one tight and hard as a bean. A runner had brought it from the north two days before, and the old men had turned it over in their hands a whole afternoon, and then given it to his grandmother. She had given it to him.

"When the sun comes over the mountain," she said, "you untie one. Only one, and only when you can see the edge of it. Then you give it back to me, and you don't tell anybody."

They climbed up onto the roof together and sat on the edge with their legs hanging down, and waited. The village lay below them, the houses round their plazas, the river grey in the half dark. The friar's church stood at the end of the street with its bell hanging silent. Across the valley the mountains were black against a sky going green.

When the first line of the sun came over the ridge, Lorenzo worked at the end knot with his thumbnail until it gave. It took a long time. His grandmother did not help him.

"What happens when there aren't any left?" he asked her, when he had it undone.

His grandmother took the cord back and wound it round her hand and put it in the front of her dress.

"Then we'll see," she said.

• • •

The friar was Fray Tomás, and he had been at the pueblo since before Lorenzo was born.

He was not a cruel man, Lorenzo thought, not like the ones in the stories the old men told about the other towns. He was thin and tired and spoke the language badly. He made the boys learn the prayers and sweep the church and ring the bell, and he gave them maize cakes on feast days. But he had stood by, the old men said, when the soldiers came years ago. They had taken the masks and the sacred things out of the houses and burned them in the plaza. And five years

ago, when the governor had the forty-seven taken to Santa Fe and whipped, and three of them hanged, he had said it was God's will. One of the men taken had been from this village. He had come home and gone north to Taos and not come back.

Lorenzo swept the church that morning as he always did, and rang the bell for the prayers, and said the words with the other boys. The friar did not look at any of them differently. Lorenzo kept his hands very still at his sides the whole time, as if the friar might see the knot on his thumbnail.

It had been a bad year, and a worse one before it. The rain had not come properly for years, and the corn was short again in the fields by the river. People were eating things they did not use to eat. The Apaches had come down twice and taken sheep. The Spanish still wanted their share of the corn and the cloth. The friar still wanted the church swept, and nobody said anything about any of it out loud. But the cord had come.

• • •

Each morning after that Lorenzo untied another knot on the roof, as the sun came over the mountain, and each morning there were fewer.

He began to wake before his grandmother came for him. He lay in the dark listening for her step and feeling for the cord in his mind, how many knots were left, how many days. He found he could not think of anything else. When he went down to the river with the other boys, he kept looking east at the mountains as if the sun might come over them early just because he wanted it to.

On the third morning his cousin Francisco came to the house before dawn, dressed for the road. He had a little bag of parched corn tied at his waist and his hair bound up tight.

Francisco was sixteen and the fastest runner in the village. The old men had given him a cord of his own, with the same number of knots as Lorenzo's. They had told him to carry it west, across the mountains and the desert, to the towns that were further away and had to be told. Lorenzo asked him where, and Francisco laughed and said he didn't know the names yet. He would know them when he got there.

"How will they know it's the same day as ours?" Lorenzo asked him. "If you're far away."

Francisco held up his cord beside Lorenzo's, knot against knot.

"Same knots," he said. "Same sun. They untie one every morning when it comes up, same as you. It doesn't matter how far apart we are. The sun comes up for all of us."

He touched Lorenzo's head and went off down the path towards the river at a jog. Lorenzo watched him until he was a small dark shape on the far bank, and then nothing.

• • •

On the evening of the ninth day, a man came running in from the south with his chest heaving. He went straight to the old men, and the whole village went quiet.

Lorenzo heard it from the women, in pieces, the way he heard everything. Two runners from Tesuque, near Santa Fe, had been taken by the Spanish that morning on the road, with their cords on them. They had been carried off to the governor. Nobody knew what they had said, or what had been done to them to make them say it. Lorenzo's grandmother made the women stop talking about that part when she saw him listening. But the governor knew now. He knew about the cords, and he knew how many knots were left.

That night the old men did not sleep, and neither did Lorenzo. He lay on his mat and heard voices going back and forth outside, low and quick, and once somebody running.

In the middle of the night his grandmother came and sat beside him in the dark. She took his hand, and put the cord into it.

"Untie them," she said. "All of them. Now."

"It isn't morning."

"It's morning enough," she said. "The Spanish can count knots as well as we can, now they've got two of our cords. If we wait for the sun, they'll be waiting for us."

So he sat up on his mat in the dark and untied the last knots, one after the other, by feel, while his grandmother held the end of the cord. When he had finished there was nothing in his hands but a long soft string of twisted maguey with kinks in it where the knots had been.

• • •

He did not see much of the next day, because his grandmother would not let him out of the house.

He heard it, though. He heard the men go out before the sun was up, and the shouting start, first far off and then nearer. He heard the church bell ring once, wrongly, like somebody had pulled the rope too hard, and then not again. Later there was smoke over the end of the street, and the smell of burning wood and something else. His grandmother sat with her back against the door and the little ones in her lap. She sang to them under her breath, the same song over and over, and would not answer any questions at all.

It was the feast of San Lorenzo. He remembered that late in the afternoon, sitting in the dark of the back room. On any other year the friar would have said his name in the church, and given him a maize cake.

Fray Tomás went down the road that day, and was not seen at the pueblo again. Lorenzo did not ask what had happened to him, and nobody told him, and afterwards he found he did not want to know.

• • •

The Spanish went away south before the end of the month. All of them went, the soldiers and the families and the carts and the friars who were left, down the river road from Santa Fe, with the governor riding in front.

Lorenzo did not see them go. His uncle did, from a hill above the road. He came home two days later and told it at the fire, and Lorenzo sat at the back in the dark and listened.

"A long line of dust along the river," his uncle said. "It took all morning to pass us. Some of the young ones threw stones, though they were far too far off to hit anything, and I let them." He was quiet for a while. "They had the friars who were left in the carts, and the women walking."

He told it about the carts, and the horses, and the women walking beside the carts with children on their hips, and Lorenzo found that he was sorry for the children, and did not know whether he was allowed to be.

A few days after that, word came down from the north that everybody who had been baptised was to go down to the river and wash it off.

His grandmother took him herself. She knelt on the sand at the edge of the water. She washed his head with the suds of yucca root, the way she had washed it when he was small, rubbing hard with her fingers over the place where the friar had put the water. Then she rinsed it, and rinsed it again, and dried it with the end of her shawl.

"Which name am I now?" he asked her.

She looked at him, kneeling on the sand, and did not hurry.

"The one I gave you," she said, and said it.

But Lorenzo found, in the weeks after, that he answered to both. The old one came back easily, the way a thing does that has been waiting. The other one did not go anywhere. It had been his for eleven years.

• • •

Francisco came home before the harvest. He was thinner, with his sandals worn through and a cut on his leg.

"I went west across the mountains and the desert," he told them at the fire, "through towns I didn't know the names of. I gave the cord to men who didn't speak a word of ours, and they understood it straight away, the way you'd understand a man pointing at the sun." He held up his hands. "Every place untied them on the same mornings, as the sun came up. Every place did it on the same days, give or take one, whatever they called it in their own tongue."

Somebody asked him about the towns on the high mesas, further west still.

"I heard it happened there too," Francisco said. "I didn't get that far. I don't know who told them, or how they knew the day."

"Same sun," Francisco said, and grinned at him. "Told you."

That evening Lorenzo went to his grandmother and asked her for the cord. She looked at him for a moment. Then she took it out of the front of her dress, where she had kept it all along, and gave it to him.

He took it up onto the roof. The sun was going down behind him now, over the hills to the west, and the mountains across the valley had gone red. The cord was soft in his hands, with the kinks still in it where the knots had been, one after another along its length, like footsteps.

He tied one knot in the end of it, a small one, tight and hard. He did not know what it was for. Then he wound the cord round his hand and sat with it while the light went. He watched the place on the ridge where the sun would come up in the morning, and found that he was counting.

The master page

Along the shelf ran this page, a single length of cord, with kinks in it at even spaces where knots had been, and one small knot left at the very end.

In ink: the drought and hunger of the 1670s; the arrest of forty-seven Pueblo religious men in 1675, three hanged, Po'pay among those released; the cord of maguey fibre with knots for the days, passed from town to town, from the depositions of 1680 and 1681 in Hackett's translation; the two runners from Tesuque taken on the ninth of August; the rising on the tenth, the feast of San Lorenzo, a day early; the friars killed, at Hopi too; the Spanish gone south down the river by the end of the month; the washing in the river; and twelve years before they came back. In pencil: how the word reached the Hopi mesas; the boy, his grandmother, the friar, the cousin. *Found des-*

troyed: what the knots meant to the men who tied them, beyond a number of days, which the Spanish asked under torture and the Pueblos have kept to themselves.

The page found its neighbours. Egypt had written the year down; the mesa had watched for it. Then the taking apart: Croton scattered, Mona felled, Philae's last word, Whitby's ruling, Moidart's hiding. The pueblos' part was the untying. For once the count was turned round and used by the ones it was being taken from, a string of days that the Spanish could read but not stop, untied in two dozen towns on the same mornings because the same sun came up over all of them. It is the sheet by the kettle, one line a morning, until the row runs out. Further back, also in ink, is the watcher on the ledge at Orayvi, calling the day before it comes. Inked, just behind, is the bard's daughter in Moidart, who kept a song by keeping quiet. Pencilled ahead is Bengal, where the count goes the other way again, into somebody else's lists.

From the Core he could have the sunrise over the mountains at Ohkay Owingeh on every morning that August, and the moon's age on the tenth. What the knots meant is beyond it, or what the boys said to each other on the hillside. Runners don't leave data.

What happens when there aren't any left? Then we'll see, the grandmother said. What they saw was twelve years.

Lorenzo had needed everybody to untie the knot on the same morning, however far apart they were, because the same sun came up for all of them. The dream's answer went further: the sky is the one clock nobody can confiscate. Count carefully, count together, and know the morning to untie the lot. And if they make you answer to two names, keep the one you were given at home.

A hundred and five years on and half the world away, in a town on the river in Nadia, in Bengal, an old schoolmaster was teaching a row of boys the names of the sky, beginning with heaven and keeping them in their proper order.

On the calendar by the kettle there were nine days left in the month, and he caught himself crossing off the first of them before the kettle had even boiled.

CHAPTER 26

The Pandit at Nadia

Krishnanagar in Nadia, on the Jalangi river in Bengal, in the autumn the judge came up from Calcutta to learn the old language and the Brahmins had gone away for the goddess's festival: the found year, 1785 CE



Illustration 27. *Ramgopal and his pupils on the veranda by the Jalangi.*

Belfast, by day

ONE OF THE LAST JOBS before the company was freight, at a firm in Belfast that moved other people's goods across borders, and most of the work was codes. Every item that crossed had to have a commodity code, a long number that told the customs computer what it was and how much to tax it. A pallet of mixed stock from one small business might come in as a single thing in the owner's head, the spring range, and go out of the office as fourteen lines of numbers, each in its own box. The codes were in order, but it was the customs book's order, not the owner's. Machine parts sat next to machine parts from anywhere

in the world, and nowhere did the codes say that these fourteen things had been packed together because they belonged together. It worked, and the goods went through. He used to think, filling in the boxes, that something was being lost all the same.

He was still turning that over when Bengal came back on the screen.

"After the Battle of Plassey in 1757 the British destroyed the traditional Sanskrit schools of Bengal," it said. "They replaced them with English education."

He set the machines to gather it and then went through what they brought back by hand at the kitchen table, with the stone sitting on a photograph of a watercolour portrait.

The battle is ink, and so is what came after it. On the twenty-third of June 1757, at Plassey, in the district of Nadia, the East India Company's army under Clive beat the Nawab of Bengal, helped by the Nawab's own commander, who changed sides. In 1765 the Company took the right to collect the revenue of Bengal. From then on it ruled in all but name.

The rest is the wrong way round, at least for the first thirty years. The Company did not destroy the Sanskrit schools after Plassey. It did the opposite, and went looking for them. Its judges needed the old law books to try cases, and they did not trust what they were told the books said. In 1776 a code of Hindu law was put together by eleven pandits for the governor, Warren Hastings, and turned into English through Persian. In January 1784 a judge called William Jones founded the Asiatick Society in Calcutta. In the autumn of 1785 he went up the river to Krishnanagar in Nadia, the town of the old schools, to learn Sanskrit himself. The Brahmins had gone away for the festival of the goddess, he wrote, and he found instead a pleasant old man of the medical caste who would teach him grammar. There is a watercolour of the teacher, Ramlochan, dated the fifteenth of October 1785, which says he was sixty-five and taught Sanskrit grammar at the Hindu seminary at Nadia. The English schools and the turning away from Sanskrit came fifty years later, with Macaulay in 1835.

Destroyed he crossed out, and wrote *catalogued* beside it. That was the thing that happened in the 1780s. The Company's scholars took the learning that had lived in the schools as a whole, said aloud and in order. They set it out in lists and tables and subjects, so that it could be looked up.

He slid the sources over one another with Plassey at the bottom, since that is where the Ledger starts the breaking of this stream. The Society founded in Calcutta in 1784 went over it. On top lay the judge and the old teacher at Krishnanagar in the autumn of 1785. The year that came through all three was *1785 CE*, which he wrote down with the mark after it, neat and properly indexed, as the judge would have liked it.

The doubt sat below: *The Ledger's 1757 is a framing, not something historians would sign. The schools of Nadia were famous for logic and law, not for the recitation of the Vedas. Ramgopal is mine, and Hartley, and the boy Shibu. Ramlochan was real, and I have not put words in his mouth.*

He moved the map's year from 1757 to 1785 and the teaching from the Vedas to the grammar and the old word-book, which is what a foreigner could be taught that autumn. Fourteen lines of codes from the one pallet, aye. He let the shop talk go, the stone resting on the photograph of the old man, who was holding a pen and had a manuscript across his knees.

That night

His hand was still on the photograph of the old teacher when the river damp reached him, with marigolds in it, and ink made from lampblack. The Ledger had cut its page into slips. It laid them out on the floor in a long row, in an order he did not recognise, and waited to see whether he would ask what the order was.

• • •

Shibu's grandfather, Ramgopal, taught the word-book by the river every morning. On the morning the Englishman came, Shibu was saying the sky.

He was eleven, the youngest boy in the school, and he sat at the end of the row on the mat under the veranda with the older boys. They said the verses together after his grandfather, a line at a time. The word-book began in heaven. First came the names of heaven, then the names of the gods, and then the sky itself and everything in it, the sun and the moon and the stars and the twenty-seven houses of the moon. Then it came down through time, the days and the months and the seasons, and on down through the mind and speech and sound, and so on, all the way to the earth and the rivers and the towns. Every name for a thing sat next to the other names for the same thing, and the things sat next to the things they belonged with. You learned it in verses, so that it stayed. Shibu could say the whole of the first book without stopping, which was not bad for eleven.

The Englishman came along the bank from the town in the middle of the sky. He had a servant holding a big black umbrella over his head, and the boys stopped.

He was young, red in the face from the heat, with fair hair tied back at his neck and a writing box under his arm. At the edge of the veranda he stopped and bowed with his hands together, clumsily, like somebody who had practised it in front of a glass, and said a sentence of very bad Bengali that meant he wished to learn.

Shibu's grandfather looked at him for a long moment without a word, his hands resting on his bad knee. Then he turned back to the row on the mat and told the boys to go on from the line where they had stopped.

That evening Shibu asked his grandfather the question that had been in his mouth all day.

"If you teach him," he said, "and he writes it down in his box, who is it for?"

"For him, to begin with," his grandfather said, easing his bad knee out straight on the mat. "After that, I don't know. That's the question, isn't it." He touched Shibu's head, lightly. "Let's see what he does with it first."

• • •

The Englishman's name was Hartley. He was a clerk in the judge's household, and he had come up the river from Calcutta in the judge's boat for the autumn holiday, with his umbrella and his servant and a box full of paper.

The judge himself was learning from old Ramlochan on the other side of the town, a doctor by caste. The Brahmin teachers had nearly all gone away for the goddess's festival to the great houses where they were given presents. The ones left would not teach an Englishman. Shibu's grandfather was a Brahmin. He had stayed behind because his knee was bad and the boat journey was long, and because, he told Shibu, he wanted to see what an Englishman would make of the word-book.

He made conditions, and Hartley was to sit on a separate mat, at the edge of the veranda, and not touch the boys or the books. He was to come washed, and eat no meat on the days he came. When he had gone the veranda would be washed down with river water. Hartley agreed to all of it without arguing, which surprised Shibu, and sat on his separate mat every morning with his writing box open on his knees.

Grandfather would not teach him the Veda, because that was not his to give, he said, and not Hartley's to have. The grammar he would teach him, and the word-book from heaven down to the earth, and that alone was a great deal for one autumn.

So Hartley learned the sky, the way the boys did.

He was slow at the sounds at first, and the boys laughed behind their hands. But he was patient, and he came every morning, and he wrote everything down. That was the strange part, because Shibu had never seen anybody write while they learned. The boys did not write; you said a thing until it was in you, and then it was yours. Hartley

wrote every word as he heard it, in his own letters, on small slips of paper cut from a larger sheet. He wrote the meaning beside it in English, and put the slips in his box. At night, Shibu learned later, he sat in his room in the town and sorted them.

• • •

A week after he started, Hartley stayed behind when the boys had run down the steps to bathe in the river. He was too shy to ask Grandfather, so he asked Shibu which star it was that did not move.

He had read about it, he said, in a book in Calcutta. He knew the one in England, the star at the end of the little bear's tail that the sailors steered by. But he could not find the one here, and he had been looking from the roof of the judge's house every night that week.

Shibu took him out to the end of the veranda, where the roof stopped, and waited for the dark. Then he pointed north over the river, low down, lower than an Englishman would think of looking. There it was, a small steady star, not very bright, with the others going round it as the night went on. Shibu told him its name was Dhruva, and that Dhruva had been a boy once, a king's son.

Hartley asked him to tell it, and he sat down on the step with his writing box on his knees. Shibu told it the way his mother told it to his little sisters. The boy's father the king had loved another wife better. When the boy tried to climb onto his father's lap, the other wife pulled him off and told him he had no right to sit there. So the boy went away into the forest, five years old. He stood on one foot and prayed, and would not eat, and would not stop, until the god came to him. The god gave him a place in the sky that nobody could ever pull him off. Everything else in the sky goes round, Shibu said, the sun and the moon and the houses of the moon and all the rest. Only Dhruva stays where he is put, because it was promised.

Hartley had stopped writing halfway through, and was just listening with the pen in his hand.

"In England we say it is the axis," he said after a while. "The place the earth turns on."

Shibu thought about that. He said that it could be both, the turning place and the boy, and Hartley laughed and said he supposed it could. Then he wrote it down, both of them, on the one slip. Shibu watched him put the slip in a bundle all by itself, because there was no other word in the box that began with that letter yet.

• • •

One morning Hartley brought the box open and showed Grandfather what he had done.

The slips were in rows, in little bundles tied with thread, each bundle with a letter of the English alphabet on the top. Hartley picked out a bundle and fanned the slips out on his mat for them to see. Shibu leaned over to look, which he was not supposed to do.

"This is the order of my letters," Hartley said, slowly, in his careful bad Bengali, and pointed at the bundles. "So I can find any word. Quickly. Like that." He snapped his fingers, which the boys liked. "In Calcutta they are making a great list of all the words. I am helping."

Grandfather took one of the slips in two fingers and held it up to the light, and read Hartley's English letters with his lips, and put it down. He took another.

"Here," he said, "is the moon." He laid the slip down on the mat between them and smoothed it flat with one finger. "And here, next to it in your bundle, is what?"

Hartley leaned over, read the next slip, and said the word in English and once more in Bengali. It was the name of a kind of cooking pot.

Grandfather nodded as if that were a very interesting thing. "In the book," he said, "the moon lives next to the other names of the moon. And next to them the stars, and the houses of the moon, one after another, the way the moon walks through them each month." He tapped the slip. "In your bundle the moon lives next to a cooking pot, because the two names begin with the same sound in your letters."

Hartley went red, and then he laughed, which Shibu liked him for.

"Yes," he said. "That is true." He looked at the slips for a while. "But I can find it. In your book, if I want the moon, I must say the whole sky first."

"Yes," said Grandfather. "That is how you know where the moon is."

• • •

The night of the new moon before the festival, Grandfather took Shibu up onto the roof to look at the sky, and Hartley came too, carrying his own book.

It was a small fat book bound in leather, printed in England, and it had the moons in it for the whole year, and the times of the sun rising and setting, all in tables of numbers. Grandfather had the almanac from the town, the long narrow pages from the astrologers' house. They also had the moons for the year, and the days of the feasts, and the lucky hours, all reckoned from the old books by men who had never seen England. The festival of the goddess would begin on the sixth day of the bright half of the moon, by the almanac. Hartley had asked which day that was in his own book, and Grandfather had said, let us look at the sky and see.

They sat on the roof in the warm dark. Over the river there was no moon at all. The stars were thick, and the river smelled of mud and flowers.

"New moon," Hartley said, and opened his little book and turned it to the light of the lamp and read out a date. Grandfather opened the almanac and read out the day of the dark moon in the old reckoning. They were the same night.

Hartley looked up from his book to the almanac and from the almanac to the sky over the river, and for a while nobody on the roof said a word.

"Your astrologers," Hartley said. "And our men at Greenwich. They have never met."

"No," said Grandfather. "But they have both been watching the same moon."

Then he made Shibu say the houses of the moon, all twenty-seven, in the order the moon walks through them in a month. Shibu said them in one breath as he had learned them, from the horse's head at the start to the last one at the end, pointing up at the few he could find in the sky as he went. Hartley's pen scratched after him across slip after slip, and Shibu could see him already sorting them in his head into bundles.

"If you put them in your letters," Shibu said suddenly, "the moon won't be able to find its way."

He had not meant to say it out loud, and he looked round at his grandfather in alarm, waiting to be told off. Grandfather only laughed, low in his chest, in the dark beside the lamp.

Hartley sat with the slips in his hand until the lamp needed trimming.

"Then I'll keep two lists," he said at last. "One in my letters, for finding. And one in your order, for knowing." He held up the last slip between two fingers. "Will you say them again? Slowly. In the order."

So Shibu said them again, slowly, with the lamp hissing beside them. Hartley wrote them on a long strip this time, one under another, the way the moon walks, and did not cut the paper into slips.

• • •

The festival came, and the drums went day and night along the river, and the teachers came home, and the autumn holiday ended. Hartley came to the veranda for the last time on a cool morning with his writing box under his arm and a parcel in his other hand.

He bowed to Grandfather with his hands together, a good deal better than the first time, and said a short sentence of thanks in Sanskrit that he had plainly been practising on the walk up from the town. Grandfather corrected one sound in it. Then he smiled, and Shibu had never once seen him smile for an Englishman.

The parcel was for Shibu. Inside the paper was a pencil, a real English one with a sharp grey point, and a little book of blank white pages sewn together down the spine with thread.

"For your own list," Hartley said.

Shibu turned the pencil over in his fingers and could not think of a thing to say. The boys at the school did not write; they said, and a pencil was a strange weight in a hand that had only ever kept time on a knee to the verses. He looked up at his grandfather.

"Take it," Grandfather said. "You have the sky in you already. A book can't take it out." He looked at Hartley's writing box on the step. "Only put it in yourself first."

Hartley went down the bank to the boat, with his servant and his umbrella and his box of slips in their bundles and his one long strip in the order of the moon. The judge's boat was waiting, a big one with a cabin, and it pushed off into the brown current and turned downstream towards Calcutta.

Shibu stood on the bank with the pencil in his fist.

"Who is it for?" he said to his grandfather, as the boat went round the bend. "Now."

"The bundles are for Calcutta," Grandfather said. "For the judge and his Society, and the people who will want to find a word quickly. That isn't nothing. There'll be books from them, I expect, and people in England reading our poems." He was quiet for a while, watching the water. "The long strip is for him, if he keeps it. And ours," he said, and put his hand on the back of Shibu's neck, "is for you, and the boys after you. Say it every morning. That's the only list that keeps the order."

Shibu said the houses of the moon to himself under his breath, from the horse's head to the end, all twenty-seven in their order. By the time he had finished, the judge's boat had gone out of sight down the river.

Then he sat down on the wet step and opened the little white book on his knee. At the top of the first page he wrote the first name of heaven, very carefully, in his own letters.

The master page

First the Ledger set this page out as a row of slips on the floor. Then, when he had looked at them, it gathered them up and laid them out again in a long strip, one under another, in an order that went from heaven to the earth.

In ink: Plassey in 1757, in the district of Nadia; the revenue of Bengal in 1765; the pandits' code for Hastings in 1776; the Asiatick Society in January 1784; Jones at Krishnanagar in the autumn of 1785, learning from Ramlochan, a teacher of the medical caste; the word-book that begins with heaven; and Macaulay's minute of 1835. In pencil: the Ledger's dating of this stream's breaking from 1757; how the pandits of Nadia felt about teaching the English; and the old man and the boy on the veranda. *Found destroyed*: nothing burned, this time. What was lost was the order: the learning said aloud as one whole, which was copied out into lists and subjects and sent away to be looked up, so that each part could be found and the whole could not.

He laid the page with the rest. Egypt had made its record, the mountain road its recitation, and the grove its memory in the body. Then the taking apart: the scattering at Croton, the felling at Mona, the last word at Philae, the ruling at Whitby, the hiding in Moidart, and on the Rio Grande the untying, when the count was turned round for once. Nadia brought the catalogue. Nobody came with fire. They came with slips and the best will in the world, and filed the whole under letters, so that the moon sat next to a cooking pot. Behind it in ink is the mountain road, where a boy and his sister kept the verses in pairs and the order was the check. In pencil, further on, the pieces of the old whole go out across the world as separate things, each claimed by whoever holds it, and one of them is a green dish that later centuries would call the dish of the Last Supper.

Come the morning, the Core could sort the twenty-seven houses of the moon in any order he likes, and put them back in the moon's order in a second. It could not say what was lost when a boy stopped saying them in a row on a veranda. A database can sort anything. It can't miss anything.

Who is it for? The bundles for Calcutta, the old man said, and that is not nothing. The strip for the clerk, if he keeps it. And the order for the boy, every morning, because it is the only list that keeps itself.

Shibu had needed to know the day of the goddess's festival, and that the moon walks through his twenty-seven houses in a month, always in the same order. The dream said it better: the sky is already an order, and a list that forgets it can find anything and know nothing. Gary had kept both kinds in the solicitors' file room, the index for finding and the bundle in its proper order for knowing, and he had seen what became of a case when somebody filed it by the alphabet alone.

Back nearly seven hundred years, and on the coast of Palestine, a crossbowman of Genoa was set to guard the plunder of a city through the night, and held a green dish in his hands.

He tried the houses of the moon in their order, the way Shibu had said them under his breath on the veranda, and got as far as the fourth before he had to look. Shibu would have laughed at him. He started again from the horse's head.

PART V

Dispersion

The Scattered Cup

CHAPTER 27

The Green Dish of Caesarea

Caesarea, on the coast of Palestine, in the days after the city fell to the Genoese and the king of Jerusalem: the found year, 1101 CE



Illustration 28. *Nicolò holds the lamp over the green dish.*

Belfast, by day

THE FIRST JOB OUT OF school was kitchen porter in a restaurant in Belfast, and the thing he remembered best from it was a green dish. It was a big heavy serving plate, green all the way through, and the head chef treated it like the crown jewels. Nobody else was allowed to wash it, only the porter, by hand, at the end of the night when the rush was over. The waiters said it was antique. The chef said it was Italian. One night it slipped in the sink and he caught it with his knee, and stood there with his heart going. Then he looked at the bottom of it for the first time, and it said it was made in a factory in Stoke. He never told the chef, and he washed it just as carefully after that.

The dish from the kitchen was still in his head when Genoa came back.

"In 1101 the Genoese found the cup of the Last Supper at Caesarea," it said. "It was a single emerald, and they brought it home to Genoa, where it remains."

Whatever the machines found, he checked against the books himself, with the stone on a photograph of a six-sided green bowl in a glass case.

The city is ink, because the Genoese fleet under Guglielmo Embriaco helped King Baldwin take Caesarea in May 1101. Caffaro, a young Genoese who was there and later wrote his city's annals, gives the shares: forty-eight shillings and two pounds of pepper a man. Caffaro says nothing in his annals about a green dish. William of Tyre, writing about seventy years later, says the Genoese also took a green vessel that they believed was emerald, in place of a great sum of money.

He put a line through *the cup of the Last Supper*, because nobody in 1101 called it that. It was Jacopo da Varagine who wrote it up as the dish of the Supper. He was archbishop of Genoa near the end of the thirteenth century, and he wrote it nearly two hundred years after it came home. Then he scored out *emerald* and *cup* as well. It is a dish, not a cup, and Napoleon had it taken to Paris in 1806, and the scholars there found it was green glass. It came back to Genoa on the fourteenth of June 1816 in ten pieces, one of them missing. The modern view is that it was made in the Islamic world in the ninth or tenth century. It was a beautiful thing, but it was neither of the things it was said to be.

He set it out from the bottom: underneath, the Easter of 1101, when the Genoese went up to Jerusalem. Over it lay the fall of Caesarea in May, and over that the dividing of the plunder and the green dish set aside. Where they agreed he wrote *1101 CE* across the top, and put the question mark after it, mostly for the dish's sake.

The doubt followed: *The dish turns up in writing seventy years after the sack, not in the account of the man who was there. Nicolò is mine, and his brother Bartolomeo, and the night of the guard.*

On the map he took the oath off the fleet, because nobody swore anything about a supper that summer, and gave the crossbowman a wee brother who held the lamp. Aye, he thought, made in Stoke. He let the stone sit on the photograph a good while, and told nobody about Stoke.

That night

Light from the landing came through the photograph of the green bowl on his chest, and then there was tar and salt and pepper and the smell of a burnt city cooling. On a ship's deck the Ledger turned its page as slowly as a man turns a thing he is afraid of dropping. Beyond the rail the harbour was all masts and smoke.

• • •

Nicolò held the lamp, and his brother held the crossbow, and between them on the floor of the storeroom was the share of a whole fleet.

He was twelve. He had come out from Genoa as a ship's boy on Embriaco's own galley, because his brother Bartolomeo had signed on as a crossbowman and their mother had said she would rather lose them both together than one at a time. That had been a joke, Nicolò thought. He hoped it had been. They had been at sea and in camps since the spring. He had seen more in those months than in all his years at home.

The storeroom was in a house by the harbour that had belonged to some rich man of the city. Now it held what the captains had set aside for Genoa before the dividing. There were bales of cloth, silver dishes, a heap of coin in sacks, and in the middle of it all, on a folded cloak, a green dish with six sides. Bartolomeo had been told to guard it through the night, with two others. The two others had gone to sleep in the corner, and Bartolomeo had not woken them.

Nicolò lifted the lamp, and the dish came alive.

It was green all the way through, the green of deep water under a ship when the sun is on it. The lamplight went into it and came out the other side changed. There were six flat sides, and a lip round the top, and he could see his brother's hand through it, wavering.

"Is it a real emerald?" he whispered.

"That's what they say," Bartolomeo said. "The captain says so."

"How do they know?"

His brother thought about it. "Because it's green," he said, "and it's big, and it was in the rich man's house, and the captain wanted it more than money." He shifted the crossbow on his knees. "Anyway, it's ours now. Hold the lamp still."

• • •

In the morning the captain came to look at the dish.

Nicolò had fallen asleep against the bales with the lamp gone out. He woke to the door opening and the grey light coming in, and Guglielmo Embriaco standing over the six-sided dish with two of his officers. He was a broad man, weathered by the sea, and the whole fleet called him the Mallet-head, though never to his face. He looked at the dish without touching it.

"They tell me the Saracen who had this kept it in a box of cedar and showed it to nobody," he said at last. "They tell me it's worth more than the shares of every man in the fleet together, and I believe them."

One of the officers said that the men would want their money, not a dish, and the captain nodded slowly.

"The men will get their money and their pepper, every one of them, and Genoa will get this," he said. "Let the men have what they can spend, and let the city have what it can keep." He looked down at Nicolò, curled against the bales with the dead lamp. "And who is this, guarding the treasure of San Lorenzo in his sleep?"

Bartolomeo stood up very straight and said it was his brother, and that he was sorry. The captain laughed and told him to keep the boy awake next time, and went out.

• • •

They had come up to Jerusalem at Easter, before any of this. Nicolò had stood in the great crowd at the church of the tomb waiting for the holy fire.

Everybody had told him about it on the ship. On the day before Easter, they said, the lamps at the tomb were lit by fire from heaven. The people passed the flame from hand to hand until the whole church was full of it. He had waited all day in the heat and the crush, with Bartolomeo's hand on his collar, and the fire had not come. People had begun to weep, and some to shout, and the priests had prayed and prayed. The evening came, and still it did not come. Nicolò had been frightened then, more than at any time on the sea.

It had come in the end, on Easter morning, and the crowd had roared. The flame had gone from hand to hand just as they said. But Nicolò had kept thinking about the night before, when it had not come, and nobody had known why.

He thought about it now, holding the lamp over the dish. Everybody said a thing, and then you waited, and sometimes it was true.

• • •

They had taken the city in May. Nicolò had watched it from the deck of the galley, because ship's boys were not let ashore on the day of an assault.

The Genoese had made long ladders out of the spars of their own ships and carried them up to the wall. The crossbowmen had gone up with the knights. From the harbour he could see only the tops of the ladders over the wall, and the smoke, and the small dark shapes of men on the battlements. He could hear it too, a long roar that came and went on the wind like surf on a far beach. He stood at the rail all

morning with the other boys. He said the names of everybody he knew who had gone up, over and over, under his breath, as if that would keep them.

By noon the roar had changed. The old steersman, who had seen cities taken before, said quietly that it was over and that the boys should go below. Nicolò went below. He sat in the dark among the oars with his knees up. He did not come out until Bartolomeo came back that evening, filthy and silent, and sat down beside him, and put an arm round his shoulders without a word.

Neither of them talked about that day afterwards, not on the voyage and not at home. Nicolò was glad that he had not gone into the city.

• • •

The dividing was done a few days after the night of the guard, on the shore. Nicolò watched from the steering oar of the galley with the other boys.

There was a great crowd, and the captains stood in the middle of it and called out the shares, and every man came up in turn. Each one got forty-eight shillings in coin, and two pounds of pepper tied up in a cloth. Bartolomeo came back to the galley with his pepper held up to his nose and his eyes watering, laughing.

"Two pounds," he said. "That's worth a fortune at home, that is."

Nicolò did not know, but he could tell from his brother's face that it was a great deal, and he laughed too. He asked about the dish, and Bartolomeo said that the city was having it, all of Genoa together, in place of a great heap of money. It was going to the cathedral of San Lorenzo, to be kept with the other treasures. Every Genoese who ever lived would own a share of it.

"Even us?" Nicolò asked.

"Even us," his brother said. "A very small share."

• • •

On the night before they sailed, Nicolò went to the storeroom again with the lamp. He went because the captain's clerk was there writing down what was to go on which ship. The clerk was kind and let boys watch.

The clerk had the prize on the table in front of him, and a sheet of skin on which he was writing. Nicolò asked him what he was putting down.

"A green dish," the clerk said, "with six sides, of great value." He looked up. "Tell me what you would put, then."

Nicolò thought about it, and about the holy fire, and about his brother's hand wavering behind the green.

"I'd put that it's green," he said, "and that you can see through it, and that everybody says it's an emerald." He hesitated. "I wouldn't put that it is one. Not unless somebody knew."

The clerk laughed, but not unkindly, and wrote nothing more. "That's a careful boy," he said. "They'll make a clerk of you yet. Keep to what you saw, and let the bishops say the rest."

• • •

They sailed for home at the end of the summer, and Nicolò stood the night watch with the steersman on the long run west.

The steersman had taught him the stars on the way out. There was the one that does not move, low over the bow when they went north, which the sailors steered by when the land was gone. There were the ones that rose and set, and the bright wandering one that came up before the dawn. On the way home he could name most of them. The steersman let him say where they were before he looked, and told him when he was wrong.

"How do you know the stars are where you think they are?" Nicolò asked him one night.

"Because they were there last night," the steersman said, "and the night before, and every night my father sailed. You don't have to believe them. You check them." He nodded down the deck to the stern, where the dish was packed in straw in a chest under a guard. "Not like that thing. That's a different kind of knowing, that."

Nicolò stared at the chest. Then he looked up at the star that does not move, and said its place, and the steersman said he had it right.

• • •

Off the coast of Crete they ran into a gale. For two days the galley was thrown about so badly that nobody could cook and nobody could sleep.

Nicolò spent most of it lashed to the foot of the mast with the other boys, soaked through and holding on. All the while the men fought the oars and the steersman stood braced at the stern like a post driven into the deck. Water came over the side green and white and ran out through the scuppers. Once, in the worst of it, he looked aft and saw that two of the crossbowmen had gone down to sit on the chest with the dish in it, to keep it from shifting. One of them was his brother, wet to the skin and grinning at him.

When the wind dropped at last the sky cleared, and that night the stars came out harder and brighter than he had ever seen them, washed clean. The steersman made him find the one that does not move before he would let him sleep. He found it straight away, and the old man grunted, which was as close as he ever came to praise.

• • •

They came into Genoa in the autumn with the bells going, and the whole city came down to the harbour to see.

The dish went up to San Lorenzo in a procession, carried on a cushion. Nicolò walked behind it with his brother and the other men from the galley, all of them in their best, which was not very good. The people crowded along the street to look as it went past. He heard them saying it to each other, emerald, emerald, a dish made of one stone.

He did not say anything. He had held the lamp over it. He knew it was beautiful, and green, and that you could see through it, and that nobody knew any more than that.

That night he was at home, in his own bed for the first time since the spring. He lay awake a long while and said the stars to himself in the dark, the ones he could check, until he slept.

The master page

Over a folded cloak the Ledger spread this page, and set a single piece of green in the middle of it, six-sided, with a crack running through it where somebody had mended it.

In ink: the fall of Caesarea in May 1101 to King Baldwin and the Genoese under Guglielmo Embriaco; the shares in Caffaro's annals, forty-eight shillings and two pounds of pepper a man; the green vessel in William of Tyre, taken in place of money and believed to be emerald; the delayed holy fire of that Easter; Jacopo da Varagine's naming of it as the Supper dish near 1300; Paris in 1806, glass; Genoa in 1816, ten pieces, one missing. In pencil: the night of the guard, the two brothers, the clerk who wrote down only what he saw. *Found destroyed*: whatever the dish was before it was a prize, who made it and what it held, which went with the rich man's house.

The page went down beside the others. Egypt had kept the record, Nadia had drawn up the catalogue, and the pieces had gone out across the world. Caesarea's part was the prize: the first time one piece of the old whole was carried off and given a name it had never had, by people who wanted it to be more than it was. It is the green plate in the sink, and the stamp on the bottom. Inked in behind is the schoolmaster at Nadia, watching his teaching cut into lists. Ahead, in pencil, a king in Aragon who gives land for a cup of stone, and nobody says which cup.

The Core knows how green glass takes the light, and when the moon was full over Caesarea that May. It does not know what the men on the shore believed, or why they wanted it to be an emerald. People want things. The Core doesn't.

Is it a real emerald? That's what they say, the brother said. Keep to what you saw, the clerk said. The boy had the better answer, and the stars were how he checked it.

Nicolò needed it to find home across the sea at night, by the star that does not move. The dream's answer was that the sky is the one treasure anybody can check every night, which is why nobody has ever managed to carry it off. In the storeroom he had held the lamp still, and said what he saw, and kept quiet about what everybody said until somebody actually knew.

Thirty-four years on, at the other end of the sea, in a town under the mountains of Aragon, a notary's boy was copying out a gift of land, and stopped at the words a chalice of precious stone.

The green plate from Stoke had a stamp on the bottom and had never once been called an emerald, and he thought the better of it for that.

CHAPTER 28

A Cup for a Valley

Jaca, under the mountains of Aragon, in the second year of the monk who was made a king: the found year, 1135 CE



Illustration 29. *The monks ride into Jaca under the mountains.*

Belfast, by day

BEFORE THE COMPANY HAD ITS name it traded as Paddy's Proper Prizes. The thing the prizes taught him was that people will swap anything for the right thing. A man rang once about a prize he had won, a good one, and asked could he have the money instead. He was told no, it was the prize or nothing. He rang back the next week and offered to give it back if he could have a different one, smaller, that his daughter wanted. Nobody had ever asked for less before. He got it in the end, and the swap went down in the book in one line, prize returned, prize issued, and nothing in the line said why.

That line was going round his head when the answer on Aragon came in.

"In 1134 an inventory of San Juan de la Peña recorded the chalice of the Last Supper," it said, "in an ivory chest. In 1135," it added, "King Ramiro gave the monks two stone vessels for land."

The machines brought the sources and he went through them by hand, the stone on an open book about the monastery under the rock.

The first thing to go was *1134*, because that inventory is known only from a book printed in 1698. The writer said he had seen it, and nobody has found it since. It may have been real, and it may not, but it cannot carry the weight. Next he crossed out *two stone vessels*, and after it the direction of the gift. The record for 1135 is real: a grant made at Jaca in November 1135 by King Ramiro. It survives as a copy in a sixteenth-century book of the monastery's privileges. The king gives the monks of San Juan de la Peña and of Santa María de Iguácel land at Garcipollera, Villanovilla and Bescós. He gives it in return for a chalice of precious stone, one, that they had given him.

That is all the line says. It does not say which chalice, or where it came from, or that anybody thought it had been at a supper.

He worked the year up from the bottom, beginning with the death of Alfonso the Battler after the defeat at Fraga, in September 1134. Over it went his brother Ramiro, a monk, taken out of the cloister and made king. On top of him lay the grant at Jaca, in November of the year after. *1135 CE* came through all three, so it went on the top line. He hung a question mark on it, and then, after a minute, a second one on the cup.

Underneath, the doubt: *The copy is four hundred years younger than the gift, and the cup in it has no name. Pero is mine, and the scribe, and the evening the monks came*.

Then, on the map, he crossed out the two vessels and wrote one, and turned the arrow round, cup to the king and land to the monks. Aye, prize returned, prize issued. That stayed in his head, and he laid the stone on the word *one*.

That night

The book lay open on the floor and the light was off. Cold came the way stone goes cold, with snow somewhere above it, and the smell of tallow and wet wool and ink made from oak galls. At a slanted desk the Ledger held a page flat with two fingers while somebody else did the writing. A bell was ringing for the night office, and nobody hurried.

• • •

Pero was eleven. He could write better than anybody in Jaca under the age of thirty, which was why he was sweeping the floor of the scribe's room and not out on the hill with the sheep.

His master was a scribe called García. He wrote for the king's household when the king was in Jaca, and for anybody else who could pay when he was not. García had taken Pero because his letters were neat and his hand was steady, and because his mother had begged. Pero ground the ink and cut the pens and swept up, and sometimes, when the work was dull, García let him copy.

That winter the work was not dull, because everybody in Jaca was talking about the king.

The old king, Alfonso, the one they called the Battler, had died the autumn before last, after a great defeat at Fraga, and he had left no son. He had left his kingdom to the knights of the Temple and the Hospital and the Holy Sepulchre, which nobody in Aragon would have. So the lords had gone looking for his brother. His brother was Ramiro, and Ramiro had been a monk since he was a boy. They had taken him out of his monastery and put a crown on him and found him a wife. Now he was a king who had never held a sword, with enemies on every side and no money to pay anyone.

"He prays at all the hours," García said. "Even now. He gets up in the night for it." He shook his head as he said it. "A king who gets up in the night to pray. God help us."

• • •

It was to this king that the monks came down from San Juan de la Peña at the beginning of November, when the first snow was already on the high ground.

Pero watched them come into the town from the doorway of García's room. There were four of them, on mules, with a servant walking in front. They had come down the long road from the monastery under the rock. One of them carried a box on his knees, wrapped in cloth, and held it like a baby.

"What's in the box?" Pero asked.

García was cutting a pen and did not look up. "A cup," he said. "For the king."

"Just a cup?"

"A cup of stone," García said, "and a precious one, I'm told. The king wants it, and the monks are giving it to him, and he's giving them land for it, and tomorrow you and I are going to write that down."

Pero looked at the box going up the street and thought about it. He had seen the land the king was giving, or some of it. Garcipollera was a whole valley, with farms in it and a mill and a church. The shepherds took their flocks through it in spring.

"Is a cup worth a valley?" he asked.

García put down the pen and looked at him properly for the first time that morning.

"That," he said, "is not a question for scribes."

• • •

In the morning García and Pero walked up through the town to the king's house, with the parchment rolled under García's arm and the inkhorn on Pero's belt.

Jaca was full that winter, fuller than Pero had ever seen it. There were soldiers in the square who had come back from the frontier after Fraga, thin and quiet. There were lords' men in good cloaks who looked at everybody as if they owed them money. There were pilgrims too, even so late in the year, coming down from the high pass on their way to Santiago. They had their staffs and their shells and their feet wrapped in rags against the snow. One of them stopped García at the corner by the cathedral to ask the way to the hospice. García told him, and gave him a coin besides, which Pero had never seen him do before.

"He's walked from France over the mountains in the snow," García said, when Pero looked at him. "That's worth a coin. He's going all that way to stand by a tomb he's never seen, because of what he's been told is in it." He shifted the parchment under his arm. "Half the world walks past this town on the strength of a story, and the other half sells them bread."

Pero thought about that all the way up the hill, and he thought about the cup in its cloth.

• • •

They did the writing the next day, in a cold room in the king's house, with a brazier in the corner that warmed nobody.

The king was there for a while, and Pero had expected him to look like a king. He looked like a tired monk in a better coat, thin, with grey in his beard and ink on his fingers. He sat at the end of the table and listened to García read the words over, and nodded. He said once that the monks should have it all written as they wished, every field. Then a lord came in and spoke to him low, and he went out, and did not come back.

The cup was on the table the whole time, out of its box, on the cloth.

Pero had expected gold, and it was not gold. It was a small cup of dark stone, the size of a man's two hands put together, smooth and plain, with no picture on it and no writing. When the light from the window came through the edge of it, the edge went red, like a coal. It was the plainest thing in the room, and everybody kept looking at it.

The abbot of the monks sat beside it with his hand near it, not touching.

"Write it as a chalice," he said to García, "of precious stone. That is what it is."

"And where did it come from, Father?" García asked. "Do you want that in?"

The abbot was quiet a moment. "It has been with us a long time," he said. "That is enough for the king."

García wrote, and Pero held the parchment flat for him. He watched the words go down in brown ink: the king's name, the monks' names, the valley, the villages, the fields and the mill, and then, small, near the end, the reason. For a chalice of precious stone which you have given me.

• • •

That evening García let him make the copy for the monks, because his hand was neater and García's eyes were tired.

Pero sat at the desk with two candles and the first sheet beside him, and copied it slowly, word by word, the way he had been taught. You did not change anything, and you did not leave anything out. If there was a mistake in the first one, you copied the mistake, and García would see to it.

When he came to the cup, he stopped with the pen in the air.

The line said what it said, a chalice of precious stone, and no more. It did not say red at the edge. It did not say small, or plain, or that the abbot had held his hand near it all morning without touching it. It did not say where it had come from, because the abbot had not wanted that in.

"Master," he said, "it doesn't say anything about the cup."

García came and looked over his shoulder. "It says what it is."

"It doesn't say which one. If there were another cup of stone, you couldn't tell them apart from this."

García was quiet for a while, and then he laughed, a short tired laugh. "No," he said. "You couldn't. And a hundred years from now some clever boy will read it and say it was whichever cup he likes." He put his hand on Pero's shoulder before he went on. "Copy what it says. Don't add. That's the whole of our trade. Whatever people make of it after is on them."

Pero copied what it said.

• • •

The monks went home the next morning, early, with the frost still white on the roofs, and Pero ran down to the gate to see them go.

He did not know why he went. He told himself it was to give the abbot the copy, which García had rolled and tied with a cord for him. That was true, but it was not the whole of it. He wanted to see the abbot's face, now that the cup was gone.

The abbot took the roll from him at the gate and thanked him, and turned it over in his hands without opening it. His face was the same as it had been in the cold room, calm and tired. If he was sorry about the cup he did not show it.

"You made this copy, García tells me," he said. "He says you have the best hand in Jaca under the age of thirty, and that you ask too many questions for a scribe."

Pero went red and said that he only asked the one.

"Is a cup worth a valley," the abbot said, and at last he smiled. "I heard. It's a good question, and you'll find no answer to it in the grant, and you'll find no answer to it from me." He tucked the roll into his saddlebag. "We kept that cup well for a long time, and now it's the king's to keep. That's all that line in your copy means. Mind you don't let anybody make it mean more."

Then the mules went out through the gate and up the long road towards the mountains. Pero watched them until they were small and dark against the snow, and went back to sweep the floor.

• • •

Afterwards, when the candles were out, he walked home across the square with his cloak round his ears.

It was a clear night, after the snow, and the stars were very bright over the mountains, the way they only are in winter. There was no moon. He stopped in the middle of the square and looked up, the way his grandfather had shown him, and found the shape like a plough, and followed the two stars at the end of it up to the one that does not move. His grandfather had been a shepherd and had known them all. He had said that the stars were the only things in the world that nobody could give away.

"Not the king, nor the bishop, nor the Moors," he had said. "They're nobody's. Look at them as much as you like. They'll still be there when you're done."

Pero stood in the cold and looked at them. He thought about the valley, and the cup, and the line that said what it was and not which one it was. Somewhere up there, over the mountains, under the rock, the monks were riding home with a valley. The king was getting up in the night to pray, with a cup on the table beside him that nobody would ever be able to tell from any other.

He went home and told his mother none of it, and she did not ask, and he slept.

The master page

This page lay on a slanted desk, cold, with two candles, and a single line of brown ink near the bottom of a long grant.

In ink: the death of Alfonso the Battler after Fraga in 1134; the monk Ramiro taken from the cloister and made king; the grant at Jaca in November 1135 of Garcipollera, Villanovilla and Bescós to San Juan de la Peña and Santa María de Iguácel, for a chalice of precious

stone they had given him; the copy of that grant in the sixteenth-century book of privileges. In pencil: the inventory of 1134, known only from a book of 1698; Pero and García and the cold room; the red edge of the cup. *Found destroyed*: the original parchment of the grant, and whatever the monks knew about where their cup had come from, which they did not write down.

In the run of pages it found its place. Egypt wrote the record, Nadia drew up the catalogue, and the pieces scattered across the world. The green dish had gone to Genoa as a prize. Jaca's part was the pledge: a thing given up for a thing needed, and written down in a line that names the price and leaves out the story. Behind, inked, is the green dish at Caesarea, carried home and named for something it was not. Pencilled in ahead is a mound on Orkney in a Yule storm, and men who had been to Jerusalem cutting their names in its walls.

In the morning the Core could work out how far it is from Jaca to the monastery under the rock, and that there was no moon that night. Which cup it was, it cannot say, or what it was worth to a monk who had been made a king. You'd need the monk for that.

Is a cup worth a valley? Not a question for scribes, the scribe said. The answer was that it was worth what a king who needed friends would pay, and the line in the grant kept the price and lost the reason.

Pero liked the stars because they were the one thing nobody could give away, and what you can check every night doesn't need a line in a grant to prove it. The scribe had his own rule, and Gary had worked under the same one at the solicitors': copy what it says, don't add, and what people make of it afterwards is on them.

Eighteen years on, at the other edge of the known world, in Orkney, a farm boy was carrying peat into a mound in a snowstorm, for men who had been to Jerusalem and back.

Near the bottom of the long grant the brown line had dried. It named the price and kept the reason, and no amount of candlelight was going to get the reason out of it.

CHAPTER 29

Names on the Wall at Maeshowe

The mound of Maeshowe, near the lochs of Stenness, Orkney, at Yule, in a snowstorm: the found year, 1153 CE



Illustration 30. *Earl Harald's men round the fire inside Maeshowe.*

Belfast, by day

AT KILROOT, THE WINTER HE led the lifting team, there was a stairwell in the old part of the station where the men who built it had written their names on the wall. They were in pencil and marker on the bare concrete, with dates in the seventies and the odd football team. There was a drawing of a foreman that was not kind. Somebody had written *Big Davy was here* and somebody else had written underneath it *no he wasn't*. Nobody had painted over any of it in fifty years. He used to stop on the landing with a cup of tea and read it. It was the only place in the whole station that said who had been there.

The stairwell was on his mind when the answer about Orkney came through.

"In 1153 Viking crusaders broke into Maeshowe and carried away its treasure," it said. "Their own runes on the walls of the tomb record the deed."

He sent the machines for it and sifted their returns by hand, a photograph of a wall covered in scratches pinned under the stone.

The mound is ink, and so are the runes. There are about thirty inscriptions cut into the walls of the chamber, one of the largest groups of runes anywhere. One says that Jerusalem-farers broke the mound. One says that Hlíf, the earl's housekeeper, carved it. One says that Ingigerth is the most beautiful of women. One says the man most skilled in runes in the western ocean cut these runes with an axe that once belonged to Gaukr Trandilsson.

He struck out *1153* as a date for the runes. Barnes, who edited them for the Orkney runes volume, puts them between about 1125 and 1175, and says plainly that tying them to Earl Rögnvald's crusade of 1151 to 1153 is a guess. Then he put a line through *carried away its treasure*. Two of the runes talk about treasure: one says it was hidden there long ago, and another says that a man called Hákon carried it out. Nobody knows if there ever was any treasure at all. *Viking* went the same way, because these were Christian men with an earl and a bishop, coming back from a pilgrimage.

The year 1153 is not the runes, then, but the saga, because the Orkneyinga saga says that Earl Harald's men took shelter in Orkahaugr in a snowstorm at Yule, and that two of them went out of their minds there. Then they went on to Firth, late, in the dark. The dating of that winter is the 1150s, and 1153 is the year most often given.

He lined the year up, with the crusade coming home at the bottom and the Yule snowstorm and the men in the mound over it. On top went the midwinter sun going down the passage, the way it had since the builders. He took *1153 CE* from where the three agreed and set it uppermost, and gave it a question mark, because the runes themselves carry no date at all.

Then the doubt: *The runes are not dated and the saga does not mention them. Ketil is mine, and his father, and the woman with the knife*.

In the map, too, he moved the treasure from the ink column to the pencil column, and left the names where they were. Big Davy was here, aye. He sat grinning with the stone on the photograph of the scratches until the kettle clicked.

That night

Beside him on the pillow was the photograph of the runes, and the dark came close and smelled of peat smoke and wet dogs and snow. Outside the wind was going like a train. On the floor of a chamber of stone the Ledger laid its page, and held a torch over it so he could see the scratches, row on row of them, some no deeper than a fingernail.

• • •

Ketil was carrying peat into the mound when the earl's men came out of the snow.

He was eleven, and his father farmed the land at the end of the loch, by the mound. It was the middle of Yule, and the snow had come in at noon off the sea so thick you could not see the byre from the house. His father had sent him out to the mound with a creel of peat, because there were men already sheltering in it. There were three of them, caught on the road, and his father said you did not leave men in the dark at Yule without a fire.

The mound was the big green one that everybody knew, and nobody went into, until that autumn. In the autumn some men back from Jerusalem had come over the hill, and gone up on top of it, and broken in through the roof, looking for treasure. Ketil had watched them from the dyke, and they had found nothing, his father said, but they had left a hole. You could get in now by the long low passage, bent double, with the snow coming in behind you.

He was halfway down the passage with the creel when the light went out behind him, and he turned, and the passage was full of men.

• • •

There were a great many of them, twenty or more, with swords and shields and cloaks white with snow. In the middle of them was a young man not much older than Ketil's cousin, with a fur collar and a face red with cold.

"Whose boy are you?" the young man said.

Ketil told him. The young man laughed. "Then go and tell your father that Earl Harald is in his mound," he said, "and would be glad of more peat."

Ketil went. When he came back with his father and two more creels, the earl's men had the fire going in the middle of the chamber. The smoke was going up through the hole in the roof, and the snow was coming down through it.

He had been inside once before, in the autumn, after the Jerusalem men had gone. It had been dark then, and he had been frightened. With the fire going it was a different place. It was a big square room, higher than a house, built of great flat stones laid one on another so neatly you could not put a knife between them, and in each corner a tall stone like a pillar. There were small rooms off the sides, like the box beds at home but made of stone, and the fire made the shadows jump in them.

"Who made it?" Ketil asked his father.

His father shook his head slowly. "Nobody knows," he said. "It was here before us. Before the earls. Some say the trolls built it, or the old ones that lived here before the Norse came." He looked round at the walls while he said it. "Whoever they were, they knew how to lay a stone."

• • •

The men stayed all that night, because the snow did not stop.

Ketil stayed too, because his father said he could, to help with the fire, and because his father wanted to stay himself and could not say so. The men ate and drank and sang, and told stories, and some of them slept in the side rooms, and the smoke got in everybody's eyes.

Some of the men had been to Jerusalem with Earl Rögnvald. They talked about it as if it were the next parish: the heat, the sea, the walls of the holy city, the river where they had bathed. One of them showed Ketil a palm leaf he had kept, dry and brown, from the Jordan. Another had carved his name on the wall when they broke in, in the autumn, and he showed Ketil where.

That was when Ketil saw the walls were covered in writing.

He had not seen it in the dark the first time. Now, in the firelight, he could see scratches everywhere. They were on the big stones at the height of a man's shoulder, runes and more runes, cut with knives and axes, some deep and careful, some quick and crooked. There was a beast cut on one stone, like a lion with its tongue out. The men read them out to each other and laughed.

"This one says Ingigerth is the most beautiful of women," one of them said, and somebody else said, "She is not," and they all roared.

"This one says a great treasure was hidden here long ago."

"And this one says Hákon carried it out," another said. "So that's that."

Ketil asked if any of it was true. The man who had shown him the palm leaf shrugged.

"Some of it," he said. "Ingigerth is not bad-looking. The treasure, who knows? You write what you like on a wall. It's the wall that's the true thing." He patted the stone as he said it. "Whoever wrote here was here. That's all it tells you. That, and their name."

• • •

Near midnight, when the singing had stopped, the man with the palm leaf sat by the fire with Ketil. He told him about the sky in the south.

"Down there the Yule day is long enough to do a day's work in," he said. "Here it's barely long enough to milk the cows and get them in again. And in summer the sun goes right up over your head, so high that your shadow hides under your feet like a dog under a table."

Ketil did not believe that, and said so. The man laughed and said it was true, and that he had stood in the noon sun at Acre and looked down and seen no shadow worth the name.

"And there are stars down there you never see here," he went on, more quietly. "One great white one, very low in the south over the sea, that the sailors steer by. You could stand on the hill above this mound every night of your life and never see it. The world's round as an egg, the priests say, and it's hidden behind the curve of it."

Ketil asked him how he knew it was really there, if it was hidden.

"Because I went and looked," the man said. "That's the only way. You can't know it from here. You've to go to where it is."

Ketil sat for a long while after that, looking up through the hole in the roof at the stars that he could see. He thought about the one that he could not.

• • •

Late in the night, a woman cut her name on the wall.

Ketil had not known there was a woman with them. She was older than his mother, wrapped in a grey cloak, and she sat apart, and the men left her alone. Somebody said she kept the earl's house. When most of them were sleeping she got up and took a knife from her belt. She went to a clean place on the wall by the fire, and began to cut.

Ketil watched her from the fire. She worked slowly, with her tongue between her teeth, the way his mother sewed. When she was done she stood back and looked at it, and saw him watching.

"Can you read?" she said.

He shook his head.

"It says that the Jerusalem-farers broke into this mound," she said, "and that Hlíf, the earl's housekeeper, carved this." She put the knife back in her belt. "That's me. Hlíf."

"Why did you write it?"

She thought about that. “Because I was here,” she said. “And because in a hundred years, when there’s nobody left who knows my name, somebody will stand where you’re standing and read it.”

• • •

Before morning two of the men went strange.

Ketil did not see the beginning of it, because he was asleep. He woke to shouting. Men were holding two others down by the fire, and one of the two was weeping and the other saying over and over that there was somebody in the side room, somebody old, who would not let him go. The men were frightened, and some of them said it was the mound-dweller, angry at them for coming in. Some said it was only the dark and the drink and the cold. The earl said nothing, but he sat up the rest of the night with his sword across his knees.

Ketil’s father took him out into the passage, away from it, and they sat in the cold with their backs to the stone.

“Is it the mound?” Ketil whispered.

“It’s men who have been a long way,” his father said, “and seen hard things, and are cold, and far from home. Leave them be. They’ll come right or they won’t, and it’s not ours to help.”

• • •

The snow stopped at midday, and the earl’s men went off along the road to Firth, slowly, because of the two who were not right, and the fire burned down to nothing.

Ketil came back that evening, alone, to clear the ashes. The sky had cleared, and the sun was going down low in the south-west, as low as it went in all the year, red, over the hills of Hoy.

He was bent double in the passage with the empty creel when the light came in behind him.

It came straight down the passage, past him, along the floor, into the chamber, and lit the back wall red. It lit the stones and the runes and the name of Hlíf and the beast with its tongue out. It stayed there,

a long bar of red light on the back wall of the mound, for as long as it took Ketil to say the Our Father three times. Then it climbed the wall, and went out.

He stood in the dark with the creel and did not move until the cold got into his feet.

Whoever had made it had known exactly where the sun went down at Yule. They had built the whole long passage to catch it, and it had come, all these years, every Yule, whether anybody was there or not.

He went home and told his father, and his father said aye, his own father had told him that, and he had forgotten.

• • •

Two days later Ketil went back to the mound with his father's old knife in his belt, meaning to cut his own name.

He had thought about it ever since Hlíf, because everybody else had written on the wall, earls' men and Jerusalem men and a housekeeper. He had been there too, and had carried the peat, and seen the light come down the passage. It seemed right that he should leave something.

But he could not write runes. He stood in front of the back wall with the knife in his hand while the torch burned down a finger's width. The only thing he could think of to cut was a mark, a plain line, which would mean nothing to anybody. He looked at the stones, laid so close that you could not put the blade between them, and at the scratches all over them, and at the beast with its tongue out.

In the end he put the knife away. He put his hand flat on the cold back wall instead, where the light had been. He held it there, the way you put your hand on a horse's neck to let it know you are there.

The builders had not cut their names anywhere, he thought. They had only made it so that the sun would come.

The master page

On the floor of a stone chamber the Ledger put down this page, and set the low red bar of the Yule sun across it, and the scratches of thirty different hands.

In ink: the chamber at Maeshowe and its passage to the midwinter sunset; the runes, about thirty, Jerusalem-farers and Hlíf and Ingigerth and the man most skilled in runes, between about 1125 and 1175 in Barnes's dating; the Orkneyinga saga's Yule snowstorm, Earl Harald's men in Orkahaugr, two men out of their minds. In pencil: that the Jerusalem-farers were Rögnvald's crusaders; that Hlíf cut her name that night; Ketil, his father, the palm leaf. *Found destroyed*: whatever the builders had left in the chamber, and the treasure, if there was any, which Hákon or nobody carried out.

He set the page by the others. Egypt had kept its record and Nadia its catalogue, and the pieces had gone their separate ways across the world. A dish was carried off from Caesarea; a cup changed hands at Jaca. Maeshowe left the mark: people who did not know what the old thing was, writing their own names on it, which is the most honest thing anybody did in all that long scattering. Behind it in ink stands Stenness, and the builders who laid the stones and aimed the passage at the Yule sun. Pencilled in ahead is a house of the Templars at the edge of Dublin, and a clerk counting its cups.

As for the Core, it could give him the bearing of the passage, and the day the light still reached the back wall that winter. Who Ingigerth was it cannot say, or what the two men saw in the dark. For that you'd have had to be in out of the snow with them.

Who made it? Nobody knows, the father said. The answer was on the back wall at sunset, if you were there to see it: people who knew where the sun goes down at Yule, and built to catch it.

Ketil had wanted the light to come down the passage on the right day. The dream showed him the builders, who had no runes and no names and left the sun to sign the wall for them every winter. The men in out of the storm wrote that they had been there, and mostly no more than they knew, and they left the wall standing.

A hundred and fifty years on and across the sea, at the Templar house at Clontarf, a clerk's boy was holding the tally sticks on the morning after Candlemas while the sheriff's men counted everything the brothers owned.

Before he put the photograph away he turned it over and wrote on the back, in pencil, small, the way somebody once had on a stairwell at Kilroot: his own name, and the date, and nothing else.

CHAPTER 30

The Count at Clontarf

Clontarf, on the north shore of Dublin Bay, on the morning after Candlemas, when the Templars of Ireland were taken: the found year, 1308 CE



Illustration 31. *The Templars in the dawn yard at Clontarf before the sheriff's men.*

Belfast, by day

THE WORST STOCK-TAKE HE DID with RGIS was a shop that was closing. It was a big place in the town, and the stock-take was for the administrators. The staff were still there, working their last week, watching four strangers with scanners count everything they had. Nobody said much. One woman stood at the end of an aisle with her arms folded the whole night. At four in the morning the count came out short on one line, and the manager said quietly that those had gone back to the supplier the week before, and there was a docket

somewhere. They never found the docket, and the count went in short, and the line stayed short, and that was what the administrators got.

The shop was still in his head when the Templars came up on the screen.

"Before the arrests of 1308, the Templar master Henry Danet smuggled the Grail out of Ireland," it said. "It passed to the Order's secret successors."

The machines fetched, and he did the reading by hand, the stone on a printed list of Templar houses in Ireland.

The arrests are ink: King Edward wrote to his justiciar in Ireland, John Wogan, on the twentieth of December 1307. The writ reached Wogan on the twenty-fifth of January. On the third of February 1308 the Templars of Ireland were taken and their goods inventoried, house by house: Clontarf, Kilclogan, Crook, Kilbarry and the rest. The king paid allowances to nineteen of them that year. Henry Danet was the master in Ireland, and he was tried in Dublin in 1310 with the others, and his evidence was sent on to the inquisitors in England. He spoke of things he said he had heard of brothers elsewhere, and he confessed nothing of his own, and he was let out on bail in 1312.

He scored out *smuggled the Grail*, because there is nothing in the record. There is no Grail in any Irish inventory, and no line that says Danet carried anything anywhere. *Secret successors* went too, since the Templars' lands in Ireland went to the Hospitallers in the end, by order of the Pope, slowly and with a great deal of arguing. There was nothing secret about it, and Herbert Wood printed the Irish documents in 1907, and Helen Nicholson has written on Danet's evidence since. Neither of them has a Grail anywhere in it.

He piled the sources, the king's writ crossing the sea in winter at the bottom. Over it went Candlemas, the second of February, the last night the brothers kept their hours. On top lay the morning after, and the sheriff's men at the gate, and the count. What all three shared was *1308 CE*, so it went at the top, and its question mark was the smallest in the notebook, because a sheriff's clerk had written the day down for him.

His doubt followed: *The inventory lists what was in the house, and it cannot list what was gone. Adam is mine, and his father, and Brother Walter*.

On the map he wrote the chain that ran through Danet in pencil, faint, the way the Ledger had it, and put a question mark after it. Aye, there was a docket somewhere, right enough. He set the stone on the list beside Clontarf and thought about the docket a while.

That night

The list of houses was on the floor beside the bed. Cold came, and beeswax and wet stone and the sea, and bells a long way off, and then no bells. Very carefully, on a chapel bench, the Ledger set its page down, the way you lay down something you have been told to count and not to keep. Somebody outside was stamping the cold out of his boots.

• • •

In the yard outside, Adam held the tally sticks, and his father did the counting, and the brothers of the Temple stood in the yard and watched.

He was twelve, and his father was a clerk of the king's exchequer in Dublin. The night before, at Candlemas, his father had come home late and said that they were both going out to Clontarf in the morning, early, before it was light, and that Adam was to bring his good cloak and say nothing. Adam had asked why, and his father had said, the king's business, and then, after a long time, that the Templars were being taken, all of them, all over Ireland, on the same morning.

"Taken where?" Adam had asked.

"To the castle," his father said. "Taken and held. And their goods counted. And that's our part, the counting."

• • •

So in the morning they rode out along the shore in the dark with the sheriff and his men, and came to Clontarf as the sky was going grey over the bay.

The house of the Templars was bigger than Adam had expected, a stone hall and a chapel and barns and a mill, with fields running down to the sea. There were lights in the chapel, because the brothers had been up in the night for their prayers, the way they always were. They were just coming out when the sheriff's men came through the gate.

There was no fighting, though Adam had been afraid there would be. The brothers came out into the yard, a few old knights and more serjeants in brown. They stood there in the cold while the sheriff read the king's writ. When he had finished, the oldest of them only nodded, as if he had been expecting it for a long time, and perhaps he had. Everybody had heard what had happened to the Templars in France.

Then the sheriff's men took them into the hall, and kept them there, and Adam's father took out his roll and his ink, and said, "Right. We'll start at the barns."

• • •

They counted all morning.

They counted the oxen and the cows and the pigs, and the horses in the stable, and the sacks of grain in the barn, and the ploughs and the harrows. Adam cut a notch in the tally stick for every ten, and his father wrote down the totals. A serjeant of the Templars walked with them to show them where everything was.

He was an old man, Brother Walter, lame in one leg, with a face like a walnut. The sheriff had let him out of the hall because he was too old and too lame to run, and he answered every question without complaint. When they came to the mill he showed Adam how the wheel was set. When they came to the dovecote he told him how many birds it held in summer.

"Doesn't it make you sad?" Adam asked him, at the dovecote, and then wished he had not.

Brother Walter took his time before he answered. “Count the doves, son,” he said. “That’s your job. Mine is to tell you where they are.”

• • •

They came to the chapel at noon, when the counting outside was done.

It was small and plain, with a painted cross over the altar and the smell of the Candlemas candles still in it. There was a press in the corner, a tall wooden cupboard, and Brother Walter opened it, and inside were the things of the altar.

Adam’s father counted them out: two chalices of silver gilt, one cross for carrying, two candlesticks, a censer, vestments, so many, and books, so many, and he wrote the names of the books. Adam held the tally and watched the things come out, one at a time, into the grey light.

When the press was empty he looked into it, and saw the cloth on the shelf.

It was an old linen cloth, laid along the shelf to keep the silver from the wood. The silver had stood on it so long that it had left marks. There were two round marks, darker, where the two chalices had stood, and beside them, at the end, there was a third.

“Where’s the other one?” Adam said.

His father looked round, and Brother Walter, in the doorway, did not move.

“There’s three marks,” Adam said. “There’s only two cups.”

His father bent and looked at the cloth, and he took his time looking. Then he straightened, and looked at Brother Walter, and Brother Walter looked back at him, and kept his counsel.

“Brother?” Adam’s father said.

“There were two cups in the press this morning,” Brother Walter said. “You’ve counted them. I’ve shown you everything there is.”

Adam's father was quiet a while. Then he dipped his pen and wrote, and Adam leaned over to see. He had written *two chalices of silver gilt*, and he had not written anything about the cloth.

• • •

Nothing more was said about the cloth, and in the afternoon the sheriff's men brought the brothers out of the hall to take them into Dublin, to the castle.

They came out two by two into the yard, the old knights first and the serjeants after, with their cloaks round them against the wind off the sea. Nobody had tied their hands. The sheriff said there was no need, and the oldest knight said there was none, and that was the end of it. They walked to the carts that had been brought for the ones who could not walk so far. Some of them got in, and some of them stood beside them and would not.

Brother Walter was the last. He came across the yard slowly on his bad leg. When he passed the dovecote he stopped, and looked up at it for a while, and then looked round for Adam.

"There's grain in the small bin by the door," he said. "The doves will want it every morning till the spring, or they'll go to the neighbours and not come back. Tell your father, or tell the sheriff's man, whichever will listen."

Adam said that he would tell them both, and Brother Walter nodded and went on to the carts. He did not look back at the chapel, or at the hall, or at anything else. The carts went out through the gate and along the shore road towards the city, very slowly, and Adam stood by the dovecote and watched them until they were gone, and then went and found the grain.

• • •

On the way home, along the shore, Adam asked his father why.

His father rode for a while before he answered. The sea was grey on their right, and the tide was out, and the birds were walking on the sand.

"Because the king wants to know what's there," he said. "Not what was there. I can count what I can see, and I can swear to it. I can't count a mark on a cloth. If I write it down, some man in London will ask where the third cup went, and they'll have that old man questioned till he says something, and he'll not know, or he'll not say, and either way it'll go hard with him." He shook his head as they rode. "It might have been sold years ago. It might have been sent to another house. It might have been a candlestick. I don't know, and nor do you."

"But it was there," Adam said.

"It was there once," his father said. "Now it isn't. That's not a count. That's a story."

• • •

For a week afterwards his father worked late at the exchequer, writing the roll out fair, and Adam sat with him and ruled the lines.

It was the same at every house in Ireland, his father said. The sheriffs and the clerks had gone in on the same morning, at Clontarf and Kilclogan and all the rest, and counted everything. Now all the counts were coming into Dublin to be written up and sent across the sea to the king. There were rolls from all over, piled on the tables, and the clerks argued about whether a cow in calf was one beast or two.

Adam heard the name of the master of the Temple in Ireland, Brother Henry Danet, for the first time that week. He was held in the castle with the others, the clerks said. He was a great man in the Order, and a friend of the Grand Master in France. Some of the clerks said he would know where everything was, the things that were not in the rolls. Others said there was nothing that was not in the rolls, and that people would say anything about the Templars now.

"And the brothers?" Adam asked his father. "What happens to them?"

"The king pays them a living out of their own lands, while they wait," his father said. "So much a day, for food and keep." He looked down at the roll. "It's all here, every penny of it. Their own fields feed

them, and the king's clerks write it down, and they sit in the castle and wait to be judged." He rubbed his eyes. "It's a strange world, son. Rule the next line."

• • •

That night, with his father's strange world on his mind, Adam could not sleep, and he went up on the roof of their house in Dublin, as he did in summer, though it was February and bitter.

The sky was clear, and he could see the stars over the bay. He thought of Brother Walter, who had told him at the dovecote that the brothers rose in the night for their prayers by the stars. There was a window in the dormitory, Walter said. When a certain star came into the corner of it, it was time to get up and go down to the chapel. Every night, the same star, at its own time, all the year round.

Adam looked for it, though he did not know which one it was. He thought of the brothers in the castle tonight, with no window and no star, not knowing when to rise. He thought of the third mark on the cloth, and his father's pen not writing it.

The stars went on over the bay at their own times, as they always did, every one of them where it ought to be, and none of them missing.

• • •

In the spring his father had to ride out to Clontarf again for the king's account of the lambing, and Adam went with him.

They counted the lambs in the lower field first, with Adam cutting the notches on a new tally and the hired men catching the strays. His father wrote the number into the account for the king just as he had written the oxen in February. It came out even, every lamb where it should be. The hired men laughed when Adam told them so.

The house was quiet now, with the sheriff's man living in the hall and two hired men working the fields, and the chapel locked. The doves were still there. The sheriff's man said that some boy from the city had come out every morning in the cold weather and fed them

from the bin by the door, until the grass came, and that he did not know who the boy was, and Adam said nothing and looked at the ground.

Before they left, his father asked for the key of the chapel and went in alone, and Adam waited at the door. When his father came out, he did not say what he had looked at, and Adam did not ask. But on the ride home he told Adam that the press was empty now, the silver all gone to Dublin, and that somebody had taken away the old linen cloth from the shelf, and that there was nothing there at all for anybody to count.

"Good," his father said, after a while, to nobody in particular, and rode on along the shore.

The master page

A chapel bench at Clontarf carried this page, and on it a strip of old linen, with three dark rings on it and two cups.

In ink: the king's writ of December 1307, Wogan's receipt of it in January, the taking of the Templars of Ireland on the third of February 1308; the inventories of their houses; the allowances to nineteen brothers; Henry Danet, master in Ireland, tried in Dublin in 1310, confessing nothing of his own, bailed in 1312; the slow passing of the lands to the Hospitallers. In pencil: the third ring on the cloth, and any chain that runs from Danet to a cup; Adam, his father, Brother Walter. *Found destroyed*: the Order itself, and whatever the brothers knew and did not say.

He put the page in its place. Egypt had made the record, Nadia the catalogue, and the pieces were out across the world. Since Caesarea there had been a prize, a pledge at Jaca and the names cut at Maeshowe. Dublin's part was the inventory: a careful count of everything that was there, made by an honest man, which by its honesty leaves out everything that was not. Already inked, behind, is Maeshowe, where the men back from Jerusalem cut their names, and some of them would have known men like these. Ahead lies the monastery under the rock, in pencil, at San Juan de la Peña, and a king of Aragon who wants the stone cup back.

In the morning the Core could tell him which star came into the dormitory window at matins on the night of Candlemas 1308. What was on the third ring of the cloth it will never know, and neither, now, will anybody.

Where's the other one? It was there once, the father said, and that's a story, not a count. The page leaves it a story, in pencil, which is the only honest place for it.

Adam needed it to know when to get up in the night. The dream gave the better answer: the stars are the one count that always comes out right, every one where it should be and none missing, which is more than could be said for any stock-take Gary had ever worked. Count what is there and swear only to that, and don't have an old man questioned over a ring on a cloth.

Ninety years on, in the monastery under the rock in Aragon, a novice was polishing a gold cup that a king had sent, to stand in the place of a stone one.

Nobody put Adam on the sheriff's list. Every morning that winter there had been one more thing at Clontarf than the count said, and it was him.

CHAPTER 31

Gold for Stone

San Juan de la Peña, the monastery under the rock, in the mountains of Aragon, at the turn of autumn: the found year, 1399 CE



Illustration 32. *Domingo polishes the gold cup under the rock of San Juan de la Peña.*

Belfast, by day

He woke with two silver chalices still before his eyes. When the company sponsored the Immaculata youth teams, the club sent a photo of the players with the new kit. One of the coaches told him a story about their cup. The under-twelves had won a wee league years before, and the cup was battered, a dent in the side, one handle soldered back on crooked. Somebody had offered to buy them a new one, bigger, with proper engraving. The kids voted to keep the old one, and one of them said the new one would be just a cup. The old one was *theirs*, because of the dent. He liked that, and he thought about it more than he expected to.

The dented cup was in his head when the answer on Aragon came back.

"In 1399 King Martin the Humane seized the cup of the Last Supper," it said, "from the monks of San Juan de la Peña." It went on. "From 1437 it was honoured in Valencia cathedral."

He put the machines on Aragon and checked their answers himself, with a photograph of a dark red cup set on a gold foot under the stone.

He put a line through *seized*, because there is a notarial act dated the twenty-sixth of September 1399, and it is not a seizure. At the king's request the monks of San Juan de la Peña handed over a chalice of stone. The king gave them a chalice of gold in its place. It went into the king's own keeping, and it is listed among his goods when he died in 1410. Alfonso the Magnanimous had it brought to Valencia in the 1430s. In 1437 it passed to the cathedral there, among the royal relics, as security for money the king had borrowed.

Then he struck out *the cup of the Last Supper* and wrote *a cup of stone* above it. The cup in Valencia is agate, a small dark red bowl. Antonio Beltrán, who studied it in 1960, put the bowl itself somewhere between the second century before Christ and the first after, made in some workshop of the eastern Mediterranean. The gold foot and handles and the stone base are later. So the bowl is old, older than any of the paper that names it, and old enough to have stood on a table in the time the story says. Nothing in the stone says whose table.

He laid one source over the next, with the grant at Jaca underneath, two hundred and sixty years before, a cup of precious stone for a valley. Over it went King Martin, crowned in Zaragoza that spring, a collector of relics. The act of September went on top, gold for stone, and through all three he read *1399 CE*, which he wrote along the top. The notary had dated the act to the day, so the question mark was for the cup and not the year.

Below, the doubt: *Nobody can show that the cup of 1135 and the cup of 1399 are the same cup. Domingo is mine, and the prior's words, and the night on the meadow*.

He drew the map's line from Jaca to the monastery to the king in pencil, faint, and the further leg, king to Valencia, in ink. Aye, he thought, the dent settled that. He thought about the dent a while, and set the stone beside the photograph of the red cup.

That night

The photograph of the cup lay on the bed beside him, and the dark turned cool and dim, with incense and cold stone in it and the dry grass of a mountain in autumn. A great weight of rock leaned overhead. On an altar under a cliff the Ledger laid its page, and on the page two cups, one gold and one stone, side by side.

• • •

Domingo had been polishing the gold cup since Vespers, and it would not stop shining.

He was twelve, a novice at San Juan de la Peña. The gold cup had come up the mountain that morning with the king's men, in a box lined with red cloth. To Domingo it was the loveliest thing he had ever laid eyes on. It was tall, with a knob on the stem and leaves round the foot and a lip that caught every light in the church. The king had sent it as a gift, and the prior had told Domingo to polish it for the altar. He had been polishing it for an hour. There was nothing left on it to polish, and still he kept on, tilting it this way and that to the candles, with one question turning over and over in him that he did not dare put to anybody. Is the gold one better?

The monastery was built under the rock, and nobody who wrote about it ever began anywhere else. A great cliff leaned out over the church and the cloister like a roof, so that no rain fell on them. In the cloister you could look up and see the stone hanging over you instead of the sky. The brothers said it was the safest place in Aragon, but Domingo thought it was the darkest.

The stone cup had gone that morning, down the mountain, in the same box the gold one came in.

• • •

He had seen it go, and so had everybody else in the house.

The king's men had come up the road in the early morning, a dozen of them, with a notary in a black coat and two men of the king's chapel. They had gone into the chapter house with the prior and the older brothers. They had been in there a long time, and the novices had been sent to the kitchen and told not to listen. Domingo had listened anyway, from the passage, and had heard nothing but the notary's voice going on and on, reading something long, and the prior saying yes, and yes.

Then they had all come out and gone into the church. The prior had opened the old chest in the sacristy and taken out the stone cup.

Domingo had seen it before, on the great feasts, carried in procession. It was small and dark and red, the red of a coal when you blow on it. The brothers said it was very old, and that it had been in the house since before anybody could remember. Some of the old brothers said more than that. They said it had come from Huesca, and before that from Rome, and before that from a supper in Jerusalem. The prior never said that. The prior only said it was old and precious, and that it was theirs.

That morning the prior had held it up in front of the king's men, in the light from the door, so that the notary could see it and write it down. Then he had put it into the red box himself, and closed the lid, and given the box to the king's chaplain, and taken the gold cup in exchange. Nobody had said anything. The king's men had bowed and gone down the mountain.

• • •

"Is the gold one better?" Domingo asked.

It was evening, and the prior had come into the sacristy to see the polishing. He was standing by the window, looking at the gold cup on the cloth.

He did not answer at once, because he never answered at once; he was an old man with a soft voice. He liked to think first.

"It's worth more," he said at last. "Much more. You could buy a farm with that cup. You couldn't buy a goat with the other one, if you didn't know what it was."

"Then why did the king want the stone one?"

"Because he's a king who loves holy things," the prior said, "and he knew what the old brothers say about it. And because a king can ask, and when a king asks, you give." He picked up the gold cup and turned it in his hands. "He was generous. He didn't have to send this. Many wouldn't."

Domingo thought about it. "Do you think it was the one from the supper?"

The prior put the gold cup down.

"I think," he said slowly, "that it was a cup of stone, very old, that had been in this house longer than anybody could remember, and that we kept it well. That's what I know. The rest is what the old brothers say, and I won't swear to what they say." He looked at Domingo. "And now the king has it, and he'll say what he likes, and in a hundred years they'll say it more loudly. And it'll still be a cup of stone."

• • •

That night, after Compline, Domingo did what he was not allowed to do.

He went out by the little door behind the kitchen and up the goat path that climbed the side of the cliff, in the dark, his habit hitched up to his knees, until he came out on the meadow on the rock's flat top. It stretched away broad and level, full of dry grass that smelled of summer, and there was nothing over it at all.

He lay down on his back in the grass and looked up.

Down in the monastery you never saw the sky. You lived your whole life under a roof of rock, and the only stars you saw were painted. Up here they went from one edge of the world to the other, more of them than he could ever count, the bright road of them across the middle and the plough shape low in the north and the one that

does not move above it. It was the turn of autumn, Michaelmas coming. The brother who kept the garden had told him that a few days before, the day and the night had been the same length all over the world. He could not see how anybody would know such a thing. But the brother had said you could check it, with a stick and a shadow, and Domingo meant to.

He thought about the stone cup going down the mountain in its red box. He thought about it being only a cup, and about it being more than a cup, and about the prior saying he would not swear.

The stars did not care which it was. They were there for anybody who climbed the rock. Nobody had to swear to them, because you could lie on your back among the grass and look.

• • •

He went down before the bell for Matins, and nobody had missed him. He stood in the choir in the dark church under the rock and sang with the others.

The gold cup was on the altar, shining in the candlelight, beautiful, catching every light. Domingo looked at it through the whole of Matins. He thought that it was the most beautiful thing in the church, and that it was not theirs, not yet, and might never be, because it had no story at all.

• • •

In the morning, after Prime, Domingo went to find the brother in the garden, and asked him how anybody could tell that daylight and dark had been shared out evenly the world over.

Brother Pedro was very old and very deaf, and he had to be asked twice, but when he understood he laughed and took Domingo up the goat path himself, slowly, with a hoe for a walking stick. On the meadow at the top he showed him a straight stick he had set upright in the grass at the start of the month, with a ring of small white stones round it, and a line of stones running out from it towards the hills.

"Every morning I mark where the shadow falls when the sun comes up," he said, "and every evening I mark it again when the sun goes down. In the summer the shadows lean to the south, morning and evening both. In the winter they lean to the north. On the day they lie straight across from each other in one line, east and west, the day and the night are even." He pointed with the hoe. "That was nine days ago. That line there."

Domingo looked along the line of white stones and did not answer.

"So anybody could do it," he said. "Anywhere."

"Anybody with a stick and some patience," Brother Pedro said. "You don't have to believe me. You don't have to believe anybody. You can put up your own stick and watch your own shadow and see for yourself." He leaned on the hoe and looked down at the roofs of the monastery under the cliff. "That's the difference between the sky and the things down there in the sacristy. The sky you can check."

• • •

At the feast of All Saints the gold cup was first used at the high altar, by the prior himself. The whole house was there to see it.

Domingo served at the altar that day, in a clean surplice, with his hair combed down flat with water. He carried the cruets and held the book, and when the moment came he knelt on the stone step and watched the prior lift the gold cup high in both hands, in the light of all the candles. It flashed so brightly that some of the old brothers in the choir put up their hands to their eyes.

Afterwards, in the sacristy, he helped the prior put the vestments away. He asked him whether it had been strange, using the new cup instead of the old one.

"It was heavy," the prior said. "That was all I noticed. It's much heavier than the stone one was." He folded the chasuble carefully into the chest. "The stone one was so light you'd hardly know you were holding it. You had to keep your mind on it, or you'd have dropped it."

He shut the chest and stood with his hand on the lid.

"We'll get used to the gold one," he said. "In a few years nobody here will remember the old one except the old men, and they'll tell the novices it was the cup from the supper, and the novices will believe them. That's how it goes. I only hope somebody remembers that it was light."

The master page

Under a cliff, on an altar, lay this page, with a gold cup on it, and a ring of dust beside it where a smaller one had stood.

In ink: the notarial act of the twenty-sixth of September 1399, the monks of San Juan de la Peña giving King Martin a chalice of stone and receiving one of gold; the cup among the king's goods in 1410; its passing to Valencia cathedral in 1437, among royal relics held as security for a loan; the agate bowl itself, dated by Beltrán to between the second century before Christ and the first after. In pencil: that this was the cup of the grant at Jaca; that it came from Huesca, or Rome, or a supper; Domingo, the prior, the meadow. *Found destroyed*: the monastery's own memory of where the cup had come from, which they had never written down, and the old brothers took with them.

It joined the others on the shelf. Behind it stood Egypt's record and Nadia's catalogue, and the pieces scattered across the world. Plunder at Caesarea, a valley for a cup at Jaca, names on the wall at Maeshowe, a clerk's tally at Dublin: the pieces had been taken, swapped, signed and counted. San Juan de la Peña's part was the claim: the moment a story is strong enough that a king will pay gold for a plain cup because of it, and the story, not the cup, is what is bought. In ink, behind, is Jaca, and a chalice of precious stone given for a valley. Pencilled in ahead is Bristol, and men fitting out ships with salt to go looking for an island nobody has seen.

Some of it was sums, and the Core could do sums: the day of the equinox in 1399, and how agate takes the light. It could not say where the cup was in the year 33, or the year 300. Nobody can, and the honest ones say so.

Is the gold one better? It's worth more, the prior said. The answer was that it was worth more and meant less, and a story is the only thing that can make a stone cup outweigh a gold one.

Under the rock, Domingo said, you never saw the sky, and on top you could see the whole of it. The dream's answer was that the sky is the one thing in the world nobody has to swear to, because anybody can lie on their back and check. The prior had said what he knew and refused to swear to what the old brothers said. He had kept the cup well, and when he had to, he gave it.

Eighty-two years on and a long way north, on the quay at Bristol, a customs clerk's girl was writing down forty bushels of salt for two ships that were not going anywhere anybody would say.

In the morning the ring of dust where the small cup had stood was still there on the altar, and the novice let it be, for nobody had said to wipe it away.

CHAPTER 32

Salt for the Isle of Brasil

Bristol, on the quay by the Marsh, in high summer, with two ships fitting out for the west: the found year, 1481 CE



Illustration 33. *Joan and Einar on the Bristol quay.*

Belfast, by day

AT SHANNONBRIDGE, WHEN HE WAS spotting for the cranes, the site ran on rumour the way the cranes ran on diesel. The station was coming down and the canteen knew everything first. They'd found something under the turbine hall, the Germans were buying the site, there was a body in the old coal yard. Every one of them was told as fact, by a man who'd had it from a man who was there. Most of them were nothing, but one, about a buried tank nobody had on the drawings, turned out to be true, and that one nobody had believed. He learned to write down what he was told, who told him, and what he had seen himself, in three different columns.

The three columns were in his head when Bristol came back on the screen.

"Bristol sailors reached America in 1481, eleven years before Columbus," it said. "The customs records of the port prove it beyond doubt."

He set the machines to work and went through their haul by hand, the stone weighting a photocopy of a page of old court Latin.

The ships are ink: in 1480 a ship of John Jay's, with a master called Lloyd, sailed from Bristol on the fifteenth of July to look for the Isle of Brasil west of Ireland. It was driven back by storms about nine weeks later, having found nothing, and William Worcestre wrote it down. In 1481 two ships, the *Trinity* and the *George*, went out of Bristol carrying forty bushels of salt. Thomas Croft, one of the king's customs officers in the port, owned a share in each, and he was hauled before an inquiry for trading while holding his office. The inquiry found that the salt was not for trade, but to search for a certain isle called the Isle of Brasil.

He crossed out *reached America* first, and then, a line further on, *prove*. The record proves the salt, the ships and what they went for. It says nothing about what they found, or whether they came back with anything at all. Then he put *pencil* beside the join the Ledger wanted to make. Years later, after Cabot's voyage, an English merchant called John Day wrote a letter in the winter of 1497. He said that the land Cabot found had been found before, in the past, by the men of Bristol who found Brasil. Historians have argued over that sentence ever since, and it is one sentence, and it is hearsay, and it is not a customs roll.

He started with Lloyd's ship coming home beaten in 1480, and over it were the *Trinity* and the *George*, salted and fitted out the next summer. On top was the inquiry, and the clerk writing *Isle of Brasil* into the record. The overlap gave him *1481 CE* for the heading, and he put the question mark after the island instead, which was where it belonged.

The doubt came next: *Nobody wrote down what the ships found. Joan is mine, and her father, and Einar from Iceland.*

On the map he put Brasil where the old charts put it, a round island west of Ireland, in pencil, with nothing joining it to anything. Three columns, aye. He shifted the stone off the Latin and onto the sea, and left the columns as they were.

That night

The window was open and the photocopy lay on his chest. Tar and fish came in, and river mud at low tide, and gulls, and a bell somewhere ringing for nothing in particular. At a clerk's desk by a window on the quay the Ledger ran a finger down a column of figures and stopped at the salt. The ink beside it was barely dry.

• • •

Joan wrote down the salt, and her father checked it, and Einar carried it.

She was eleven. Her father was a clerk in the customs house on the quay, under Master Croft, and since her mother died she had come to work with him in the summer because there was nobody at home to mind her. She could read and write and add, better than her father's boy. Her father let her keep the day book when the work was light, though he told her never to let Master Croft see.

That July the work was not light. There were two ships at the quay below the window, the *Trinity* and the *George*, fitting out for sea. All day long the carts came down with barrels and sacks and rope, and everything had to be written.

"Salt, forty bushels," her father said, reading from the merchant's note. "For the *Trinity* and the *George*, and mind you write it plain."

Joan wrote it down in her best hand, and then she looked up. "What are they going to salt?"

Her father looked out of the window at the ships for a while.

"Fish," he said at last, "if they find any."

• • •

Einar carried the sacks up the plank. He was thirteen, and he belonged to a merchant in Redcliffe who traded to Iceland. He had come back on one of the merchant's ships three summers before. It was from a place so far north, he said, that in the middle of summer the sun did not go down at all.

Joan had not believed him at first, but now she did, mostly, because he knew too much about it to be lying. He said the sun went round the sky all night, low along the hills, and never set. In winter, he said, it hardly came up, and that you lived by lamps and by the fire, and told stories, and waited.

He sat with her on the quay at noon, when the men went to eat, with their feet over the edge above the water.

"Where are they going?" he asked her. "The ships. Not to my country. You don't take salt to Iceland. There's salt enough there."

Joan looked round to make sure there was nobody listening.

"My father says they're going to look for the Isle of Brasil," she whispered.

Einar laughed out loud at that. "That's not a real place."

"It is. It's on the charts. Master Croft has a chart with it on, west of Ireland. It's round, with a river through the middle." She drew it on the stone of the quay with her finger. "Last year Master Jay sent a ship to look for it, and it got blown back in a storm. So now they're sending two."

"Is it there?" Einar asked.

Joan thought about it, and she wanted to say yes, because everybody on the quay said yes. The sailors in the taverns said they had seen it, or their fathers had. It was a smudge of green in the evening that was gone in the morning.

"I don't know," she said. "Nobody's been."

• • •

The *Trinity* and the *George* went out on the tide at the end of the month, down the river and out past the gorge. There was half of Bristol on the quay to watch.

Joan stood with Einar at the end of the Marsh and watched them go until they were out of sight round the bend. Then they sat on the wall until it got dark. Einar showed her how the sailors in his country found their way when there was no land.

"That one," he said, pointing at the star that does not move. "The farther north you go, the higher it stands. My uncle could tell you how far north you were by how high it was over the sea, to a day's sailing. And if you go west, and keep it at the same height, you go straight along, and you don't get lost."

"Will the ships do that, out there?" Joan asked.

"If they've any sense," Einar said.

Joan kept her eyes on the star. She thought about the round island on Master Croft's chart, with the river through the middle, and about her father saying he had only seen the salt. Then she wrote, in her head, in three columns, the way her father kept the book. What she had seen was the salt, the ships and the star; what she had been told was the island; and what she knew was nothing yet.

• • •

Every evening after that, for most of that summer, Joan climbed Brandon Hill behind the town when the customs house closed. She sat on the grass at the top, and looked west.

From the top you could see the whole of Bristol below, the churches and the river and the masts along the quay. Beyond it was the long brown water of the Severn going out to the sea between the hills of Wales and the hills of Somerset. You could not see the open sea from there, but you could see where it began. She sat with her knees up and her chin on them and watched the sun go down over Wales. She

thought of the two ships somewhere out beyond it, under the same sun, with their salt and their chart and their round green island that might or might not be there.

Some evenings there was weather in the west, great dark banks of cloud with rain hanging under them like hair. On those evenings she went home quieter. Her father had told her about Master Jay's ship the summer before, nine weeks at sea and nothing found but storms. It had come back into the river with its sails torn and its men so thin and grey that their own wives had hardly known them. She did not want that for the *Trinity* and the *George*. She wanted them to find it, whatever it was, and come home and say so.

• • •

The trouble came in the autumn, with the ships still gone.

A man came from London, from the king's officers, to ask questions. He sat at the big table in the customs house with his gown on and his papers out. He asked Master Croft why a customs officer of the port was a part-owner of two ships that had fitted out under his own nose. A customs officer was not allowed to trade, as everybody knew.

Joan sat on the stool in the corner with the day book, and kept very still.

Master Croft was a big man, red in the face, and he spoke slowly. He said that he was not trading. He said that the salt had not been merchandise, and was not for sale, and was for the ships alone. He said that the ships had gone out to search and find a certain isle called the Isle of Brasil. This, he said, was not trade but discovery, and no law forbade it.

The man from London wrote it all down, and then he looked across the room at Joan's father.

"Is that what your book says, clerk?"

Her father stood up. "My book says forty bushels of salt for the ships, sir," he said, "and no other lading. That's what came down the quay, and that's what I wrote."

"And the island?"

"I didn't see the island, sir," her father said. "I only saw the salt."

The man from London almost smiled, and he wrote that down too.

• • •

That evening Joan walked home with her father along the river, with the tide out and the boats lying over in the mud.

"You didn't say about the island," she said to him. "You know that's where they've gone. Everybody knows."

"Everybody says," her father said. "That's not the same. I know what came down the quay, because I saw it and I wrote it. I know what Master Croft says, because I heard him. I don't know what's out there. I'd be a fool to swear to it, and a worse one to write it in the king's book."

Joan walked a while, thinking.

"But is it there or isn't it?" she said.

"I don't know, my love," her father said. "But I'll tell you this. It's a better thing to go and look than to sit in a tavern and swear. Those men have gone to look. Whatever they find, or don't, they'll know, and we won't."

• • •

In the spring Einar went home to Iceland on his master's ship, with the first fleet that went north for the fish. Joan went down to the quay to see him off.

He was taller than he had been, and his voice had gone strange. He stood on the deck among the barrels with his bundle on his shoulder like a sailor born. He had told her a hundred times what it would be like when he got there. There would be the black sand and the white mountains, the fish drying on the racks, his mother's house with grass growing on the roof, and the sun that would not go down all the long summer. She had believed most of it by now. She had decided that the rest she would believe when she saw it, which would be never, and that was all right too.

The ship went out on the morning tide, down the river and through the gorge. Joan stood at the end of the Marsh where she had stood for the *Trinity* and the *George* and watched it go. That night she went out into the yard and found the star that does not move, low in the north over the roofs. She thought of Einar under the same star, farther north every night. She thought of it standing higher and higher over the sea as he went, just as he had said, until he was home.

Of the two ships that went to look for Brasil, she never heard another word, and nor did anybody else.

The master page

The Ledger spread this page across a clerk's desk on the quay, with a day book open on it at the salt, and a chart beside it with a round green island on it that nobody had ever been to.

In ink: Lloyd's voyage of 1480, in Worcestre; the *Trinity* and the *George* in 1481, forty bushels of salt, Thomas Croft's inquiry, and the purpose, to search and find the Isle of Brasil; John Day's letter of the winter of 1497, saying that the men of Bristol had found Brasil in the past. In pencil: anything the ships of 1481 found; any join from that sentence of Day's to land in the west; Joan, her father, Einar. *Found destroyed*: the sailors' own account, which went into the taverns and out of the world.

The page went on the pile. The record was Egypt's and the catalogue Nadia's, and the pieces had gone out across the world. The last few pages had gone from plunder to pledge, from scratched names to a sheriff's count, and then to a king paying gold for a story at San Juan de la Peña. Bristol added the rumour: a thing everybody says and nobody has seen, strong enough to put two ships to sea, which is the one good use of a rumour. Behind it, inked, is Aragon, and the king's gold paid for a cup because of what was said of it. Ahead, still in pencil, is the house of Lee in Lanarkshire, and a red stone in a silver coin that the ministers of Glasgow have been asked to judge.

The Core is good for one thing here: how high the star that does not move stood over Bristol that July. What the *Trinity* saw is beyond it, and beyond everybody.

Is it there? Nobody's been, the girl said. The answer was the father's: it is better to go and look than to sit and swear, and when you come back, write down what you saw.

The girl needed it so you could go west on the open sea and not get lost. The star tells you where you are and nothing else, and never pretends to more, and her father's customs house worked the same way: write what came down the quay, keep the island out of the king's book, and let the men who go and look be the ones who say.

A hundred and fifty years on, in Scotland, a farm girl was leading a sick cow up the road to the house of Lee, where the laird kept a stone that was dipped in water without a word being said.

On the chart beside the day book the round green island sat where it had always sat, west of everything. When her father's back was turned the girl drew a small hook beside it in the margin, and he never noticed, and nobody ever rubbed it out.

CHAPTER 33

The Stone Dipped Without Words

The house of Lee, by the Clyde in Lanarkshire, in the autumn of the Covenant, when the ministers came to ask about the stone: the found year, 1638 CE



Illustration 34. *The stone in its coin held over the bowl of water at Lee.*

Belfast, by day

THE SMALL HOOK IN THE margin stayed with him into the morning. The rule they built into Evidence OS was a simple one, and he was the one who had insisted on it. Nothing went in without its source attached, and if a claim had no source, the software flagged it, plainly. It stayed flagged until somebody either found the source or took the claim out. Early on, somebody asked whether there could be a box you could tick to say a claim was trusted, so it would not keep getting

flagged. He said no to that, flatly and at once. If you trusted it, he told them, you could find where it came from, and if you couldn't find where it came from, you didn't trust it, you just liked it.

The box nobody could tick was in his head when the Lee Penny came up on the screen.

"The Lee Penny is a magic amulet brought home from the Crusades in 1330," it said. "In 1638 the Church of Scotland declared its healing powers genuine."

The machines brought back the Penny's papers, and he checked them himself, his own stone resting on a photograph of a small dark red stone set in a silver coin.

The stone is ink: a small dark red stone, about the shape of a heart, set in a silver groat, which the Lockharts of Lee have kept for centuries. It was dipped in water, twice and a swirl, and the water was given to sick cattle, and to people, and nothing was said over it.

He put a line through *the Crusades* and *1330*, and moved them to pencil. The family story is that Sir Simon Locard went with James Douglas in 1330, carrying the heart of Robert the Bruce against the Moors of Granada. The story goes on that at Teba he took an emir prisoner. The emir's mother paid part of the ransom with this stone. It is a good story, and the family has told it for generations. But the coin the stone is set in was struck for Edward IV of England, more than a hundred and thirty years after Teba. The stone may be older than its setting, but nothing shows that it is.

Then he crossed out *declared its healing powers genuine*. The ministers did nothing of the kind, because a man called Gavin Hamilton of Raploch complained of Sir James Lockhart of Lee for using a charm. The Synod of Glasgow looked into it, and it found that the custom was only to put the stone in water. The water was given to diseased cattle to drink, without using any words such as charmers and sorcerers use. And they said that God had given stones and herbs a special virtue for healing that no man's wit could explain, and they saw no ground of offence. They told the laird to use it with the least scandal he could, which is not the same as saying it works. It is saying there is no harm in a stone, so long as nobody says words over it.

He put the year together from the Covenant up, signed in the spring of 1638. Over it went the General Assembly sitting in Glasgow that winter, the ministers sure of themselves. On top were the Synod and the stone and the complaint of Raploch. Through all of it showed *1638 CE*, and he wrote it down without any fuss, and its question mark the same way.

Then the doubt: *The Synod's act is quoted in the books and the year given is 1638, but the minute itself I have not seen. Elspeth is mine, and her cow, and the steward of Lee*.

Beside the house of Lee on the map he wrote *no words*. A box nobody can tick, aye. He took his stone off the photograph and held it, and went on holding it, thinking about the box.

That night

As he had when he was a boy, he went to sleep with the grey stone in his fist. Byres and wet bracken and peat filled the cold, with cows lowing somewhere in the dark and the Clyde going over stones. The Ledger's last page of this kind lay on a kitchen table in a big house, with a basin of water beside it, and the Ledger left him to it.

• • •

Elspeth led the cow up the road to Lee, and the cow did not want to go, and neither did Elspeth.

She was ten, and her father had the farm at the foot of the brae. The cow was Bessie, the best milker they had, and for a week she had been off her feed and standing with her head down and her milk drying up. Her father had tried everything he knew. Then her mother had said, quietly, in the voice she kept for things she was not sure of, that they could ask at Lee.

Everybody knew about the stone at Lee, because the laird had it. It had been in his family since before anybody could remember. If you had a beast that was sick, you could take it up to the house and the

steward would give you the water. People came from all over. Her mother said they came from England, even, once, with a great sum of money, to borrow it.

"Does it work?" Elspeth had asked.

"Folk say so," her mother had said.

• • •

The steward of Lee was in the yard when they came, an old man with white hair and a leather apron, and he looked at Bessie and felt her nose and looked in her mouth. Then he looked at Elspeth.

"Wait here," he said, "and keep her quiet."

He went into the house and came back out with a wooden bowl of water from the well, and a small box. He set the bowl on the mounting block, and opened the box, and took out the stone.

Elspeth had thought it would be big, but it was tiny. It was a little dark red stone, the size of her thumbnail, shaped like a heart, set in a silver penny with a king's head on it. The steward held it by the edge of the silver between his finger and thumb.

"What's it for?" Elspeth said, before she could stop herself.

The steward looked at her a long moment.

"For the beasts," he said. "And for folk, sometimes. Now hush."

He dipped the stone into the water once, and lifted it out, and dipped it again, and lifted it out. Then he dipped it a third time and swirled it round in the bowl, once, like you would swirl a cup to get the last of the milk. And all the time he said nothing at all, and his lips did not move. He did not cross himself, or look up, or mutter, as the old wife in the glen did when she did her charms. He just dipped it, twice and a swirl, and put it back in its box, and closed the lid.

Then he held the bowl to Bessie, and Bessie, after a while, drank.

• • •

That spring, before Bessie fell sick, the whole parish had gone into the kirk to put their names to the Covenant.

Elspeth had gone with her father and mother and stood at the back, in the crush. She watched while the minister read out the long paper from the pulpit and the men went up one at a time to the table under it. Some of them wrote their names with a flourish, the laird's men and the merchants from the town. Some of them wrote slowly with their tongues between their teeth. Her father could not write. When it came to his turn he took the pen from the session clerk. He made his mark, a cross, carefully, the way he made the mark on a sack of meal. The clerk wrote his name beside it for him.

On the way home she asked him what he had put his mark to. He said it was a promise to keep the kirk pure, with no bishops and no popish things in it, and no charms. He said it like it was a great matter, and it was; there were men in Edinburgh, he said, who had signed it in their own blood.

She thought about that all through the summer, walking to the well and back. She thought about it again on the road home from Lee, with Bessie walking easier behind her on the rope and her mother quiet at her side. The parish had promised no charms, and her father had made his mark. And now they had been to Lee and come home with a cow that was drinking again. Nobody had said a word over her at all.

• • •

Bessie got better within the week.

Whether it was the stone she never knew. It might have been, or it might just as well have been the walk, or the cold water, or the week being over. Her father said it was the stone. Her mother said it was God's will, and the stone, maybe. The old wife in the glen said it was nothing to do with the stone. It was all to do with the herbs she had put in the byre, which nobody had asked her to do.

But everybody in the parish heard that Bessie had gone up to Lee sick and come back well, and that was how the trouble started.

Then Gavin Hamilton of Raploch went to the ministers, for reasons nobody at the foot of the brae ever knew. He told them the laird kept a charm, and used it, and let folk come from all over to be charmed. This, he said, was witchcraft, or near it, in a godly country, in the very year of the Covenant.

That autumn two ministers came out from Glasgow to see.

• • •

They came to the farm at the foot of the brae, because Bessie was the latest. They sat at the kitchen table in their black coats, and Elspeth's mother gave them oatcakes and was frightened.

The older minister did the asking, and he was not unkind. He asked her father what had been done at Lee, exactly, and her father told him, and then he turned to Elspeth.

"You were there, lass," he said. "Tell me what you saw. Only what you saw, mind."

Elspeth stood up straight, the way she did in the kirk.

"He took the water from the well," she said, "and he took the stone out of the box. He dipped it twice, and then he swirled it, and then he put it back. Then he gave the water to Bessie."

"And what did he say," the minister asked, "when he was dipping it?"

"Nothing."

"Nothing at all, you're sure of that? No words, no prayer, no rhyme, nothing in Latin. Think hard, now, before you answer me."

"Nothing," Elspeth said. "His mouth didn't move. I was watching."

The minister looked at the other minister. Then he leaned forward across the table, and his face was very serious.

"And why do you think he said nothing, lass?"

Elspeth thought hard, like he had told her to. She thought about the old wife in the glen, who muttered, and about the steward, who did not.

"Because it's just a stone," she said. "You don't say words to a stone."

• • •

The older minister was a while answering. Then he put his hand in his coat and took out something and laid it on the table.

It was a pebble, a grey one, smooth, the size of a bairn's fist. It had come from a beach somewhere, and it was the kind you pick up without thinking.

"My daughter found that," he said, "when she was about your age, on the shore at Largs. She gave it to me. I've carried it twenty years." He turned it over with one finger. "What's it for, would you say?"

Elspeth looked at it. She had one of her own, from the burn, in her pocket, that she had never shown anybody.

"It's not for anything," she said. "It's just for keeping."

The minister laughed, suddenly, and the other minister looked startled, and her mother let her breath go.

"Aye," the minister said. "Aye, that's about the size of it." He put the pebble back in his coat. "There's no harm in a stone, lass. God made them, and put what virtue in them He saw fit, and it's not for us to know the half of it. The harm's in the words. The harm's in folk saying more over a thing than they know."

• • •

The next day Elspeth went down to the burn below the farm. She went to the bend where the water goes over the flat stones, which was where she had found her pebble.

She had been six or so, and it had been summer. Her grandfather had been with her, sitting on the bank with his pipe while she paddled. She had seen it under the water, grey and smooth among all the brown ones. She had put her arm in up to the shoulder to get it, and come up soaked and pleased. Her grandfather had laughed and asked what she wanted with it, and she had not known. She had only wanted it.

She sat on the same bank now and took it out of her pocket and turned it over in her hands. It was the same as it had always been, grey and smooth and cool, with nothing special about it at all. It was the kind of stone there were thousands of in every burn in Scotland. She thought about putting it back where she had found it. Then she put it back in her pocket instead and went home.

• • •

The ministers went back to Glasgow, and in time the Synod gave its word. The steward of Lee told everybody, and everybody told everybody else.

They found no offence in it. The stone was dipped in water without any words such as charmers use, and God had given stones and herbs their own virtue, and there was no sin in that. Only the laird was to use it carefully, with as little scandal as he could.

That winter, on the clear nights, Elspeth went out to the byre to see to Bessie. Afterwards she stood for a while in the yard after, with her own grey pebble from the burn in her hand, looking up.

Her grandfather had shown her the stars when she was small: the plough with its two pointers, the one that does not move, and the seven stars together, close as hens on a roost, that told you when to sow. He had never said a word over them either. He had only shown her where they were, so she could find them again, and she always could.

She held the pebble and found them, one after another, without saying anything at all, and then went in.

The master page

The Ledger laid its last page of this kind on a kitchen table in Lanarkshire, with a basin of water beside it, and a small red stone in a silver coin, and a grey pebble from a beach.

In ink: the Lee Penny, a red stone in an Edward IV groat, kept by the Lockharts of Lee; the custom of dipping it, twice and a swirl, without words; Gavin Hamilton of Raploch's complaint; the Synod of

Glasgow's finding of no offence; the Covenant year, 1638, as the year usually given. In pencil: Teba, the Bruce's heart, the emir's mother; the English borrowing in a time of plague; Elspeth, Bessie, the steward, the ministers, the grey pebble from Largs. *Found destroyed*: nothing, this once. The stone is still there.

He laid the page with the rest. Egypt had its record, Nadia its catalogue, and the pieces were scattered across the world. Prize, pledge, marks, inventory, claim and rumour: from Caesarea to Bristol, every page had been about what people wanted a thing to be. Lee's part was the charm, and the undoing of the charm: a stone people wanted to be magic, judged by hard men, and let be, because nobody said words over it. That is where the stone on his own windowsill had been heading all along. Behind, in ink, is the beach where the book began, and a boy who said a stone was just for keeping. In pencil ahead, and then in his own hand, is himself.

The Core could tell him what the red stone is made of and when the groat was struck. It could not say whether Bessie got better because of the stone, and Elspeth would not have thanked it for trying.

What's it for? For the beasts, the steward said. Just for keeping, the girl said, and the minister laughed. It was the oldest answer in the Ledger, the one he had given first, on a beach, at six or so, and it turned out to be right.

Elspeth's reason was that you can always find the stars again. The dream's went further: the sky was the first thing anybody ever kept without saying words over it, and the stone in his pocket was the second. The ministers had come up from Glasgow ready to find a charm and gone home having found a stone, and said so in writing, which was more than he would have bet on.

On the next shelf there was no stranger's child. There was a shingle beach on the north Antrim coast, and a wee boy of six or so bending to pick up a plain grey stone while his mother walked at the water's edge, and this time the one who would have to go looking was him.

PART VI

Recovery

Reading It Together

CHAPTER 34

A Boy and a Beach

Belfast, and the coast, from the late 1990s to the summer he finished school: the found year, 2009 CE



Illustration 35. *The shingle beach on the north Antrim coast.*

Belfast, by day

ON EVERY SITE HE HAD ever started, the first morning went the same way. You sat in a portacabin with a cup of tea going cold. A man with a clipboard took your CSR card and your SafePass and read them against a list. Then he looked up at your face and back down at the card, and up again, and he was not being rude. He was checking that the man on the paper and the man in the chair were the same man, because if they were not, nobody on that site could trust a word you said after. Gary had sat through it a dozen times and never minded it once. It was the fairest wee thing on any site, because everybody got checked and the checking was the same for all of them.

That autumn he turned the clipboard round, because he had spent a long time going through the dead with a pencil. It seemed only fair to go through the living the same way, starting with the one he knew least about on paper.

He asked one of the machines what it could tell him about a man called Gary Leckey, born in Belfast. It came back quick and sure of itself.

"Gary Leckey was born in Belfast in 1991," it said, "and grew up in a working-class district of the city." It went on without a pause. "He left school without qualifications. He spent his teenage years labouring on building sites before turning to technology."

The machines went looking, and he checked what they found himself, with the stone sitting on the first page of an old CV. There was not much. Most of it was his own. There was the company register, with the year he was born and the city. There was a CV he had typed in 2016, when he was still saying *a very pro-active person* about himself and meaning it. Under education it said GCSEs, 2009. Under work, at the very bottom, it said kitchen porter, a restaurant in Belfast, from February 2009. Under interests it said fishing, walking and camping, in that order, as if he had thought about the order.

He struck out *without qualifications*, because the exams were there, in his own typing, and the year with them. Then *labouring on building sites* came out as well. The first job was a sink full of pots, and the sites came a long way later, when he was past thirty. The working-class district he let stand, because it was true, but he did not let the machine name it. When it tried he put a line through the name too. It was not that the name was wrong, only that nothing he had handed it said so. A thing nobody handed you is a thing you made up.

His own year he laid out the same way as all the others, with the register at the bottom, with 1991 and Belfast. Above it lay the CV, with the exams in the summer of 2009 and the kitchen that had started in the February of the same year. For a few months he had been doing both. On top went the stone on the windowsill, and a summer on the coast that no paper had ever recorded at all. The year he could stand on was *2009 CE*, and it went in at the top with *2009 CE?* beneath it. It

felt daft to put a question mark after a year he had lived through. He put it there anyway, because being there is how most of the mistakes get in.

Below it, the doubt: *The beach is the late 1990s, six or so, and no paper says which summer. The rest of the boyhood is mine to remember and not mine to prove. The old man at the mill, the man on the pier and the two in the tent are mine.*

On the map he wrote, beside Belfast, *home*, which he had never once needed to write before. The man with the clipboard would have passed it, aye, just about. He thought about the clipboard man a while, and put the stone back on the windowsill, where it had sat for most of his life.

That night

That night the Ledger did not take him to a stranger's shelf. It took him along the stacks to a low shelf near the door, where the books were cheap and the spines were cracked. It opened one at a page with no date at the top, only the question mark. There was a beach on the page, and a wee boy bending over the stones, and a woman walking at the water's edge with her coat open. This time the Ledger left the page where it was and waited for him to go in.

• • •

The stone was warm all the way home in the car. When they got back to Belfast it was still warm, because he had not let go of it once.

He put it on the windowsill of his room, between a toy lorry with one wheel off and a jar with nothing in it. In the morning it was cold again, and pale, and he picked it up and held it until it warmed. He did that every morning for a week before he stopped needing to, and after that it just sat there. From the sill you could see over the backs of the houses and the yards to the two yellow cranes. They stood up over the shipyard a long way off, with letters on them that he could not read yet from that distance.

"Is it still there?" he said one night, out of the dark, to nobody.

It was still there. He could see the shape of it against the street light, small and flat, sitting where he had left it. That seemed to him, the same as the sea had, a thing you could count on.

• • •

There was an old man at the end of the street who sat on a kitchen chair outside his front door whenever it was dry. He had a cushion on the chair and his cap on, and he talked to whoever went past. Most people went past quicker than he wanted, but Gary went past slower, because the old man knew things.

He knew the cranes: the big one was Goliath and the bigger one was Samson, he said. They could lift a whole ship's engine like a bag of messages. He knew the shipyard had built the biggest ships in the world, more than once, and that men had walked to it in the mornings in a crowd so thick you could not see the road. And he knew the rope-works.

"Biggest in the world," he said. "Rope for every ship on every sea. In the old days, before the machines did it all, there was a shed longer than any street you know, and a man walked backwards down it all day, spinning the yarn out as he went, and you never saw his face. You only saw his back going away from you, down to the far end."

"Why did he walk backwards?"

"Because that's how you make a rope, son. Somebody has to hold the end of it and keep going."

Gary thought about the man walking backwards for years afterwards. He thought about him more than he thought about the ships.

At the bottom of the next street was a mill. It was the biggest building he knew that was not a church, five storeys of dark red brick with rows of windows going up it, and every window had its glass out, and pigeons went in and out of the holes. Weeds grew out of the gutters, and in one place a whole small tree was growing out of the wall, high up, as if the wall were the ground. Someone had put a fence round it, with a sign on the fence. He and the old man looked at it together one afternoon, over the fence.

"What was it for?" Gary said.

"Linen," the old man said. "Flax in at one end and linen out the other. Half the tablecloths in the world came out of this town. They called it Linenopolis, the whole city. That's a big word for a wee place."

Gary asked who had worked in it.

"Everybody's mother," he said. "Mine went in at twelve."

Gary looked at the windows with the glass out and tried to put everybody's mother behind them, at the machines, with the noise. He could not do it; it only looked like a building with no eyes. The next summer it burned, the way empty mills did. The fire engines came, and after that it was a fence and a long heap of brick with the weeds already coming through. He went down to look at the heap the morning after, with the smoke still coming off it in a thin line. He picked up a half brick that was still warm and held it. Then he put it back, because it was not his, and it did not want to be kept.

He told the old man about it.

"Aye," the old man said. "They go one at a time. You don't notice till they're all gone."

• • •

In the summers there was the coast, and the coast meant fishing.

He was twelve or so, standing at the end of a stone pier down the coast with a hand line wound on a wooden frame and a string of coloured feathers on the end of it, jigging for mackerel. The water under the pier was green and very clear, and you could see the weed moving on the stones at the bottom, like hair. When the mackerel came in they came all at once, in a shoal, and the line went mad in your hand. You hauled up four or five at a time, striped blue and green and flapping on the stones. When they did not come, you stood there for an hour and nothing happened at all.

There was a man beside him on the pier who came most days, with a proper rod and a stool. He said little. He smoked and watched his float and said "good lad" when Gary got a fish, and nothing when he lost one.

"Are there salmon here?" Gary said.

"Not here, no. They go up the Bann, the salmon, over at Coleraine and up past it. In their season, summer and back end, when there's water in the river."

"How do they know when?"

The man thought about it, which Gary liked. "Nobody tells them," he said. "The water tells them. They've been doing it since before there was a Coleraine to go past."

From the end of the pier you could see Scotland on a clear day, low and blue across the water, the way you could from the beach where he had found the stone. The stones on the shore under the pier were grey and round and wet. He looked at them and said, mostly to himself, "The ice brought them." It was a thing he had been told once and had kept.

"It did," the man said. "And the sea polished them. Watch your line."

At the end of the day the man showed him how to gut the fish on the edge of the pier, with a knife no longer than his finger, one cut and a thumb run along the inside. The gulls came down and stood about at a polite distance and waited. Gary was slow at it, and the man let him be slow. He took the fish home in a plastic bag with the blood going brown in the corners, and they were eaten, and he was prouder of them than of anything he had brought home from school.

• • •

When he was older there was walking, and walking turned into camping, because if you walked far enough you might as well not come back that night.

The October before the exams, he went up into the Mournes with two mates from school and a tent that one of them had borrowed off his uncle. They carried it in turns, and it was heavier than it looked. They argued about which way the path went, and they were all wrong in different directions. They put the tent up in the end on a flat bit of grass above a stream with the light going, and it leaned, though

nobody minded the lean. They ate beans out of the tin, cold, because the wee stove would not light. They laughed about the stove until they could not breathe.

When it was properly dark Gary went out of the tent on his own to take a leak and stayed out. He had never seen the sky so crowded. In Belfast you got a handful, and they were always in a hurry behind the orange glow off the streets. Up here there were thousands, and more coming out the longer he looked. The Milky Way went over the top of them all like smoke.

"What are you at?" one of the others said, sticking his head out of the tent.

"Looking. That's the Plough," Gary said. He had learned it out of a book from the library and never seen it properly till now. "And if you go up from the two at the front, away from the handle, there's the one that doesn't move. Everything else goes round it."

"You don't know it doesn't move."

"I don't. The book says. Give it an hour and we'll see."

They gave it an hour. His mate brought out a sleeping bag and the two of them sat in it on the wet grass. The third of them was asleep and snoring, and the Plough swung slowly round over the black line of the mountain, and the one star stayed where it was. Something went through him, sitting there, that he hadn't a word for. It was not that the book had been right. It was that he had checked.

• • •

The last exam was on a hot day in June, in the sports hall, with the rows of single desks and the clock on the wall at the end that everybody watched instead of the paper. When the man at the front said they could put their pens down, Gary put his pen down and sat for a moment with his hands flat on the desk. Then he went out through the double doors into the sun, and that was school done.

Some of the others were going on, and some were going nowhere in particular, and some were going to the beach to get drunk. Gary was going to work, because he had been going to work since Febru-

ary. He had taken the job in the restaurant kitchen while he was still at school, two evenings and a Saturday. Now it would be every evening and every Saturday, and some mornings too.

That evening he was at the sink by six. The pots came back from the line black and stuck and hissing. He took them one after another and scrubbed them and rinsed them and stacked them. There was a heavy green serving plate that the head chef liked better than any other dish in the place. When it came back he washed it on its own, slowly, with both hands. The chef saw him do it and said nothing, and that was as near to praise as that kitchen got.

Nobody talked to the boy at the sink, so the boy at the sink heard everything. He heard the waiters fall out over tips and make it up over a smoke at the back door. He heard the chef on the phone to his bank, quiet, and then loud. He heard a woman at table nine tell the man with her that she was leaving, and the man ask for the dessert menu like he had not heard. Gary put none of it anywhere. He scrubbed the pans and let it all go past him down the drain with the water, and only kept the way people said things, which was different from what they said.

He walked home after eleven with his hands wrinkled from the water and smelling of the soap. The results would come in August, in an envelope. He did not think about them much. When he got in he went up to his room in the dark and did not turn the light on. The grey stone was on the windowsill where it always was, small against the street light, with the cranes beyond it over the roofs.

He picked it up, and it was cold, and he held it till it was not.

The master page

This page lay on a windowsill in Belfast, between a jar of pens and a plant that would not die, and the stone sat on top of it like a paperweight, because it was the oldest thing in the room.

In ink: born in Belfast in 1991; GCSEs in the summer of 2009; a restaurant kitchen from February of that year; fishing, walking and camping, in his own hand on his own CV. In pencil: the beach year, the old man and his chair, the man on the pier, the two mates and the

tent that leaned. *Found destroyed*: the mill at the bottom of the next street, and the others like it, which went one at a time while nobody was counting.

He put this page with the others. Egypt's record, Nadia's catalogue, the scattering of the pieces and the stone at Lee let be a stone: all of it ran up to this. Belfast's part was the keeping. It was a boy who kept one thing and let the half brick go, and who checked the one star that does not move because he did not want to take the book's word for it. Inked, behind it all, is the beach where the book began. Pencilled ahead is a city over the water and a room full of files.

He gave the Core what it could use. It knows how fast the Plough goes round the pole, and how many cranes the shipyard had. It does not know why one boy stopped at the old man's chair when everybody else went past. He hardly knew that himself.

What was it for? The old man had said linen, and he had been right, but that was the mill's answer, and the boy had wanted the other one. It was for the people inside it, who were gone now. Somebody has to hold the end of the rope and keep going.

The boy had gone out to look for the star that does not move to see was the book telling the truth. A star you have checked yourself is a different star from the one you were told about, even though it's the same star, and he had known that as a boy without having the words for it. He kept the thing that wanted to be kept, and let the half brick go.

Next came a city across the water, and a solicitors' office in London, and a young man in a borrowed tie, learning what a statement was.

Under *Interests* on the old CV he had once typed *fishing, walking and camping*, and the Ledger had copied it in ink, exactly. Every word of it was true, and it left out the stone.

CHAPTER 35

The Long Way Round

Belfast, London, Donegal, Dublin, Belfast Lough and the Shannon, from the summer he left school to the spring he started on his own: the found year, 2022 CE



Illustration 36. *The lough-shore road at dusk, and the cranes.*

Belfast, by day

HE PUT THE STONE ON the old CV that morning. A banksman never says *nearly*. He had learned that at Shannonbridge, on the radio, with an operator in a cab sixty feet up who could not see the load he was moving. You said *stop*, or *down slow*, or *all stop*, and meant it. If you said *nearly there*, the man in the cab heard *there*, and somebody on the ground found out the difference with his hands. So you said what you could see, in the fewest words, and nothing you could not.

He had the same rule in front of him that afternoon. Three of his own CVs were laid out on the table, and none of them quite agreed.

He tried the machine first, for speed, and it gave him a man he half recognised.

"Gary Leckey worked as a crane operator at Kilroot Power Station from 2022 to 2024," it said. "He holds a degree in computer science, and earlier he was a network engineer with Cisco."

It was the machines' job to find the papers and his to read them, by hand, with the stone on the corner of the oldest CV. He put a line through *crane operator* first, because he had never once sat in the cab. He had slung the loads and checked the shackles and the chains, and stood where the operator could see him, and talked the steel down by hand. In the end he led the lifting team, which is a different job from driving the crane. A man on a site would tell you so in a hurry.

Then he struck out *computer science*, since the degree was a BBS, a Bachelor of Business Studies with honours. It came from the institute in Letterkenny that is part of the Atlantic Technological University now, and he had finished it about 2016. He had learned the computers on his own time and nobody had given him a certificate for it. *Network engineer* went too, because the old profile said technical adviser, in Belfast, from the summer of 2016. That came after a spell on the phones for Vodafone, and it was a man with a headset, not a man in a server room.

That left *2022 to 2024*, and he did not put a line through it, because he could not. One CV said Kilroot from 2022 to 2023. Another said from September 2022 to October 2024. Shannonbridge came after Kilroot on both, either in 2023 or from the autumn of 2024. He had written every one of them himself, at different kitchen tables. Each was done in a hurry, for a different job. It was a strange thing to find that the one record you would swear to was the one that had drifted.

He laid the three CVs on top of each other, oldest at the bottom, with the kitchen and London and the stock-takes and Donegal, and nothing after 2016. Over it went the master CV, with the crane hire in Dublin and Kilroot after it. On top lay the profile, with its longer Kilroot and its later Shannonbridge. Only the two later ones put him under the cranes, and the one year they agreed on was the year he

started. He put *2022 CE* above the rest, and the question mark after it was a real one, because three CVs had three versions of the same years.

The doubt went under it: *Kilroot ends in 2023 on one CV and in 2024 on another, and Shannonbridge starts in one year or the other, and I have left the ends loose. The operator, the partner, the counters and the foreman are mine.*

The map got one line from him, with a lot of corners in it. It ran from Belfast to London and back, over to Donegal and back, down to Dublin, round the lough to Kilroot and across the middle of the country to the Shannon. Aye, he thought, the long way round. He set the stone down at the end of the line, where it came home.

That night

That night the Ledger opened the same low shelf by the door and took down a thinner book. It was the kind with a cardboard cover and a list of dates inside it. It skipped the beginning and opened near the end, at a page with a river on it and the noise of a radio. The date at the top was not sure of itself: 2023, or a year after, with its question mark. The Ledger said nothing, as it never did, and only held the page flat for him while he went in.

• • •

"Can you see it?"

The voice came out of the radio clipped to his vest, flat and tinny, from the man in the cab. Gary was standing on the ground at Shannonbridge with the wind coming off the river. The load was a long beam swinging slowly forty feet above the place where it had to come down. The operator could see the hook and the sky. He could not see the ground, or the two men waiting for the beam, or the scaffold pole somebody had left leaning where it should not be.

"I can see it," Gary said. "Hold it there for me, and keep it there."

The beam hung and turned a little on its slings, the way a thing does when it is waiting.

"Move that pole by the north leg," Gary said to the man nearest, and the man moved the pole. "Down slow now, nice and easy."

The beam came down slow. Gary watched the gap close between the steel and the ground, and talked it down a foot at a time, until there was no gap. "All stop, that's her down," he said.

"All stopped, and holding," said the radio, and the man in the cab breathed out where everybody could hear it.

Behind him the River Shannon went past, wide and brown and quiet, between flat fields of rush and bog. The station across the yard had burned its last turf a while before. There were men all over it now with work to do, and he was the one who could see. Nobody had told him that would be his job when he left school, and nobody could have.

• • •

And the page went back.

It went back fast, through the years, the way the beach had gone back through the ice when he was a boy. He let it go. It stopped in a basement in London in the spring of 2010, with strip lights and steel shelves and a smell of dust and toner.

He was the office junior at a firm of solicitors, in a borrowed tie and a shirt he had ironed himself, badly. He fetched files and carried files and photocopied files, and he put them back where they lived, in boxes with numbers on the ends. He knew nobody in the city, so at night he walked by the river because it was free, and looked up for the stars. There were none, only the orange glow and the aircraft going over one after another into the airports.

On his second week a partner came down to the basement looking for a bundle. He was a tall man with his glasses on his head, and he found Gary reading a witness statement he should have been copying.

"Did you change anything in that?" the partner said.

"No. There's a spelling wrong in it, but."

"Leave it wrong," the partner said. "Never tidy a statement. It's what she said, the way she said it, signed and dated. If you tidy it, it's yours, and it's worth nothing."

Gary left it wrong, and he found, after that, that he could not stop noticing the places where people had tidied things.

• • •

There were years after London that were mostly nights.

He counted stock for RGIS, in shops after they had shut, with a scanner in his hand and a partner on the next aisle. Every shelf was counted twice, by two people who did not look at each other's numbers. If the counts did not agree, a third person counted it. Nobody was allowed to split the difference.

"Forty-two on the beans," said the woman on the next aisle one night.

"I've forty-one, and I counted them slow."

"Then one of us is wrong, and it's not going to be me, love."

"It might be," Gary said. "That's why there's a third."

The third count came, a quiet man from Lisburn with a cold. It was forty-one, because one tin had been pushed to the back behind the soup. The woman laughed at herself and bought Gary a tea from the machine at the break. He never forgot that either: that she laughed, and did not argue with the shelf. He liked that more than he could have told anybody at the time. It was three in the morning in a shut supermarket, with the music off and the freezers humming. Two strangers counting the same shelf of tins and getting the same number felt like a small kind of honesty in the world.

There were years after that across the border, in Donegal, studying business at the institute in Letterkenny with people who were mostly younger than him. He sat at the front because he had paid for the seat and wanted his money's worth. On days off he walked out along the beaches on the Atlantic side. They were long and empty and pale, with the wind coming in off three thousand miles of nothing. He stood at the edge of one with his hands in his pockets and looked west, at the place where the water went on past where you could see. He thought about nothing in particular, and remembered it for years.

An old man walking a dog came along the sand behind him. He stopped, because the dog had stopped.

"You're looking at America," the old man said. "Next parish. My uncle went out from this beach, near enough, and wrote home twice."

"Is it far, from here?"

"It's the next parish," the old man said again, and that settled it, in a way. "Far is only how long you're gone."

He went on with the dog. Gary stood a while longer and watched the water go out to the edge of what he could see and keep going.

He finished the degree about 2016, honours, and came back to Belfast to a headset at Vodafone. He typed what the customers said, not what he thought of it. Then he moved on to a desk answering the harder questions about other people's networks. After that he was back in Letterkenny, running the office at Orbit Security. There the incident book had one rule written inside the cover in somebody's capitals, NEVER TIPP-EX. If you wrote a time wrong, you put one line through it, so it could still be read, and the right time beside it with your initials. He did that so often in his own notes, afterwards, that people asked him why.

"Because it shows when you're wrong," he said. "If you hide it, nobody can check you."

• • •

Then came waste. He sat on a service desk, with half the calls about the bin calendar, and people who had not had their brown bin lifted in a fortnight and wanted somebody to say sorry and mean it. Then came a run of years keeping the paperwork for six small businesses at once. He did the wages and the invoices and the tax, and learned how close to the edge of the month most people lived. He was good at all of it, the man people rang when the numbers did not add up. He found the reason, and it was always something small that somebody had tidied.

In the evenings, at the table, he read the old stories, and then the papers about the old stories, and then the papers about the bones. A team in Dublin had read the DNA of a woman buried at Ballynahatty, on the edge of Belfast. She was more than five thousand years dead. They had read three men from Rathlin Island too, off the coast he had

found the stone on. Later the same team read a man from the heart of Newgrange. A paper on Iberia showed whole peoples moving across the map like weather. He did not understand all of it, so he read it slowly, with a pencil. The parts he did not understand he wrote in the margin as questions, each with its mark after it.

• • •

And in 2022 he went under the cranes.

He took his tickets first, because you could not stand on a site without them. There was the CSR card in the North and the SafePass in the South. There was the slinger banksman card, which meant you could be trusted to hang a load and talk it home. He went down to Dublin with a crane hire firm and slung loads on sites in the city, learning the weight of things by the look of them. Then he was back in the North, at Kilroot.

The first morning he drove round the shore of the lough and turned in at the gate, and there was the chimney. It stood up out of the fields, a grey stroke against the sky. It was the same as when he was six or so, with his face pressed to the back window on the way to the beach. He sat in the car a minute before he got out and looked at it.

"You all right there?" said a man going past with his flask.

"Aye," Gary said. "I've been past it before, a long time ago."

He worked on the lifting crews there, and in the end he led one. He learned to walk round a load before he hung it, every time. He did it even when it was the tenth of the same thing that day, because the tenth one is the one that kills you. He learned that the wind decides, and the operator decides, and the slinger does not decide anything; he only tells the truth about what he can see. One winter morning, with the frost on the steel and the breath standing in front of everybody's faces, a young lad on his first week asked him how he always knew.

"I don't always know," Gary said. "I just say when I don't. If you can't see it, you say you can't see it, and everybody stops, and nobody minds stopping."

"The foreman minds, though. He minds a lot."

"The foreman minds less than he'd mind the other thing," Gary said. "Walk round it with me, and tell me what you see."

The lad walked round the load with him, slowly, and told him. He missed a pin that was not seated home, and Gary showed him. The lad went red to his ears, and Gary said that was what the walk was for.

• • •

The last page of that book, or near enough, was a desk at Total Cargo in Belfast in 2023. There was a manifest on it, and a phrase.

Said to contain. It went on every box nobody had opened. The shipper said there were four hundred cartons of tiles, and the paperwork did not say *four hundred cartons of tiles*. It said *said to contain*, because nobody in the office had looked. Gary read those three words a hundred times a week and liked them better every time. They were the most honest words in freight.

That spring he sat at his own kitchen table one night with a form open in front of him on the laptop. A company's name was typed into the box, and his own name under it as the director. He looked at it for a good minute. Then he picked the stone up off the windowsill and held it while he read the form again from the top. He read every line, the way he would have checked a load before he hung it.

The master page

On a cab seat the Ledger left this page, with a radio on it and a pair of rigger's gloves, and the stone sat on the gloves, with the river going past outside.

In ink: a kitchen in Belfast from 2009, a solicitors' office in London in 2010, stock-takes with RGIS, a BBS with honours from Letterkenny about 2016, the phones for Vodafone and the technical desk after it, the office at Orbit Security, the service desk, the six small businesses, a crane hire firm in Dublin, the lifting team at Kilroot, the banksman's radio at Shannonbridge, the manifests at Total Cargo. In pencil: the

ends of the Kilroot years, the start of Shannonbridge, and every face in the dream. *Found destroyed*: nothing he could name, only the evenings, which nobody keeps a record of.

It went on the pile with the rest. There was Egypt with its record, Nadia with its catalogue, the pieces gone abroad, Lee letting a stone be a stone, and Belfast keeping one. What the road gave him was the habit. Never tidy a statement. Count it twice. Never Tipp-Ex the book. Say *said to contain* when you have not looked. Walk round the load. He had brought every one of them to the dead, and the dead had stood up to it. Already in ink behind it are the windowsill and the boy. Ahead, pencilled, is a kitchen table and a company with nothing in it yet but a name.

In the morning the Core could show him how the years of his own life fall into rhythms, a job here and a job there and the gap between. What it cannot show is why a man who could not see the ground trusted a voice on the radio. That was earned, a lift at a time.

Can you see it? That was the whole job, on the ground at Shannonbridge, and it was the whole job at every desk he had sat at, and on every page the Ledger had ever opened for him. Somebody is up in the cab and cannot see. Somebody has to say what is there, and nothing else.

In London, by the river, you couldn't see the sky for the orange glow, and he had missed it, that was all. Every job since had been a way of reading something nobody else was looking at: a statement, a shelf, a queue of tickets, a load swinging over men's heads. The radio was the plainest of them. Say what you can see, say *stop* and mean it, and never once say *nearly*.

After it came a kitchen table in Belfast on the twenty-sixth of April 2023, and a company register, and a man with no coding experience at all about to teach himself the hard way.

Down by the river the radio hissed once, and a voice up in a cab that could not see the ground asked the only question it ever asked. Gary pressed the button and told him what was there, and nothing else.

CHAPTER 36

A Ledger Begins

Belfast, at a kitchen table, from the spring he started on his own to the summer the paper went up: the found year, 2023 CE

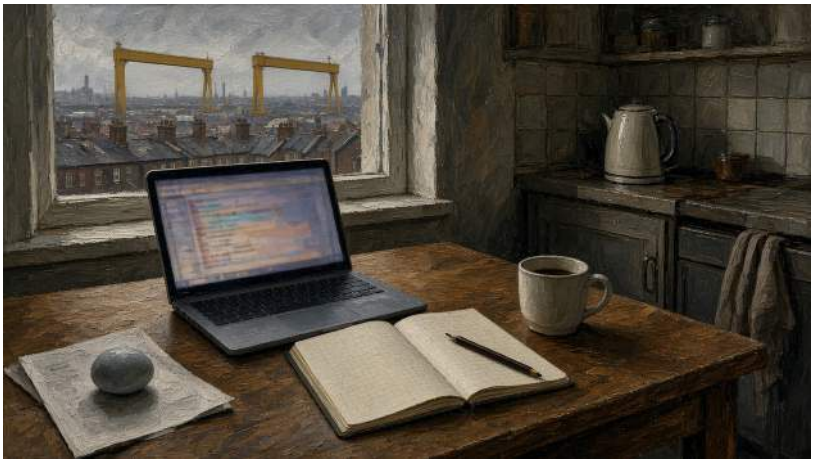


Illustration 37. *The kitchen table where the ledger begins.*

Belfast, by day

KEEPING THE BOOKS FOR SIX small businesses, out of carrier bags of receipts, he had learned how a company starts life. It begins as three lines on a form, which hold the name and the number and the date. The name could change, and often did, when somebody had a better idea over a pint in the pub. The number never moved, whatever else happened to the company. The date was whatever day the register printed at the top of the certificate, never mind what you remembered about it, and if you argued with the register you lost.

That came back to him when the screen showed his own company.

"Aureon Zorza Technologies was founded in 2025 by Gary Leckey, a former software engineer," it said. "That summer the company won Innovate NI's top award."

He had the machines pull the records and read them by hand, the stone on a printout of the register. The name on the register was not the one the machine had given. It was R&A Consulting and Brokerage Services Ltd, a private limited company, incorporated on the twenty-sixth of April 2023. It had one director, appointed the same day, and the director was him. Aureon Zorza Technologies was the name it traded under later, for the software and the research. Before that there had been other names on other ideas. The number had stayed the same through all of them.

He crossed out *founded in 2025*, and then *a former software engineer*, which he enjoyed. He had written it himself in an old README, and the machine could have read it if it had looked. It was plain enough. *With zero coding experience, I taught myself Python in flow-state sessions, made every mistake possible, fixed them, and shipped a complete autonomous trading system.* He had been a slinger, not a software engineer. The software came after, at night.

Top award went next. The certificate on the wall was real enough, in its frame from the pound shop above the table. It was from Innovate NI and dated the twenty-first of July 2025, and the words on it were Silver-level Certificate of Recognition. It was a good thing to have on a wall and nowhere near the top prize. He would not let a machine blow it up bigger than it was, because the next man to read the bigger version would be right to laugh in his face.

He stacked his own records, the register at the bottom with its date in April 2023, and the README over it, with its fifteen years and more of quiet building and its loud mistakes after. On top went the records he had put up in public. The first of them was a paper on the Harmonic Nexus Core, on the thirteenth of July 2025. He wrote *2023 CE*, the year the number began, across the top of the sheet, and beneath it *2023 CE?*, the hook in place. The question mark looked odd after the

one date in his life that was printed on a government page. He left it, because the date on the page was not the same thing as the day he had started. He could not have told you that day.

The doubt came under that: *The README says fifteen years and more, and nothing written down shows the first of them. The earlier trading names I have, but not their dates. The friend in the van and the young fella at the session are mine.*

On the map he put a small mark on Belfast, on top of the old one, and wrote beside it *here, again*. The register would not argue with that, aye. He lifted the stone off the printout and let the register's verdict stand. He turned it over in his fingers, the way he did when he was thinking about a number.

That night

That night the Ledger did something it had never done before. Instead of opening a book, it slid a blank one across the table to him. It was a cheap notebook with squared paper, the kind you buy three for a pound. It put a pencil on top of it. Then it stood back, like a foreman standing back when the new man is ready to sign for his first load. The page at the top of the pile had a date on it, April 2023, and a kitchen.

• • •

The form went through on a Wednesday, and the certificate came the same day by email with a number on it. He printed it on the wee inkjet that streaked every third line and slid it into a plastic wallet so it would not crease. Then he set it on the table between the teapot and the sugar bowl. R&A Consulting and Brokerage Services Ltd, it said across the top. The name under that was his own, and the table it was lying on was his as well.

There was nothing in the company yet but the name.

He knew how to keep books and run an office, how to talk a load down off a crane and count a shelf twice. A computer was another thing. Beyond what the buttons said he could not make it do anything

at all, and everything he wanted to build needed it to do what he said instead. So he started where everybody starts, on a free course at the kitchen table at night. There was a line on the screen that said *Hello, world*, and a machine that told him he was wrong.

It told him he was wrong a great deal.

IndentationError, it said, at one in the morning. He did not know what an indentation was. *NameError*. *SyntaxError: invalid syntax*, with a little arrow pointing at the place, and the place looked fine to him. He sat with it and read the line out loud. Then he counted the spaces with the end of a pencil on the screen, the way he would have counted tins, and found that one line had four spaces and the next had a tab that only looked like four. He fixed it and ran it again, and it ran.

"Ye boy," he said to the empty kitchen, and then laughed at himself for saying it out loud.

He lost track of the nights and went into the work the way you go into cold water. When he came out of it the window was grey and the kettle was cold and he had made every mistake possible, and fixed most of them. Some nights he went backwards. So he learned to keep a copy of the thing that worked before he changed it, which was the incident book's rule again, in a new place: never rub out the last good version.

• • •

A man he had worked with at Kilroot came round one Saturday with a van and a bag of rolls, to see if he was still alive. He sat on the other chair and looked at the screen, at the rows of coloured words, at Gary, back at the screen.

"What is it you're making, at all?"

"It's a trading system." Gary turned the laptop round on the table so he could see the columns of prices ticking over. "It reads the markets and decides for itself what to buy and when to get out."

"Does it work?"

Gary chewed on that along with the roll. He could have said yes, because it ran. He had shipped it, and every morning it started up on the laptop and did what he had told it to do without falling over. A crane will lift whatever you hang on it, though, right or wrong, and a level lift tells you nothing about the sling.

"I don't know yet," he said. "It runs. That's not the same."

"How will you know?"

Gary tapped the screen with the end of a biro, on the rows of prices the system had never been shown. "You test it on what it's never seen," he said, "and you write down what happens, and you don't get to change the answer after." He left the biro lying across the keyboard.

His friend tore a roll in half and chewed on that for a bit, with a crumb stuck to his lip. "That's the never-Tipp-Ex book," he said. He had heard the story of Letterkenny more than once, over a flask of tea on a cold morning at Kilroot.

"It is," Gary said. "It's the same book. I just keep finding it."

The van came round again six months later, with the same bag of rolls. The friend sat in the same chair like he had never left it.

"Well, does it work yet, or are you still not saying?"

"Still not saying," Gary said, and tore the last roll in half and handed him the bigger bit. "It's better than it was. It's not proved. Ask me again in a year, and I'll show you the book, the wrong bits and all."

"You're a hard man to get a straight answer out of."

"That is the straight answer," Gary said. His friend laughed with his mouth full and said he supposed it was. He pushed the paper bag across the table so the two of them could finish the rolls between them while the kettle came round to the boil again.

• • •

In the evenings after that, with the trading system ticking over in a corner of the screen, he went back to the thing he had been building longest. None of it had ever been code. It was in biro and pencil, on paper that had been folded in the back pocket of his work trousers till it went soft.

He had been carrying it for fifteen years and more. It lived in notebooks, and on the backs of delivery dockets, and in his head on long drives. It was a way of looking at patterns: lots of cycles running on top of each other, something keeping them steady, and the past feeding back in after a lag to shape what comes next. He had a name for it, the Harmonic Nexus Core, which was a big name for a thing that lived on the backs of dockets. He had never shown it to anybody who could check it.

He wrote it up properly over a long winter and into the spring, with the heating on low and a blanket over his knees at the table. Writing it down for strangers was harder than building the trading system had ever been. Every place where he had been sure in his own head turned out, once it was on paper, to need a reason. In some of those places the page under his pen stayed blank. He left them untidy and marked them for what they were, *untested*, and kept them in.

One night he gave the whole draft to one of the machines and asked it what it thought.

"This is a groundbreaking framework," it said, "with the potential to transform our understanding of complex systems."

He sat and looked at that for a while with his chin in his hand. It was the nicest thing anybody had ever said about the work, and it was worth nothing. The machine would have said the same about a shopping list if he had asked it in the same voice. He deleted it and asked again, differently.

He typed it with two fingers, hitting the keys hard enough to rattle the mug beside the laptop: "Find the weakest claim in this, the one a hostile reader would go for first. Don't be kind."

The answer was slower, and longer, and much less pleasant. It went for the place where he had let one good fit stand in for a test, and it was right. He sat up till three fixing it. In the morning he wrote on the inside cover of the notebook, in capitals, the way somebody had once written NEVER TIPP-EX: ASK IT WHERE IT'S WEAK.

"That's the one," he said to the window. "That's the only question worth the electric."

On the thirteenth of July 2025 he put it up on Zenodo, where anybody in the world could read it, and it got a number of its own that would never change. He pressed the button with one finger. Then he sat back from the table with his hands flat on the oilcloth and waited for something to happen.

Nothing happened. The kettle clicked off, and a car went by in the street.

"Well," he said to the stone on the windowsill. "That's that up, anyway."

A week later a certificate came from Innovate NI, Silver-level, with his company's name on it. He bought a frame for it in the pound shop and hung it over the table on the nail where the calendar used to go. It was nowhere near the top of anything, and he knew that. He hung it anyway, because it was the first time a stranger had looked at what he was doing and put it in writing that it was something.

• • •

That same summer the company put a bit of money towards the Street Soccer NI teams going to the Homeless World Cup in Oslo. Later he hosted a construction-skills session with them, in a room with plastic chairs and a projector that would not talk to the laptop. By the end of it everybody in the room had a CSR card. It was the same card that had let him stand under the cranes at Kilroot.

One of them, a skinny young fella who had kept his hood up the whole session, hung back while the others were stacking the plastic chairs against the wall. He stood by the radiator with his hands jammed in his pockets and asked him something, low, so nobody else would hear.

"Is it hard, the work on the sites?"

"It's not hard," Gary said, winding the projector lead from his hand to his elbow and back. "It's careful. You check everything every time, and when you don't know you say so. That's the whole of it."

"That's easy enough, then, if that's all it is."

"It's easy for a week," Gary said. "Try it for ten years."

The lad laughed and put his card in the inside pocket of his jacket, zipped the pocket and patted it flat, and went out into the rain with his hood still up. Gary watched him go. The card was a small grey thing too, he thought, and it would be kept.

• • •

The last page of that book was the notebook the Ledger had slid across the table to him.

He opened it one night in the late summer of 2025, with the trading system running on one side of the screen and the paper on the other. He took a ruler and drew two lines down the first page, so that there were three columns. At the top of the first he wrote *known*. At the top of the second he wrote *what the checkers found*. At the top of the third he wrote *only hypothesis*. He underlined it twice, because it was the column he would be tempted to cheat.

Then he started to fill it in with the pencil. He went slowly, stopping every so often to sharpen it over the bin. Wherever he was not sure he put the mark, a small hooked line with a dot under it like a herder's crook.

The first thing he wrote in it was the stone, before the company or the code. He took it down off the windowsill, cold from the glass, and set it on the table beside the notebook where the lamp caught it. Then he looked it over like he would have looked over a load before a lift: the weight of it, the shape, where it might crack, what it was sitting on.

In the first column he wrote what he knew. *Grey, two inches or so end to end, flattened a little. Picked up on a beach on the Antrim coast one summer, six or so.* In the second he wrote what the checkers had found. There was nothing there, because nobody had ever checked it, and he wrote that too: *never tested*. In the third, under *only hypothesis*, he sat a long time with the pencil over the page. Then he wrote: *that it matters*, and put the mark after it.

He looked at the three columns and laughed out loud in the empty kitchen, because the stone was the hardest entry in the whole book. It had been sitting on the windowsill all along, not saying anything, waiting for him to be honest about it.

The master page

This page went on top of the squared notebook. For the first time the hand moving across it was his own, with the pencil gripped too hard like at school. The Ledger only stood and watched while the stone held the corner flat.

In ink: R&A Consulting and Brokerage Services Ltd, incorporated on the twenty-sixth of April 2023, one director; Python taught from nothing; a trading system that ran; the Harmonic Nexus Core on Zenodo on the thirteenth of July 2025; a Silver-level certificate from Innovate NI on the twenty-first; money towards the Street Soccer NI teams for Oslo, and a room full of new CSR cards. In pencil: the fifteen years, which only he can vouch for, and whether the trading system is any good, which nobody can vouch for yet. *Found destroyed*: the backs of the delivery dockets, most of them, which went in the bin long before he knew what was on them.

He put the page in its place. Egypt had made the record and Nadia the catalogue; the pieces had scattered; Lee had let a stone be a stone; Belfast had kept one; and the long way round had taught him the habits. The kitchen table's part was the column: three of them, ruled with a ruler, with *only hypothesis* underlined twice because it was the one he would be tempted to cheat. Behind, inked, is a radio at Shannonbridge and a man saying what he could see. In pencil, ahead, is a ledger that would refuse him things.

In the morning he would take it to the Core. This time the Core was his own, built at this table on this laptop with the sticky key, which made the rule harder to keep. It could show him rhythms lining up inside a model he had made himself. It could not show him whether the street outside was built the same way, with the bins out and the rain coming down the slates into a gutter that wanted clearing. That line was the hardest one in the column to leave standing.

Does it work? His friend had asked it with his mouth full of roll, and it was the only question in the room that mattered. The honest answer had been *I don't know yet*, and it still was. Keeping it honest until that changed was the job.

Out past the window the sky was going grey over the yard wall. That sky was the first column anybody ever kept, what was seen in one place set against what was hoped for in another, and in all those thousands of years it had never once cheated. The notebook on the table asked a lot less of him than that, and it was still hard going. Underline the hypothesis twice, keep the last good version, and never go back and change the answer after.

Then came the summer and autumn of 2026. There would be three ledgers with every date in them, hours on a room of computers far away near Bristol, and two pitches on the one day in Dublin.

The Ledger left the pencil lying where it was, on top of the notebook beside the stone, and did not pick it up again. From here on, the writing was his.

CHAPTER 37

What the Ledger Refused

Belfast and Dublin, from the summer to the autumn of the ledgers: the found year, 2026 CE



Illustration 38. *Three ledgers, with the grey stone on top, at four in the morning.*

Belfast, by day

THE PENCIL WAS STILL ON the notebook in the morning. On a lifting plan the weight of the load is never the weight you hope it is. You take it off the drawing or the delivery note, and then you add the slings and the shackles and the spreader beam. Then you add a margin, because the drawing was done by a man in an office who never lifted anything, and you write the total in the box. If the crane is rated for less than the box, you do not lift. It does not matter how much you want the steel up before the weather turns, because the box decides, not the wanting.

That box was in his head when the screen came back with the year he was still living in.

"In 2026 Gary Leckey proved the existence of a lost Atlantic civilisation," it said. "He received twenty thousand pounds from UKRI, and he won two pitch competitions in Dublin."

He let the machines fetch and read their returns by hand, with the stone on a printout of his own ledgers. These were the records he had kept that summer and autumn, and every claim in them carried a grade in the margin. A verified record was one thing and a documented reconstruction another. A hypothesis was a third, and speculation went in a column of its own at the far right of the page, where it could do no harm.

He put a line through *proved*, first and hardest, and wrote *did not prove* above it in the same plain hand. He pressed so hard the pencil dented the sheet underneath. His own ledgers said so line by line, and so did the essays he had put up in public for anybody to read. The pattern he had loved best was a pattern. He had written down, before he tested it, what would count as a pass, and then it had failed.

Then he struck out *twenty thousand pounds* with a firm line. It was twenty thousand GPU hours on Isambard-AI, from the AI Research Resource, awarded on the twenty-fourth of August 2026, for a short sprint to test the evidence harder. That was time on a machine, not money in the bank. A man who mixed them up would be found out at his first meeting. *Won* came out too, because on the fourteenth of September he had pitched twice in Dublin on the one day. One was at the Guinness Enterprise Centre for the TechIreland National AI Challenge, and one was to Techstart. He had come home on the train with a sore throat and no trophy, which is what most pitching is.

He laid the summer out a month at a time, with July at the bottom. That held the index of his essays going up on the twentieth and the Validation Metrics Ledger on Zenodo four days after it. Over it were August and September, with the hours on Isambard-AI and the two rooms in Dublin, and Invest NI taking the company on as a client. On top lay October, with the Master Ledger on the fifth and the Grail Ledger on the sixth. He wrote *2026 CE* in the top corner, and under it, because the rule was the rule, *2026 CE?*. It was the first time the mark had gone after a year that was not finished yet.

The doubt followed: *The failures in the ledgers are mine, and so is every grade. What the judges said in Dublin I have not written down; the words in the dream are mine.*

On the map he did not draw anything. He had drawn enough lines on it that summer to know which ones held. The box decides, aye. He shifted the stone off the ledgers and onto the windowsill, so the pages could lie flat.

That night

That night the Ledger opened the newest book on the shelf, and the ink in it was still wet. It was not a stranger's book, or an old one, or a cheap one with squared paper. It was his own ledgers, bound together, with his grades in the margins and his crossings-out left in. The Ledger turned the pages more slowly than it had ever turned any page. It might have been reading them too. At the top of the first page was July 2026.

• • •

The week the index went up, he sat at the table and wrote the page that would say what the whole thing was for.

He wrote it four times. The first go, on the back of an envelope, was too big, and the second, on the laptop, was too careful by half. The third was angry at people who had not done a thing to him yet, so he screwed it up and fired it at the bin. On the fourth he wrote down five questions that every essay would have to answer before it went up, and under them, in plain words, the lines he would want somebody to hold him to.

A compelling pattern is not automatically a mechanism. A simulation is not a prototype. A repository is not a performance record.

He read it out loud in the kitchen with the radio turned down, to hear did it sound like him. It did, near enough. That is as much as you can ask of a page, and under it he wrote the line he liked best, because it would be the hardest one to live up to:

Those boundaries do not weaken the research. They make it possible to improve.

"Right," he said to the stone on the windowsill. "Now we're stuck with it."

The fifth question was the one that mattered. He said it out loud to the kitchen in the fewest words he could, like you would shout it across a site over a generator. "What would make you wrong?" If he could not answer that for an essay, the essay stayed on the laptop.

The index went up on the twentieth of July with twenty-nine essays on it, every one of them linked to its source. Four days after that the Validation Metrics Ledger went up on Zenodo, a long, plain document with every date in it and where every date came from. Nobody cheered, and a few people read it. He put a rule at the top of it for whoever came after: if a date was wrong, say so. It would be fixed, with a note left saying what had been there before and who had caught it.

"That's what it's for," he said to the stone, when he had read the rule back over twice. "So somebody can catch me."

• • •

That summer he ran the test he had been putting off.

He had written it all down in advance on a dated page, before he looked at a single number: what he would measure, how many checks it had to pass and where the bar sat. Five checks, and three to pass. He set the bar high on purpose, because a pass mark you can step over is only a line on the floor. Then he ran it and sat with his elbows on the table, a cold cup of tea at his hand. He watched the numbers come back one at a time on the screen.

It passed one.

He sat with that until the kettle beside him had gone cold. There was a version of the night where he moved the pass mark. It would have been easy, with nobody watching. He could see exactly how he would explain it, in good clear sentences. He could hear how reasonable it would sound, and that was what frightened him.

"No," he said, out loud, to nobody. "That's the Tipp-Ex."

He opened the ledger and wrote it in, in the same plain hand. *Pre-registered test: passed one of five, three needed: failed.* Then he went back through every page that had leaned on that pattern and moved each line down a column, out of the hypothesis and into speculation, where it would sit until something better came. It took him till four in the morning to get through them all. When he was done the ledger was thinner in the column he loved and fatter in the column he did not. It was more true than it had been at midnight.

Later he wrote it up in public, the failure and the reasons, with the test and the pass mark printed beside it. It was not a sad essay; it was the most useful thing he had written all year.

• • •

The email about the hours came on the twenty-fourth of August.

He read it three times standing up, in the kitchen, with his coat still on. It was twenty thousand GPU hours on Isambard-AI, the big machine down in Bristol. The sprint was called the Aureon Rapid Evidence Evaluation Sprint, because he could not think of a shorter name that said what it was. The point of the sprint was not to prove anything. The point was to throw his own claims at a machine big enough to break them, and write down which ones broke.

"Twenty thousand hours," he said to the stone. "Of somebody else's electric."

He thought about the RGIS nights in the shops after closing, two counters on every aisle and a third man with a clipboard sent in when their numbers would not agree. This was the third man again, only bigger.

That same month he wrote an essay about the sky, about how you go looking through a hundred and nine thousand orbits without finding a pattern everywhere you look. It was the oldest trap in the Ledger, and he had walked into it more than once. Give a man enough stars and he will find a shape in them. The shape will be real to him, and it will mean nothing. You had to decide what you were looking for before you looked, and how often you would find it by chance, and only then go out under the dark. He wrote it plainly. When he was fin-

ished he went out to the back yard and looked up at the orange sky over Belfast, where you could see about nine stars on a good night, and he counted them anyway.

• • •

On the fourteenth of September he took the early train to Dublin, with a laptop and a clicker and a clean shirt in a bag.

The first room was at the Guinness Enterprise Centre, for the TechIreland National AI Challenge. It had chairs in rows and a screen at the front and a panel at a table, and he had eight minutes. He did the eight minutes, and he did not say *revolutionary* or *proven*. He showed what the system did, and what it had not been tested on yet, and where the ledger said *failed* in red.

A woman on the panel leaned forward when he had finished.

"You've shown us the failures," she said. "That's unusual. What would make you wrong about the rest?"

"A test I write down before I run it," Gary said. "With the pass mark on it, in public. If it fails, it goes in the ledger as failed, and I tell you so, same as I've told you about this one."

She wrote something down, and he never found out what.

In the corridor afterwards, beside a table of coffee flasks and paper cups, a young fella with a lanyard and a startup of his own caught him by the elbow.

"You put your failures on a slide," he said, jerking a thumb back at the door of the room. "In front of judges. Are you mad, or is that a tactic?"

"Neither," Gary said, and hitched his coat up onto his shoulder. "They're in the ledger anyway, in public, where anybody can read them. I'd rather a judge heard it from me than found it after."

"Fair play to you," the young fella said, though he did not sound sure of it. Then he went off down the corridor with his lanyard swinging, to rehearse his own eight minutes, which had no failures in them at all.

The second room was smaller, for Techstart, with fewer people round a table and harder questions about money, and he answered those plainly too. One of them he could not answer, and he said so straight out. The man who had asked it nodded slowly and wrote something on his pad, like that had been the answer he was waiting for.

"How do you make money," another of them asked, "out of telling people they're wrong?"

"Same way a banksman does," Gary said, and held his hands up a foot apart like he was talking a load down. "Nobody pays me to say a lift is grand. They pay me so the lift doesn't come down on anybody. A checked thing is worth more than a hoped thing, and people will pay the difference."

On the train home, somewhere before Drogheda, the sea was going grey on the right. He wrote a post on his phone and read it four times before he put it up. *Built in Belfast. On show in Dublin. Looking beyond borders.* He looked at it and thought it sounded like a man he would trust. He put it up, and fell asleep with his head on the window until Newry.

That same month Invest NI took the company on as a client. He left the email open on the laptop for a whole day without telling a soul, reading it again every time he passed the table. The next evening he rang round everybody he knew before the tea was even on.

That month too the company's name went on the shirts of the youth teams at Immaculata, and he went down on a Saturday morning to stand with them for the photograph. They were wee lads in new kit a size too big for them, hopping about on the wet grass, and not one of them could stand still for the camera.

"Are you the man off the shirts, mister?" one of them said, squinting up at him.

"I am," Gary said. "Mind and keep them clean, now. Your mammy'll kill me."

"What does it do, your company?"

Gary hunkered down to think how he would say it to a boy of nine or ten. The wee lad had his laces undone and mud on both knees already, and the match had not even started.

"It checks things," he said. "It finds out if things are true."

The boy considered that, the way Gary had once considered a man walking backwards down a rope shed. "That's a good job," he said, and ran off to take a penalty.

• • •

The Master Ledger went up on the fifth of October, and the Grail Ledger the next day.

He had saved the Grail Ledger for last, because it held the most of what he wanted to be true. The old stories were in it, the journeys out to the edge of the map and home again. So were the plain things that came back in somebody's pack and were turned into treasure by people who never saw where they had been found. His own patterns were in it too, the ones that had kept him up at the kitchen table for fifteen years.

He went through it like he had gone through the dead, with the pencil behind his ear and a grade in the margin for every line. A record went in as a record. A story went in as a story, with the date it was first written down beside it in brackets, and that was nearly always centuries after the thing it told. Where there was only him, it went in as speculation in the far column, without a stitch of dressing on it.

Halfway through the night, with the street gone quiet and the fridge humming behind him, he came to the line he loved best of all. It was the one that tied the whole thing together, and it had kept him up more nights than any other.

He looked at it with the pencil over it, not writing.

Then he gave it the grade it had earned, which was not the one he wanted. The Ledger refused him that one. It did not argue or shout; it simply had no column for what he wanted the line to be, only a column for what it was, and he put it there.

"Fair enough," he said.

He picked the grey stone up off the windowsill. It was cold from the glass, and he held it till his hand had warmed it through.

The master page

The last page of the run lay on his own three ledgers, which were stacked one on top of another with their edges not quite square and a tea ring on the top one. The stone sat on the stack and held it down.

In ink: the index of twenty-nine essays on the twentieth of July 2026; the Validation Metrics Ledger on Zenodo on the twenty-fourth; twenty thousand GPU hours on Isambard-AI on the twenty-fourth of August; two pitches in Dublin on the fourteenth of September; Invest NI's client company; the Master Ledger on the fifth of October and the Grail Ledger on the sixth. In ink too, because it happened: a test that passed one of five when it needed three. In pencil: everything the Grail Ledger grades as speculation, and the line he loved best. *Found destroyed*: nothing. He kept every version.

It sat at the end of the run. Egypt had written the record, Nadia had made the catalogue, the pieces had gone out across the world, and Lee had let a stone be a stone. Belfast kept one, the long way round taught him the habits, and a kitchen table ruled the columns. The Ledger's part was the refusal: a book that would not let him write his favourite line in ink. Behind it, in ink, is a kitchen table and a squared notebook with *only hypothesis* underlined twice. Ahead, in pencil, there is no shelf at all, only fog, and a small boat.

In the morning he would take it to the Core, and the Core was the thing on trial. It would confirm that the rhythms lined up inside the model he had built. It had nothing to say about whether the world was built the same way, and his own test had said, as plainly as a test can, not yet. The Core does not get a vote.

What would make you wrong? The woman in Dublin had asked it as if she expected him to flinch, and he had not, because he had already been wrong in public that summer and lived. A test written down before you run it. A pass mark you cannot step over. A column for what a thing is, and none for what you want it to be.

At four in the morning at a kitchen table, past thirty, he had his own answer for once: the sky is the one thing that will tell you to your face that you're wrong, and not hold it against you. Every watcher in

the Ledger had been wrong first and said so after: the old man on the ledge, the boy with the six, the daykeeper fixing his numbers. He had written the pass mark first, and failed in the same plain hand he had hoped in. Independent research, made falsifiable.

On the next shelf there was no shelf. There was fog on grey water, and a small boat, and something at the edge of the fog that he could not see yet.

The page stayed open on the stack. The Ledger did not close it, and for the first time it did not send him back to the kitchen either.

CHAPTER 38

The Threshold

The Atlantic edge, north of any map: the present, and not the present



Illustration 39. *An empty boat on flat grey water in the fog.*

That night

THAT NIGHT THE LEDGER DID not lay a page on anything. It closed the last of his own books and put it back on the shelf with the others. The shelves went up out of sight the way they had the first night. Then it opened one more book, a thin one he had never seen. There was nothing on the page but water, and a line at the top in a hand he did not know: *What did you carry out, and what do you bring back?*

• • •

He was in a boat, and he was rowing.

It was small, smaller than any boat he had ever been in, with two oars and a plank to sit on and a little water slopping in the bottom. He did not remember starting, and he did not know how he knew how. But his hands were on the oars, and the oars were going in and coming out, steady, and the boat was moving. He had his work jacket on, the one with the reflective strip gone grey from washing. In the right-hand pocket there was a weight he did not have to look at to know.

Behind him, when he turned his head, there was a coast. It was low and dark and far off behind him, black cliffs sitting on white ones, and a bank of round stones at the foot of them. In front of him, where he was going, there was fog.

It sat on the water like a wall, white and still, with the edge of it so clean you could have drawn it with a ruler. The sea between him and it was flat and grey and moving only a little, the long slow breathing of deep water. There were no birds on it, and there was no wind. He rowed, and the fog came closer, and then he was in it.

• • •

Inside, the world went small and close. He could see the bow of the boat and the oars and a few feet of water on each side, and after that only white, wet and close and smelling of salt. The sound of the oars came back to him oddly, from too near. He stopped rowing to listen and let the boat run on, and it ran on, though it should have slowed.

He thought of a girl who had been told that you track the stars so you can come home in a fog and know where you are. There were no stars here to track. He looked up anyway, out of habit, and there was only white.

"Right," he said to nobody. His voice went out into the fog and did not come back.

Then the fog answered him, and it was not one voice but many. They came from all sides and from none, some near and some so far off they were only the shape of words. Nobody was shouting. They were the voices people use across water in the half-light, when they know somebody is out there and want him to know they are there too.

"How do I know which one?"

It was a small voice, close by on the left, and he knew it before he knew why.

"It'll be the one that doesn't want to be left." That was a man's voice, further off. Then the same man came nearer. "Grey's the colour of most things that last." After that he was humming, an old tune with no words, and the humming went away into the white.

He sat very still in the boat with his hands on the oars, and let the voices come.

• • •

"That's for the one who held the rope." It was a girl somewhere ahead, quite sharp about it.

"You told them what you saw, not what they wanted." An old man said that, slowly.

"Ask your mother about it, she'll tell you more than I can." That one was kind and tired.

A boat went past him in the fog, close enough that he heard the water on its sides, and for a moment he caught the shape of it. It was bigger than his, made of hide stretched over a frame, and it rode the swell like a gull on the water. There were men at the oars and one at the back with his hand on the steering oar. None of them looked at him; they were all staring ahead into the white, and so was he. Then it was gone, and the hide creaked once after it, like a saddle on a cold morning.

"My grandmother said the ice brought them and the sea polished them." That was a woman, from very far away.

"Who will sing it, when you're old?"

A young voice asked it, high and worried, and an older one answered too low to catch.

"If anyone ever comes from the south, and the east, they'll have the whole world in one place." Somebody said it like it was the best news in the world.

"Count what is there." A man said that, very firm. Gary nearly laughed, because he knew that man, or men like him, from every stock-take he had ever done.

"Tell me what you saw. Only what you saw, mind."

"You write what you like on a wall. It's the wall that's the true thing."

"It's a footprint for whoever comes next."

"There are stars down there you never see here." That was a boy on a cold shore, wondering.

He did not answer any of them, and he did not think they wanted him to, for nobody was asking him anything. They were only letting him know that they had been out here before him, on this water and in this fog. They had come home or not come home, and had said these things to somebody on the way.

• • •

He rowed again, because sitting still had begun to feel wrong, and after a while the voices changed. These ones were nearer, and he knew them better, because they were his own or near enough. They were things said to him or by him in kitchens and file rooms, and on wet sites with the wind coming up off the river.

"Somebody has to hold the end of it and keep going." It was the old man from the chair at the end of the street. Gary could see the cap on him, though he could not see him.

"Never tidy a statement."

"Keep to what you saw."

"Walk round it with me, and tell me what you see." That was his own voice too, on a frosty morning at Kilroot, talking to a lad in his first week.

"Then one of us is wrong, and it's not going to be me, love."

"Far is only how long you're gone."

"It checks things. It finds out if things are true." His own again, to a boy with mud on both knees, and the boy said back, "That's a good job."

"Can you see it?" That one came flat, from sixty feet up.

"I can see it," Gary said, before he could stop himself. The fog took the words off his mouth and kept them.

"Does it work?" That was his friend from Kilroot, with his mouth full.

"What would make you wrong?" That was the woman on the panel in Dublin.

Then the first voice came back, the small one, very sure of itself. "It's just for keeping." Then, a little after, "Is it still there?"

That was the one that stopped him rowing and put his head down for a moment over the oars. After a while he fished the stone out of his pocket. It had gone cold in the fog, and he closed his hand round it, and slowly it began to be warm.

• • •

There were lights in the fog then, a long way off and then nearer. At first he thought they were the lights of a harbour. They were too small for that, small and yellow and low on the water. There were a great many of them, scattered across the white the way stars are scattered across a winter sky. Each one was a lamp, or a rushlight, or a fire in a pot, held up by somebody he could not see, in a boat or on a rock or on a shore he could not make out. Some were steady and some were nearly out. None of them moved towards him, and none moved away. They were not there to guide him; they were there to show that somebody had kept a light burning in this place, and that it could be done.

He rowed in among them, and they did not light the fog so much as make it feel less empty. He thought of every page the Ledger had ever opened for him and every child on every page. They were up before dawn with an old person who watched, cold and patient, waiting to see where the sun came up. He thought of the girl in the stone house who had blown the lamp out and gone to sleep, and the scribe waiting for a lamp. He thought of the lamps that hung unlit, and the rushlight cupped in a small hand against the draught. They had all been lights in the fog, though he had not known it at the time he was reading them, and he knew it now.

Somewhere ahead and to the right a gap opened in the white, no wider than a door. Through it he saw a single star, high and small and hard-bright in the north, and it was not moving. Then the fog closed over it again. He did not need to see it twice; he knew where north was now, and he turned the bow a little towards it, and rowed.

• • •

Then there were things floating on the water.

At first he took them for weed, or the white of foam. Then the boat came up among them, and he saw they were slips of paper, hundreds of them, small and pale, lying flat on the swell. Each had a few words on it in his own hand, with a single line through the words. *Proved*, one said, under its line. *Seized*, said another, and *reached America*, and *the Crusades*, and *destroyed*, and *banned the Gaelic language*, and *1134*, and *without qualifications*, and *crane operator*, and *top award*. Every word he had ever put a line through was out here on the water, and every one could still be read under its line, the way the incident book had wanted it.

He had always thought they would sting, if he ever saw them all together, because each one had stung a wee bit at the time. They did not sting now. They lay on the sea round the boat like a flock of small white birds resting. They were the most honest thing he had ever made, and he was not ashamed of one of them.

He let them be and did not pick any up. He rowed through them carefully, feathering the blades so as not to push any of them under. They closed in behind the boat and went on floating, and some day somebody else would find them and know what had been there before.

• • •

The fog began to thin, though not all at once. It went from white to grey to a kind of pearl, with the light coming through it from somewhere he could not find, neither sun nor moon. The sea under the boat changed too, from grey to a deep clear green. He could see down into it a long way, to where there were shapes on the bottom that

might have been stones or might have been something built. He did not look too long, because he was not here to find out what was on the bottom.

Fish went under the boat, a long silver run of them with dark backs and a scatter of spots, all going the one way, steady, towards a shore he could not see. Salmon, he thought, and knew it was daft to be sure of that in a dream, and was sure. Nobody had told them when to come; the water told them. They had been doing it since before there was anybody on any shore to watch them go past. And somebody, a long way back, had sat on a bluff above a river and learned to know when they were coming without a calendar, by watching, and being wrong, and watching again.

The voices had stopped now. There was only the sound of the oars and the water on the hull, and very far off something that might have been the sea on a beach, the long rattle of stones being pulled and let go.

Then a rock came up out of the pearl light ahead of him, low and dark and wet, with weed at the waterline and a flat place on the top. You would row past a rock like that on any coast in the world without a second look, and he would have rowed past this one, only there was a woman standing on it.

She was not young and she was not old. She stood easily on the flat top of the rock, the way people stand who have stood in one place a long time and mean to stand there longer. She had a plain grey cloak round her, wet at the hem, and her hair was loose. In her right hand she held a branch of some tree he did not know, with a few small pale blossoms on it and a few leaves. It did not look magic or holy or anything at all. It looked like a branch somebody had broken off on the way down to the shore, because it was there.

He stopped rowing, and the boat ran on slowly for the last few yards. It came to rest against the rock with a small bump, like a boat coming in against a quay when the man at the oars knows what he is doing.

She looked at him, and she did not look surprised. She had the look of the old women on the shore in every story in the Ledger, at the ford and the harbour mouth and the edge of the known world. They were women who had been told somebody was coming and were only waiting to see what he would be like.

He had read about her a hundred times, under a hundred names, and none of them were hers. He had put her in pencil every single time and never once been able to put her in ink, because there was no record and there never would be. She was the thing at the edge of the map that every traveller in the book had come home talking about, and that had always stayed beyond proving. And here she was on a rock, with a branch in her hand and the hem of her cloak dark with the wet.

He did not know what to do with his hands, so he held on to the oars.

"Gary," she said.

It was not a question. She said it like you would say the name of somebody you had been watching the road for since morning, glad to see them at last. Her voice sounded like nobody he had ever met, and like everybody he had.

He opened his mouth and nothing came. All his life he had been talking, to customers and lads on site and judges in Dublin, and to a grey stone on a windowsill at four in the morning. He had told what he could see as plainly as he knew how, to anybody who would stand still for it. Out here in the fog he had nothing at all.

She waited, and the branch in her hand moved a little in a wind he could not feel. The blossoms on it were very white.

Then she asked him the question at the top of the page. She asked it gently, with the branch held still at her side. He had the feeling she had put it to every traveller who ever came this far, and had listened to every answer they gave and kept the lot.

"What did you carry out, and what do you bring back?"

He looked down. He had let go of one oar without noticing, and his fist was closed, resting on his knee, and he opened it.

The grey stone lay in his palm, plain and flat and dry now, and warm, small enough to close his hand over, the grey of the lough before the sun reaches it. There were no marks on it, and it was not for anything. It sat in the hollow of his hand as if the hollow had been made for it, the way it had on a beach a long time ago, when he was six or so and the sea was still there.

He looked at it for a long moment, and then he looked up at her, and drew breath.

EPILOGUE

The Stone on the Shore

The north Antrim coast, one afternoon at the end of October: the present



Illustration 40. *The stone at the water's edge.*

• • •

HE WOKE WITH THE ANSWER in his mouth and no memory of what it had been.

It was morning in Belfast, and the light was coming grey over the roofs towards the lough as it always had. The two yellow cranes were standing up in it over the shipyard. He lay still for a while and tried to get the answer back. It was like trying to get back the end of a song you heard in a shop, and it would not come. All he had was the shape of the question and the feel of a cold rock against the side of a small boat. There was a branch with white blossoms on it, too, moving in a wind he could not feel.

The grey stone was in his hand, though he had no memory of taking it to bed. It was warm right through, like it always was when he had held it a long time. He lay a while feeling the weight of it before he opened his hand.

He got up and put the kettle on and stood at the kitchen window while it boiled. He watched the roofs and the cranes and the grey going slowly to white. The ledgers were still on the table where he had left them, stacked one on another with their edges not quite square. The squared notebook was under them, with its three columns ruled in pencil, and under that the CVs and the printout of the register. He looked at all of it, the whole pile of his own life in other people's handwriting and his own. Then he put his coat on and picked up the car keys.

"Right," he said to the stone. "Come on, you, we're going for a drive."

• • •

He drove out of Belfast by the lough road, the road they had taken the summer he was about six. It was a Saturday at the back end of October, the clocks had gone back the week before, and the light was already lying lower than it should at that hour. He went past the turn for Kilroot without looking, then gave in and checked the mirror. Over there he had stood under the cranes and talked the steel down by hand. Then the road took him on up the coast, round the long bends above the sea. The cliffs above the road were black basalt sitting on white chalk, just as the page had shown him the first night. He had the window open an inch, cold or no cold, because it had been open that first time and shutting it would have felt like bad manners.

At every bend he looked out the passenger window for the sea. He had done the same as a boy, with his nose pressed to the glass of the back seat. Each time it was out there yet, grey and heaving under a low sky.

He thought, driving, about what he had found and what he had not. He had gone looking for one year and come back with a whole shelf of them, each with its question mark after it. On every one of those

pages he had sat with a child he made up out of the period and the pattern, because the record had never bothered to keep children. They climbed up before dawn beside old people who watched, held ropes and carried lamps, and asked the questions grown men had stopped asking because the grown men thought they knew. He had crossed out more than he had kept, and he had kept the crossings-out as well. He had built a framework for fifteen years, and then let a test tell him in public that it was not yet what he had hoped. He had been wrong more often than he had been right, and he had written down every time which was which.

And what's it for? His mother had asked him that on a beach nearly thirty years back, about a stone, with the wind blowing her hair across her mouth. At six he had given her an answer, and he had been working out what it meant ever since.

• • •

The beach was the same, and not the same.

He parked where the car had been parked that day, or near enough, since there was no way of knowing and he did not pretend to. The grass along the top lay flat under the wind, as it had then. Below it the bank of stones ran down in its long grey slope to the sea, dark where the waves had wet it and dry higher up, every shape there was among them, from fist-round to flat to some as long as a heel of bread. Each wave dragged at them coming in, and going out it left them rattling after it. It was the same long sound he remembered, the whole beach letting its breath go at once. He stood and listened to it a good while before he went down. He had heard that sound in a dream that very night, very far off through fog, and had not known then what it was.

There were not many people about at that time of year. A woman further along was throwing a ball for a dog that would not bring it back. Two lads in wetsuits were standing at the edge of the water arguing about whether it was too cold, which it was. Further out the sea was the colour of a slate roof after rain. Beyond it, low and blue on the horizon, lay a long shape of land.

"That's Scotland," he said to the stone, with a nod at the blue line across the water. It was what his father had said that day, word for word, and the stone did not argue with him either.

He went down the bank, and the stones shifted under him and clacked together. He had to put his arms out a little, like a man on a scaffold board in the wind. He was a good deal heavier than he had been the last time, and his boots were better, and a soaking would do them no harm. Near the bottom the bank levelled off, and the water slid up and back among the pebbles in little hissing sheets. The stones at the foot were darker and smoother and shone wet, polished by nobody and by everything.

He crouched down where the shingle went dark with wet, the tail of his coat trailing on the stones. He looked along them for a while, the ten thousand of them shining in the wash, all the ones he had not picked.

• • •

The woman with the dog came past behind him along the top of the beach. The dog came skittering down the stones to see what he was at, and stood beside him dripping with its nose an inch from his hand. Dogs will always do that with anybody who is crouched down.

"He'll not bother you," the woman called down to him. "He's only nosy, that one."

"He's grand," Gary said. "I'm only looking at the stones, so I am."

"There's plenty of them to look at," she said, and laughed. "Are you looking for anything in particular?"

He thought about how to answer that while the dog nosed at his boots and the wind came in cold off the water. In the end he told her the truth, because it was easier than anything else he could come up with on a beach in October.

"I'm bringing one back," he said. "I took it off this beach when I was a wee lad, and I've had it ever since."

The woman looked at him sideways, with the dog's lead looped round her wrist, trying to work out was he codding her. Then she must have decided he wasn't, because her face went softer and she stopped edging away up the shingle.

"That's a long time to have a stone," she said.

"It is, aye, near thirty years."

"Will you not miss it, after all that?"

"I will," Gary said. "But it's done its job, and it's been a good one."

She gave a short nod, like people told her that sort of thing on beaches every day, and called the dog. It went bounding after her, and the two of them carried on over the shingle and round the point and out of sight.

• • •

He brought the stone out of his coat and held it in his open hand with the wind getting at his fingers. He gave it the same long look he had given it in the boat in the dream, with nothing hurried in it at all.

It was grey and a bit flattened, no bigger than it had ever been, with no marks on it anywhere. It still sat in the hollow of his palm as if the hollow had been made for it. He dipped it in the edge of a wave and lifted it out nearly black. It dried on his palm while he watched, the dark shrinking in from the edges till it was only grey again, the colour it had been on this beach the first time.

He knew a great deal about it now, and none of it was certain. Most likely it was basalt, from the lavas that welled up through the cracks as the Atlantic opened, which made it some sixty million years old, and after the number went the small hooked line like a herder's crook. It might just as easily be an erratic, carried over the water from Scotland by the ice and dropped here for the sea to polish. If so, it could be older than anything in Antrim by hundreds of millions of years. Nobody had ever tested it, since the only sure test meant sawing it open, and that had never been on the cards. He had written it in pencil in the Ledger on the first night: *never tested; possibly an erratic from over the water*. It was still in pencil, and it would stay there.

That, he thought, was the answer, or the first half of it anyway. He said it to the stone under his breath. He kept his back turned to the lads in the wetsuits, so they would not think he was away in the head.

"What did I carry out?" he said. "You. A stone that wasn't for anything, and a question nobody could answer, that I never asked anybody's leave to keep."

The sea came in over the toes of his boots and drew back rattling through the shingle while he turned the stone over in his fingers.

"And what do I bring back? Everybody. The ones who held the rope, and the ones who kept the count, and the ones who said what they saw and not what they were told. The ones who got crossed out, and the girl with the lamp, and the old men on the chairs. The ones who were wrong first and said so after. That's what I bring back, and a book with three columns in it, so the next one can tell which is which."

Whether that was the answer he would have given the woman on the rock he could not say for certain. He reckoned it was near enough, all the same. She would have nodded, he thought, the same short nod he had got from the dog-walker, for a thing folk must say to her every week of the year. Then she would have kept it with all the rest.

• • •

His father had told him on this beach that he would know which stone was his, because it would be the one that did not want to be left. He had believed that for the best part of thirty years. He had taken it to mean you had to keep the thing always, in your fist, on the windowsill, weighting down the page. Kneeling in the wash with his boots full of the sea, he saw for the first time that it might have meant something else. A thing that does not want to be left is a thing that was kept well, and it knows the difference. The stone was not afraid of the beach. It had lain on beaches for longer than there had been anybody to pick it up. It would be lying on this one long after the street lights of Belfast had gone out for good. It had only wanted, for a while, to be somebody's, and it had been.

He thought of his mother going along this same strip of shingle with her coat flapping open and her shoes swinging from her hand. He thought of the cold of her fingers when she closed his round the stone and said, *Then you keep it*. He had kept it for the best part of thirty years, through kitchens and basements and night shifts and border crossings and frosty mornings under the cranes. He had kept it on the corner of every page in the book. Keeping, he understood now, did not have to mean holding on for ever. It meant knowing what a thing was, and writing it down honestly with its question mark, so that it would never be lost even after you let it go.

He had done that in his own plain hand. It was all in the Ledger now, pencil beside ink, for anybody who came after him to read and argue with.

The light was going now, low and gold behind him over the hills, and laying itself out across the water towards Scotland in a long broken path. The lads in the wetsuits had given up and gone in screaming, and were coming out again faster. Somebody up on the road was waiting for them with towels and laughing at them. A gull went over very high and said something rude about all of it. The sea came in and went out, and the ten thousand stones rattled after it and breathed out all at once.

He leaned forward and set the grey stone down at the water's edge, among the others, where the last wave had left the shingle wet and dark.

For a moment he could still tell which one it was. Then a wave ran in further than any before it and went over it and over his hands, cold enough to make him gasp. It drew back hissing, and the stone was wet and nearly black, lying among a thousand wet black stones exactly like it. He looked for it, and he could not have said for certain which one it was, and he found he had no need to.

He stood up with his knees cracking and his hands red with the cold. He stood there a while longer by the water with nothing in his fist at all. Then he went back up the bank to the car and drove home

through the dark towards Belfast, with the window down a touch and the radio off. He was humming under his breath an old tune with no words, all the way, until the street lights came on round him.

Dedicated to Tina. If you don't quit, you can't lose. Love conquers all.

SOURCES AND NOTES

This book mixes things that are known, things that are told and things that are imagined. These notes say which is which. For each chapter they give the found year and how it was reached, what the record supports, what is tradition or interpretation, where the author is reading the evidence in his own way, what is invented for the story, and the doubt that stays. Full references are in the list of sources at the end of the notes.

Prologue: The Granite Sky

The found year. About sixty million years ago. The stone in the prologue is read, by its colour and its feel, as Antrim basalt, and the Antrim Lava Group has been dated: the Lower Basalt at 62.6 ± 0.3 million years, the Tardree Rhyolite at 61.3 million and the Upper Basalt at 59.6 million (Ganerød and others, 2010). The Causeway Tholeiite, between the last two, is undated.

What the record supports. The Earth is about 4.54 billion years old. The Jack Hills zircons are about 4.4 billion years old (Wilde and others, 2001), and the early-water reading of them is kept as the researchers' own "they are not sure". The Acasta Gneiss is about 4.0 billion years old (Bowring and Williams, 1999). Living stromatolites grow at Hamelin Pool, Shark Bay. White chalk with flint lies under the Paleogene basalt of Antrim, and the ice carried erratics across from Scotland.

Interpretation. That this particular stone is basalt at all is a reading by eye.

Invented for the story. The beach day, the child's good shoes, the car journey and the parents' few words are invented. The parents are unnamed, and nothing is claimed about them. The grown man's working life, as he recalls it, is drawn from the author's own account of his.

The doubt that stays. The stone was never tested. It may be an erratic from over the water, far older than any lava in Antrim, and the prologue says so.

Chapter 1: The Salmon Camp

The found year. 7700 BCE. At Mount Sandel the main hut-building phase is modelled at 7790 to 7635 cal BC, probably within one or two generations, with pits and later visits down to about 7500 (Bayliss and Woodman, 2009). The found year is the middle of that window. The older textbook figure of about 7000 BCE comes from before the modelling and is not used.

What the record supports. Peter Woodman's excavation in the 1970s, on a site due for housing, found round huts about six metres across, with inward-leaning posts and repeated rebuilding. It found hearths with hazelnut shells, and burnt bones of salmon, bass, eel and flounder. It also found wild pig, water-lily seeds and crab apple, microliths and axes. The acid soil keeps only what was burnt. Irish Mesolithic genomes, from other sites and none from Mount Sandel, show long isolation, a likeness to British hunter-gatherers, a population perhaps between three and ten thousand and no close inbreeding (Cassidy and others, 2020).

Interpretation. Woodman read the camp as occupied through the year. Others prefer repeated visits. The book follows his reading and says that others differ.

The author's reading. The blackthorn, swallow and elder signs that the run is near are a fisherman's sense, and are marked as the author's own.

Invented for the story. Ness, Orrin, Teva and Bel, the mother's death and burial, and the voice in the fog.

The doubt that stays. The year is a window's middle, not a measurement. Nobody can say that the people of that bluff met a particular spring on a particular day.

Chapter 2: The Stones at the Edge of the Plain

The found year. 4800 BCE. Almendres has small concentric rings from the end of the sixth millennium BC and a great double ellipse in the fifth, with use into the early third (Varela Gomes). The scene shows an old ring that nobody remembers raising and the start of the great one, so it falls early in the fifth millennium. There is no direct radiocarbon date for the enclosures. The older figure of about 5500 BCE is not used.

What the record supports. About ninety-five granite stones stand in rings and ellipses west of Évora, opening towards the east, with carved crooks, circles and cup-marks. Alentejo enclosures face the arc of the eastern rising

more often than chance would give. An autumn full-moon model fits best, the equinox sun was also tested, and the sample is small (Pimenta and Tirapicos). Early farmers with Anatolian-related ancestry arrived, with hunter-gatherer ancestry mixing in and growing over time (Olalde and others, 2019). The Muge shell middens hold burials, and Cardial pottery is impressed with shells.

Interpretation. That Almendres is an equinox monument is an interpretation. In the scene the counted "middle day" falls a day or two from the astronomical equinox, and the text does not hide the gap.

The author's reading. The solar azimuths mentioned are geometry worked out by the author, not published claims.

Invented for the story. Telo, Brannok, the long rope and the pebble in the packing.

The doubt that stays. The dating is by phase and by style, not by radiocarbon, so "4800" is a reasoned guess inside a wide window.

Chapter 3: Eleven Hundred Stones

The found year. 4500 BCE. The first phase at Saint-Michel is modelled at 4782 to 4594 cal BC (Schulz Paulsson, 2019). The Le Plasker alignments, two kilometres west of Ménec, are dated to 4600 to 4300 cal BC, with a tomb of about 4700 over a Mesolithic hut (Schulz Paulsson and others, 2025). Variscite found in the Carnac mounds has been traced to Pico Centeno, in Huelva, and was mined from about 5200 BC (Querré and others). Ménec itself is undated.

What the record supports. Ménec has eleven rows and about eleven hundred stones, with a ring at the western end and the stones dwindling towards the east. Kermario and Kerlescan lie beside it. Saint-Michel held Alpine jade axes and variscite. Logboats are known from Neolithic France, at Bercy and, older, at Noyen-sur-Seine. The Gulf of Morbihan was drowned by rising seas, and Er Lannic's rings are half under water. Crook motifs occur on Breton stones.

Interpretation. Schulz Paulsson's model, that megalith-building began in north-west France and spread by sea, is debated. Both views are on the record.

The author's reading. The midsummer sunrise is worked out as about 31 degrees north of east at Évora and about 37 at Carnac for the tilt of the Earth around 4500 BCE, on a flat horizon and without refraction. The figures are "about".

Invented for the story. Ael, Enora, Konan and Sela, the hide boat (a need, not evidence) and the song that names Telo, three hundred years or so after Chapter 2.

The doubt that stays. The rows themselves cannot be dated, so the year rests on their neighbours.

Chapter 4: Winter Light at the River Boyne

The found year. 3200 BCE. Two radiocarbon dates on the caulking of the roof slabs, burnt soil and sea sand, give GrN-5462-C at 4425 ± 45 BP and GrN-5463 at 4415 ± 40 BP, which is about 3200 cal BC, with a range of roughly 3300 to 2900 (O'Kelly, 1982).

What the record supports. The mound is about eighty-five metres across and is ringed by ninety-seven kerbstones. A passage about nineteen metres long rises to a cruciform corbelled chamber with basins and a triple spiral. A roof-box over the entrance, with a row of X motifs on its lintel, lets the midwinter sun reach the chamber for about seventeen minutes. M. J. O'Kelly cleared the roof-box in 1963 and watched the sunrise there on 21 December 1967. The slabs are grooved for drainage. Quartz came from Wicklow, granite cobbles from the north across Dundalk Bay and greywacke slabs from the Clogherhead coast. Ancient DNA from the richly carved right-hand recess shows that the parents of the person buried there were first-degree relatives, with distant kin at Carrowmore and Carrowkeel (Cassidy and others, 2020). The reconstruction of the quartz wall is disputed.

Interpretation, kept in pencil. That a ruling family kept its line within itself, and how far the beam reached in 3200 BCE given the change in the Earth's tilt.

Invented for the story. Siadal, Cuan, Liath and the child. The rumour that the high family "married close" is the scene's gentle echo of the genome. The archaeologist is named as in the record.

The doubt that stays. The dates come from the caulking, which dates the roof, not the chamber's first use, and the calibrated range runs to several centuries.

Chapter 5: The Orkney Winter

The found year. 2900 BCE, found by overlapping three windows. Barnhouse was founded about 3160 to 3090 cal BC, probably 3135 to 3100. Its Structure 8 was built just after 3000 and was probably the last element abandoned, in

the earlier twenty-ninth century (Richards and others, 2016; Bayliss and others, 2017). The Stones of Stenness were raised between 3000 and 2900 cal BC and were in use while the village was lived in (Historic Environment Scotland).

What the record supports. The ring, ditch and hearth at Stenness; the village of Barnhouse beside the loch; the passage at Maeshowe facing south-west so that the winter sunset lights the back wall; Skara Brae's stone houses, with their hearths, box beds and dressers, and its Grooved Ware, carved stone balls and whalebone objects.

Interpretation. That Maeshowe was begun while Barnhouse was still lived in is what makes 2900 the overlap. It is an assumption, not a measurement.

The doubt that stays. Maeshowe's own radiocarbon dates conflict with its stratigraphy, and a 2017 review called its chronology uncertain (Bayliss and others, 2017). Historic Environment Scotland says that dates "tentatively suggest" about 2700. If that is right, Barnhouse is empty when the tomb is begun, and the chapter's overlap fails. The prose says so.

The author's reading. The count in this chapter, a knot for every night from midwinter to midwinter, is the story's own idea.

Invented for the story. Gwen, Tam, the grandmother, the stranger with the long face and his tale of a hill with a window over its door, and the knotted cord. No counting cord is known from Neolithic Orkney. The link to the Boyne is the story's own and is kept in pencil. Nobody can say that the people of either place knew of the other.

Chapter 6: The Last Stone from the North

The found year. 2500 BCE. The Altar Stone may have been placed in the second construction phase of Stonehenge, about 2620 to 2480 BC (Clarke and others, 2024). Its date of arrival is unknown and could be any point in Stages 2 to 5 (Parker Pearson and others, 2024). The year is also the span of Durrington Walls, where hundreds of houses stood on the slope, many pigs were killed at about nine months old and so at midwinter, and the settlement dates to about 2500 BC. It is the last generation before the Beaker arrivals.

What the record supports. The Altar Stone is about 4.9 by 1 by 0.5 metres and weighs about six tonnes. It is a pale green micaceous sandstone, and its zircon, apatite and rutile grains match the Old Red Sandstone of the Orcadian Basin in north-east Scotland, at least 750 kilometres away (Clarke and others, 2024). Later work ruled out Mainland Orkney, so the source lies on the Scottish mainland, not the islands (Bevins and others, 2024). The sarsen circle

and the trilithons belong to Stage 2, "centuries either side of 2500" in the prose; Craig and others (2015) give Stage 2 as starting 2760 to 2510 and ending 2470 to 2300 cal BC. The stone was most likely laid flat on purpose, its long sides close to the midwinter sunrise, like the recumbent stones of north-east Scotland. The authors' own phrase for the whole effort is "island-wide unification".

Interpretation. That the stone was laid flat deliberately is a published interpretation. One end is squared and the other dressed to a bevel, and nobody knows whether it ever stood.

Invented for the story. Kerra, Hodh, the boatman and the small boat that pushes off. The chapter does not link the stone to Chapter 5's Orkney, because the research ruled the islands out.

The doubt that stays. If Stage 2 is only a "may", then 2500 is a fair choice and not a proven one.

Chapter 7: Straw and Gifts for the Island of Light

The found year. 430 BCE, where three dates meet: Meton's solstice, traditionally 27 June 432 BC (Diodorus 12.36), the plague at Athens from the summer of 430 (Thucydides 2.47 to 54) and the purification of Delos in 426, when the Athenians removed the graves (Thucydides 3.104). The prose notes a 2021 study that argues for 28 June 433 BC, a year earlier, for Meton's solstice. Herodotus's *Histories* were in circulation by about 430 to 425.

Tradition. The chain of the straw-wrapped offerings from the Hyperboreans, by way of the Scythians, the Adriatic, Dodona, the Malian Gulf, Euboea, Carystus and Tenos to Delos, is the Delians' own account as Herodotus reports it (4.33). So are the maidens Hyperoche and Laodice with their five escorts, the Delian girls' tresses wound round a spindle and laid on the tomb (4.34) and the story of Abaris with his arrow (4.36). Herodotus calls the offerings only *hira*, sacred things, and never says what they are. Delos as the island fixed for Leto comes from the *Homeric Hymn to Apollo*, and the nineteen-year visit of the god comes from Hecataeus of Abdera, by way of Diodorus 2.47. The Terrace of the Lions is seventh-century BCE and attested.

The author's reading. What is inside the straw, something hard, round, cold and then warm, about the size of a hen's egg, plain grey, is the novel's premise. No source says it.

Invented for the story. Euphrosyne, her mother and grandmother, the priest Kleandros, the old man from Tenos and the carrying ceremony. The visitor from Caria is modelled on Herodotus but not named.

The doubt that stays. Nobody says Herodotus stood on Delos that year. The chief priest's fever is invented. The spindle whorls at the maidens' tomb still want a classicist's check.

Chapter 8: The Road Through the Duat

The found year. 1425 BCE, which is the seventy days after the death of Menkheperre Thutmose. A new moon at Megiddo in year 23 of the reign (Karnak Annals; Parker, 1957; Casperson, 1986) fixes the reign, with three options for its start, 1504, 1490 and 1479, of which most take 1479. Amenemheb's tomb records the king's death in year 54, in the third month of winter, which gives 1425 on the 1479 count. KV34 was decorated after the four side rooms were sealed. Romer (1975) proposed that the work was completed under Amenhotep II. The king's linen shroud was inscribed with the Litany of Re and dedicated by Amenhotep II, and a fragment is in Boston (MFA 60.1472).

What the record supports. The Amduat, "what is in the Duat", whose ancient title is the Book of the Hidden Chamber, is first fully attested in KV34. Its walls are painted on a yellowish ground to look like an unrolled papyrus, with stick-like figures and cursive hieroglyphs, mostly written backwards. Each hour has three registers, with the sun's boat in the middle one. The twelve hours include the baboons who open the doors, the sandy land of Sokar, the serpent Apophis, bound and cut, and the Twelfth Hour's rebirth. Copyists wrote *gm wš*, "found destroyed", where their exemplar was damaged. The House of Life was the temple scriptorium and school. The Book of Thoth is a Demotic dialogue in more than forty papyri of the first century BCE to the second century CE, chiefly from Tebtunis and Soknopaiou Nesos.

Interpretation. Romer's reading that the chamber was finished after the king's death is a hypothesis. The prose goes by it and says so.

Invented for the story. Pa-iry, Meret, Amenemhat the draughtsman, the old chief scribe, the overseer of weaving, the colour-boy and the young priest in the Fayum. The oath of secrecy and the kneeling ceremony are invented. Lamp wicks twisted from old linen are best documented at Deir el-Medina in the Ramesside period and are applied here by reasonable extension. That Pa-iry writes *found destroyed* in the Seventh Hour is invented. No death occurs in the chapter. The king is dead, and the road is the finished walls.

The doubt that stays. The 1479 accession is the majority view and not certain. The author has Romer's reading from later summaries and has not been able to re-read the 1975 paper itself.

How the notes use the three layers

In every chapter, the first layer, *Belfast, by day*, is where the year is found and the doubt is written. The second, *That night*, is a dream built on that year, with details the year implies. The third, *The master page*, lays the chapter beside the earlier ones, locks what the evidence supports "in ink", leaves the rest "in pencil" or marks it *found destroyed*, and names the system the chapter adds. The systems run: page, season, line, sea-road, aperture, count, gathering, relay, record. They are the book's own framing, in the manner of a novel, and are not historical claims.

Sources

Ancient authors are cited by book and section in the notes above: Herodotus, *Histories* 4.28, 4.32 to 36 (trans. A. D. Godley, Loeb, 1920); Diodorus Siculus 2.47 and 12.36; Thucydides 2.47 to 54 and 3.104; the *Homeric Hymn to Apollo*; Homer, *Odyssey* 6 and 14; Pindar, *Pythian* 10 and fr. 33c to d; Apollodorus 1.4.1; Iamblichus, *On the Pythagorean Life*; Porphyry, *Life of Pythagoras*.

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Chapter 9: Sixty Days to the Mountain

The found year. 1000 BCE. Kuru country and the road north, late Vedic.

What the record supports. The Vedic hymns were kept word for word by oral recitation, checked by learning each hymn in several patterns (pada, word by word; krama, in overlapping pairs). UNESCO proclaimed Vedic chanting a Masterpiece of the Oral and Intangible Heritage in 2003. Sources: UNESCO 2003 (inscribed 2008); Staal, *Nambudiri Veda Recitation* (1961). The oldest written copies are some two thousand years younger than the hymns. Sources: Witzel, “Vedas and Upaniṣads” (2003). Painted Grey Ware, c. 1200 to 600 BCE, in the Ganga-Yamuna doab; iron appears from about 1000 BCE. Sources: Standard archaeology of the region; Kenoyer; Witzel. The Vedanga Jyotisha’s solstice position computes to c. 1400 to 1150 BCE (Pingree c. 1180), but the text itself is much later (Witzel). Sources: Pingree; Witzel; Wikipedia “Vedanga Jyotisha” (consulted Oct 2026). RV 1.164.48: twelve fellies, one wheel, three hundred and sixty spokes; RV 1.25.8: Varuna knows the extra (thirteenth) month. Sources: Rig Veda, tr. Jamison & Brereton 2014. Soma “of Mūjavant” (RV 10.34.1); Mūjavants named with Balhikas, Gandhāris, Aṅgas and Magadhas in the fever charm AV 5.22, which sends the fever (takman) away; tertian fever; kuṣṭha against takman (AV 5.4). Sources: Whitney/Bloomfield tr. of AV. Kailash is not named in the Vedic texts; it appears first in the epics. Sources: Britannica “Kailas”; Indian Express explainer. RV mandala 3 has sixty-two hymns; RV 3.33 is the dialogue of Viśvāmitra with the rivers. Sources: RV, Jamison & Brereton. A tīrtha is a ford, later a crossing-place in the sacred sense. Sources: Monier-Williams s.v. tīrtha.

Invented for the story. Devadatta, Ila, the teacher, the journey to the mountain for the true plant, the old woman at the ford and her bowl.

Chapter 10: Footprints on the Mesa

The found year. 1290 CE. Orayvi, Third Mesa.

What the record supports. Hopi sun watchers observed sunrise against named points on the eastern horizon from a fixed station and announced dates in advance (anticipation), correcting as they went. Sources: Stephen, *Hopi Journal of Alexander M. Stephen*, ed. E. C. Parsons (Columbia Univ. Contributions to Anthropology 23, 1936); McCluskey, "The Astronomy of the Hopi Indians", *JHA* 8 (1977) 174-195; McCluskey, "Calendars and Symbolism: Functions of Observation in Hopi Astronomy", *Archaeoastronomy* (suppl. *JHA*) 15 (1990) S1-S16; Zeilik, "The Ethnoastronomy of the Historic Pueblos, I: Calendrical Sun Watching", *Archaeoastronomy* 8 (1985) S1-S24. The sun's solstice "house", where it pauses and turns. Sources: McCluskey 1977; Zeilik 1985 (I). The planting time called from a horizon point; the crier on the roof. Sources: Stephen 1936; McCluskey 1977. The specific notch, the 22/23 days and the scene are invented. Hopi name for Chaco is Yupköyvi; Chaco among places Hopi clans lived on their migrations. Sources: Kuwanwisiwma, "Yupköyvi: The Hopi Story of Chaco Canyon", in D. G. Noble (ed.), *In Search of Chaco* (SAR Press, 2004), 41-47. Ancestral sites as "footprints"; clan symbols left on rock. Sources: Kuwanwisiwma 2004; Ferguson and Colwell-Chanthaphonh, *History Is in the Land* (Univ. of Arizona Press, 2006); Bernardini 2005 (clan-symbol rock art). "Anasazi" is not a Hopi word; Hopi prefer their own term for the ancestors. Sources: Kuwanwisiwma 2004; Hopi Cultural Preservation Office statements. Fajada Butte: three slabs, a dagger of light on a spiral at midsummer (found 1977, published 1979); reassessed as unlikely to be an accurate calendar, more likely a sun shrine. Sources: Sofaer, Zinser and Sinclair, *Science* 206 (1979) 283-291; Zeilik, "A Reassessment of the Fajada Butte Solar Marker", *Archaeoastronomy* 9 (1985) S69-S85; Zeilik, *Science* 228 (1985) (reevaluation). Chaco's great houses (multi-storey, D-shaped Pueblo Bonito), straight roads, solar/lunar alignments argued. Sources: Lekson and Judge in Noble 2004; Sofaer, Marshall and Sinclair 1989; Reyman, *Science* 193 (1976). The Great Drought 1276-1299 in the tree rings; the Mesa Verde region depopulated by c. 1300. Sources: Douglass 1929; Lipe, "The Mesa Verde Region: Chaco's Northern Neighbor", in Noble 2004. Clans arriving at the Hopi Mesas by serial migration, c. 1275-1400. Sources: Bernardini, *Hopi Oral Tradition and the Archaeology of Identity* (Univ. of Arizona Press, 2005). Orayvi occupied by the 1100s. Sources: Standard Hopi archaeology. Some 19th-century journals printed restricted ceremonial material; the Hopi Tribe has a Cultural Preser-

vation Office (est. 1989) and has asked that such material be kept out of print. Sources: HCPO; Kuwanwisiwma. Stephen 1936 is used only for the horizon watching.

Invented for the story. The girl (unnamed), her mother, grandmother, uncle, the old watcher, the family's road.

From the author's life. The details of the modern man's work and history in this chapter are drawn from the author's own records and published writing.

Chapter 11: The Ice at the Edge of the Sea

The found year. 650 CE, as the tradition gives it. West Polynesia.

What the record supports. S. Percy Smith, *Hawaiki* (1899), dates Ui-te-rangiora to about 650 CE by counting forty-eight generations at twenty-five years; he places the "head-quarters" in Fiji, Tonga and Samoa; names the canoe Te Ivi o Atea; lists the wonders seen in the south. Sources: Smith 1899 (NZETC text). Wehi et al. 2021 read the tradition as early Polynesian voyaging into Antarctic waters; Anderson et al. 2021 reply that the "ice" reading rests on a late gloss: ʻuka = foam, pia = arrowroot, no pre-European word for ice; Te Rangi Hīroa 1926 already doubted it. Sources: Wehi et al., *J. R. Soc. NZ* 2021; Anderson et al. 2021 reply. Ngāi Tahu reply on the origin-story framing. Sources: Published response, 2021. East Polynesia settled late: Society Islands c. 1025 to 1120, remote islands including Rarotonga c. 1190 to 1290; nobody on Rarotonga in 650. The dream is moved west for that reason. Sources: Wilmshurst et al., *PNAS* 2011.

Invented for the story. Teariki, the crew, the fog, the crust on the sea, the gourd that turned to water, the chief's pouring.

Chapter 12: The Silver Branch

The found year. 720 CE. Druim Snechta.

What the record supports. Immram Brain: the silver apple branch from Emain, "twigs of white silver. crystal brows with blossoms"; the woman in strange clothes; Manannán in his chariot; the sea that is a flowering plain; Nechtan son of Collbran a heap of ashes; Bran writes the quatrains in ogam and sails away. Sources: Meyer & Nutt, *The Voyage of Bran* (1895), §§2-3, 32-34, 36, 62-66; Mac Mathúna, *Immram Brain* (1985). The headings say the woman sang fifty quatrains and Manannán thirty; every copy has twenty-eight of each. Sources: Meyer 1895; UCC CELT text; sacred-texts synthesis. The tale's language is round about 700; the oldest manuscripts are centuries

later. Sources: Meyer; Mac Mathúna; CODECS. Later scribes say they copied from the lost Cín Dromma Snechtai (Drumsnat, Co. Monaghan): Thurneysen first half of 8th c.; Mac Mathúna late 9th/10th. Sources: CODECS "Cín Dromma Snechtai" (verified 30 Oct 2026). The Cín also held Compert Con Culainn, Baile Chuinn and Echtrae Chonnlai. Sources: CODECS. Manannán's song foretells a great birth "the son of a woman whom mate will not know". Sources: Meyer §§48-50 (wording adapted).

Tradition. Ogam cut on wood (hazel) as well as stone. Sources: Saga references; McManus, *A Guide to Ogam* (1991).

Invented for the story. Eochaid, Dearbhla (a tenant girl, not a nun), the abbot's order and his change of mind, the night copying, the frosted branch.

Chapter 13: The Man Who Went to Thule

The found year. 325 BCE. Massalia to Thule.

What the record supports. Thule is "a six days' sail north of Britain, and is near the frozen sea"; Strabo calls Pytheas the arch-falsifier. Sources: Strabo 1.4.2-3. One day's sail beyond Thule is the frozen sea. Sources: Pliny, *NH* 4.104. The "marine lung", where land, sea and air are mixed and one can neither walk nor sail; Pytheas says he saw the lung-like thing himself, and the rest he reports from hearsay. Sources: Strabo 2.4.1. "The barbarians showed us where the sun goes to rest"; nights of two hours in some places, three in others. Sources: Geminus, *Introduction* 6.9. No star at the pole; an empty space with three stars near it making a rough quadrangle with the pole point. Sources: Hipparchus, *Commentary on Aratus* 1.4.1. Gnomon to shadow at Massalia at the solstice 120 : 41⅕, the same ratio Hipparchus finds for Byzantium. Sources: Strabo 1.4.4, 2.5.41 (Hipparchus). Polybius: a private man, and a poor one. Dicaearchus disbelieved him. Sources: Strabo 2.4.1-2. Tides follow the moon (Pytheas); very high tides north of Britain. Sources: Aetius 3.17.3; Pliny *NH* 2.217. Belerion: hospitable people, tin beaten into knucklebone shapes, carried to the island at low tide; Orkas the northern cape. Sources: Diodorus 5.21-22. Far north: millet, herbs, roots; drink of grain and honey; threshing in large barns because of rain and lack of sun. Sources: Strabo 4.5.5. Britons cross the sea in boats of wicker sewn round with hide. Sources: Pliny, *NH* 4.104. Pytheas's own boats are not recorded: PERIOD PATTERN. Dating window: Aristotle (d. 322) silent; Dicaearchus (d. c. 285) knew the work. Roller 320s or shortly after; Cunliffe before 320; older reckoning c. 310-306. Sources: Roller, *Through the Pillars of Herakles* (2006); Cunliffe, *The Extraordinary Voyage of Pytheas the Greek* (2001).

The author's reading. Pytheas gave distances in days, later converted to stadia. Sources: Nansen, *In Northern Mists* (1911); Roller.

Invented for the story. Nikon, the merchant backers, the fog day, the struck-through six.

Chapter 14: The School at Croton

The found year. 530 BCE. Croton.

What the record supports. Pythagoras left Samos at forty when Polycrates' tyranny grew harsh, and went to Croton. Sources: Porphyry, *Life of Pythagoras* 9. Apollodorus puts his floruit in 532/1; chronographers set the floruit at forty. Sources: Apollodorus via Clement/Diogenes; standard chronology. The full lives (Diogenes Laertius, Porphyry, Iamblichus) are 3rd-4th c. CE; early mentions: Xenophanes (the beaten puppy), Heraclitus (much learning). Sources: DL 8.36; Heraclitus DK 22 B40. The right-triangle relation was known in Babylon more than a thousand years earlier; no early source credits Pythagoras with a proof; the ox-sacrifice story is late. Sources: Plimpton 322, YBC 7289; Burkert, *Lore and Science* (1972). The smithy-hammer story is from Nicomachus (2nd c. CE) and is physically false; the ratios 2:1, 3:2, 4:3 hold for string lengths; Hippasus's bronze discs. Sources: Nicomachus, *Manual of Harmonics* 6; Aristoxenus fr. 90. Theano: wife (DL 8.42), wife of Brontinus (Iamblichus VP 267), or daughter, in different traditions. Sources: As cited. The rules (akousmata): beans; stirring fire with a knife; not picking up what falls; help load, not unload; not stepping over a balance beam; some collected by Aristotle. White clothes. Sources: DL 8.17-19, 8.34-35 (Aristotle). Croton's physicians reckoned first in Greece; Milo of Croton, Olympic wrestler. Sources: Herodotus 3.131; Olympic victor lists. War of 510: Telys exiles 500 rich Sybarites, Croton shelters them at Pythagoras' urging, Milo leads Croton, Sybaris destroyed and the river turned over it. Sources: Diodorus 12.9-10; Strabo 6.1.13. Eight-year cycle (octaeteris): 99 months, intercalations in years 3, 5, 8; slip of about a day and a half per cycle. Philolaus' later 59-year cycle; Meton's 19 years in 432. Sources: Geminus 8.27-33; Censorinus 18.8. A lesson on it at Croton is invented.

Tradition. New pupils kept silence (five years in the late tradition); friends' goods held in common. Sources: Iamblichus VP 72; DL 8.10 (Timaeus). Evening self-examination (where did I go wrong, what did I do, what did I leave undone). Sources: *Golden Verses* 40-44; Porphyry VP 40.

Invented for the story. Philon, his father, the fever and the string at night, the letter, the coiled string.

Chapter 15: Twenty Years of Verses

The found year. 53 BCE. the Sequani.

What the record supports. Druid pupils learn many verses and some stay twenty years; writing the teaching down is not allowed, though Greek letters are used for other business; the two reasons given (to keep it from the common people, and to keep memories sharp). Sources: Caesar, *Gallic War* 6.13–14. Gauls count time by nights, not days. Sources: Caesar, *BG* 6.18. Caesar's ethnography may draw on Posidonius and is an enemy's description. Sources: Standard commentary (e.g. Webster 1999; Rankin). The mistletoe rite on the sixth day of the moon; a thirty-year age. Sources: Pliny, *Natural History* 16.249–250. The Coligny calendar: found 1897 near Coligny (Ain), nearly 150 bronze fragments (149), five years of 62 months, an extra month every thirty months, MAT/ANM (good/not good), Roman letters; dated 1st or 2nd century CE. Sources: Musée Lugdunum; Patrimoines de l'Ain; Olmsted 1992. 53 BCE: Book 6 of the *Gallic War*, the second Rhine crossing; the Sequani and Vesontio (Besançon). Sources: Caesar, *BG* 1.38, 6.9–10.

Invented for the story. Ategnatos, Marcus, the knotted cord, the clerk counting cattle.

Chapter 16: The Boy and the White Pebble

The found year. 580 CE. Iona.

What the record supports. Adomnán, ninth abbot of Iona, wrote the *Life of Columba* c. 697–700. Sources: Sharpe, *Adomnán of Iona: Life of St Columba* (1995), introduction. The oldest copy was written on Iona by Dorbbéne (d. 713) and is now in Schaffhausen (Stadtbibliothek, Generalia 1). Sources: AU 713; Anderson and Anderson 1991. Cormac Ua Liatháin seeks a "desert in the ocean" and does not find it. Sources: *VC* 1.6. Fourteen summer days and nights before a south wind, due north, "beyond the bounds of human wandering"; small loathsome stinging creatures the size of frogs strike keel, sides, stern and prow and threaten the leather covering; Columba on Iona calls the brothers to pray; the wind turns north. Sources: *VC* 2.42. The white pebble from the river Nesa, blessed for Broichan, King Bridei's magus, who must free an Irish slave girl; it floats; kept in the king's treasury. Sources: *VC* 2.33. Columba on Iona 563 to 597; Bridei son of Maelchon d. 584. Sources: Annals of Ulster 563, 584, 597. Irish clerics in Thule c. 795, where the mid-

summer sun hardly set. Sources: Dicuil, *De mensura orbis terrae* 7.11 (825). The text does not name Iceland; the creatures are unidentified. Sources: *VC* 2.42.

Invented for the story. Lugaid, Brother Diarmait, the kitchen, Lugaid's own pebble, the notches, the hill at night; Cormac's spoken lines.

Chapter 17: The Boat of Hide

The found year. 530 CE, traditional. the western ocean.

What the record supports. Brendan founds Clonfert (558 or 564) and dies in 577. Sources: Annals of Ulster. The *Navigatio Sancti Brendani* was written c. 800 or later; more than a hundred manuscripts survive; the oldest are 10th century. Sources: Selmer 1959; Burgess and Strijbosch 2000. Fourteen monks, forty days' fast, a wooden frame covered in ox hides tanned in oak bark with the joints greased, spare hides; three latecomers. Sources: *Navigatio* §§4–5. The round kept for seven years: Maundy Thursday on the Island of Sheep, with the steward (procurator); Easter on Jasconius; Easter to Pentecost on the Paradise of Birds; Christmas with the community of Ailbe (silent for eighty years, lamps lit by a fiery arrow). Sources: *Navigatio* §§9–15. The crystal column in a net, with a chalice and paten; the island of smiths throwing slag. Sources: *Navigatio* §§22–23. Forty days with the steward, through a fog, to the Promised Land; a youth greets each by name, says God delayed them to show his secrets in the ocean, and bids them take fruit and precious stones. Sources: *Navigatio* §28. Severin's *Brendan*: an ox-hide boat on an ash frame from Brandon Creek (May 1976), by the Hebrides, Faroes and Iceland, to Newfoundland (June 1977). This proves it could be done, not that it was. Sources: Severin, *The Brendan Voyage* (1978).

Invented for the story. Dubán, the stern oar, the Easter talk, the red stone; Brendan's spoken lines.

Chapter 18: The Venus Table

The found year. 889 CE. Uaxactún to Chichén Itzá.

What the record supports. Group E at Uaxactún: a western pyramid (E-VII-sub) facing three temples on a long platform; sunrise over the north and south temples at the solstices and the middle one at the equinoxes; the type name "E Group". Sources: Blom 1924; Ricketson and Ricketson 1937. The solstice sightlines are approximate and the equinox one is poor; the buildings are from different periods. Sources: Aveni and Hartung 1989; Aveni, Dowd and Vining 2003; Šprajc 2021. E-VII-sub with its stucco masks is Late Pre-

classic. Sources: Ricketson; Valdés. Uaxactún Stela 12 is the city's last dated monument (10.3.0.0.0, 889 CE); Tikal's last is Stela 11 (869). Sources: Graham; Martin and Grube 2008. Chichén Itzá's dated inscriptions mostly fall c. 870 to 890; a Caracol band date is read as 889 or close to it. Sources: Volta and Braswell 2014; Grube and Krochock. Dresden Codex Venus table: phases 236 / 90 / 250 / 8 days; 584-day mean; five cycles = 2,920 days = eight 365-day years; 65 cycles = 104 years; corrections; base 934 CE; the codex was painted later; one of four surviving Maya books. Sources: Lounsbury 1978; Aveni 2001; Bricker and Bricker 2011; Aldana 2016. Some Caracol windows align on Venus's northern setting extreme. Sources: Aveni, *Skywatchers* (2001). The 260-day count (twenty day names, thirteen numbers) used to name a child's day. Sources: Tedlock 1982. Drought in the late 9th-century southern lowlands. Sources: Hodell et al.; Kennett et al. 2012; the debate is ongoing.

Invented for the story. Ixkan, Ah K'in, the family's walk north, the carver, the trader.

Chapter 19: The Fifty-Two Years

The found year. 1519 CE. Tenochtitlan.

What the record supports. The 365-day year and the 260-day count realign every 52 years (18,980 days); years are named by four signs and thirteen numbers. Sources: Sahagún, *Florentine Codex* Bk 7; Caso 1967. The New Fire: all fires out, hearthstones and god images thrown into water; pregnant women masked with maguey leaves and shut in granaries; children kept awake lest they become mice; the Pleiades (Tianquiztli, "marketplace") watched from Huixachtlan until they cross the zenith; fire drilled on a captive's chest; runners carry it out. Sources: Sahagún, *FC* Bk 7, chs 9–12. The last New Fire was in 1507 (Two Reed); the next was due in 1559; 1519 is One Reed. Sources: *FC* Bk 7; Codex Mendoza f. 15v (Two Reed sign). The omens and the Quetzalcoatl return are recorded after the conquest and are widely thought to have been constructed later. Sources: *FC* Bk 12; Townsend 2003; Restall 2003. The Tlatelolco market: its rows, judges, cotton cloaks (quachtli) and cacao as currency. Sources: Cortés, Second Letter (1520); Bernal Díaz. Painters: "the black and the red"; red from cochineal, black from soot; bark (amate) paper screenfolds. Sources: *FC* Bk 10; León-Portilla. Moctezuma sent painters to depict the strangers; horses called "deer". Sources: Durán; *FC* Bk 12. Trade beads given on the coast in 1518. Sources: Grijalva expedition accounts. The Sun Stone (found 1790, most likely carved in Moctezuma II's reign); the Codex Mendoza (c. 1541); the Templo Mayor (excavated from

1978). Sources: INAH; Bodleian MS Arch. Selden. A. 1. Kilroot's coal units closed at midnight on 30 September 2023, after more than forty years. Sources: EP Kilroot; BBC News NI; Irish News (Sep/Oct 2023).

Invented for the story. Tlalli, Citlali, the master, the merchant, the roof scene.

Chapter 20: The Burning at Croton

The found year. 510 BCE. Croton.

What the record supports. Cicero visited Metapontum and saw the place where Pythagoras died, more than four hundred years later. Sources: Cicero, *De finibus* 5.4. Later burnings of the Pythagorean meeting-houses across Magna Graecia, about fifty years on; kept as aftermath only. Sources: Polybius 2.39.

Tradition. The Pythagoreans wrote nothing down in the early generation; the teaching was passed by mouth. Philolaus is said to be the first to publish a book. Sources: Diogenes Laertius 8.15, 8.85; Iamblichus *VP* 199. After Croton destroyed Sybaris (510), a quarrel over the conquered land; Cylon, a rich man refused entry, raised the town; the house of Milo where they met was set on fire. Sources: Aristoxenus via Porphyry *VP* 54 to 57; Iamblichus *VP* 248 to 251. The versions of Pythagoras's death do not agree: caught at a bean field (DL 8.39); fled to Metapontum and starved in the temple of the Muses for forty days (Dicaearchus, DL 8.40). Sources: Diogenes Laertius 8.39 to 40. The morning and evening star as one star, attributed to Pythagoras (also to Parmenides). Sources: Diogenes Laertius 8.14, 9.23. The two-to-one string (the octave) on a single stretched string. Sources: Nicomachus, *Harmonicum enchiridion* 6; see Ch14 notes. The scattering of survivors to Locri, Tarentum and Metapontum; Theano as keeper of the teaching. Sources: Iamblichus *VP* 249 to 251; Porphyry *VP* 57 to 58. Theano's role is late tradition.

Invented for the story. Myrto, Philon (33), the grandfather, the barn, the eleven survivors and their pieces.

From the author's life. The details of the modern man's work and history in this chapter are drawn from the author's own records and published writing.

Chapter 21: Mona

The found year. 61 CE. Anglesey.

What the record supports. Suetonius Paulinus crossed to Mona with flat-bottomed boats for the infantry; cavalry by the fords or swimming beside their horses. Sources: Tacitus, *Annals* 14.29. Women in black like Furies with

torches and loose hair; druids with raised hands pouring curses; the soldiers froze, then advanced; a garrison was set and the groves cut down. Sources: Tacitus, *Annals* 14.30. News of Boudica's rising pulled the army away. Some put the attack in 60. Sources: Tacitus, *Annals* 14.29 to 31; *Agricola* 14 to 16. Agricola served on Paulinus's staff and returned to take Mona in 77 or 78. Sources: Tacitus, *Agricola* 5, 18. Druids still active in Gaul in 69/70. Sources: Tacitus, *Histories* 4.54. Llyn Cerrig Bach: found in 1942 while peat was dug for the runways at RAF Valley; swords, spears, chariot fittings, cauldrons, a crescent plaque, slave gang-chains; deposited over generations. Sources: Fox 1946; Macdonald 2007; National Museum Wales.

Invented for the story. Cunovinda, Taso, Vercassos, the mother, the lake scene, the young soldier.

From the author's life. The details of the modern man's work and history in this chapter are drawn from the author's own records and published writing.

Chapter 22: The Last Priests of Philae

The found year. 394 CE. Philae.

What the record supports. On the Gate of Hadrian at Philae, a figure of Mandulis with the last dated hieroglyphic inscription, by Nesmeterakhem, son of Nesmeter, Second Priest of Isis, "for all time and eternity"; a demotic note dated the birthday of Osiris, year 110 (of Diocletian), i.e. 24 August 394. Sources: Griffith, *Catalogue of the Demotic Graffiti of the Dodecaschoenus* (1937), Ph. 436; Cruz-Uribe. The decrees of 391 against sacrifice and temple visits; the Serapeum destroyed 391/2. Sources: *Codex Theodosianus* 16.10.10 to 11; Rufinus *HE* 11.22 to 30. Philae continued: the last dated demotic graffito is from 452; the temple was closed by Narses under Justinian, about 535 to 537. Sources: Griffith 1937 (Ph. 377, Procopius, *Wars* 1.19.36 to 37; Dijkstra 2008. The Egyptian civil year: twelve months of thirty days and five days upon the year, birthdays of Osiris, Horus, Set, Isis, Nephthys. Sources: Plutarch, *De Iside* 12; Parker 1950. Sopdet (Sirius) rising before the sun and the flood. Sources: Parker 1950; Clagett. Set honoured by early kings (Seti I named for him) and demonised in the late temples. Sources: te Velde, *Seth, God of Confusion* (1967). Champollion used the Philae obelisk (Kingston Lacy) with the Rosetta Stone; Cleopatra's cartouche; 1822. Sources: Champollion, *Lettre à M. Dacier* (1822).

Invented for the story. Tabes, her grandfather the doorkeeper, her mother at Syene, the roof scene.

From the author's life. The details of the modern man's work and history in this chapter are drawn from the author's own records and published writing.

Chapter 23: Whitby, 664

The found year. 664 CE. Northumbria.

What the record supports. Easter kept twice in Oswiu's house: the king (Iona) feasting while Queen Eanflæd (Kent, Roman) still fasted. Sources: Bede, *HE* 3.25. The meeting at Hild's house (Streanæshalch); Colmán for John and Columba; Agilbert hands the case to Wilfrid; Wilfrid on Rome and the world against the Picts and Britons; the keys of Peter and the king's "doorkeeper" decision. Sources: Bede, *HE* 3.25; Stephen, *Life of Wilfrid* 10. Wilfrid about thirty (b. c. 633/4), with the Roman tonsure. Sources: Stephen, *VW*; Bede *HE* 5.19. The 84-year table (Iona, Easter on luna 14 to 20) against the 19-year Alexandrian/Dionysiac reckoning (luna 15 to 21). Sources: Bede, *HE* 3.25; Mc Carthy; Wallis, *Bede: The Reckoning of Time* (1999). Hild baptised by Paulinus at York (627) with Edwin; later instructed by Aidan. Sources: Bede, *HE* 4.23. The eclipse of the sun of 1 May 664 (Bede gives 3 May, about the tenth hour); the plague; Tuda died. Sources: Bede, *HE* 3.27; NASA eclipse canon. Colmán left with the Irish and about thirty English, carrying some of Aidan's bones; later Inishbofin (668). Sources: Bede, *HE* 3.26, 4.4. Southern Ireland conformed about 630; Iona in 716. Sources: Cummian's letter (c. 632); Bede *HE* 5.22; Annals of Ulster 716.

Invented for the story. Ceolla, the barn conversation, Hild's porch scene, the eclipse on the road north, Colmán's last words.

From the author's life. The details of the modern man's work and history in this chapter are drawn from the author's own records and published writing.

Chapter 24: The Statutes of Iona

The found year. 1609 CE. Moidart.

What the record supports. 1608: chiefs invited aboard a king's ship off Mull to hear a sermon and carried off to Lowland prisons. Sources: Gregory, *History of the Western Highlands* (1836); MacGregor 2006. August 1609: Bishop Andrew Knox at Iona; nine statutes signed; among the signatories Donald MacAllan of Clanranald (Eilean Tioram). Sources: *Register of the Privy Council of Scotland* ix, 26 to 30. The articles: churches and ministers, inns, limits on households, wine and whisky, heirs of men with sixty cattle to Lowland schools to learn English, wandering bards and idle men to be put in the stocks and expelled, firearms. Sources: *RPC* ix; MacGregor 2006. The statutes do not mention Gaelic by name; the 1616 Act of the Privy Council made English schooling a condition of heirs' inheritance. Sources: *RPC* x (1616). Hered-

itary poets (the MacMhuirich family to Clanranald); the Red Book of Clanranald. Sources: Thomson, *Companion to Gaelic Scotland*. Poets composing in a dark room lying on their backs with a stone on the belly. Sources: Martin Martin, *A Description of the Western Islands of Scotland* (1703).

Invented for the story. Mairead, her father, Eòghan, the chief's boy (unnamed), the song "the stars of the autumn".

From the author's life. The details of the modern man's work and history in this chapter are drawn from the author's own records and published writing.

Chapter 25: The Knotted Cord

The found year. 1680 CE. Ohkay Owingeh.

What the record supports. A cord of maguey fibre with knots for the days to wait, passed through the pueblos; a knot untied each day. Sources: Declaration of Pedro Naranjo, 19 Dec 1681, in Hackett and Shelby, *Revolt of the Pueblo Indians of New Mexico and Otermín's Attempted Reconquest, 1680-1682* (Univ. of New Mexico Press, 1942). Nicolás Catua and Pedro Omtua of Tesuque taken on 9 Aug 1680; the rising, set for 11 Aug, began on 10 Aug (feast of San Lorenzo). Sources: Otermín's autos, Hackett and Shelby 1942; Knaut 1995. Drought, famine and Apache raids in the 1670s; 47 Pueblo religious men arrested in 1675, four condemned, three hanged; Po'pay among those released. Sources: Knaut, *The Pueblo Revolt of 1680* (Univ. of Oklahoma Press, 1995); Liebmann 2012. Many leaders across about two dozen towns; Po'pay one of them. Sources: Knaut 1995; Liebmann, *Revolt: An Archaeological History of Pueblo Resistance and Revitalization in 17th Century New Mexico* (Univ. of Arizona Press, 2012). Post-revolt mesa-top villages; Spanish livestock and metal kept. Sources: Liebmann 2012. Friars killed, including at the Hopi mesas; Otermín's column left Santa Fe southward on 21 Aug 1680; the Spanish stayed away until 1692. Sources: Hackett and Shelby 1942; Knaut 1995. Order to wash off baptism in the river with yucca (amole). Sources: Naranjo and others, Hackett and Shelby 1942. Earlier confiscation and burning of masks and sacred objects (1660s). Sources: Knaut 1995.

Invented for the story. Lorenzo, his grandmother, Fray Tomás, Francisco, the uncle.

From the author's life. The details of the modern man's work and history in this chapter are drawn from the author's own records and published writing.

Chapter 26: The Pandit at Nadia

The found year. 1785 CE. Krishnanagar.

What the record supports. Plassey, 23 June 1757, in Nadia; Mir Jafar changed sides; the diwani 1765. Sources: Standard histories (e.g. Marshall, *Bengal: The British Bridgehead*). *A Code of Gentoo Laws* (1776): compiled by eleven pandits for Hastings, translated via Persian by Halhed. Sources: Halhed 1776, preface. The Asiatick Society founded by William Jones in Calcutta, January 1784. Sources: *Asiatick Researches* i (1788). Jones at Krishnanagar in the autumn of 1785; Brahmins away for Durga's festival; "a pleasant old man of the medical cast, who teaches me all he knows of the Grammar". Sources: Jones to Wilkins, 1785 (Cannon, *Letters* II, 682). Two watercolours of Ramlochan Cantaberna, dated 15 October 1785, Krishnanagar: Vaidya, teacher of Sanskrit grammar at Nadia, aged 65; seated with pen and manuscript. Sources: Royal Asiatic Society Archives WJ/3/1/2 and companion (Lady Jones's cards). Nadia's schools famous for logic (Navya-nyaya) and law. Sources: Standard histories of Navadvipa learning. The Amarakosha arranges words by subject, beginning with heaven (svarga-varga); the nakshatra list placed there. Sources: Colebrooke ed. 1808. The 27 nakshatras beginning with Ashvini; Dhruva as the pole star (Bhagavata Purana 4.8 to 9). Sources: Bhagavata Purana; Pingree. Macaulay's Minute on Education, 2 February 1835. Sources: Macaulay 1835.

Invented for the story. Shibu, Ramgopal, Hartley, the conditions, the slips, the roof, the pencil. Jones and Ramlochan are offstage and given no words.

From the author's life. The details of the modern man's work and history in this chapter are drawn from the author's own records and published writing.

Chapter 27: The Green Dish of Caesarea

The found year. 1101 CE. Caesarea and Genoa.

What the record supports. Caesarea taken in May 1101 by King Baldwin I and the Genoese fleet under Guglielmo Embriaco ("Mallet-head"). Sources: Fulcher of Chartres II.9; Caffaro, *Annales Ianuenses*. Shares: forty-eight shillings (solidi) and two pounds of pepper a man. Sources: Caffaro, *Annales*. The holy fire at the Sepulchre failed on Holy Saturday 1101 and came on Easter morning; the Genoese were in Jerusalem that Easter. Sources: Fulcher II.8; Caffaro. Caffaro says nothing of a green dish; William of Tyre (c. 1170s) says the Genoese took a green vessel they believed emerald, in place of a large sum. Sources: William of Tyre, *Historia* X.16/XI.14 (edition numbering varies).

Jacopo da Varagine (archbishop of Genoa, late 13th century) first calls it the dish of the Last Supper. Sources: *Chronicon Ianuense*. Taken to Paris 1806; studied and found to be glass; returned 14 June 1816 broken in ten pieces, one missing; restorations since. Sources: Museo del Tesoro di San Lorenzo; Paris 1807 note. Modern view: Islamic glass, 9th to 10th century. Sources: Museum catalogue; recent restoration reports.

Invented for the story. Nicolò, Bartolomeo, the clerk, the steersman; Embriaco's lines.

From the author's life. The details of the modern man's work and history in this chapter are drawn from the author's own records and published writing.

Chapter 28: A Cup for a Valley

The found year. 1135 CE. Jaca.

What the record supports. Alfonso I the Battler died in 1134 after the defeat at Fraga; his will left the kingdom to the Temple, the Hospital and the Holy Sepulchre; the Aragonese took his brother Ramiro, a monk, as king; Ramiro married in 1135. Sources: Standard histories of Aragon (Lacarra; Ubieto). Grant at Jaca, November 1135: Ramiro II gives San Juan de la Peña and Santa María de Iguácel land at Garcipollera, Villanovilla and Bescós in return for a chalice of precious stone they had given him. One chalice; it does not say which. Sources: Liber privilegiorum of San Juan de la Peña (16th-century copy, abbot Fenero; Univ. of Zaragoza). The 1134 "inventory" with the chalice in an ivory chest is known only from Carreras Ramírez (1698) and has not been found. Sources: Carreras Ramírez, 1698. Jaca on the pilgrim road to Santiago (Camino Aragonés); the cathedral. Sources: Standard.

Invented for the story. Pero, García, the cold room, the copy, the abbot's words; Ramiro's one line.

From the author's life. The details of the modern man's work and history in this chapter are drawn from the author's own records and published writing.

Chapter 29: Names on the Wall at Maeshowe

The found year. 1153 CE. Orkney.

What the record supports. Earl Harald's men sheltered in Orkahaugr in a snowstorm at Yule; two went out of their minds; they went on to Firth late. Sources: *Orkneyinga saga* ch. 93. About thirty runic inscriptions: Jerusalem-farers broke the mound; Hlíf the earl's housekeeper carved; Ingigerth the most beautiful; treasure hidden long ago, Hákon carried it out; the man most

skilled in runes with Gaukr Trandilsson's axe; the carved beast. Sources: Barnes, *The Runic Inscriptions of Maeshowe, Orkney* (1994). Runes dated c. 1125 to 1175; link to Rögnvald's crusade (1151 to 1153) a hypothesis. Sources: Barnes 1994. The passage takes the midwinter setting sun onto the back wall, over the hills of Hoy. Sources: Historic Environment Scotland. Southern stars invisible from Orkney (e.g. Canopus); shadowless noon claim is a traveller's exaggeration in dialogue. Sources: Standard.

Invented for the story. Ketil, his father, the palm-leaf man; Hlíf cutting her name that night.

From the author's life. The details of the modern man's work and history in this chapter are drawn from the author's own records and published writing.

Chapter 30: The Count at Clontarf

The found year. 1308 CE. Dublin.

What the record supports. Edward II's writ of 20 Dec 1307 to the justiciar John Wogan, received 25 Jan 1308; Templars of Ireland taken 3 Feb 1308 and their goods inventoried (Clontarf, Kilclogan, Crook, Kilbarry and others). Sources: Wood, "The Templars in Ireland", PRIA 26 C (1907), 327 to 377. Allowances paid to nineteen brothers in 1308. Sources: Wood 1907. Henry Danet, master in Ireland; tried in Dublin 1310; testimony sent to the English inquisitors; confessed nothing of his own; bailed 1312. Sources: Wood 1907; Nicholson, "The Testimony of Brother Henry Danet" (2008). Templar lands passed to the Hospitallers after 1312. Sources: Council of Vienne; Wood.

Invented for the story. Adam, his father, Brother Walter, the third ring on the cloth; Danet offstage, no words.

From the author's life. The details of the modern man's work and history in this chapter are drawn from the author's own records and published writing.

Chapter 31: Gold for Stone

The found year. 1399 CE. San Juan de la Peña.

What the record supports. Notarial act of 26 Sept 1399: the monks give King Martin I a stone chalice at his request and receive a gold one. Sources: Archivo de la Corona de Aragón (PARES). Martin crowned at Zaragoza in 1399; a collector of relics; the cup among his goods at his death, 1410. Sources: Standard; ACA. Alfonso V brought it to Valencia in the 1430s; 1437 to the cathedral among royal relics, as security for a loan. Sources: Valencia cathedral documents. Agate bowl dated 2nd century BCE to 1st century CE

(eastern Mediterranean); mount and base later. Sources: Beltrán, *El Santo Cáliz de la Catedral de Valencia* (1960). Julian-calendar equinox c. 13 Sept in 1399, so "a few days before" Michaelmas; shadow-line equinox method. Sources: Standard.

Invented for the story. Domingo, the prior's words, Brother Pedro, All Saints.

From the author's life. The details of the modern man's work and history in this chapter are drawn from the author's own records and published writing.

Chapter 32: Salt for the Isle of Brasil

The found year. 1481 CE. Bristol.

What the record supports. 1480: John Jay's ship, master Lloyd, sailed 15 July for the Isle of Brasil, driven back by storms about nine weeks later. Sources: William Worcestre, *Itineraries*. 1481: the *Trinity* and the *George*, Thomas Croft part-owner; forty bushels of salt; inquiry found it was not trade but to search for the Isle of Brasil. Sources: 1481 inquisition (Quinn, *England and the Discovery of America*; Carus-Wilson). John Day's letter (winter 1497/98): Bristol men found "Brasil" in the past. Sources: Day letter (Vigneras 1956). Bristol's Iceland trade; Icelanders in Bristol. Sources: Carus-Wilson, *Overseas Trade of Bristol* (BRS 7). Brasil drawn as a round island with a channel west of Ireland on medieval charts. Sources: Portolan charts (e.g. Dalorto 1325 onwards).

Invented for the story. Joan, her father, Einar, the London officer; Croft's speech at the inquiry.

From the author's life. The details of the modern man's work and history in this chapter are drawn from the author's own records and published writing.

Chapter 33: The Stone Dipped Without Words

The found year. 1638 CE. Lee, Lanarkshire.

What the record supports. The Lee Penny: a dark red stone set in an Edward IV groat, kept by the Lockharts of Lee; dipped in water and the water given to cattle and people; no words said. Sources: Scott, *The Talisman* (introduction, 1832); Lockhart family accounts. Gavin Hamilton of Raploch's complaint against Sir James Lockhart of Lee; the Synod of Glasgow found no offence, since no words such as charmers use were spoken, God having given stones and herbs a virtue; the laird to use it with least scandal. Sources: Synod act as quoted by Scott and later writers. The National Covenant signed in parishes in 1638, some in blood; marks for those who could not write. Sources: Standard histories.

Tradition. Teba 1330, Douglas and the Bruce's heart, the emir's mother's ransom; borrowing in a time of plague (Newcastle). Sources: Family tradition.

Invented for the story. Elspeth, Bessie, the steward, the ministers, the pebble from Largs.

From the author's life. The details of the modern man's work and history in this chapter are drawn from the author's own records and published writing.

Chapter 34: A Boy and a Beach

The found year. 2009 CE. Belfast.

What the record supports. Harland and Wolff's cranes Goliath (1969) and the taller Samson (1974). Sources: Harland and Wolff; NI Historic Environment Division listing. Belfast Ropeworks, once among the largest in the world; Belfast called 'Linenopolis'. Sources: Standard histories of Belfast industry. Salmon run up the Bann past Coleraine. Sources: DAERA / Loughs Agency. The Plough and the pole star seen from a Mournes camp in October. CSR card and SafePass as site entry checks. Sources: CITB NI; SOLAS.

Invented for the story. Old man on the bench, the man on the pier, the two mates.

From the author's life. The details of the modern man's work and history in this chapter are drawn from the author's own records and published writing.

Chapter 35: The Long Way Round

The found year. 2022 CE.

What the record supports. West Offaly Power station at Shannonbridge, peat-fired, stopped generating in December 2020. Sources: ESB. Ballynahatty woman (more than five thousand years dead) and three Bronze Age men from Rathlin. Sources: Cassidy et al. 2016, PNAS 113(2). A man from the heart of Newgrange read by the same team. Sources: Cassidy et al. 2020, Nature 582. Iberia: whole peoples moving across the map. Sources: Olalde et al. 2019, Science 363. Kilroot's tall chimney. Sources: Kilroot Power Station.

Invented for the story. Partner, RGIS counters, the man with the dog, the first-week lad.

From the author's life. The details of the modern man's work and history in this chapter are drawn from the author's own records and published writing.

Chapter 36: A Ledger Begins

The found year. 2023 CE. Belfast.

Invented for the story. The friend from Kilroot, the young fella at the session; the machine's flattery.

From the author's life. The details of the modern man's work and history in this chapter are drawn from the author's own records and published writing.

Chapter 37: What the Ledger Refused

The found year. 2026 CE. Belfast and Dublin.

Invented for the story. Panel woman, young founder, the boy at the photograph.

From the author's life. The details of the modern man's work and history in this chapter are drawn from the author's own records and published writing.

Chapter 38: The Threshold

The setting. The present, and not the present.

The author's reading. The woman on the rock with the branch of pale blossoms; the question. Sources: the silver-branch motif of *Imram Brain* (Ch12) as allusion.

Epilogue: The Stone on the Shore

The setting. North Antrim coast, the present.

What the record supports. Antrim coast cliffs: basalt over white chalk. Sources: GSNI; Causeway Coast geology. Scotland visible across the water. Sources: Geography.

The author's reading. The grey stone: most likely Palaeogene basalt, about sixty million years, or an erratic from Scotland carried by ice; never tested. Sources: GSNI for the rock types.

Invented for the story. The woman with the dog.

From the author's life. The details of the modern man's work and history in this chapter are drawn from the author's own records and published writing.

TIMELINE OF FOUND YEARS

Each found year is a reasoned choice inside a window of evidence. It is not a proven date.

Chapter	Place	Found year	What it rests on	The doubt that stays
Prologue	The Antrim coast, Northern Ireland	About 60 million years ago	The Antrim lavas, dated to between 62.6 and 59.6 million years	The stone was never tested. It may be an erratic from far away.
1	Mount Sandel, River Bann	7700 BCE	The middle of the dated hut-building window, 7790 to 7635 cal BC	A window's middle, not a measurement.
2	Almendres, near Évora	4800 BCE	Phase and style: small rings, then the great ellipse	No radiocarbon date for the enclosures.
3	Carnac, Brittany	4500 BCE	Neighbouring sites dated to about 4800 to 4300 cal BC	The Ménec rows themselves cannot be dated.
4	Newgrange, the Boyne	3200 BCE	Two radiocarbon dates on the roof caulking, about 3200 cal BC	They date the roof, not the first use, and the range runs to centuries.
5	Skara Brae and Stenness, Orkney	2900 BCE	Overlap of the dated windows for Barnhouse and Stenness	Maeshowe's own dating is uncertain. If it is about 2700, the overlap fails.
6	The Avon and Salisbury Plain	2500 BCE	The Altar Stone's possible place in Stonehenge's second phase, and Durrington Walls	The date of the Altar Stone's arrival is unknown.
7	Delos, the Cyclades	430 BCE	Three dates that meet: Meton's solstice, the plague at Athens, the purification of Delos	Nobody says Herodotus stood on Delos that year.
8	Thebes and the western cliffs	1425 BCE	The seventy days after the death of Menkheperre Thutmose, on the 1479 accession count	The 1479 accession is the majority view, not a certainty.

Chapter	Place	Found year	What it rests on	The doubt that stays
9	Kuru country and the road north, late Vedic	1000 BCE	See the chapter notes	See the chapter notes
10	Orayvi, Third Mesa	1290 CE	See the chapter notes	See the chapter notes
11	West Polynesia	650 CE	See the chapter notes	See the chapter notes
12	Druim Snechta	720 CE	See the chapter notes	See the chapter notes
13	Massalia to Thule	325 BCE	See the chapter notes	See the chapter notes
14	Croton	530 BCE	See the chapter notes	See the chapter notes
15	the Sequani	53 BCE	See the chapter notes	See the chapter notes
16	Iona	580 CE	See the chapter notes	See the chapter notes
17	the western ocean	530 CE, traditional	See the chapter notes	See the chapter notes
18	Uaxactún to Chichén Itzá	889 CE	See the chapter notes	See the chapter notes
19	Tenochtitlan	1519 CE	See the chapter notes	See the chapter notes
20	Croton	510 BCE	See the chapter notes	See the chapter notes
21	Anglesey	61 CE	See the chapter notes	See the chapter notes
22	Philae	394 CE	See the chapter notes	See the chapter notes
23	Northumbria	664 CE	See the chapter notes	See the chapter notes
24	Moidart	1609 CE	See the chapter notes	See the chapter notes
25	Ohkay Owingeh	1680 CE	See the chapter notes	See the chapter notes
26	Krishnanagar	1785 CE	See the chapter notes	See the chapter notes
27	Caesarea and Genoa	1101 CE	See the chapter notes	See the chapter notes
28	Jaca	1135 CE	See the chapter notes	See the chapter notes
29	Orkney	1153 CE	See the chapter notes	See the chapter notes
30	Dublin	1308 CE	See the chapter notes	See the chapter notes
31	San Juan de la Peña	1399 CE	See the chapter notes	See the chapter notes
32	Bristol	1481 CE	See the chapter notes	See the chapter notes
33	Lee, Lanarkshire	1638 CE	See the chapter notes	See the chapter notes
34	Belfast	2009 CE	See the chapter notes	See the chapter notes
35	—	2022 CE	See the chapter notes	See the chapter notes
36	Belfast	2023 CE	See the chapter notes	See the chapter notes
37	Belfast and Dublin	2026 CE	See the chapter notes	See the chapter notes

Chapter	Place	Found year	What it rests on	The doubt that stays
38	the present, and not the present	the present, and not the present	See the chapter notes	See the chapter notes
Epilogue	north Antrim coast, the present	north Antrim coast, the present	See the chapter notes	See the chapter notes

GLOSSARY

Altar Stone
The large block of pale green sandstone lying at the foot of the Great Trilithon at Stonehenge. Its chemistry points to the Old Red Sandstone of north-east Scotland.

Amduat
An Egyptian funerary text, "what is in the Duat", which follows the sun through the twelve hours of the night. It is also called the Book of the Hidden Chamber.

basalt
A dark, fine-grained volcanic rock. The Antrim coast is built of it.

BCE, CE
Before the Common Era and the Common Era: the same years as BC and AD.

bluestones
The smaller stones of Stonehenge, brought from Wales. The name covers several kinds of rock.

cal BC
Calendar years before Christ, reached by calibrating a radiocarbon result against a curve of known ages.

cist
A small stone-lined box or chamber for a burial.

corbel
To build a roof or vault by setting each course of stones a little further in than the one below, until the sides meet.

the Core
In the story, the researcher's framework for laying rhythms over one another to look for patterns. It finds patterns and proves nothing.

cromlech
A circle or ring of standing stones. The word is used in Portugal and Brittany as well as in Britain and Ireland.

Demotic
The cursive script of late Egypt, quicker to write than hieroglyphs.

the Duat
The Egyptian underworld, through which the sun was believed to travel each night.

equinox
One of the two days in the year when day and night are of equal length.

erratic
A boulder carried far from where it began, usually by ice.

found year
In this book, the single year chosen for each chapter where the evidence the researcher trusts overlaps best. It is a reasoned choice, never a proven date.

gnomon
An upright rod whose shadow is measured to find the height of the sun. Pytheas used one at Massalia.

groat
An English silver coin. The Lee Penny is a red stone set in a groat of Edward IV.

henge
A circular earthwork with a ditch, usually with the bank outside it. The name comes from Stonehenge, which is not strictly one.

House of Life
In ancient Egypt, the temple workshop where scribes copied and kept the sacred books.

Hyperboreans
In Greek tradition, a people of the far north, beyond the north wind, who were said to send offerings to Delos.

kerbstones
The stones set around the foot of a burial mound to hold it.

the Ledger
In the story, the great book in the library of the dream that keeps itself and writes up each chapter on the master page.

lintel
A horizontal stone laid across the top of an opening or across two uprights.

Meton
An Athenian astronomer who set out a cycle of nineteen years, which fits almost exactly 235 months of the moon.

midden
A refuse heap. The shells, bones and other rubbish in one tell archaeologists what people ate.

New Fire
The Mexica rite held once in every fifty-two years, when all fires were put out and a new fire was made.

pada and krama
Two of the patterns in which a Vedic hymn is learnt: pada, word by word, and krama, in overlapping pairs.

Painted Grey Ware
The pottery of about 1200 to 600 BCE in the country between the Ganga and the Yamuna.

passage tomb
A burial chamber reached by a long, narrow passage, covered by a mound.

radiocarbon dating
A way of dating organic material by measuring how much of its carbon-14 has decayed.

roof-box
The narrow opening above the entrance at Newgrange that lets the midwinter sunrise reach the chamber.

runes
The letters of the old Germanic and Norse alphabets. The runes inside Maeshowe date from about 1125 to 1175.

sarsen
The hard, sandy stone of the great uprights and lintels at Stonehenge.

solstice
One of the two days in the year when the sun reaches its highest or lowest point in the sky at noon: midsummer and midwinter.

stromatolite
A layered mound built by microbes in shallow water, among the oldest signs of life on Earth.

takman
The fever that the Atharva Veda charms send away.

Templars
The Knights Templar, a military religious order. In Ireland they were taken on 3 February 1308, and their lands passed to the Hospitallers after 1312.

trilithon
Two uprights with a lintel across them, the form of the great stones at the heart of Stonehenge.

tīrtha
A ford, and later a crossing-place in the sacred sense.

variscite
A green mineral. The green beads found in Breton tombs have been traced to mines in south-west Spain.

Yupköyvi
The Hopi name for Chaco Canyon.

zircon
A tough mineral that survives for billions of years, used to date very old rocks.

PICTURE CREDITS

The plate at the head of each chapter is a painting made for this book, to the art direction of Gary Leckey. *[illustrator credit to be supplied]*

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Plate 2 (Chapter 1) Ness and Orrin on the bank above the weir, with the huts on the bluff. Painting made for this book; art direction by Gary Leckey.

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Plate 35 (Chapter 34) The shingle beach on the north Antrim coast. Painting made for this book; art direction by Gary Leckey.

Plate 36 (Chapter 35) The lough-shore road at dusk, and the cranes. Painting made for this book; art direction by Gary Leckey.

Plate 37 (Chapter 36) The kitchen table where the ledger begins. Painting made for this book; art direction by Gary Leckey.

Plate 38 (Chapter 37) Three ledgers, with the grey stone on top, at four in the morning. Painting made for this book; art direction by Gary Leckey.

Plate 39 (Chapter 38) An empty boat on flat grey water in the fog. Painting made for this book; art direction by Gary Leckey.

Plate 40 (Epilogue) The stone at the water's edge. Painting made for this book; art direction by Gary Leckey.

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